

Geography

Teaching tables, Confused Pairs, Common Traps, Current Affairs, and other high-yield blocks.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

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Topic 1 — Indian Physical Geography: Mountains & Hills

Current Affairs (this topic)

Year	Fact	Why asked	Source
Mar 2024	Sela Tunnel (BRO) — all-weather Tezpur– Tawang (Arunachal); longest twin-lane tunnel above ~13,000 ft	Border tunnel / pass CA	PIB / inauguration
2025	Z-Morh / Sonamarg Tunnel — all-weather Srinagar–Ladakh (लद्दाख) axis	Pair with Zoji La (ज़ोजिला)	PIB
2020 / PYQ 2025	Atal Tunnel qualifier only: longest highway tunnel above 10,000 ft , not unqualified world longest		PIB / BRO

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q84

The Shiwalik range is primarily composed of which type of material?

A. Igneous rocks B. Consolidated rocks C. Unconsolidated sediments D. Metamorphic rocks

Q. UKPCS Prelims 2025, Q90

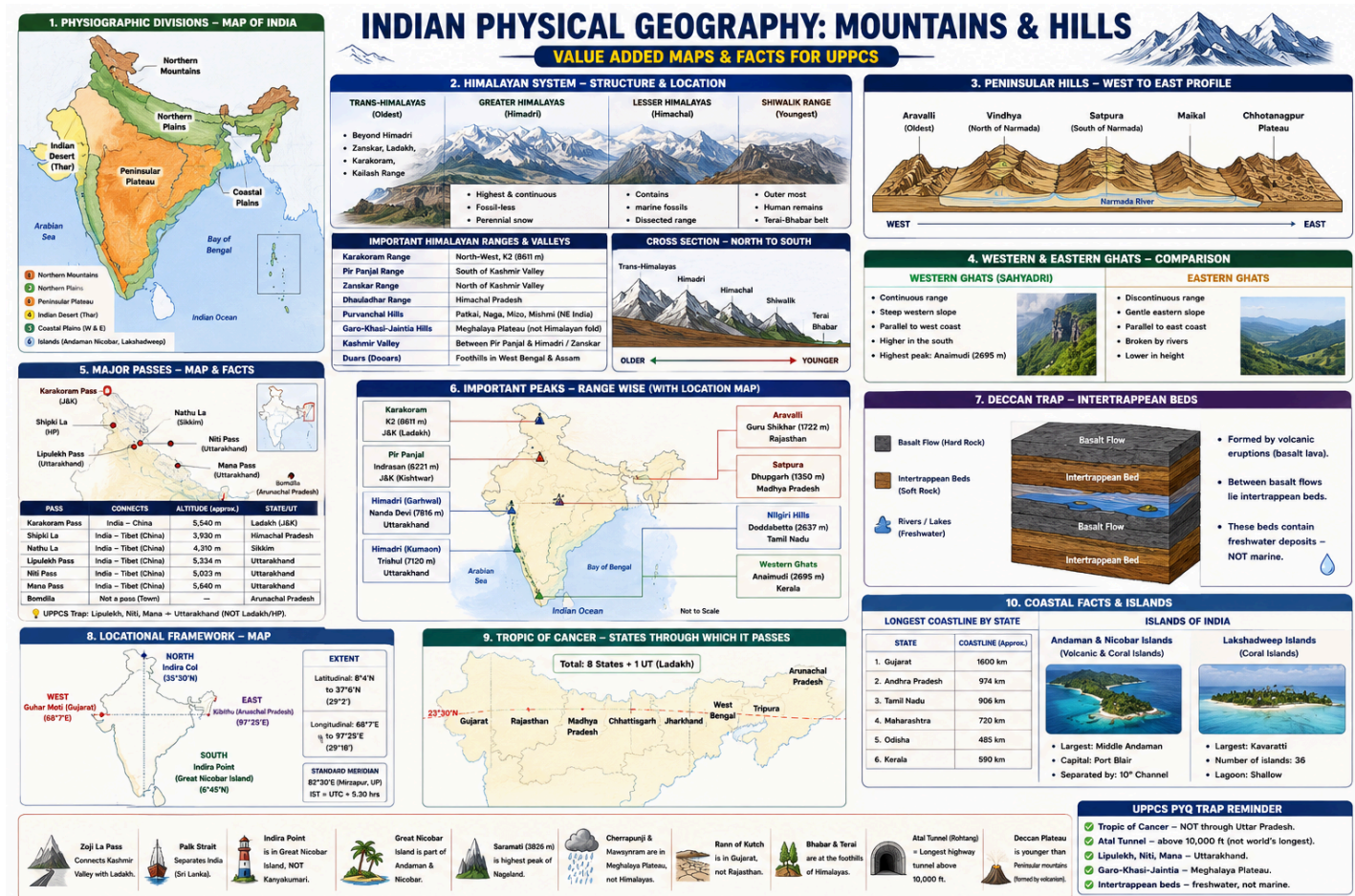
Which is the largest glacier of the Trans-Himalayas?

A. Biafo B. Siachen (सियाचिन) C. Baltoro D. Hispar

Confused Pairs

A	B	Correct	Hindi
Himadri	Himachal	Fossil-less crystalline vs marine fossils	हिमाद्रि / हिमाचल
Himachal	Shiwalik	Marine fossils vs human remains; outer youngest	हिमाचल / शिवालिक
Bhangar	Khadar	Older kankar upland vs newer flood-renewed, more fertile	भांगर / खादर
Bhabar	Teraï	Streams sink (pebbles) vs streams re-emerge (marsh/forest)	भाबर / तराई
Western Ghats	Eastern Ghats	Continuous high wall vs discontinuous eroded hills	पश्चिमी घाट / पूर्वी घाट
Vindhya	Satpura	North of Narmada vs south of Narmada	विंध्य / सतपुड़ा
Andaman–Nicobar	Lakshadweep	Volcanic (Bay of Bengal) vs coral atolls (Arabian Sea)	अंडमान / लक्षद्वीप
Deccan Plateau	Deccan Trap	Southern tableland vs basalt lava cover on it	दक्कन पठार / दक्कन ट्रैप
Purvanchal Hills	UP Purvanchal	NE hill system vs eastern UP region	पूर्वांचल पहाड़ियाँ / पूर्वांचल (UP)
Indira Point	Kanyakumari	Southernmost territory vs southernmost mainland	इंदिरा पॉइंट / कन्याकुमारी
K2	Kanchenjunga	Karakoram (PoK) vs highest peak fully in India (Sikkim)	के2 / कंचनजंगा
Dun	Karewa	Shiwalik longitudinal valley vs Kashmir lacustrine terrace	दून / करेवा
Marusthali	Bagar	Sandy desert core vs semi-arid eastern fringe	मरुस्थली / बागड़
10° Channel	9° Channel	Andaman–Nicobar vs Minicoy–rest of Lakshadweep	दस डिग्री / नौ डिग्री
Coast of emergence	Coast of submergence	Coromandel / Kerala raised vs Konkan drowned	उद्गमन / निमज्जन तट
Marwar Plateau	Marwar Plain	East of Aravalli vs west (Thar)	मारवाड़ पठार / मैदान

A	B	Correct	Hindi
Malnad	Maidan	Karnataka forested hill tract vs rolling granite plain	मलनाड / मैदान
Kangra Valley	Kulu Valley	Strike / longitudinal vs transverse valley (HP)	कांगड़ा / कुल्लू
Standard Meridian	Tropic of Cancer	82°30' E N-S time line (Mirzapur; 5 states) vs 23°30' N E-W line (8 states; not UP)	मानक याम्योत्तर / कर्क रेखा
Indira Col	Indira Point	Northern extreme (Siachen / Ladakh) vs southernmost territory (Great Nicobar)	इंदिरा कोल / इंदिरा पॉइंट



1.0 Physiographic framework

Structural unit	What it is	Note
Peninsular Block	Oldest stable crust; gneiss, granite, schist	Faulted and eroded, not recently folded like the Himalaya
Himalayas and extra-peninsular mountains	Young fold belt from India-Eurasia collision	Still rising in zones; includes Karakoram and Purvanchal
Indo-Ganga-Brahmaputra (ब्रह्मपुत्र) Plain	Foredeep filled with Himalayan alluvium	One of the world's largest alluvial (जलोढ़) tracts

1.1 Physiographic Divisions of India

Division	Where	Age	One-line identity
Northern Mountains	Northern frontier	Tertiary (young)	Himalaya + Karakoram + Purvanchal

Division	Where	Age	One-line identity
Northern Plains	Between Himalaya and plateau	Recent	Indus–Ganga–Brahmaputra alluvium
Peninsular Plateau	South of Narmada–Son	Precambrian	Oldest stable block
Indian Desert	Western Rajasthan	Pleistocene–recent	Thar / Marusthali (मरुस्थली), leeward of Aravalli
Coastal Plains	East and west margins	Recent	West narrow ; East broad and deltaic
Islands	Arabian Sea and Bay of Bengal	Recent	Lakshadweep coral ; A&N volcanic

Northern Plains

Sub-region	Location	Fact
Punjab Plain	Punjab, Haryana	Doabs; chos gullying on Shiwalik edge
Haryana Tract	Ghaggar–Yamuna	Water-divide; Ghaggar (घग्गर) as Saraswati (सरस्वती) successor
Ganga Plain	Delhi to Kolkata	Rohilkhand, Awadh (अवध), Mithila, Magadh
Brahmaputra Plain	Assam	Braided channel with bils (ox-bows) and heavy floods

Doab	Between
Bist	Beas (ब्यास) and Satluj
Bari	Beas and Ravi (रावी)
Rachna	Ravi and Chenab (चिनाब)
Chaj	Chenab and Jhelum (झेलम)
Sind (सिंध) Sagar	Jhelum–Chenab and the Indus

Coastal Plains

Western strip	Stretch
Kachchh–Kathiawar	Gujarat
Gujarat Plain	Narmada–Tapi–Mahi (माही)–Sabarmati (साबरमती) fill
Konkan	Daman to Goa
Kanara (Canara)	Goa to Mangaluru
Malabar	Kerala
Eastern strip	Stretch
-----	-----
Utkal Coast	Odisha coast with the Mahanadi (महानदी) delta and Chilika
Northern Circars	Mahanadi to Krishna (कृष्णा)
Carnatic (कर्नाटक)	Krishna to Kaveri (कावेरी)
Coromandel / Payan Ghat	False Divi Point to Kanyakumari

1.2 Himalayan Mountain System

North to south (parallel ranges)

Range	Typical height	Rock / fossil fact
Trans-Himalaya	3000–6500 m	Granitic cold desert north of the Indus
Himadri (Greater)	generally >6000 m	Crystalline, fossil-less , with permanent snow

Range	Typical height	Rock / fossil fact
Himachal (हिमाचल) (Lesser)	3700–4500 m	Sedimentary; marine fossils
Shiwalik (शिवालिक) (Outer)	900–1100 m	Youngest unconsolidated rocks; human remains

Region	Local name of Shiwalik
Jammu	Jammu Hills
Uttarakhand	Dhang / Dundwa
Nepal	Churia Ghat Hills
Arunachal	Dafla, Miri (मीरी), Abor, Mishmi (outer hills)

Himalayan peaks (match home)

Peak	m	Note
K2 (Godwin Austin)	8611	Karakoram (PoK) — not the main Himalayan arc
Kanchenjunga	8586	Highest peak fully in India (Sikkim, Himadri)
Nanda Devi	7816	Uttarakhand; highest entirely in Uttarakhand
Kamet (कामेत)	7756	Uttarakhand
Cho Oyu	8188	Himalaya (Nepal–Tibet) —
Lhotse	8516	Himalaya (Nepal–Tibet) —
Nanga Parbat	8126	Western syntaxial bend (Gilgit–Baltistan / PoK)
Namcha Barwa	7782	Tibet / China — not in India
Gurla Mandhata	7694	Tibet , north of Mansarovar — not an Indian peak

1.3 Peninsular Mountains, Hills & Plateaus

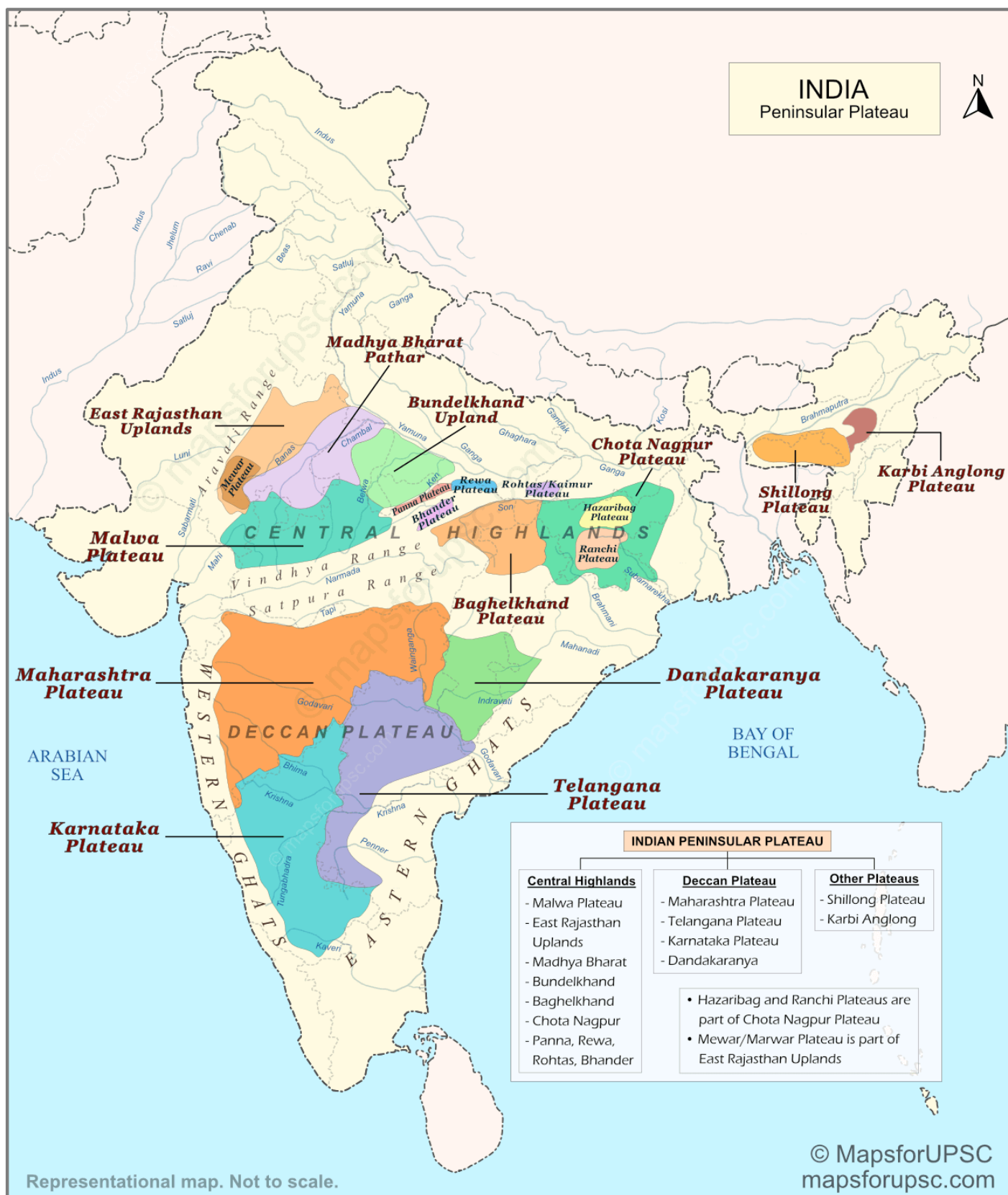
Range	Trend	Location	Highest	Fact
Aravalli	SW–NE	Gujarat–Rajasthan–Delhi	Guru Shikhar 1722 m	Oldest fold mountains; Rajasthan highest
Vindhya	E–W	North of Narmada , MP	Sadbhawna Shikhar	Aryavarta / Dakshinapatha (दक्षिणापथ) divide
Satpura	E–W	South of Narmada	Dhupgarh 1350 m (Pachmarhi)	Madhya Pradesh highest
Western Ghats (पश्चिमी घाट)	N–S	Tapi gap to Kanyakumari	Anaimudi 2695 m	Continuous ; Sahyadri in Maharashtra
Eastern Ghats	N–S	Odisha to Tamil Nadu	Jindhagada ~1690 m	Discontinuous
Nilgiri (नीलगिरि)	—	Tamil Nadu–Kerala	Doddabetta 2637 m	Junction of both Ghats
Rajmahal	—	Jharkhand	—	Rajmahal Trap (volcanic)
Shevaroy	—	Salem, Tamil Nadu	—	Eastern Ghats; not Tirupati

Deccan Plateau and Deccan Trap

Trap layer	Depth fact
Upper	about 450 m
Middle	about 1200 m
Lower	about 150 m

Trap layer	Depth fact
Intertrappean beds	Sediment between lava flows; land / freshwater fossils — not sea plants and animals

Central Highlands and north-eastern plateaus



- The Central Highlands are the **northern** Peninsular Plateau, **north of the Narmada–Son**.
- **Marwar / Mewar (मेवाड़) Plateau** is **east** of the Aravalli (sandstone–shale–limestone, Banas toward Chambal). **Marwar Plain** is the Thar **west** of the Aravalli. Do not swap them.

- **Madhya Bharat Pathar** (Central Highland) is the Chambal basin with ravines / badlands.
- The **Malwa Plateau** is lava-covered and **north-sloping**. It has **dual drainage**: Narmada–Tapi–Mahi to the Arabian Sea, and Chambal–Betwa–Ken to the Yamuna / Bay of Bengal.
- The **Bundelkhand Plateau** lies in south-west Uttar Pradesh and Madhya Pradesh. It is **granite–gneiss**, drought-prone, and includes Jhansi (झांसी), Banda, and Chitrakoot (चित्रकूट). Betwa, Dhasan, and Ken cross it.
- The **Baghelkhand Plateau** lies **east** of Bundelkhand, **north of Maikal**. The Son bounds it on the north. It divides Son drainage from Mahanadi drainage.
- The **Chotanagpur Plateau** lies mainly in Jharkhand (with Odisha–West Bengal edges). Average height is about **700 m**. **Damodar** (दामोदर) flows in a **rift** through Gondwana coalfields.
- **Hazaribagh Plateau** lies north of Damodar. **Ranchi Plateau** lies south of Damodar.
- The **Rajmahal Hills** are the north-eastern basalt edge of Chotanagpur.
- The **Meghalaya Plateau** is Garo–Khasi–Jaintia. The **Malda / Garo–Rajmahal Gap** is a down-fault filled by Ganga–Brahmaputra alluvium.
- **Mikir (Karbi) Hills** are listed with this north-eastern block on the relief map.
- **Jharia** and **Bokaro** are the Damodar-rift coalfields. Iron and mica are the other Chotanagpur matches.
- Chotanagpur is the easternmost unit of the west-to-east central-India hill sequence.

1.4 Passes, Peaks & Sacred Geography

Mountain passes

Pass	State / UT	Route / note	Trap
Burzil	J&K / Ladakh	Greater Himalaya (pass table)	—
Thaga La	Uttarakhand	Greater Himalaya with Niti / Lipulekh	—
Zoji La (ज़ोजिला)	Ladakh / J&K	Srinagar–Kargil–Leh (लेह)	Pair with Z-Morh tunnel
Banihal	J&K	Jammu–Srinagar; Jawahar Tunnel	Not a Tibet pass
Rohtang La	Himachal Pradesh	Kullu–Lahaul; Atal Tunnel below	Not Karakoram
Bara Lacha La	Himachal Pradesh	Manali–Leh	—
Kunzum La	Himachal Pradesh	Lahaul–Spiti	—
Shipki La	Himachal Pradesh	India–Tibet; Sutlej (सतलुज) valley	—
Niti Pass	Uttarakhand	Garhwal (गढ़वाल)–Tibet	2023 Q57 correct pair
Mana Pass	Uttarakhand	Near Badrinath (बद्रीनाथ)	Not HP
Lipulekh (लिपुलेख)	Uttarakhand	Kailash–Mansarovar; India–Nepal–China	Not Ladakh
Nathu La	Sikkim	India–Tibet trade (पण्थाध्यक्ष) (reopened 2006)	—
Jelep La	Sikkim	Alternate to Nathu La	—
Bomdi La	Arunachal Pradesh	Near Bomdila town	Bomdila ≠ pass name
Sela Pass	Arunachal Pradesh	Tezpur–Tawang; Sela Tunnel (2024)	—
Bum La	Arunachal Pradesh	Tawang–Tibet	—

Pass	State / UT	Route / note	Trap
Pangsau Pass	Arunachal Pradesh	India–Myanmar (Stilwell Road)	—
Diphu / Aghil	Karakoram	Not Arunachal / not “Ladakh town” traps	2023 wrong options
Khardung La	Ladakh	Leh–Nubra road	High motorable pass; not a Tibet trade pass of the Nathu type

State-wise highest peaks (match-list home)

State / UT	Peak	m	Range
Sikkim	Kanchenjunga	8586	Himadri
Uttarakhand	Nanda Devi	7816	Himadri
Himachal Pradesh	Reo Purgyl	6816	Kinnaur Himalaya
Arunachal Pradesh	Kangto	7090	Eastern Himalaya
J&K / Ladakh	Nun / Kun	7135 / 7077	Zaskar
West Bengal	Sandakphu	3636	Singalila
Nagaland	Saramati	3841	Naga Hills
Manipur	Tenipu / Mt Iso	2994	Manipur Hills
Mizoram	Phawngpui	2165	Mizo Hills
Meghalaya (मेघालय)	Shillong Peak	1965	Khasi
Kerala	Anaimudi (अनाइमुदी)	2695	Western Ghats
Tamil Nadu	Doddabetta	2637	Nilgiri
Karnataka	Mullayanagiri	1930	Western Ghats
Maharashtra	Kalsubai	1646	Western Ghats
Rajasthan (थार)	Guru Shikhar	1722	Aravalli
Madhya Pradesh	Dhupgarh	1350	Satpura
Andhra Pradesh	Arma Konda	~1680	Eastern Ghats
Odisha (ओडिशा)	Deomali	1672	Eastern Ghats
Gujarat	Girnar	1117	Kathiawar
Jharkhand	Parasnath	1366	Chotanagpur
Uttar Pradesh	Amsot (Sonbhadra)	~941	Kaimur
Chhattisgarh	Gudla Maina	~1270	Maikal extension

Sacred geography

Site	State	Hill / fact
Tirupati (Venkateswara)	Andhra Pradesh	Tirumala / Mallamalla , Eastern Ghats — not Shevaroy
Kedarnath (केदारनाथ)	Uttarakhand	Garhwal Himalaya; Mandakini (मंदाकिनी)
Badrinath	Uttarakhand	Garhwal Himalaya; Alaknanda (अलकनंदा)
Gangotri	Uttarakhand	Bhagirathi source glacier
Yamunotri	Uttarakhand	Yamuna source glacier
Amarnath	J&K	Lidder Valley (Greater Himalaya, not Pir Panjal)
Vaishno Devi	J&K	Trikuta Hills , Jammu

Site	State	Hill / fact
Sabarimala	Kerala	Western Ghats
Palitana (पालिताना)	Gujarat	Shatrunjaya Hills
Parasnath	Jharkhand	Chotanagpur
Girnar	Gujarat	Junagadh (जूनागढ़)
Ajmer Sharif	Rajasthan	Aravalli foothills
Chamundi	Karnataka	Mysuru
Konark (कोणार्क)	Odisha	Coastal plain (not a hill shrine)

1.5 Locational Framework

Latitude, longitude, area

Item	Note
Latitude	8°4' N – 37°6' N (about 29°; about 3214 km N–S)
Longitude	68°7' E – 97°25' E (about 29°; about 2933 km E–W)
Area	about 3.28 million km² = 2.4% of world land
Rank	Conventionally 7th largest (after Russia, Canada, USA, China, Brazil, Australia)

Standard Meridian

Step	Easy fact
Earth	360° in 24 h means roughly 15° of longitude per hour, or 1° ≈ 4 minutes
Offset maths	82.5 ÷ 15 = 5.5 hours = 5 h 30 min ahead of GMT
Why this line	It sits near the middle of India's longitude span (~68° E to ~97° E), so west (Gujarat) and east (Arunachal / Andaman) share one national clock

State	Place / belt fact	trap
Uttar Pradesh	Mirzapur (near Prayagraj belt)	Not Lucknow / Delhi; Tropic does not hit UP
Madhya Pradesh	Eastern MP / Singrauli belt	Not the whole of MP
Chhattisgarh	Central–east belt (Bilaspur–Korba class)	Not Jharkhand
Odisha	Western Odisha	Not the Bhubaneswar coast as the “IST city”
Andhra Pradesh	Northern / coastal AP strip	Not Telangana (TG lies west of 82.5° E)

Tropic of Cancer (23°30' N)

Order (west → east)	State on the Tropic
1	Gujarat
2	Rajasthan
3	Madhya Pradesh
4	Chhattisgarh
5	Jharkhand
6	West Bengal
7	Tripura

Order (west → east)	State on the Tropic
8	Mizoram

Extreme points

Extreme	Place	Easy fact
North	Indira Col , Siachen / Ladakh	High pass on the northern frontier — not a city
South (territory)	Indira Point , Great Nicobar	Southernmost India (islands included)
South (mainland)	Kanyakumari (Cape Comorin)	Southernmost peninsula only
East	Kibithu , Anjaw, Arunachal Pradesh	Near the Myanmar–China corner; not Walong as the usual key
West	Guhar Moti (also Ghuar Mota), Kutch, Gujarat	Westernmost village belt of Kachchh

Coastline

Rank (longest → shortest)	Coastal state
1	Gujarat
2	Andhra Pradesh
3	Tamil Nadu
4	Maharashtra
5	Kerala
6	Odisha
7	Karnataka
8	West Bengal
9	Goa (shortest state coast)

Common Traps — Don't Fall For These

- Lipulekh is in Ladakh** — FALSE. **Uttarakhand** (India–Nepal–China trijunction).
- Mana Pass is in Himachal** — FALSE. **Niti and Mana** are both **Uttarakhand**.
- Bomdila is a pass** — FALSE. Bomdila is a **town**; the pass is **Bomdi La** (Arunachal).
- Tirupati is on Shevaroy Hills** — FALSE. **Tirumala / Mallamalla (Eastern Ghats)**.
- Kanyakumari is India's southernmost point** — FALSE. **Indira Point (Great Nicobar)** = southernmost territory.
- Aravalli is the youngest range** — FALSE. **Himadri** is youngest among usual options; Aravalli is **oldest**.
- India = 6th largest / wholly tropical** — FALSE. **7th; 2.4%**; Tropic through the middle; **not** wholly tropical.
- Intertrappean beds have sea fossils** — FALSE. **Freshwater/land** fossils.
- K2 is in the main Himalaya** — FALSE. **Karakoram**.
- Kashmir Valley = Kangra + Dhauladhar** — FALSE. **Pir Panjal + Himadri**.
- Annamalai / Sirumalai are Himalayan** — FALSE. **Tamil Nadu peninsular**.
- Atal Tunnel is the world's longest highway tunnel (unqualified)** — FALSE. **Pir Panjal**; longest **above 10,000 ft**.
- Both Ghats are continuous** — FALSE. **Western = continuous; Eastern = discontinuous**.
- Lakshadweep is volcanic like Andaman** — FALSE. Lakshadweep = **coral**; A&N = **volcanic**.
- Vindhya lies south of Narmada** — FALSE. **Vindhya = north; Satpura = south**.
- Tropic of Cancer crosses UP / Ladakh** — FALSE. **Eight states only** (GJ–RJ–MP–CG–JH–WB–Tripura–Mizoram).
- Standard Meridian = Prayagraj / Lucknow / "only eastern states use IST"** — FALSE. Place = **Mirzapur**; path = **UP–MP–CG–Odisha–AP**; IST = **one zone for all India**. Uttarakhand / Jharkhand / Telangana are **not** on the line.

18. **Karewa = dun** — FALSE. Karewa = Kashmir **lacustrine** terrace (saffron). Dun = **Shiwalik** longitudinal valley.
19. **Ten Degree Channel separates Minicoy** — FALSE. **10°** = Andaman–Nicobar. **9°** = Minicoy–rest of Lakshadweep. **8°** = Minicoy–Maldives.
20. **Palghat is the Mumbai–Pune gap** — FALSE. Palghat / Palakkad = **Kerala–Tamil Nadu**. Bhor Ghat = Mumbai–Pune.
21. **Gurla Mandhata is an Indian Trans-Himalayan peak** — FALSE. It is in **Tibet**. Namcha Barwa is also **not** in India.
22. **Whole west coast is submerged** — FALSE. **Konkan** is drowned; **Kerala / Malabar** is emerged.
23. **Marwar Plateau is west of Aravalli** — FALSE. Plateau = **east**; Thar plain = **west**.

Topic 2 — Climate of India

Current Affairs (this topic)

Year	Fact	Why asked	Source
2026	IMD SW monsoon (दक्षिण-पश्चिम) below normal (~90% of LPA)	Seasonal rainfall category	IMD LRF Apr/May 2026
2026	ENSO: neutral → El Niño during Jun–Sep 2026	Weak-monsoon tendency	IMD / MMCFS
2026	IOD neutral (no +IOD (धनात्मक) offset of El Niño)	Contrast with 2019 +IOD rescue	IMD Jul 2026
2026	Onset over Kerala 4 June (slightly after normal 1 June)	Onset date trap	USDA / IMD advance
Ongoing	Mission Mausam (MoES/IMD) — improved monsoon & severe-weather forecast	Scheme/institution CA	IMD–MHA

El Niño **tendency** ≠ guaranteed all-India drought (सूखा) — still the 2026 -relevant pairing.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q96

According to Koppen’s climatic classification, the (Amw) climate is found predominantly in which part of India?

A. North-Western India B. Western coastal region C. Thar (थार) desert D. Northern plain

Q. UKPCS Prelims 2025, Q98

Which local wind affects some parts of Kerala and Tamil Nadu (नाडु) during the pre-monsoon season?

A. Loo B. Mango Shower C. Kal Baisakhi (बैसाखी) D. Mistral

Confused Pairs

A	B	Correct	Hindi
SW monsoon	NE monsoon	Jun–Sep most of India vs Oct–Dec SE coast only	दक्षिण-पश्चिम / उत्तर-पूर्व मानसून
Retreating monsoon	Western Disturbance	Withdrawal/Oct heat vs winter extra-tropical NW rain	लौटता मानसून / पश्चिमी विक्षोभ
Loo	Nor’wester	Sustained hot dry wind vs violent pre-monsoon thunderstorm	लू / काल बैसाखी
Anabatic	Katabatic	Day upslope vs night downslope	आरोही / अवरोही पवन
STWJ	TEJ	Winter westerly (WD) vs summer easterly (SW monsoon)	पश्चिमी जेट / उष्णकटिबंधीय पूर्वी जेट
El Niño	La Niña	Warm E Pacific, weak monsoon vs cool E Pacific, strong monsoon	अल नीनो / ला नीना
+IOD	–IOD	Helps Indian monsoon vs suppresses it	धनात्मक / ऋणात्मक IOD
Köppen	Thornthwaite	T+P letter codes vs moisture/vegetation index	कोपेन / थॉर्नथ्वेट
Absolute humidity	Relative humidity	Mass of vapour vs % of saturation (RH falls as T rises)	निरपेक्ष / सापेक्ष आर्द्रता
Active monsoon	Break monsoon	Trough on Ganga plain (plains wet) vs trough on Himalaya (central India dry)	सक्रिय / विराम मानसून

A	B	Correct	Hindi
Doldrums	Horse latitudes	Rainy ITCZ calm belt vs dry subtropical high near 30°	डोलड्रम / हॉर्स अक्षांश
Am (Köppen)	As (Köppen)	Short dry spell / wet SW coast-NE vs dry summer Coromandel (NE rain)	Am / As
Willy-willy	Brickfielder	Australian cyclone vs Australian hot local wind	विली-विली / ब्रिकफील्डर
Bora	Chinook	Adriatic cold vs Rockies warm dry	बोरा / चिनूक
Harmattan	Khamsin	West Africa dust vs Egypt heat	हरमट्टन / खामसिन

1. INDIA'S CLIMATE – AT A GLANCE

- Tropical Monsoon Climate
- Not wholly tropical (Himalaya temperate)
- 4 IMD Seasons
 - Cold Weather (Dec-Feb)
 - Hot Weather (Mar-May)
 - SW Monsoon (Jun-Sep)
 - Retreating Monsoon (Oct-Nov)

CLIMATE OF INDIA – VALUE ADDED MAPS & FACTS

★★ UPPCS REVISION SHEET ★★

2. SEASONS – MONTHS, FEATURES & IMPACT

Season (IMD)	Months	Key Features	Impact / Phenomena
Cold Weather	Dec-Feb	Western Disturbances in NW; fog, frost	Rabi crops benefit
Hot Weather	Mar-May	High temperature; low pressure	Loo, Nor'westers, Mango & Blossom showers
SW Monsoon	Jun-Sep	Main rainfall season (75–90%)	Kharif crops
Retreating Monsoon	Oct-Nov	Withdrawal of SW monsoon	Oct heat, BoB cyclones
NE Monsoon (Part of Retreating)	Oct-Dec	NE winds over Bay → SE coast rain	TN, South AP, Puducherry rain

4. PHYSIOGRAPHIC & OROGRAPHIC IMPACT

6. WESTERN DISTURBANCES (WD)

7. LOCAL WINDS & STORMS (INDIA)

Wind/Storm	Region	Season	Character
Loo	NW & N plains	May–Jun	Hot, dry, dust
Nor'wester / Kal Baisakhi	E & NE India	Apr–May	Thunderstorm, hail, lightning
Mango Showers	Kerala, Karnataka	Apr–May	Pre-monsoon
Blossom / Cherry Showers	TN, Kerala	Apr–May	Coffee blossom
Andhi	Rajasthan	Pre-monsoon	Dust-storm
Anabatic	Mountains	Day	Upslope wind
Katabatic	Mountains	Night	Downslope cold wind

3. SOUTHWEST MONSOON – FLOW & BRANCHES

5. NORTHEAST MONSOON – OCT TO DEC

8. CYCLONES – BASIN & FACTS

Bay of Bengal: More frequent & intense; High storm surge risk (Odisha, AP, WB, Saurashtra); Peak: Apr–Jun & Oct–Dec.

Arabian Sea: Fewer & generally weaker; Moderate surge risk (Gujarat, MH, Kerala); Peak: Apr–May & Oct–Nov.

Cyclone needs: SST ≥ 26–27°C	Moisture	Coriolis force	Low vertical wind shear
Depression	31–49 km/h	50–61	62–88
Deep Depression	62–88	89–117	118–166
Cyclonic Storm	89–117	118–166	167–221
Very Severe Cyclonic Storm	118–166	167–221	≥222
Extremely Severe Cyclonic Storm	167–221	≥222	

9. LOCAL WINDS – WORLD (MATCH MAP)

10. ENSO & IOD – MONSOON CONNECTION

El Niño (Warm E. Pacific): Weak monsoon; Drought tendency; Less upwelling → Less plankton.

Neutral / Normal: Average conditions.

La Niña (Cool E. Pacific): Strong monsoon; Good rainfall; More upwelling → More plankton.

+IOD (W Indian Ocean warmer): Helps Indian monsoon (Good rainfall).

-IOD (E Indian Ocean warmer): Suppresses Indian monsoon (Less rainfall).

11. JET STREAMS OVER INDIA

Subtropical Westerly Jet (STWJ): Winter (Dec-Feb); Position: South of Himalaya (~25–30°N); Direction: Westerly (W → E); Role: Steers Western Disturbances.

Tropical Easterly Jet (TEJ): Summer (Jun-Sep); Position: ~14°N over peninsular India (~25–30°N); Direction: Easterly (E → W); Role: Steers Western Disturbances.

12. ATMOSPHERIC LAYERS – STRUCTURE & ROLE

Layer	Height (approx.)	Key Feature / Role
Troposphere	0–8 km poles (0–18 km equator)	All weather; T.L. with height
Stratosphere	~8–50 km	Ozone layer (UV shield)
Mesosphere	~50–80 km	Meteors burn up
Thermosphere	~80–400 km	Ionosphere, aurora
Exosphere	>400 km	Outer most layer

Pressure Belts:

Pressure Belt	Height (approx.)	Character
Equatorial Low (ITCZ)	~0°	Low pressure, rain, convergence
Subtropical High	~30°	High pressure, descending air
Subpolar Low	~60°	Low pressure, rain, convergence
Polar High	~90°	High pressure, cold, dry

13. MAJOR RAINFALL DISTRIBUTION (ANNUAL)

Location	Rainfall (approx.) & Reason
Mawsynram / Cherrapunji	>1100 cm (BoB & orographic)
Korikan / Malabar	300–400 cm (Arabian Sea & W Ghats)
Ganga Plain (UP/Bihar)	100–150 cm (BoB branch & depressions)
Interior Deccan	50–75 cm (Rain shadow)
Chennai	~120 cm (NE monsoon)
Jaisalmer	<25 cm (Far from branches)
Leh / Ladakh	<10 cm (Rain shadow)

All-India Mean = 118 cm

15. TEMPERATURE – INDIA (JAN & JUN)

16. QUICK MEMORY HOOKS

- TN dry in SW, wet in NE → 2023 Q54
- WD from the West, rain falls East → 2019 Q83
- Winter NW = WD, not monsoon → 2021 Q36
- Thornthwaite talks vegetation → 2018 Q25
- El Niño = less fish, not more → 2023 Q47
- Jets are westerly; TEJ is easterly exception → 2024 Q80
- Shamal Arabia, Mistral France → 2021 Q87 / 2019 Q80
- Heat from below = long wave → 2022 Q138
- Snow stones most (highest altitude) → 2021 Q44
- ITCZ / monsoon trough = LOW pressure, not HIGH

17. CURRENT AFFAIRS SNAPSHOT (2026)

- SW Monsoon 2026: Below normal (~90% of LPA) – IMD
- ENSO: Neutral → El Niño in Jun–Sep 2026 (weak monsoon tendency)
- IOD: Neutral (no +IOD offset of El Niño)
- Onset over Kerala: 4 June 2026 (Normal: 1 June)
- Mission Mausam: Improved monsoon & severe weather forecasting – IMD (MoES)

2.1 Climate Basics

Factor	What it does in India
Latitude	Most of India is tropical / subtropical; the far north is cooler
Himalaya	Winter wall against Central Asian cold; summer orographic rain on southern slopes
Distance from sea	Coasts stay moderate (नरम दली); NW plains and interior Deccan have continental extremes
Altitude	Temperature falls about 6.5 °C per 1000 m (lapse rate)
Relief / orography	Western Ghats → windward wet, leeward Deccan dry; Meghalaya (मेघालय) scarp → extreme rain
Pressure & ITCZ	Summer thermal low + north-shifted ITCZ pull the south-west monsoon
Jet streams	Winter STWJ (पश्चिमी जेट) steers Western Disturbances; summer TEJ (उष्णकटिबंधीय पूर्वी जेट) supports SW monsoon
ENSO / IOD	Year-to-year monsoon strength (El Niño weakens; +IOD helps)

Temperature and altitude

Station	January (approx.)	Control
Drass / Ladakh (लद्दाख)	−20 to −40 °C	Altitude plus continentality
Agra (आगरा)	about 16 °C	Ganga plain —
Darjeeling (about 2000 m)	about 4 °C	Same latitude as Agra; lapse rate
Mumbai / Chennai	24–26 °C	Maritime
Jaisalmer	12–15 °C in January; 40–45 °C in June	Highest diurnal range among usual stations

Climate contrast	Main control
Chennai warmer than Kolkata (annual)	Latitude (Chennai farther south)
Himalayan snowfall / hill stations cooler	Altitude
Rainfall falls West Bengal → Punjab	Distance from the Bay / moisture loss westward
Sutlej (सतलुज)–Ganga plain winter rain	Western Disturbances

Heat budget, albedo, humidity

Surface	Albedo	Fact
Fresh snow	80–90%	Highest among usual surface options
Desert sand	30–40%	Moderate
Prairie (प्रेयरी)	20–25%	—
Paddy / cropland	10–20%	Lower
Forest / water (typical)	under 10–20%	Absorb more; warm more

Temperature inversion

Type	When / where	What happens
Radiation inversion	Clear, calm winter nights on the Indo-Gangetic plain	Ground cools fast; fog / smog trap over Delhi, UP, Bihar
Valley / drainage inversion	Kashmir, Doon, hill valleys	Cold air drains to the floor ; slopes stay milder
Subsidence inversion	Under high-pressure / descending air	Mixing stops; haze can persist

Pressure, wind, Coriolis

Hemisphere	Deflection
Northern (India)	to the right
Southern	to the left

Spell	Trough position	Rain
Active monsoon (सक्रिय)	Over the Ganga plain	Plains wet (UP, Bihar, MP)
Break monsoon (विराम मानसून)	Over the Himalaya	Central India dry; foothills wet

Atmosphere, pressure belts, ITCZ

Layer	Height	Fact
Troposphere (क्षोभमंडल)	0–8 km at poles / 0–18 km at the equator	All weather ; temperature falls with height
Tropopause	Top of the troposphere	Jets fly near here

Layer	Height	Fact
Stratosphere	to about 50 km	Ozone UV shield
Mesosphere (मध्यमंडल)	about 50–80 km	Meteors burn
Thermosphere	about 80–400 km	Ionosphere and aurora
Exosphere	outermost	Space

Step	Process	Surface result
1	Strong heating at the equator	Air rises
2	Rising air reduces surface weight	Equatorial low / ITCZ
3	Air aloft moves poleward and sinks near 30°	Subtropical high (horse latitudes)
4	Surface air returns from 30° to the equator	Trade winds
5	Other surface air from 30° moves toward 60°	Westerlies
6	Cold air sinks at the poles	Polar high
7	Polar air moves toward 60° and meets westerlies	Subpolar low ; polar easterlies
Cell	Span	Role
-----	-----	-----
Hadley cell	Equator to ~30°	Rise at ITCZ, sink at 30°, trades return
Ferrel cell	~30° to 60°	Mid-latitude belt of the westerlies
Polar cell	~60° to pole	Polar easterlies; rise near 60°
Nickname	Meaning	
-----	-----	
Doldrums	Rainy, weak-surface-wind belt of the ITCZ	
Horse latitudes	Dry, calm subtropical highs near 30°	
Belt	Latitude	Pressure
-----	-----	-----
Equatorial low / ITCZ	Near 0° (shifts with the Sun)	Low
Subtropical high	About 30° N / S	High
Subpolar low	About 60° N / S	Low
Polar high	About 90°	High

Season	ITCZ position	Effect on India
Summer (Jun–Sep)	Shifts north over South Asia	Forms / guides the monsoon trough; supports SW monsoon
Winter (Dec–Feb)	Shifts south	Most of India under dry NE trades
Strong El Niño years	Indian branch often weaker	Monsoon tends to weaken

Wind	Direction	Where	Cause	Effect
Trade winds	East → west (NE trades in NH; SE trades in SH)	Between ~30° highs and the ITCZ	Air leaves subtropical high toward equatorial low; Coriolis turns it	Feed ITCZ rain; after equator crossing they become India's SW monsoon
Westerlies	West → east	About 30°–60° (both hemispheres)	Air leaves 30° high toward 60° low; Coriolis turns it	Drive mid-latitude storms and help steer Western Disturbances
Polar easterlies	East → west	Polar high toward ~60°	Cold air leaves polar high; Coriolis turns it	Bring cold outbreaks; meet westerlies at the subpolar low

2.2 Monsoon System

System	Months	Where the rain falls	Trap
South-west monsoon	Jun–Sep	Most of India (~75–90% of annual rain)	TN SE coast stays dry
Retreating monsoon	Sep–Nov	Withdrawal phase with October heat and BoB / Andaman Sea cyclones	Not WD winter rain
North-east monsoon	Oct–Dec	TN, south AP, SE Karnataka fringe, Puducherry, Karaikal	Not all-India winter rain
Western Disturbances	Nov–Feb (peak)	NW India rain/snow	Extra-tropical; not a monsoon

Monsoon mechanism

Concept	Fact
Thermal	Land heats faster than sea → summer low over India pulls ocean air in
Dynamic (Flohn)	Pressure belts and the ITCZ shift with the Sun; monsoon is largely that seasonal shift
Step	What happens
-----	-----
1	Tibetan Plateau / NW India heat strongly in late spring
2	ITCZ shifts north onto India
3	Mascarene High strengthens over the southern Indian Ocean
4	SE trades cross the equator roughly between 40° E and 60° E ; Coriolis turns them SW
5	Cross-equatorial flow strengthens as the Somali Jet
6	STWJ jumps north of Tibet; TEJ sets up near 14° N
7	Onset burst at Kerala (~1 June), then the front marches north

Place	Normal onset	Withdrawal (start)
Kerala	about 1 June	Extreme south last (December for NE rains)
Mumbai / Kolkata	about 10 June	Early October
Lucknow (लखनऊ) / eastern UP	about 15–20 June	Late September
Delhi	about 29 June	Late September
Rajasthan	about 15 July (last)	September
Chennai	South-west is weak and dry	North-east monsoon October–December

Branch	Path	Wet places	Dry / weak places
Arabian Sea branch	Arabian Sea → Western Ghats → inland (also a stream toward central India / Ganga that later meets the Bay branch)	Konkan, Malabar, Goa (गोवा); some rain into Gujarat / western MP	Leeward Deccan and TN rain shadow; Chhattisgarh basin is little fed by this branch
Bay of Bengal branch	BoB → NE India → curves west along Himalaya	Assam, Meghalaya, Ganga plain, UP, Bihar	TN coast (sits parallel — little orographic lift)

Pattern	Fact
Season	Mostly June–September (SW monsoon)

Pattern	Fact
Relief	Orography governs amounts (Western Ghats, Meghalaya scarp)
Distance from sea	Rain generally declines inland / with distance from the moisture source
Northern plains	Annual rain decreases east → west (humid Bay air loses moisture westward)
Duration	Monsoon season length decreases south → north (Kerala longer than the far north-west)
Break	Rainless spells when the trough sits on the Himalaya
Rajasthan dryness	Aravalli runs roughly parallel to SW winds → little forced uplift on the western Thar

Northeast monsoon and retreating monsoon

Feature	Retreating SW monsoon	North-east monsoon
Months	About Sep–Nov	About Oct–Dec
Process	SW monsoon withdraws from north to south	Dry NE winds pick up Bay moisture and rain on the SE coast
Signature	October heat, clear spells, and Bay / Andaman Sea cyclones	Coromandel rain
Rain belt	Withdrawal rains + east-coast cyclone rains	TN, south AP, SE Karnataka, SE Kerala fringe, Puducherry, Karaikal
North India	Mostly dry	Mostly dry

Indian seasons (IMD)

Season	Months	Main weather
Cold weather	December–February	Western Disturbance rain/snow in the north-west, fog on the plains, and the rabi season
Hot weather	March–May	Loo heat waves, Nor'westers, and mango / blossom showers
South-west monsoon	June–September	Main rainy season for most of India
Retreating	October–November	Withdrawal, October heat, Bay cyclones, then NE monsoon on the SE coast

Western Disturbances

Item	Note
Type	Extra-tropical cyclone
Carrier jet	STWJ branch south of the Himalaya (not the northern Tibetan branch)
Peak season	November–February
Rain / snow belt	Punjab, Haryana, Himachal, J&K / Ladakh fringe, Uttarakhand, western UP, northern Rajasthan
Gradient	Decreases west → east
Farm link	Helps rabi wheat; also hail, fog, avalanche risk

2.3 Winds, Storms and Cyclones

Local winds and storms of India

Time	Land vs sea	Wind
Day	Land heats faster → low over land	Sea breeze (sea → land)
Night	Land cools faster → high over land	Land breeze (land → sea)
Wind	When	Direction

Time	Land vs sea	Wind
-----	-----	-----
Anabatic	Day	Upslope
Katabatic	Night	Downslope
Foehn / Chinook	After crossing a mountain	Warm and dry downslope

Name	Region	Season	Character
Loo	RJ, PB, HR, UP, Bihar plains	May–June	Hot, dry dust and heat-wave; not a thunderstorm
Nor'wester / Kal Baisakhi	WB, Assam, Odisha, Bihar, Jharkhand	April–May	Violent thunderstorm ; hail, lightning
Mango showers	Kerala, Karnataka	April–May	Pre-monsoon; mango ripening
Blossom / cherry showers	TN, Kerala (coffee belt)	April–May	Coffee flowering
Andhi	Rajasthan	Pre-monsoon	Dust-storm wall
Elephanta	Malabar / Kerala	About September	End of SW monsoon; rough sea
Sea / land breeze	Coasts	Daily	Onshore by day; offshore by night
Anabatic	Mountain slopes	Day	Upslope
Katabatic	Valleys	Night	Downslope cold drainage

World local winds (full match map)

Family	Wind	Region	trap
Warm dry downslope	Foehn	Alps	Not Australia
Warm dry downslope	Chinook	Rockies	Not a cold wind
Warm dry downslope	Zonda	Andes, Argentina	Not Alps
Warm dry downslope	Santa Ana	California	Not Spain
Warm dry downslope	Berg	South Africa	—
Cold	Mistral	S France, Rhône	Not Australia
Cold	Bora	Adriatic	Not Chinook
Cold	Pampero	Argentina	—
Cold	Buran / Purga	Siberia	—
Hot dusty	Sirocco	Sahara → Mediterranean	—
Hot dusty	Khamsin	Egypt	Not Australia
Hot dusty	Shamal	Arabia / Gulf	Not Austria
Hot dusty	Harmattan	West Africa	Not East Africa
Hot dusty	Samun	Kurdistan	—
Hot dusty	Leveche	Spain	—
Hot dusty	Ghibli	Libya	—
Hot dusty	Brickfielder	Australia	Not Mistral
Hot dusty	Haboob	Sudan	—
Hot dusty	Black roller	North America	—

Family	Wind	Region	trap
Other	Yamo	Japan	—
Other	Levanter	Gibraltar	—
Not a local wind	Willy-willy	Australia	Cyclone, not Brickfielder

Tropical cyclones

Need	Note
Sea surface temperature	About 26–27 °C or more
Moisture	Deep moist air
Coriolis	Enough spin — usually 5°–30° latitude
Vertical wind shear	Low (strong SW-monsoon shear suppresses many Bay storms in Jul–Sep)
Part	Meaning
-----	-----
Eye	Calm centre with sinking air and clearer skies
Eyewall	Ring of strongest winds and rain around the eye
Spiral bands	Outer rain bands

Basin	Frequency / strength	Why	Hit states
Bay of Bengal	More and stronger	Warm shallow shelf + river freshwater; storm surge (तूफानी महोर्मि) risk high	Odisha, AP, West Bengal, Sundarbans (सुंदरबन)
Arabian Sea	Fewer, often weaker	Higher salinity; narrower fetch	Gujarat, Maharashtra, Kerala still take hits
IMD class	Wind (km/h)		
-----	-----		
Depression	31–49		
Deep depression	50–61		
Cyclonic storm	62–88		
Severe cyclonic storm	89–117		
Very severe	118–166		
Extremely severe	167–221		
Super cyclone	222 or more		
Local storm	Scale	Fact	
-----	-----	-----	
Thunderstorm / Nor'wester	Hours; local convection	Not a cyclone	
Tropical cyclone	Days; ocean-scale	Synoptic storm	

Cyclone naming (match bank — next year can swap any pair)

Local / basin name	Where	Note
Hurricane ★	USA / Atlantic–Caribbean / E Pacific	2019–2020 PYQ fact
Typhoon ★	NW Pacific — China / Philippines / Taiwan class	2020 Q74 “Typhoons–China”
Taifu ★	Japan -facing name for NW Pacific storm	Same basin as typhoon; 2019 paired with Japan

Local / basin name	Where	Note
Baguio / Baguios ★	Philippines	2019–2020 fact
Willy-willies ★	Australia	Cyclone — not Brickfielder (hot local wind)
Cyclone	North Indian Ocean — India / BoB / Arabian Sea	IMD / RSMC New Delhi names
Cordonazo	West Mexico / E Pacific coast	Lucent-class distractor
Hurricane (also)	Caribbean / Gulf of Mexico	Same Atlantic family as USA

2.4 Classification and ocean–atmosphere phenomena

Köppen (India) versus Thornthwaite

Thinker	Method	Note
Köppen	Temperature + precipitation letter codes	India's Am / Aw / As / BWh / Cwg map
Thornthwaite	Moisture balance (P minus potential ET)	"Vegetation is the true index of climate" — 2018 Q25
Trewartha	Modified Köppen	Not Thornthwaite

Letter clue	Meaning	India example
A	Tropical	Kerala, Deccan, Coromandel
B	Dry (arid / semi-arid)	Thar, rain-shadow Deccan fringe
C	Temperate / subtropical with distinct seasons	Ganga plain
m	Monsoon (short dry spell)	Kerala / Konkan / NE hills
w	Dry winter	Interior Deccan (Aw); Ganga plain (Cwg); also the second small letter in strings like Amw
s	Dry summer	Tamil Nadu As — dry in SW monsoon; rain in NE monsoon
h	Hot	Thar BWh

Code	Type	India
Am / Amw	Tropical monsoon (short dry spell; w = dry winter in full strings)	Kerala / Konkan, north-east (Cherrapunji)
Aw	Tropical savanna (सवाना)	Interior Deccan
As	Monsoon with dry summer	Tamil Nadu / Coromandel (rain in NE monsoon)
BWh	Hot desert	Thar (थार), Kutch
BSh	Semi-arid	Punjab, Haryana, Gujarat, rain-shadow Deccan
Cwg	Monsoon with dry winter	Indo-Gangetic / Great Plains (Lucknow, Delhi, Patna (पटना)); also covers North Bihar, North Bengal / much of NE plain on class maps
H	Highland	Himalayan slopes
E	Polar / tundra (टुंड्रा)	Highest snow zones only

Jet streams

Jet	Season	Direction	Where	Role for India
Polar front jet	Year-round (shifts)	Westerly	High latitudes	Extra-tropical weather belt

Jet	Season	Direction	Where	Role for India
STWJ	Winter	Westerly	About 25–30° N, south of Himalaya	Steers Western Disturbances
STWJ north branch	Winter	Westerly	North of Tibet after the plateau splits the jet	Does not drive India's winter rain
STWJ summer position	Summer	Westerly	North of Tibet	Clears space for monsoon setup
TEJ	Summer	Easterly	Near 14° N over the peninsula	Supports SW monsoon circulation
Somali Jet	Summer	Cross-equatorial / SW	Off East Africa → Arabian Sea	Feeds SW monsoon moisture

El Niño, La Niña, and IOD

Name	Ocean	India monsoon	Note
El Niño	Pacific (warm Peru)	Often weaker	Plankton decrease
La Niña	Pacific (cool Peru)	Often stronger	Opposite of El Niño
+IOD	Indian Ocean (west warmer)	Helps	Can offset El Niño
–IOD	Indian Ocean (east warmer)	Hurts	Can worsen drought

Common Traps — Don't Fall For These

- TN gets heavy SW monsoon rain** — FALSE. Dry in SW; **NE monsoon** Oct–Dec — 2023 Q54.
- Winter NW rain = retreating monsoon** — FALSE. **Western Disturbances** — 2021 Q36.
- WD rain increases eastward** — FALSE. Decreases **west** → **east** — 2019 Q83.
- “Vegetation is the true index of climate” = Köppen** — FALSE. **Thornthwaite** — 2018 Q25.
- El Niño increases plankton off Peru** — FALSE. **Reduces** upwelling/plankton — 2023 Q67.
- Jet streams are easterly** — FALSE as a general rule. Mid-latitude jets = **westerly**; **TEJ** is the summer exception — 2024 Q60.
- Shamal = Austria / Mistral = Australia** — FALSE. Shamal = Arabia; Mistral = France.
- Atmosphere heated mainly by short-wave sun** — FALSE. **Long-wave terrestrial** — 2022 Q138.
- +IOD is bad for Indian monsoon** — FALSE. **+IOD helps**; –IOD hurts.
- Ozone UV layer = troposphere** — FALSE. **Stratosphere** — 2018 Q52 / 2023 Q52.
- NE monsoon rains all India** — FALSE. **TN, south AP, Puducherry** only.
- Mango showers in Punjab** — FALSE. **Kerala/Karnataka**.
- Highest albedo = desert sand** — FALSE. **Fresh snow** — 2021 Q66.
- ITCZ / monsoon trough = high pressure** — FALSE. Both are **lows**.
- Nor'westers = Western Disturbances** — FALSE. Nor'westers = **pre-monsoon thunderstorms**; WD = **winter**.
- Harmattan = East African coast** — FALSE. **West Africa** / Gulf of Guinea.
- Willy-willy = Brickfielder** — FALSE. Willy-willy = Australian **cyclone**. Brickfielder = Australian **hot local wind**.
- Bora is a warm Rockies wind** — FALSE. Bora = **cold Adriatic**. Chinook = warm Rockies.
- Active monsoon = trough on Himalaya** — FALSE. Active = trough on **Ganga plain**; break = trough on **Himalaya**.
- Köppen As = Kerala / Konkan** — FALSE. **As** = Tamil Nadu dry **summer**; Kerala/Konkan = **Am**.
- Retreating monsoon = NE monsoon** — FALSE. Retreating = SW withdrawal + Oct heat + BoB cyclones; NE monsoon = SE-coast rain Oct–Dec.
- Monsoon from Hindi / Spanish** — FALSE. Arabic **mausim** (season).
- IOD = west Indian Ocean vs eastern Pacific** — FALSE. West vs **east Indian Ocean** (UPSC 2017).

24. **Northern jet branch steers India's winter rain** — FALSE. Tibet splits the jet; **southern** branch (STWJ south of Himalaya) steers WD.
25. **India lies exactly between the Tropics** — FALSE. Extends to ~**37°6' N** (subtropical north).
26. **IMD rainy day = any drizzle** — FALSE. $\geq$ **2.5 mm** in 24 hours.
27. **January tropical/subtropical divide = 21 °C / 15 °C** — FALSE. **18 °C** isotherm.

Topic 3 — Drainage System

Current Affairs (this topic)

Year	Fact	Why asked	Source
Dec 2021 / Dec 2024	Ken–Betwa (बेतवा) Link: Cabinet ₹44,605 cr; PM foundation 25 Dec 2024	First ILR under implementation	PIB
2025–26	Only priority NPP link in implementation (30 identified: 16 peninsular + 14 Himalayan); NWDA nodal	“How many links operational?” trap	Rajya Sabha (राज्यसभा) / Jal Shakti (शक्ति) Aug 2026
2026	Spend ~₹12,382 cr to Jun 2026; target ~Mar 2030; Bundelkhand MP–UP; Panna TR forest	Cost / ecology (पारिस्थितिकी) pairing	LS Q Aug 2026
Ongoing	Polavaram includes Godavari–Krishna (Vijayawada) link assistance	Peninsular grid	PIB
2016	National Waterways Act — 106 NWs; core still NW-1 to NW-5	Number vs names	Act

Namami Gange (नमामि गंगे) and GAP are Ganga cleaning programmes. They are not river-origin facts.

INDIAN PHYSICAL GEOGRAPHY – DRAINAGE SYSTEM

COMPLETE SOURCE OF TRUTH – UPPCS/UPSC PRELIMS & MAINS

1 QUICK REVISION – RAATA THIS FIRST

India Drainage – 4 Major Himalayan Systems + Peninsular Rivers

- Indus – 3180 km (mostly in Pakistan)
- Ganga – 2525 km; basin ~8.6 lakh km² (largest in India)
- Brahmaputra – 2900 km (Yarlung Tsangpo in Tibet)
- Peninsular Rivers – old, graded, rain-fed

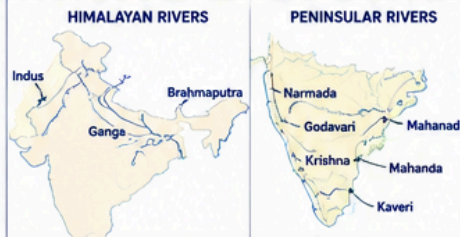
Key Contrasts

HIMALAYAN RIVERS	PENINSULAR RIVERS
Youth, vigorous	Mature, graded
Perennial	Mostly seasonal
V-shaped valleys	Shallow valleys
Meanders, floodplains	Fixed course
Large deltas	Estuaries (west flowing)

IMD Seasons (Hydrology view)

- Cold Weather: Dec – Feb
- Hot Weather: Mar – May
- SW Monsoon / Rainy: Jun – Sep
- Retreating: Oct – Nov
- NE Monsoon (SE coast only): Oct – Dec

2 MAJOR DRAINAGE SYSTEMS OF INDIA



BASIN AREA & LENGTH (High Yield)

River System	Basin Area (approx.)	Length (approx.)
Ganga	~8.6 lakh km ² (Largest in India)	2525 km
Godavari	~3.1 lakh km ² (Largest Peninsular)	1465 km (Longest Peninsular)
Brahmaputra	~5.8 lakh km ²	2900 km
Indus	~3.2 lakh km ² (in India part)	3180 km
Krishna	~2.6 lakh km ²	1400 km
Mahanadi	~1.4 lakh km ²	851 km
Narmada	~0.98 lakh km ²	1312 km (Longest West-flowing)
Tapi	~0.65 lakh km ²	724 km
Kaveri	~0.81 lakh km ²	800 km

TRANS-HIMALAYAN RIVERS: Indus, Sutlej, Brahmaputra (Tsangpo), Zaskar
MCQ = Sutlej (2020 Q52)

3 PANCH PRAYAG (HIMALAYAN CONFLUENCES)

Order (Upstream to Downstream)	Confluence	Rivers Meet	On	Forms
1	Vishnu Prayag	Dhuli Ganga + Alaknanda	Joshimath	
2	Nanda Prayag	Nandakini + Alaknanda	Nandaprayag	
3	Karn Prayag	Pindar + Alaknanda	Karnaprayag	
4	Rudra Prayag	Mandakini + Alaknanda	Rudraprayag	
5	Dev Prayag	Bhagirathi + Alaknanda	Devprayag	GANGA

Memory Trick: V → N → K → R → D

Vishnu → Nanda → Karn → Rudra → Dev
(Dev = Done → Ganga is born) (2023 Q50, 2024 Q74)

4 GANGA SYSTEM – TRIBUTARIES

LEFT BANK TRIBUTARIES (Stand facing downstream)

- Ramganga
- Gomti
- Ghaghra
- Gandak
- Kosi
- Mahananda

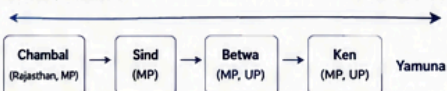
RIGHT BANK TRIBUTARIES (Stand facing downstream)

- Yamuna
- Son
- Damodar
- Punpun
- Jalangi

Jonk = Mahanadi tributary NOT Ganga (2021 Q5)

5 YAMUNA TRIBUTARIES (West → East)

Source (Yamunotri)



- Order (2020 Q68) = 1, 2, 3, 4 ✓
- Betwa joins Yamuna at Hamirpur (UP) – NOT Prayagraj (2023 Q55)
- Ramganga joins Ganga at Kannauj (UP)

6 ORIGIN OF MAJOR PENINSULAR RIVERS (2023 Q53 = A)

River	Origin
Yamuna	Yamunotri (Uttarakhand)
Krishna	Mahabaleshwar (Maharashtra)
Godavari	Trimbakeshwar, Nashik (Maharashtra)
Mahanadi	Sihawa Hills (Chhattisgarh)
Narmada	Amarkantak (MP)
Tapi (Tapi/Tapti)	Multai (Satpura range, MP) NOT Western Ghats (2018 Q102)
Kaveri	Brahmagiri Hills (Karnataka)
Son	Amarkantak (MP) – radial with Narmada

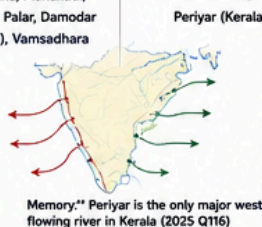
7 PENINSULAR DRAINAGE – OUTFALL

EAST FLOWING (Bay of Bengal)

Godavari, Krishna, Mahanadi, Kaveri, Pennar, Palar, Damodar (Subarnarekha), Vamsadhara

WEST FLOWING (Arabian Sea)

Narmada, Tapi, Mahi, Periyar (Kerala)



Memory: Periyar is the only major west flowing river in Kerala (2025 Q116)

8 IMPORTANT TRAPS (High Yield)

- Pranhita → joins Godavari (NOT Mahanadi) (2023 Q49)
- Hemavati → Kaveri tributary
- Srisaillam (Krishna) – dam named as per river it is on
- Luni → Rann of Kutch (inland, saline lower course) (2019 Q84)
- Ghaggar → disappears in Thar Desert (inland)
- Barak River → flows to sea (Bay of Bengal), NOT inland (2021 Q65)

9 RIVER FEATURES & PATTERNS

DELTA (Types)

Arcuate (Fan shaped) Sundarbans (Ganga-Brahmaputra)	
Bird's foot (Digitate) Mississippi (USA)	
Cuspate Mahanadi Delta	
Estuarine Narmada, Tapi, Mahi, Periyar	
Moribund (Dying) Cauvery Delta (2019 Q85)	

DRAINAGE PATTERNS

Dendritic – on uniform rocks (example: Most Peninsular rivers)	
Trellis – in folded/striped rocks (2024 Q31)	
Radial – streams diverge from a central highland (example: Amarkantak)	
Centripetal – streams flow towards a central basin (example: Imphal basin)	

10 RIVER CAPTURE, INLAND & REJUVENATION

RIVER CAPTURE (PIRACY)

- Teesta captured by Brahmaputra. A/R both true (2021 Q63)

INLAND DRAINAGE

- Luni → Rann of Kutch (saline)
- Ghaggar → Thar Desert depression

RIVER REJUVENATION

Renewed erosive power after uplift or fall in sea level → forms terraces, gorges, waterfalls.

Gomti in UP = Biological Disaster! (2018 Q99)

11 WATERWAYS, LINKING & MAJOR DAMS

NATIONAL WATERWAYS (NW)

- NW-1: Ganga – Hooghly (Allahabad to Haldia)
- NW-2: Brahmaputra (Dhubri to Sadiya)
- NW-3: West Coast Canal, Kerala (NW-3 Kollam to Kottapuram)
- NW-4: Krishna – Godavari Canal (Dellang to Pudukcherry)
- NW-5: Brahmani – Mahanadi (Talcher to Dhamra)

NEW MULTI-MODAL WATERWAYS

- NW-6: Barak (Assam)
- NW-7: Sundarbans Waterways (WB)
- NW-10: Amba – Arpa – Vainganga, MP

IMPORTANT DAMS (River wise)

Dam	River
Baglihar Dam	Chenab
Pandoh Dam	Beas
Srisaillam Dam	Krishna
Hirakud Dam	Mahanadi
Mettur (Stanley) Dam	Kaveri
Tehri Dam	Bhagirathi
Rihand Dam	Rihand (Son Basin)

RAMGANGA PROJECT (UP) (2024 Q72 = D)

- 198 MW (Garhwal)
- Irrigated area ~5.75 – 5.90 lakh ha (NOT 6 lakh)

★ RAPID FACTS (RAATA)

- ★ Largest basin in India – Ganga
- ★ Largest peninsular basin – Godavari
- ★ Longest peninsular river – Godavari (~1465 km)
- ★ Longest west flowing – Narmada (~1312 km)
- ★ Only river in Kerala flowing to Arabian Sea – Periyar
- ★ Only trellis pattern – in folded mountains
- ★ Radial pattern centre – Amarkantak
- ★ Only moribund delta – Cauvery
- ★ Teesta – captured by Brahmaputra
- ★ Jonk – Mahanadi tributary, NOT Ganga
- ★ Sutlej – Trans-Himalayan (MCQ trap)
- ★ Rann of Kutch – formed by Luni

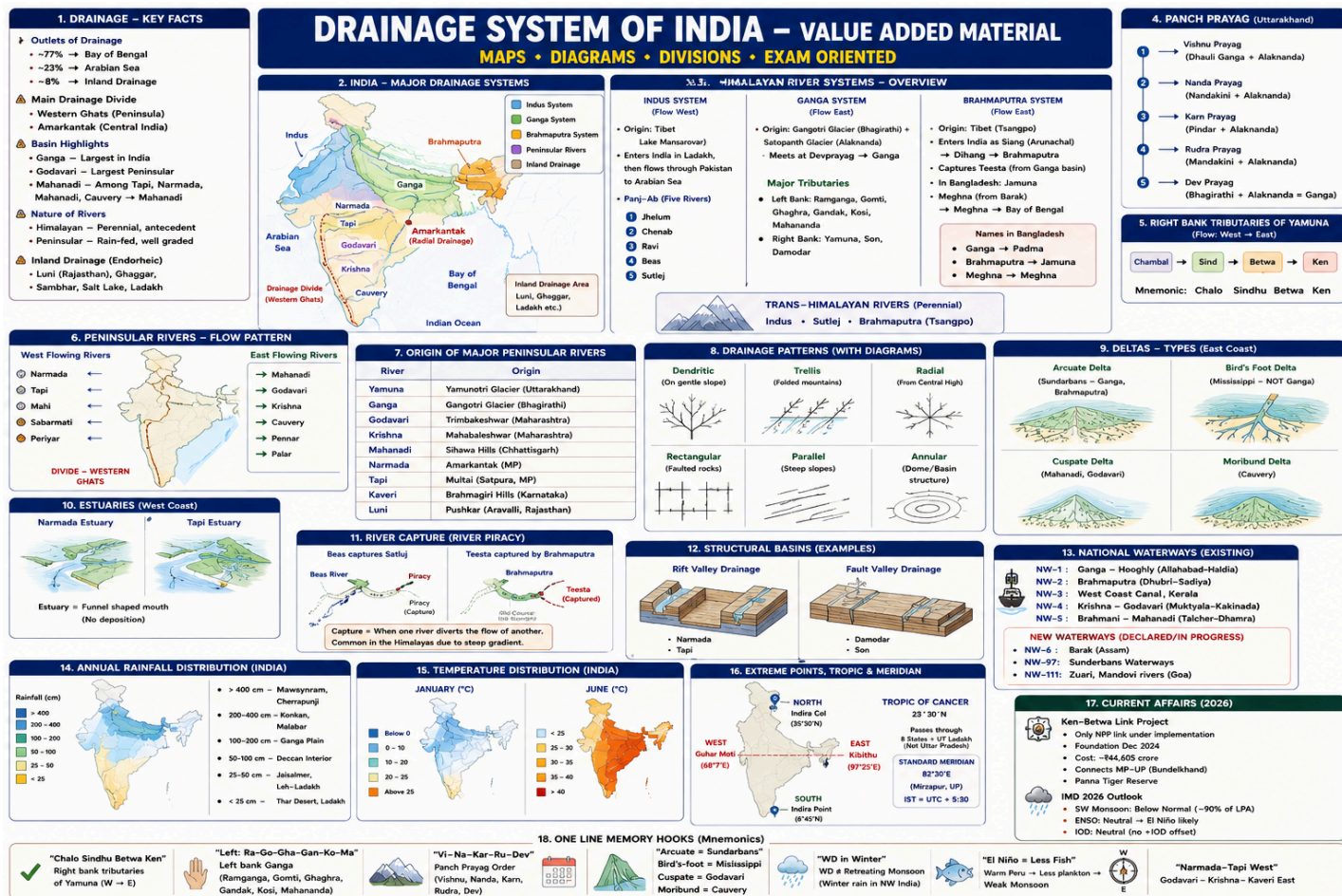
12 MVET-KNOW TERM COMPARISONS (One-line Difference)

Term	One-line Difference	Hindi	Term	One-line Difference	Hindi	Term	One-line Difference	Hindi	Term	One-line Difference	Hindi
Drainage basin	Land area drained by a river and its tributaries	जल विभाजक क्षेत्र	River capture	Stronger stream steals headwaters of weaker stream (piracy)	नदी अपक्षय	Fault valley river	Course in tectonic rift/graben (Narmada, Tapi, Damodar)	फॉल्ट घाटी नदी	River rejuvenation	Renewed erosive power after uplift / sea-level fall	नदी पुनः युवा अवस्था
Watershed / Water divide	Ridge separating two drainage basins	जल विभाजक रेखा	Inland drainage	River does not reach sea; ends in lake/plain/desert	आंतरजल जल प्रणालि	Estuary	Drowned river valley; tide-dominated, sediments removed	अंतराक्षर	Structural basin	Lowland enclosed by hills; centripetal drainage	संरचनात्मक क्षेत्र
Perennial river	Flows throughout the year (snow-fed + rain-fed)	बारम्बारगी नदी	Antecedent river	Maintains course across rising land (Indus, Brahmaputra)	पूर्व जल प्रणालि	Delta	Deposited sediment at river mouth; progrades into sea	डेल्टा / सिंगुल	Prayag	Sacred Himalayan confluence (Uttarakhand)	प्रयाग
Non-perennial / Ephemeral	Flows only after rain; dry channel in dry season	अवस्थायिक नदी	Superimposed river	Maintains course after burying relief (Chambal)	अवस्थायिक नदी	Moribund delta	Old inactive delta lobe no longer receiving sediment	मृत डेल्टा	Sangam	Confluence of Ganga + Yamuna at Prayag	संगम



Confused Pairs

A	B	Correct	Hindi
Ganga basin	Godavari basin	Largest India vs largest peninsular	गंगा / गोदावरी
Left-bank Ganga	Right-bank Ganga	Himalayan trib vs Yamuna/Son/Damodar	बायाँ / दायाँ तट
Delta	Estuary	East-coast deposition vs west-coast tidal funnel	डेल्टा / ज्वारनदमुख
Arcuate	Bird's-foot	Sundarbans vs Mississippi	चापाकार / पक्षी-पाद
Antecedent	Superimposed	Cuts rising Himalaya vs inherited course (Chambal)	पूर्ववर्ती / अध्यारोपित
Trans-Himalayan	Himalayan-origin	Tibet origin + cuts Himalaya (Sutlej) vs rises in Himalaya (Jhelum/Ravi)	पार-हिमालयी
Narmada	Tapi	Amarkantak vs Multai (Satpura)	नर्मदा / तापी
Luni	Barak	Inland saline lower vs BoB via Meghna	लूनी / बराक
Padma	Jamuna	Ganga in BD vs Brahmaputra in BD	पद्मा / जमुना
Devprayag	Rudraprayag	+ Bhagirathi = Ganga vs + Mandakini	देवप्रयाग / रुद्रप्रयाग
Trellis	Dendritic	Folded structure vs homogeneous slope	जालनुमा / वृक्षाकार
Consequent	Subsequent	Follows original slope vs later weak-belt tributary	अनुवर्ती / परवर्ती
Concordant	Discordant	Follows structure vs cuts across (antecedent Himalaya)	संगत / असंगत
Pranhita	Jonk	Godavari trib vs Mahanadi trib	प्रणहिता / जोंक
Hyderabad–Musi	Hyderabad–Krishna	City on Musi trib, not Krishna stem	मुसी / कृष्णा
E. Dhauliganga	Alaknanda Dhauli	Kali trib vs Vishnuprayag Dhauli	पूर्वी धौली
Chambal ravines	Damodar rift	Badlands vs fault-valley tag	चंबल / दामोदर
Kosi	Damodar	Sorrow of Bihar vs Sorrow of Bengal	कोसी / दामोदर



3.0 Drainage framework

River basics (must know before systems)

Outlet	Share of area	Typical rivers
Bay of Bengal	about 77%	Ganga, Brahmaputra, Godavari, Krishna, Mahanadi, Kaveri, Pennar, Palar, Vaigai, Damodar, Subarnarekha, Barak
Arabian Sea	about 23%	Indus (in Pakistan), Narmada, Tapi, Mahi, Sabarmati, Periyar , Sharavati, Mandovi
Inland	about 8% of area	Luni, Ghaggar, Rajasthan salt (खर्वाह) lakes, Ladakh

Type	Meaning	India
Consequent	Follows the original slope of the land	Godavari, Krishna, Kaveri from Western Ghats
Subsequent	Grows later along a weak belt; often joins at right angles	Chambal, Sind, Betwa, Ken, Tons, Son to Yamuna/Ganga
Antecedent	Older than uplift; keeps course by cutting a gorge through rising land	Classic set: Indus, Sutlej, Brahmaputra ; many PCS keys also tag Ganga
Superimposed	Older course inherited onto newer structure; does not follow present slope	Chambal (also Son in some notes)
Concordant	Follows structure and slope	Most peninsular rivers
Discordant	Cuts across structure	Himalayan antecedent rivers

3.1 Himalayan rivers

Ganga system

Tributary	Bank	Meets Ganga at / near
Yamuna (यमुना)	Right	Prayagraj (Sangam)
Ramganga (रामगंगा)	Left	Kannauj
Gomti	Left	Near Saidpur / Ghazipur belt (after Lucknow)
Ghaghara	Left	Chhapra / Revilganj (Saran)
Gandak	Left	Sonpur near Patna
Son	Right	Near Patna
Punpun	Right	Fatuha (Patna)
Kosi	Left	Kursela, Katihar
Mahananda (মহানন্দা)	Left	West Bengal (last left-bank in India)
Damodar	—	Joins Hooghly , not the main Ganga stem

Tributary	Bank of Yamuna	Meets Yamuna at / near
Tons (टोंस)	Left	Near Kalsi (Uttarakhand)
Hindon	Left	NCR / western UP belt
Chambal	Right	Near Etawah
Sind (सिंध)	Right	Between Chambal and Betwa (Auraiya–Kalpi belt)
Betwa (बेतवा)	Right	Hamirpur — not Prayagraj
Ken	Right	Near Banda

Panch Prayag and Uttarakhand

Prayag	Confluence
Vishnuprayag	Alaknanda + Dhauli Ganga (western / Alaknanda Dhauli — not Eastern Dhauliganga of Kali)
Nandaprayag	Alaknanda + Nandakini
Karnaprayag	Alaknanda + Pindar (पिंडर)
Rudraprayag	Alaknanda + Mandakini (Kedarnath belt)
Devprayag	Alaknanda + Bhagirathi = Ganga

3.4 River features and patterns

Types of deltas — Teaching Card

Type	Meaning	example
Arcuate	Fan-shaped front, many distributaries	Ganga–Brahmaputra (Sundarbans), Nile, Godavari–Krishna
Lobate	Rounded bulge, many mouths	Godavari (often)
Bird’s-foot	Long finger-like distributaries	Mississippi — not Ganga
Cuspate	Pointed front from opposing waves	Ebro; Mahanadi often cited for India
Moribund (Bengal subdivision stem)	Dying belt inside a large delta’s moribund–mature–active scheme	Bengal Delta (UPPCS 2019)

Type	Meaning	example
Moribund (lobe / river match stem)	Abandoned / ageing delta lobe	Cauvery in many landform keys
Estuarine mouth	Tidal funnel, little delta build-up	Narmada, Tapi

Origins, cities, dams (one home)

River	Origin
Bhagirathi	Gomukh / Gangotri
Alaknanda	Satopanth (above Badrinath)
Yamuna	Yamunotri (Bandarpunch)
Indus	Tibet, near Kailash / Mansarovar
Sutlej	Rakas Tal, Tibet (Trans-Himalayan; enters near Shipki La)
Brahmaputra	Tibet / Angsi–Chemayungdung (Tsangpo)
Godavari	Nashik / Trimbakeshwar
Krishna	Mahabaleshwar
Mahanadi	Sihawa, Chhattisgarh
Kaveri	Brahmagiri / Talakaveri, Coorg
Narmada	Amarkantak
Tapi	Multai, Satpura
Son	Amarkantak
Chambal	Mhow / Ambedkar Nagar (Malwa)
Luni	Aravalli near Ajmer
Sharavati	Western Ghats (Jog Falls)
Subarnarekha	Chotanagpur (Ranchi / Nagri class)
Jhelum	Verinag (Pir Panjal)
City	River
-----	-----
Lucknow	Gomti
Ayodhya	Ghaghara / Saryu
Prayagraj	Ganga + Yamuna (Sangam)
Kanpur	Ganga
Varanasi	Ganga
Patna	Ganga
Bhagalpur	Ganga
Surat	Tapi
Hyderabad	Musi
Vijayawada	Krishna
Ahmedabad	Sabarmati
Cuttack	Mahanadi
Bhubaneswar	Not on Mahanadi bank
Ujjain (उज्जैन)	Kshipra

River	Origin
Jabalpur	Narmada
Nasik (नासिक)	Godavari
Nanded	Godavari
Rajahmundry	Godavari
Srinagar	Jhelum
Leh (लेह)	Indus (right bank)
Ludhiana	Old Sutlej belt
Firozpur	Old Sutlej belt
Jamshedpur	Subarnarekha
Guwahati	Brahmaputra
Dibrugarh	Brahmaputra
Kota	Chambal
Gorakhpur	Rapti
Dam / project	River
-----	-----
Bhakra–Nangal	Sutlej
Pandoh	Beas (not Ravi)
Baglihar / Salal	Chenab
Ranjit Sagar	Ravi
Tehri	Bhagirathi
Farakka Barrage	Ganga (Hooghly diversion)
Hirakud	Mahanadi
Nagarjuna Sagar / Srisailem	Krishna (not Tungabhadra)
Mettur (Stanley)	Kaveri (not Krishna)
Jayakwadi	Godavari (Maharashtra)
Sriram Sagar	Godavari (Telangana)
Polavaram	Godavari (Andhra Pradesh)
Bargi	Narmada (MP, near Jabalpur)
Indira Sagar	Narmada (MP, Khandwa / Punasa)
Omkareshwar	Narmada (MP)
Sardar Sarovar	Narmada (Gujarat, Navagam) — terminal Narmada project
Ukai	Tapi (Gujarat)
Gandhi (गांधी) Sagar	Chambal
Rihand	Rihand (Son system)
Rajghat	Betwa (UP–MP)

Drainage patterns, rejuvenation, basins — Teaching Card

Pattern	Control	India
Dendritic	Uniform rock	Indus, Godavari, Mahanadi, Krishna, Kaveri plains
Trellis	Folded ridges and valleys	Alternating hard and soft beds

Pattern	Control	India
Radial	Dome or peak	Amarkantak , Girnar
Centripetal	Closed basin	Imphal / Loktak
Parallel	Steep uniform slope	Western Ghat torrents
Rectangular	Joints and faults	Himalayan stretches

3.5 Waterways and river linking

NW	Route
NW-1	Ganga–Bhagirathi–Hooghly (Haldia–Prayagraj) — longest
NW-2	Brahmaputra (Dhubri–Sadiya)
NW-3	Kerala west-coast canal / backwaters
NW-4	Krishna–Godavari canals
NW-5	Brahmani–Mahanadi–Paradip

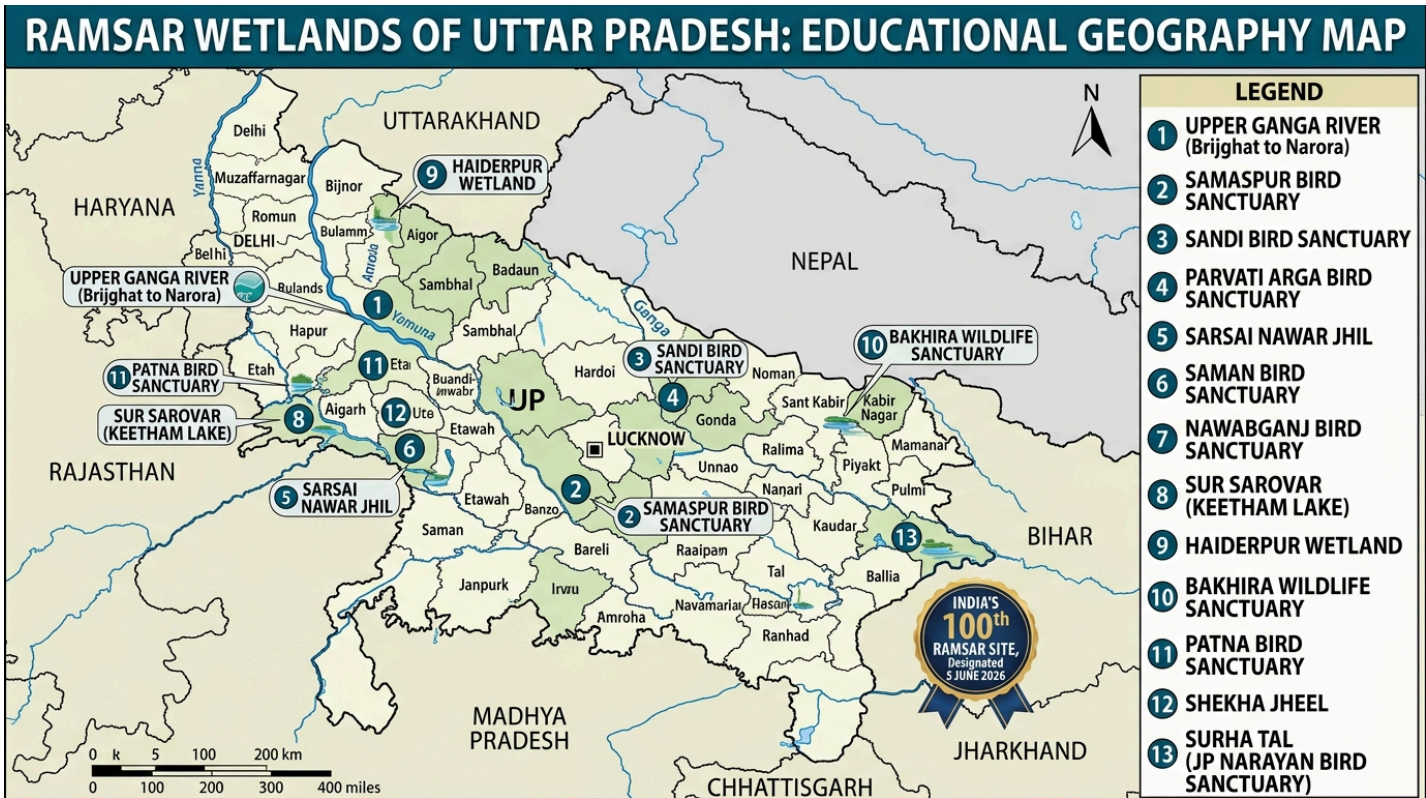
Common Traps — Don't Fall For These

1. **Pranhita** → **Mahanadi** — FALSE. **Godavari**.
2. **Betwa joins Yamuna at Prayagraj** — FALSE. **Hamirpur**.
3. **Any Prayag forms the Ganga** — FALSE. Only **Devprayag** (Alaknanda + Bhagirathi).
4. **Ganga delta = bird's-foot** — FALSE. **Mississippi**; Ganga is **arcuate**.
5. **Moribund delta subdivision = Cauvery / Krishna–Godavari** — FALSE. UPPCS 2019 keys **Bengal Delta** (moribund / mature / active). Cauvery is the usual trap.
6. **Tapi from Western Ghats** — FALSE. **Multai, Satpura**.
7. **Jonk in Ganga basin** — FALSE. **Mahanadi**.
8. **Srisailem on Tungabhadra / Pandoh on Ravi** — FALSE. **Krishna / Beas**.
9. **Pennar and Palar** → **Arabian Sea** — FALSE. **Bay of Bengal**; Periyar is west.
10. **Fault-valley tag = Chambal** — FALSE. **Damodar**.
11. **Folded structure = dendritic** — FALSE. **Trellis**.
12. **Trans-Himalayan = Jhelum or Ravi** — FALSE. **Sutlej** among usual options. Also Indus and Brahmaputra are Trans-Himalayan; Ganga is not the pick in that set.
13. **Chambal Sanctuary includes Haryana** — FALSE. **UP, MP, Rajasthan** only (2020).
14. **UP biological disaster = Yamuna** — FALSE. **Gomti**.
15. **Barak = inland saline like Luni** — FALSE. Barak → **BoB**; Luni = fresh upper / saline lower.
16. **Indus / Sutlej / Brahmaputra = superimposed** — FALSE. They are **antecedent**. Superimposed example = **Chambal**.
17. **Hemavati** → **Mahanadi** — FALSE. **Kaveri**.
18. **Largest share of river water = Ganga basin area rank** — FALSE. **Brahmaputra** ~40% of water; Ganga is the largest **Indian basin by area**.
19. **Hyderabad on Krishna** — FALSE. On **Musi** (Krishna tributary).
20. **Northern Assam Brahmaputra tribs = left bank** — FALSE when looking downstream west: north bank = **right bank** (Subansiri, Manas, Sankosh, Teesta).
21. **Eastern Dhauliganga = Alaknanda Prayag set** — FALSE. Eastern Dhauliganga → **Kali / Sharda**.
22. **Gandak flows through UP** — FALSE. Nepal → Bihar → Ganga at Sonpur.
23. **Narmada forms a delta** — FALSE. **Estuary** (rift / hard rock / little silt).

24. **Ludhiana on Ravi** — FALSE. Old **Sutlej** bank belt.

Topic 4 — Lakes, Waterfalls & Water Resources

Current Affairs (this topic)



Year	Fact	Why asked	Source
5 Jun 2026	India designates 100th Ramsar — Jai Prakash Narayan Bird Sanctuary (Surha Tal), Ballia, UP (World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस))	"How many Ramsar in India / latest UP site?"	MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) / PIB
2025–26	UP Ramsar total 13 — adds Patna (पटना) BS (Etah), Sheikha/Shekha Jheel (Aligarh (अलीगढ़)), Surha Tal (Ballia)	State-paper list + 2025 Q58 "NOT in UP"	MoEFCC LS annex
2025	India among top Ramsar countries in Asia; proposals in pipeline include Sauj Jheel (UP)	Count vs names trap	MoEFCC
Ongoing	Jal Jeevan Mission extended (rural tap water) — water-supply overlay, not a dam (दाम -)-river fact	Don't confuse with multipurpose dams	Jal Shakti (शक्ति)
2019–20	Six UP Ramsar added in the Jan 2020 window	Number question	UPPCS 2020

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q93

What is the correct sequence of lakes in descending order of size?

- A. Victoria, Superior, Chilka, Michigan B. Superior, Victoria, Michigan, Chilka C. Michigan, Superior, Victoria, Chilka D. Chilka, Michigan, Superior, Victoria

Confused Pairs

Pair	Correct	Trap	Hindi
Chilika vs Sambhar	Chilika = lagoon / often "largest saline"; Sambhar = inland saline	Pick Sambhar whenever "saline" appears	चिल्का = तटीय; सांभर = अंतर्देशीय खारा
Chilika vs Vembanad	Chilika = largest east-coast lagoon ; Vembanad = Kerala largest + India longest, west coast	Call Vembanad India's overall largest lake	वेम्बनाड = सबसे लंबा, पश्चिम
Chilika vs Pulicat vs Kolleru	Chilika largest E-coast lagoon; Pulicat 2nd; Kolleru freshwater	Kolleru as lagoon king	कोलेरू = मीठा, लैगून नहीं
Wular vs Dal vs Loktak	Wular size king; Dal tourism; Loktak = Keibul Lamjao	Keibul on Wular/Dal	लोकटक = मणिपुर, केइबुल लामजाओ
Kunchikal vs Jog	Kunchikal = UPPCS highest; Jog = famous / width	Jog = highest	कुंचिकल ऊँचा; जोग चौड़ा
Jog vs Shivanasamudra	Jog = Sharavati width ; Shivanasamudra = Cauvery volume	Swap crowns	जोग चौड़ा; शिवसमुद्र आयतन
Lodh vs Hundru	Lodh/Budha Ghagh = Burha , JH highest; Hundru = Subarnarekha	Lodh on Kanchi/Barakar	लोध = बूढ़ा
Roopkund vs Renuka	Roopkund = UK mystery/skeleton; Renuka = HP Ramsar	Roopkund in HP	रूपकुंड = उत्तराखंड
Kayal vs inland lake	Kayal = Kerala coastal lagoon/backwater	Call Wular a kayal	कायल = केरल
Dudhsagar state	Goa (Mandovi)	Karnataka-only	दूधसागर = गोवा
Chachai-Bihad	Wrong pair	Treat as correct	चाचई ≠ बिहद
Bhakra vs Pandoh	Bhakra = Sutlej; Pandoh = Beas	Pandoh on Ravi/Sutlej	भाखड़ा = सतलुज
SSP vs Indira Sagar	SSP = Narmada Gujarat ; Indira Sagar/Omkareshwar = Narmada MP	Swap states	सरदार सरोवर = गुजरात
Ban Sagar vs Narmada set	Ban Sagar = Son	Put Ban Sagar with Bargi/Omkareshwar	बन सागर = सोन
DVC year	1948 first multipurpose of independent India	1947 / Bhakra-first	DVC = 1948
Sidrapong vs Shivasamudram	Sidrapong 1897 oldest; Shivasamudram 1902 second	Call Shivasamudram oldest	सिद्रापोंग सबसे पुराना
Cauvery dispute parties	TN + KA + KL + Puducherry	Add Gujarat / drop PY	कावेरी = चार पक्ष
Gobind Sagar vs Pant Sagar	Gobind = Bhakra/Sutlej ; Pant = Rihand/Sonbhadra	Swap	गोबिंद = भाखड़ा
Chambal vs Gobind/Pant	Chambal = Gandhi/RPS/Jawahar Sagar	Gobind or Pant as Chambal	चंबल ≠ गोबिंद सागर
Tehri confluence	Bhagirathi + Bhilangana	Alaknanda / Yamuna alone	टिहरी = भागीरथी-भिलंगना
Telugu Ganga city	Krishna water to Chennai	Bangalore / Madurai	तेलुगु गंगा = चेन्नई
NBA / Medha Patkar	Opposed Sardar Sarovar height	Tag to Indira Sagar only	मेधा = SSP
Tehri river	Bhagirathi (rock-fill)	Alaknanda / Yamuna	टिहरी = भागीरथी
Srisaillam vs Tungabhadra Dam	Srisaillam = Krishna ; TB Dam = Tungabhadra	Swap	श्रीशैलम = कृष्णा
Nagarjuna Sagar vs Srisaillam	Both Krishna; NS = AP-TS iconic	Put NS on Godavari	नागार्जुन सागर = कृष्णा
Ramganga irrigation	~5.75–5.90 lakh ha	Neat 6 lakh ha	सिंचाई ≠ 6 लाख हेक्टेयर
Tulbul vs Baglihar	Tulbul = Jhelum/Wular; Baglihar = Chenab	Swap	तुलबुल = झेलम

Pair	Correct	Trap	Hindi
IGC vs Upper Ganga Canal	IGC = longest, Harike → RJ; UGC = Bhimgoda/Haridwar → W-UP	Mix offtakes	इंदिरा गांधी = सबसे लंबी
IGC vs Gang Canal	Gang Canal = oldest developed RJ system (1927 , Ganga Singh, Hussainiwala)	Call IGC the oldest	गंग नहर = 1927
Harike vs Bhakra source	IGC source = Harike Barrage (Sutlej–Beas)	Bhakra / Gandhi Sagar as IGC source	हरिके = IGC
Onega	Russia	Canada	ओनेगा = रूस
Oxbow vs lagoon	Oxbow = cut-off meander on a floodplain; lagoon = barred coast	Call Chilika an oxbow	गोखुर ≠ लैगून
Tectonic vs crater	Wular = tectonic basin; Lonar = meteorite in basalt	Lonar as tectonic	वुलर = विवर्तनिक; लोनार = उल्का
Kabartal vs Wular	Kabartal = Bihar oxbow Ramsar; Wular = J&K tectonic freshwater	Swap size kings	कबरताल = बिहार
Sharavati vs Cauvery	Jog is on Sharavati , a west-flowing Karnataka river	Wrong slip: Sharavati as Cauvery tributary	शरावती ≠ कावेरी
Kunchikal vs Nohkalikai	Kunchikal = highest total ; Nohkalikai = tallest plunge	Swap the two height titles	कुंचिकल कुल; नोहकलिकाई प्लंज
Kapildhara vs Jog	Kapildhara = Narmada ; Jog = Sharavati	Kapildhara on Godavari (UPSC 2008 trap)	कपिलधारा = नर्मदा
Shivanasamudra vs Jog	Both Karnataka; Shivanasamudra = Cauvery	Put Jog on Cauvery	शिवसमुद्र = कावेरी
Hirakud vs Indira Sagar vs Tehri	Longest earthen vs largest volume vs tallest dam	Swap the three superlatives	हीराकुड लंबा; इंदिरा सागर आयतन; टिहरी ऊँचा
Idukki vs Mullaperiyar	Both Periyar ; Idukki = arch hydro	Put Idukki on Cauvery	इडुक्की = पेरियार
Canal vs groundwater irrigation	IG plains = canals + tubewells; peninsula historically tanks ; GW now >60%	"Dominant source = canals only"	मैदान = नहर; प्रायद्वीप = तालाब
Minor vs major irrigation	Minor CCA ≤ 2000 ha (~62%); major > 10,000 ha	Swap shares	लघु ≈ 62%
Dam vs barrage	Dam stores ; barrage diverts (Farakka ~ 40,000 cusec to Hooghly)	Call Farakka a storage dam	बांध जलाशय; बैराज मोड़
Garland vs Visvesvaraya	Garland = Dastur ; first linking idea often Visvesvaraya ; later K.L. Rao	Tag Garland to Rao alone	गारलैंड = दस्तूर
Jawai Project	Rajasthan (Luni tributary)	Tamil Nadu	जवाई = राजस्थान

N.1 Lakes of India

1. Chemistry / size kings (first question type)

If the question says...	Pick	Where	Why this one wins
Largest lagoon / east-coast lagoon	Chilika	Odisha	Barred from the sea; brackish; biggest east-coast lagoon
Largest saline (लवणीय) (no word "inland")	Chilika	Odisha	Papers often treat Chilika as the saline king by area
Largest inland saline	Sambhar	Rajasthan	Desert playa; no sea link; evaporation makes it salt
Largest freshwater	Wular	J&K (Jhelum)	Usual volume fact; tectonic basin

Wrong thought	Correct
"Saline" → always Sambhar	Only if the question says inland
Chilika = inland Rajasthan lake	Chilika = Odisha coast
Wular = saline	Wular = fresh
Kolleru = lagoon king	Kolleru = freshwater delta (डेल्टा) lake; Chilika is lagoon king
Pulicat = largest lagoon	Pulicat = second after Chilika

2. Origin kings (second question type)

Origin	Meaning in one line	Indian fact	Trap
Tectonic	Basin from earth movement / warp	Wular (also freshwater king)	Calling Wular a lagoon
Oxbow	Cut-off river meander on a plain	Kabartal / Kanwar (Bihar)	Calling it a coastal lagoon
Crater (meteorite)	Hollow from a meteor hit	Lonar (Buldhana, MH)	Calling it a volcano or RJ salt pan
Glacial / high mountain	Ice-cut tarn or high closed basin	Chandra Tal, Roopkund, Tsomgo; Pangong / Tso Moriri (high, often brackish)	Calling Pangong India's freshwater size-king
Lagoon (bar)	Sea water trapped behind a sandbar	Chilika , Pulicat	Calling Chilika inland saline
Wind / playa	Desert hollow, water evaporates	Sambhar belt	Using Sambhar for every "saline" question
Artificial	Reservoir behind a dam	Gobind Sagar, Hirakud, Tehri, Pant Sagar	Treating them as natural glacial lakes

Lake–state match bank

Lake	State / UT	Note
Sala ★2020	Arunachal Pradesh	2020 Q56
Badkhal ★2020	Haryana (Faridabad belt)	2020 Q56
Loktak ★2020	Manipur	Phumdis; Keibul Lamjao
Kaliveli ★2020	Tamil Nadu	Coastal wetland
Chilika	Odisha (ओडिशा)	Largest E-coast lagoon
Pulicat	Andhra Pradesh + Tamil Nadu	Sriharikota bar
Vembanad	Kerala	Largest in Kerala; longest in India; west coast
Ashtamudi	Kerala	Kayal / Ramsar
Sasthamkotta	Kerala	Largest freshwater in Kerala (not TN)
Periyar Lake	Kerala	Artificial; not a lagoon
Kolleru	Andhra Pradesh	Freshwater; between Godavari–Krishna
Wular	J&K	Largest freshwater (volume)
Dal	J&K (Srinagar)	Tourism; not size king
Anchar	J&K	Match distractor set
Hokera	J&K	Not Punjab
Sambhar	Rajasthan (थार)	Largest inland saline
Didwana / Kuchaman / Pushkar / Nakki	Rajasthan	Nakki = Mount Abu

Lake	State / UT	Note
Pichola / Fateh Sagar	Rajasthan (Udaipur (उदयपुर))	Tourism; not Sambhar
Lonar	Maharashtra (Buldhana)	Meteorite crater
Kabartal / Kanwar	Bihar (Begusarai)	Oxbow Ramsar
Deepor Beel / Chapanala	Assam	Deepor = Guwahati Ramsar
Renuka / Nako / Ghepan / Pong	Himachal Pradesh	Renuka = Ramsar; Ghepan = Lahaul
Roopkund / Nainital / Bhimtal	Uttarakhand (उत्तराखंड)	Roopkund = Mystery Lake (not HP)
Surajkund / Sultanpur	Haryana	Sultanpur not UP
Rudrasagar	Tripura	Not UP
Pangong / Tso Moriri / Yaya Tso	Ladakh (लद्दाख)	High / brackish; Yaya Tso = BHS
Tsomgo / Gurudongmar	Sikkim	Glacial tarns
Hussain Sagar / Usman Sagar	Telangana (Hyderabad)	Artificial
Chembarambakkam / Kodaikanal	Tamil Nadu	Kodaikanal = artificial
Bhojtal (Upper Lake)	Madhya Pradesh (Bhopal)	Urban wetland
Nalsarovar	Gujarat	Bird sanctuary / Ramsar class
Harike	Punjab	Sutlej–Beas meet
Sukhna	Chandigarh	Man-made
Phulhar / Gomat Taal	Uttar Pradesh (Pilibhit)	Gomti source

World lake–country match bank

Lake	Country	tag / trap
Baikal ★2023	Russia (Siberia)	Deepest + oldest freshwater; huge volume
Michigan ★2023	USA only	Only Great Lake wholly in USA
Onega ★2023	Russia	Europe's 2nd-largest; not Canada
Maracaibo ★2023	Venezuela	Large brackish; oil belt
Ladoga	Russia	Europe's largest lake (Onega is 2nd)
Caspian Sea	Russia / Kazakhstan / Turkmenistan / Iran / Azerbaijan	World's largest lake by area ; saline
Dead Sea	Israel / Jordan	Rift; hypersaline; below sea level
Superior	USA + Canada	Largest Great Lake by area
Huron / Erie / Ontario	USA + Canada	Shared Great Lakes (unlike Michigan)
Victoria	Uganda / Kenya / Tanzania	Africa's largest by area
Tanganyika	Several E. African states	Deep rift lake
Malawi (Nyasa)	Malawi / Mozambique / Tanzania	Rift lake
Turkana	Kenya (mainly)	Rift; saline
Titicaca	Peru + Bolivia	Highest large navigable lake
Great Bear / Great Slave	Canada	Not Onega
Loch Ness / Loch Lomond	Scotland (UK)	
Eyre	Australia	Endorheic salt lake
Chad	Chad / neighbours	Shrinking inland
Aral	Kazakhstan / Uzbekistan	Shrinking saline

Crown	Lake
Largest lake by area (world)	Caspian (saline)
Largest freshwater by area (usual set)	Superior
Largest / deepest freshwater by volume	Baikal
Lowest / densest salt	Dead Sea

UP wetlands (state paper gold)

Site	District / belt
Upper Ganga River	Brijghat to Narora
Nawabganj	Unnao
Parvati Arga	Gonda
Saman	Mainpuri
Samaspur	Rae Bareli
Sandi	Hardoi
Sarsai Nawar	Etawah
Sur Sarovar	Agra
Haiderpur	Bijnor–Muzaffarnagar (Ganga)
Bakhira	Sant Kabir (कबीर) Nagar
Patna Bird Sanctuary	Etah
Sheikha / Shekha Jheel	Aligarh
Jai Prakash Narayan BS / Surha Tal	Ballia (India's 100th Ramsar, 5 Jun 2026)
Name that sounds like UP	Actual state
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Rudrasagar	Tripura
Sultanpur	Haryana
Surinsar–Mansar	J&K

N.2 Waterfalls

Superlative	Fact	Trap
Highest	Kunchikal (Varahi, Karnataka; older figure ~455 m)	Jog
Tallest plunge (newer World Waterfalls Database)	Nohkalikai (Meghalaya (মেঘালয়), ~340 m)	Using Nohkalikai to beat every Kunchikal height question
Largest by width	Jog / Gersoppa (Sharavati, Karnataka)	Calling Jog highest
Largest by volume	Shivanasamudra (Cauvery, Karnataka)	Putting volume crown on Jog
Waterfall	~Height	River
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Kunchikal	older ~455 m; newer DB ~183 m	Varahi
Barehipani	~399 / ~217 m by DB revision	Budhabalanga
Nohkalikai	~340 m	Plateau stream

Superlative	Fact	Trap
Nohsngthiang (Seven Sisters) / Mawsmat	~315 m class	—
Dudhsagar	~310–320 m	Mandovi
Barkana	~259 m	Sita (सीता) belt
Jog (Gersoppa)	~253 m	Sharavati
Khandadhar	~244 m	Korapani Nala → Brahmani
Palani	—	Kullu belt
Hundru	~98 m	Subarnarekha
Jonha	—	Rarhu
Dasam / Dasong	—	Kanchi (कांची)
Lodh / Budha Ghagh	~143 m (468 ft)	Burha
Dhuandhar	~30 m	Narmada
Kapildhara	—	Narmada
Dugdhdhara / Bheraghat belt	—	Narmada
Bhalkund (Rahatgarh)	—	Bina
Chachai	~130 m	Bihad / Beehar (Tamsa (तमसा))
Chulia	—	Chambal (चंबल)
Joranda	~150 m	Budhabalanga
Duduma	—	Machhkund
Gokak	—	Ghataprabha
Shivanasamudra	~90–98 m	Cauvery
Hogenakkal	—	Cauvery
Athirappilly	—	Chalakudy
Chitrakote	wide, low (~29 m)	Indravati
Amritdhara	—	Hasdeo (Hasdo)
Kevti / Rakim Kund	—	Rewa plateau streams

N.3 Dams, canals and hydropower

Star projects

Project	River	Where / type	Reservoir / extra
Bhakra–Nangal	Sutlej	HP dam + Punjab headworks; gravity ; joint PB–HR–RJ (command also touches HP)	Gobind Sagar ; completed 1963 ; Nehru’s “temple of modern India” tag
Hirakud	Mahanadi	Odisha (~15 km from Sambalpur); started 1948 , completed 1957	Longest earthen-dam narrative (~ 4.8 km main / ~ 26 km with dykes); ~10 lakh ha irrigation note
Sardar Sarovar	Narmada	Near Navagam , Gujarat; height ~ 163 m ; ~ 1450 MW	Beneficiaries GJ, RJ, MH, MP ; MP takes the largest Narmada water share; NBA / Medha Patkar opposed height rise
Indira Sagar (Narmada Sagar)	Narmada	Near Punasa / Khandwa , MP	Largest reservoir by volume ; ~1.23 lakh ha + ~ 1000 MW target; Harsud town submerged (2004)

Project	River	Where / type	Reservoir / extra
Nagarjuna Sagar	Krishna	AP-TS border; completed 1967	Multipurpose; Nalgonda–Khammam–Guntur–Prakasam belt; height ~ 125 m
Tehri	Bhagirathi (+ Bhilangana confluence at old Tehri)	Uttarakhand; rock-fill ; THDC	India's tallest dam (~ 260.5 m); cleared 1972 , work from 1978 ; seismic / displacement protest

Master Table — Major Dams & Multipurpose Projects in India (भारत के प्रमुख बांध एवं बहुउद्देशीय परियोजनाएं)

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Tehri Dam (टिहरी बांध)	Bhagirathi (+ Bhilangana confluence)	Uttarakhand	Tallest dam in India (260.5 m); rock-fill earth dam; THDC
Bhakra-Nangal (भाखड़ा नांगल)	Sutlej	Himachal Pradesh & Punjab	Highest concrete straight gravity dam (226 m); reservoir: Gobind Sagar ; joint venture of PB, HR, RJ
Hirakud Dam (हीराकुंड बांध)	Mahanadi	Odisha (Sambalpur)	Longest earthen dam in the world (main dam 4.8 km; total length with dykes 25.8 km); Gandhi Minar & Nehru Minar
Sardar Sarovar Dam	Narmada	Gujarat (Navagam, Kevadia)	Height 163 m; benefits Gujarat, Rajasthan, Maharashtra, MP; Statue of Unity downstream; Narmada Bachao Andolan (Medha Patkar)
Indira Sagar Dam (Narmada Sagar)	Narmada	Madhya Pradesh (Khandwa)	Largest water reservoir in India by storage volume (12.22 BCM); submerged Harsud town
Nagarjuna Sagar Dam	Krishna	Andhra Pradesh & Telangana	World's largest masonry dam; built 1955–1967; named after Buddhist scholar Acharya Nagarjuna
Srisailem Dam	Krishna	Andhra Pradesh & Telangana	Second largest capacity working hydroelectric station; located in deep gorge of Nallamala hills
Almatti Dam (Lal Bahadur Shastri)	Krishna	Karnataka (Bijapur)	Main reservoir of Upper Krishna Irrigation Project
Pandoh Dam (पंडोह बांध)	Beas	Himachal Pradesh (Mandi)	Diverts Beas waters into Sutlej via Beas-Sutlej Link for power at Dehar
Pong Dam (Maharana Pratap Sagar)	Beas	Himachal Pradesh (Kangra)	Earth-fill embankment dam; wetland Ramsar site
Baglihar Dam (बगलीहार)	Chenab	Jammu & Kashmir (Ramban)	Run-of-the-river project; cleared after neutral expert Raymond Lafitte adjudication under Indus Waters Treaty
Salal Dam (सलाल बांध)	Chenab	Jammu & Kashmir (Reasi)	Run-of-the-river hydroelectric project on Chenab rock pedestal
Dulhasti Dam (दुलहस्ती)	Chenab	Jammu & Kashmir (Kishtwar)	Run-of-the-river power project on Chandra/Chenab river
Pakal Dul & Ratle	Chenab & Marusudar	Jammu & Kashmir (Kishtwar)	Major new hydropower projects under execution
Uri Dam (Uri-I & II)	Jhelum	Jammu & Kashmir (Baramulla)	Near LoC; run-of-the-river hydro plant
Tulbul Navigation Project	Jhelum (Wular Lake outlet)	Jammu & Kashmir	Navigation barrage contested by Pakistan; suspended since 1987
Rihand Dam (Govind Ballabh Pant)	Rihand (tributary of Son)	Uttar Pradesh (Pipri, Sonbhadra)	Largest man-made artificial lake by area in India ; power source for Renukoot aluminium plant

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Matatila Dam (माताटीला बांध)	Betwa	Uttar Pradesh (Lalitpur)	Multipurpose dam; joint project with MP
Rajghat Dam (Rani Laxmibai)	Betwa	UP & MP border (Lalitpur/Chanderi)	Inter-state multipurpose project on Betwa
Parichha Dam (परीछा बांध)	Betwa	Uttar Pradesh (Jhansi)	Thermal power station water supply & canal irrigation
Chambal Project (4-stage cascade)	Chambal	MP & Rajasthan	1. Gandhi Sagar (Mandsaur, MP) 2. Rana Pratap Sagar (Rawatbhata, RJ) 3. Jawahar Sagar (Kota, RJ) 4. Kota Barrage (irrigation canals)
Idukki Dam (इडुक्की बांध)	Periyar	Kerala	Concrete double-curved arch dam between Kuravan & Kurathi hills; underground power house at Moolamattom
Mullaperiyar Dam	Periyar	Kerala (leased to Tamil Nadu)	Masonry gravity dam built 1895 by British engineer John Pennycuik; administered by TN
Mettur Dam (Stanley Reservoir)	Cauvery	Tamil Nadu (Salem)	Built 1934; chief irrigation reservoir for Cauvery delta farmers
Krishna Raja Sagara (KRS)	Cauvery	Karnataka (Mandya/Mysuru)	Designed by Sir M. Visvesvaraya ; Brindavan Gardens located below dam
Ukai Dam (Vallabh Sagar)	Tapi	Gujarat (Surat)	Second largest reservoir in Gujarat after Sardar Sarovar
Kakrapar Project	Tapi	Gujarat	Weir project for irrigation + Kakrapar Atomic Power Station
Koyna Dam (कोयना बांध)	Koyna (tributary of Krishna)	Maharashtra (Satara)	Largest completed hydroelectric power plant in Maharashtra; triggered 1967 Koynanagar earthquake
Damodar Valley Project (DVC)	Damodar & Barakar	Jharkhand & West Bengal	First multipurpose river valley project of independent India (1948); modeled on US Tennessee Valley Authority (TVA) ; dams at Tilaiya, Maithon, Belpahari (on Barakar) and Panchet (on Damodar)

Multipurpose valley cards (Ghatnachakra)

Project	River / place	Note
Telugu Ganga	Krishna → Chennai drinking water	Joint TN–AP–KA–MH
Chukha	Wang Chu / Raidak, Bhutan	India-aided; handed to Bhutan 1991 ; ~ 336 MW
Mayurakshi	Mayurakshi; Canada / Massanjore dam in JH	Shared WB–JH
Babli	Godavari, Maharashtra	Disputed with Telangana / AP
Mahakali Treaty	Sharda (शारदा) / Mahakali	India– Nepal (1996) ; Sharda, Tanakpur, Pancheshwar
Gandak Project	Gandak Barrage (1959 treaty)	India–Nepal; UP + Bihar (+ Nepal)
Kishau	Tons (टौंस) (Yamuna)	HP–UK border; benefits HP/UK (+ others)
Kalpong	Kalpong	First hydro of Andaman (अंडमान) & Nicobar (निकोबार)
Rihand / Pant Sagar	Rihand (Son tributary), Pipri, Sonbhadra	Largest UP multipurpose note; Obra power below dam
Rajghat / Rani Lakshmibai (रानी लक्ष्मीबाई)	Betwa (बेतवा), Lalitpur	Joint UP–MP
Matatila	Betwa	Joint UP–MP

Project	River / place	Note
Pochampad	Godavari	Telangana
Tapovan–Vishnugad	Dhauliganga (धौलीगंगा), Chamoli (चमोली) UK	
Dulhasti	Chenab (चिनाब), Kishtwar	
Baglihar	Chenab , Ramban	IWT / World Bank dispute note
Suil	Suil (Ravi tributary), Chamba HP	
Kol Dam	Sutlej , HP	NTPC; silt check for Bhakra note
Meja (RJ)	Kothari (Banās), Bhilwara	Irrigation — not UP's Meja
Kalpasar	Gulf of Khambhat, Gujarat	Freshwater reservoir / tidal idea
Teesta hydro	Teesta	Sikkim / West Bengal (TLDP III–IV = WB)
Malampuzha	Malampuzha (Bharathapuzha), Kerala	Largest Kerala reservoir note
Aliyar / Isapur / Kangsabati	—	All are water reservoirs

Tribunal	Year / note	Parties (count)
Mahadayi	2010	Goa, Karnataka, Maharashtra (3)
Godavari (गोदावरी) (Bachawat)	April 1969	AP, MP, Chhattisgarh, Odisha, Karnataka (5)
Narmada	6 Oct 1969	Gujarat, MP, Maharashtra, Rajasthan (4)
Mahanadi	2018	Odisha, Chhattisgarh (2)

River-wise dam map (match fodder)

Indus (सिंधु) system

- **Pandoh** is on the **Beas** (ब्यास). It is not on the Ravi.
- **Pong / Maharana Pratap Sagar** is also on the Beas in Himachal (bird sanctuary reservoir).
- **Ranjit Sagar (Thein)** is on the **Ravi** (रावी).
- **Baglihar**, **Salal** and **Dulhasti** are on the **Chenab**.
- **Nathpa Jhakri** and **Kol Dam** are Sutlej hydro works in Himachal.

Ganga–Yamuna–UP

- **Tehri** and **Koteswar** stand on the Bhagirathi (Bhilingana joins at the Tehri site).
- **Rihand** (रिहंद) dam makes **Govind Ballabh Pant Sagar** at Pipri, Sonbhadra.
- **Matatila** and **Rajghat (Rani Lakshmibai)** are Betwa projects serving Bundelkhand.
- **Narora** is a Ganga **barrage**, not a high dam.
- **Kalagarh** is the site of the **Ramganga Dam**.

Narmada–Tapi–Chambal

- Narmada cascade in MP includes **Bargi**, **Indira Sagar**, **Omkareshwar**, **Maheshwar**, then **Sardar Sarovar** in Gujarat.
- **Gandhi Sagar**, **Rana Pratap Sagar**, **Jawahar Sagar** and **Kota Barrage** are the Chambal chain (**MP–RJ**).
- **Ukai** and **Kakrapar** are on the **Tapi** (तापी) in Gujarat.
- **Ban Sagar** is on the **Son**, not the Narmada.

East-flowing peninsula

- **Srisailem** and **Almatti** and **Nagarjuna Sagar** are on the **Krishna**.
- **Tungabhadra** (तुंगभद्रा) **Dam** is on the Tungabhadra (KA–AP joint).
- **Jayakwadi** and **Pochampad** are on the Godavari.

- **Nizam (निजाम) Sagar** is on the **Manjra**, a Godavari tributary.
- **Hirakud** is on the Mahanadi. **Rengali** is on the Brahmani.
- **Mettur** (Stanley Reservoir) is on the Cauvery in Tamil Nadu.
- **Krishna Raja Sagara (KRS)** and **Shivasamudram** are on the Cauvery in Karnataka.

West-flowing Ghats and others

- **Koyna** is on the Koyna (Krishna system) in Maharashtra. The **1967** shock is the textbook **reservoir-induced seismicity** case.
- **Idukki / Cheruthoni** are on the **Periyar** in Kerala.
- **Mullaperiyar** is also on the Periyar. Kerala hosts the dam. Tamil Nadu operates it.
- **DVC** dams **Tilaiya, Konar, Maithon, Panchet** sit on the Damodar–Barakar system.

Canals

Canal	Source / offtake	Region	Tag
Gang Canal	Sutlej at Hussainiwala (Firozpur)	North Rajasthan / Bikaner command	Among world's oldest developed systems; 1927 ; Maharaja Ganga Singh
Indira Gandhi Canal	Harike Barrage (Sutlej–Beas confluence, Punjab)	Ganganagar, Bikaner, Jodhpur, Jaisalmer	India's longest (~ 649 km main); foundation Mar 1958 (G.B. Pant); also draws Beas–Ravi–Sutlej waters (Pong / Beas project support)
Upper Ganga Canal	Ganga at Bhimgoda, Haridwar (right bank)	W-UP Doab (Haridwar–Saharanpur–Muzaffarnagar–Meerut (मेरठ)–Ghaziabad–Bulandshahr–Aligarh–Etah–Mathura (मथुरा)–Firozabad–Mainpuri–Agra belt)	Father often tagged Sir P.T. Cautley ; first discharge 8 Apr 1854 ; main ~ 298 km ; system ~ 6496 km ; also feeds Delhi (~200 cusec) and Harduaganj TPP (~100 cusec)
Lower Ganga Canal	Ganga at Narora (Bulandshahr)	Central / lower Doab	Completed 1878 ; main ~ 99 km ; system ~ 8278 km
Eastern Yamuna Canal	Yamuna left bank at Tajewala; Hathinikund barrage ~3 km upstream (Haryana)	Hindon–Yamuna Doab: Saharanpur, Muzaffarnagar, Meerut, Ghaziabad	Revived functioning from 1830 ; main ~ 197 km
Agra Canal	Yamuna (Okhla)	Agra–Mathura belt	
Sharda Canal	Sharda right bank near Banbasa (Nainital / UK border)	Protective irrigation across Rohilkhand–Awadh (अवध)–Purvanchal (पूर्वांचल) (Pilibhit to Prayagraj (प्रयागराज) belt)	Main ~ 44 km ; system ~ 9961 km (longest UP canal-system length fact)
Sharda Sahayak / CAD	Sharda command	E / central UP	CADP projects also include Ramganga and Gandak
Betwa Canal	Betwa	Bundelkhand	
Bhakra / Sirhind	Sutlej system	Punjab–Haryana	
Buckingham Canal	East-coast backwaters	AP (Kakinada) → TN (Viluppuram); Coromandel	Freshwater navigation canal (~796 km), not IGC
Gandak / Saran canals	Gandak barrage near Triveni / Balmiki Nagar (~740 m)	NW Bihar	Saran irrigation canal drawn from this system
Ukai Canal	Tapti (ताप्ती) / Ukai	Gujarat	
Ghataprabha Canal	Ghataprabha	Karnataka	
Nira Canal	Nira (Bhima tributary)	Maharashtra	
Jawai Project	Jawai (Luni (लूनी) tributary)	Rajasthan	Trap: not Tamil Nadu

Canal	Source / offtake	Region	Tag
Mangalam	Bharathapuzha system	Palakkad, Kerala	One of several projects on that river

Irrigation project classes (CCA)

Class	Culturable Command Area	Share of irrigation potential
Minor	Up to 2,000 ha (wells, tubewells, tanks, lift, drip, sprinkler)	About 62%
Medium	2,000–10,000 ha	Part of the remaining ~ 38% with major
Major	Above 10,000 ha	Major + medium together about 38%

Common Traps — Don't Fall For These

1. **Sambhar = largest saline (any wording)** — FALSE unless **inland**. Else often **Chilika**.
2. **Chilika in Rajasthan / freshwater** — FALSE. **Odisha coastal lagoon**.
3. **Kolleru = largest east-coast lagoon** — FALSE. **Chilika**; Kolleru is freshwater.
4. **Keibul Lamjao on Wular / Dal** — FALSE. **Loktak, Manipur**.
5. **Jog = highest waterfall** — FALSE. **Kunchikal** (Varahi, KA) for UPPCS. Jog = renown / **width**; Shivanasamudra = **volume**. 5a. **Lodh / Budha Ghagh on Kanchi or Barakar** — FALSE. On the **Burha** (Latehar). 5b. **Roopkund in Himachal** — FALSE. **Uttarakhand** Mystery / Skeleton Lake. 5c. **Periyar Lake is a lagoon / kayal** — FALSE. It is **artificial**. Kayals are Kerala coastal lagoons. 5d. **Hokera in Punjab / Sasthamkotta in Tamil Nadu** — FALSE. Hokera = **J&K**; Sasthamkotta = **Kerala**.
6. **Dudhsagar = Karnataka-only** — FALSE. The UPPCS answer places it in **Goa**.
7. **Chachai–Bihad is correct** — FALSE. That is the **wrong** pair.
8. **Onega = Canada** — FALSE. **Russia**. Ladoga is also Russia. Michigan = USA-only; Superior is shared USA–Canada.
9. **Rudrasagar / Sultanpur in UP** — FALSE. **Tripura / Haryana**.
10. **Bhakra on Beas / Ravi** — FALSE. **Sutlej**; reservoir **Gobind Sagar**.
11. **Tehri on Alaknanda** — FALSE. **Bhagirathi**, rock-fill.
12. **Ramganga irrigates 6 lakh ha** — FALSE. **~5.75–5.90 lakh ha**.
13. **Sardar Sarovar = Indira Sagar** — FALSE. SSP = Narmada **Gujarat**; Indira Sagar = Narmada **MP**.
14. **Srisaillam on Tungabhadra / Pandoh on Ravi** — FALSE. **Krishna / Beas**.
15. **E-UP/Bihar can fully tap Himalayan hydro to their north** — FALSE. Those ranges lie largely in **Nepal**.
16. **Sharavati is a Cauvery tributary** — FALSE. It is **west-flowing** to the Arabian Sea.
17. **Kapildhara is on the Godavari** — FALSE. It is on the **Narmada**.
18. **Jog is India's highest waterfall** — FALSE. **Kunchikal** is highest; Nohkalikai is the tallest plunge.
19. **Hirakud is India's tallest dam** — FALSE. **Tehri** is tallest. Hirakud is the long earthen dam.
20. **Idukki is on the Cauvery** — FALSE. **Periyar**, Kerala, arch dam.
21. **Industry uses most of India's groundwater** — FALSE. **Agriculture ~92%**.
22. **Lonar is a volcanic caldera** — FALSE. It is a **meteorite** crater in Deccan basalt.
23. **Dominant irrigation = canals only** — FALSE. Wells/tubewells dominate (~64% together; tubewell ~48.5% in 2018–19).
24. **IGC source = Bhakra / Gandhi Sagar** — FALSE. **Harike Barrage** (Sutlej–Beas).
25. **Gang Canal = Indira Gandhi Canal** — FALSE. Gang Canal = **1927** Ganga Singh / Hussainiwala; IGC = longest / Harike / 1958.
26. **Lower Ganga offtake = Haridwar** — FALSE. **Narora**; Upper Ganga = Bhimgoda/Haridwar.
27. **Jawai Project = Tamil Nadu** — FALSE. **Rajasthan** (Luni tributary).
28. **Garland Canal = K.L. Rao** — FALSE. **Dastur**; Visvesvaraya/Rao = linking ideas.

29. **Farakka carries 75,000 cusec** — FALSE. About **40,000 cusec**.
30. **Ghaggar feeds IGC** — FALSE.
31. **Minor irrigation = smallest potential share** — FALSE. Minor (~62%) is the largest potential share.
32. **Hariyali = second Green Revolution** — FALSE. Watershed / wasteland–water storage (Jan 2003).
33. **Ban Sagar on Narmada** — FALSE. **Son**.
34. **Harsud submerged by SSP** — FALSE. **Indira Sagar** (2004).
35. **Shivasamudram = oldest hydro** — FALSE when Sidrapong is an option. Sidrapong **1897**; Shivasamudram **1902**.
36. **DVC = 1947 / Bhakra first multipurpose** — FALSE. **DVC 1948** is first of independent India.
37. **Chambal = PB–HR–RJ** — FALSE. **RJ–MP**. Gobind / Pant Sagar are not Chambal.
38. **Tehri on Alaknanda alone** — FALSE. **Bhagirathi** (+ Bhilangana confluence).
39. **Nagarjuna Sagar on Godavari / Cauvery** — FALSE. **Krishna**.
40. **Tilaiya on Konar / Damodar main** — FALSE. **Barakar**.
41. **Telugu Ganga to Bangalore** — FALSE. **Chennai** (Krishna).
42. **Rana Pratap Sagar = Madhya Pradesh** — FALSE. **Rajasthan** (Rawatbhata).
43. **Shivasamudram = Kerala** — FALSE. **Karnataka**.
44. **Babli = Andhra Pradesh project site** — FALSE. Built by **Maharashtra** on Godavari (disputed by AP/TG).

Topic 5 — Soils

Current Affairs (this topic)

Year	Fact	Why asked	Source
19 Feb 2015	Soil Health Card launched at Suratgarh, Rajasthan ; slogan Swasth Dharaa, Khet Haraa ; 2015 = International Year of Soils	Scheme–place–slogan	PIB
2022–23 onward	SHC merged into RKVY as Soil Health and Fertility	New name vs old scheme	PIB / DA&FW
Jul 2025	> 25 crore Soil Health Cards distributed; card reports 12 parameters (N,P,K,S, Zn,Fe,Cu,Mn,B, pH, EC, OC)	Numbers / parameters	PIB Aug 2025
Ongoing	WDC–PMKSY 2.0 = watershed soil–water conservation (not a soil-type topic)	Scheme vs conservation method	Jal Shakti (शक्ति) / DA&FW
Static	World Soil Day (विश्व मृदा दिवस) 5 December (UN;)	Date trap vs 5 June	UN / PYQ

Confused Pairs

Pair	Correct	Trap	Hindi
Edaphic vs climatic	Edaphic = soil	Call climate “edaphic”	मृदीय = मिट्टी
Urvara vs Usara	Ancient: Urvara = fertile; Usara = sterile	Mix with modern usar chemistry	उर्वरा / ऊसरा
Residual vs transported	Residual / sedentary = on rock (regur/red); transported / azonal = alluvium	Call alluvium residual	अवशिष्ट / वाहित
Red vs alluvial area	Alluvial is largest ; red is widespread on the peninsula	Wrong claim: “red is the largest group”	जलोढ़ सबसे बड़ा
NCERT P vs other notes on phosphoric acid	Class 11: alluvium poor in phosphorus	Do not copy “phosphoric acid adequate” over the NCERT poor-P fact	फॉस्फोरस कम (NCERT)
Pedocal vs laterite	Pedocal = arid lime-accumulating ; laterite = wet leached Fe/Al	Mix the two climate products	पेडोकल = चूना; लैटेराइट = निक्षालित
Himalaya north vs south slope	Soil on gentler north-facing forested slopes; south face often too steep	Soil equally on both Himalayan faces	उत्तर ढाल मिट्टी; दक्षिण तेज
Kari vs Kuttanad vs regur	Kari = Kerala peaty (Kottayam–Alappuzha)	Call kari Deccan black	कारी = पीट
Eluviation vs illuviation	E = wash out of A/E; I = deposit in B	Swap	ऊपरी हानि / निचला जमाव
Khadar vs Bangar	Khadar = new/fine/fertile; Bangar = old/ more kankar	Khadar has more kankar	खादर नया; बांगर पुराना
Black vs Himalaya	Regur = Deccan Trap	Black in Himalaya	काली = दक्कन
Black vs peaty black	Regur = basalt clay; peaty = organic marsh	Kerala marsh = cotton soil	पीट ≠ रेगुर
Red vs laterite	Red = Fe colour on crystalline rock; laterite = hot-wet leached Fe/Al	Every red = laterite	लाल ≠ लैटेराइट
Laterite Fe/Al	Rich in Fe/Al oxides	“Poor in Fe/Al”	लोहा–एल्युमीनियम प्रचुर
Cotton vs cashew	Cotton = black; cashew/ tapioca /tea/coffee = laterite	Cotton on laterite as default	कपास = काली

Pair	Correct	Trap	Hindi
Gypsum vs lime	Gypsum = alkali/sodic/usar; lime = acid only	Lime for usar (Usar BEO trap)	जिप्सम क्षारीय; चूना अम्लीय
Splash vs sheet	Splash = raindrop first stage; sheet = thin wash	Start ladder at sheet only	स्प्लैश → शीट
Tea vs regur	Tea prefers slightly acidic , well-drained	Tea = black cotton default	चाय = अम्लीय
Zinc vs Fe micros	Among Cu/Fe/Mn/Zn, Zn is most often deficient in Indian soils	Iron as "most deficient"	जिंक कम
Saline vs sodic	Saline = soluble salts; sodic = exchangeable Na + high pH	One remedy for both	लवणीय / क्षारीय
Shelter belt vs check dam	Shelter belt = wind /dunes; check dam = gully	Swap agents	रक्षक पट्टी = हवा
World Soil Day	5 December	5 June (Environment Day)	मृदा दिवस = 5 दिसंबर
Mitti Bachao	Hoshangabad, MP	Darbhangha / Thane / Mysore	मिट्टी बचाओ = होशंगाबाद
Terra Roxa vs regur	Terra Roxa = Brazil coffee basaltic	Indian black soil	टेरा रोक्सा = ब्राज़ील
Terra Rossa	Limestone / dolomite weathering (red)	Terra Roxa Brazil coffee	टेरा रोसा
Chernozem vs Chestnut	Steppe humus-rich vs drier less humus	Swap	चेरनोज़ेम / चेस्टनट
Capillary max	Clay (tiny pores)	Sand	केशिका
Halophyte	Saline soil	Acid / cold soil	हैलोफाइट
Bhabar vs Tarai	Bhabar = pebbles, streams vanish ; Tarai = streams re-emerge , moist fertile	Bhabar = best rice soil	भाबर पत्थर; तराई फसल
Mar vs Parua	Mar/Kabar = Bundelkhand black ; Parua/Rakar = red	Swap colours	मार काली; परुआ लाल
UP soil identity	~90% alluvial	UP = black-soil state	UP = जलोढ़
Bangar kankar vs arid kankar	Bangar = old alluvium nodules; arid = lower-horizon Ca kankar	Same origin	बांगर / मरु कंकर
Rann salt vs usar reh	Rann crust = SW monsoon salt deposit; canal usar = irrigation+poor drain	Same cause always	कच्छ = मानसून लवण
Karewa vs alluvium	Karewa = Kashmir lacustrine (saffron)	Call it ordinary Gangetic alluvium	करेवा = कश्मीर

N.1 Formation, Profile, Classification

How rock becomes soil (weathering)

Type	Simple meaning	What changes?	India note
Physical	Breaks rock into smaller pieces	Size only — chemistry stays the same	Heat cracking, frost, wind abrasion on exposed rock
Chemical	Changes minerals inside the rock	New compounds form (hydration, oxidation, carbonation, solution)	Strong under hot wet monsoon toward laterite
Biological	Life mixes and dissolves	Roots split rock; organic acids + burrowers loosen and enrich	Forest floors, earthworms

Residual vs transported

- **Residual** (mostly peninsular) soils form **in place** on the rock beneath them. They are also called **sedentary** soils.
- Regur on **Deccan Trap basalt** and red soil (लाल मिट्टी) on **granite–gneiss** are residual.
- **Transported** (extra-peninsular plains) soils are carried in by rivers, wind, or ice — **alluvium** is the classic. These are often tagged **azonal** (weak horizons, little relation to the rock underfoot).
- Same climate on different rock can still give different soil. **Basalt** weathers to **black** soil. **Granite–gneiss** weathers to **red** soil.
- Farming, irrigation, and deforestation can strip a profile **faster** than nature rebuilds it.

CLORPT

Factor	Role	India example
Climate	Weathering + leaching intensity	Laterite on hot-wet Ghats
Living organisms	Humus, mixing, nutrient cycling	Forest cover (वन आवरण) in the Himalaya
Relief	Depth, drainage, erosion	Thin soils on steep Himalayan slopes
Parent material	Mineral base of the soil	Deccan Trap weathers to regur
Time	How mature the horizons are	Mature black vs young khadar

Profile (top to bottom)

Horizon	Content	tag
O	Organic litter / humus	Forest floors
A	Topsoil: mineral + humus	Crop root zone
E	Leached zone (if present)	Eluviation (wash out)
B	Clay / Fe / Al / lime accumulation	Illuviation (wash in) / subsoil
C	Weathered parent, not fully soil	Transition
R	Unweathered bedrock	Base

ICAR / school classification

Soil	How it forms	Core belt	Star crop / use	Classic trap
Alluvial	River-borne silt (transported)	Indo-Gangetic, Brahmaputra (ब्रह्मपुत्र), E-coast deltas (NCERT ~40%; other notes ~46%)	Wheat, rice, sugarcane, jute	UP is alluvial , not a black-soil state
Black (Regur)	On Deccan Trap basalt (residual)	MH–MP–GJ–N KA–TG	Cotton	Not Himalayan
Red & Yellow	On granite–gneiss (crystalline)	TN, KA, S MH, Odisha, Chota Nagpur	Millet, pulses	Red ≠ laterite; not always barren
Laterite	Hot-wet intense leaching	Ghats, NE hills, plateau edges (Malabar /Kerala is the classic home belt)	Cashew, tapioca , tea, coffee, cinchona	Rich in Fe/Al — not poor; not RJ/UP plains
Desert / Arid	Arid climate + sand	W Rajasthan (Thar)	Millet if irrigated	Not forever barren
Mountain / Forest	Altitude + slope	Himalaya, NE hills	Orchards, terraces	Thin on slopes; ≠ regur
Saline / Usara	Salts + poor drainage	Canal plains, arid belts, coastal deltas, Rann of Kutch	Needs reclamation	More water without drainage worsens it
Peaty / Marshy	Waterlogged organic pile-up	Kerala (Kuttanad), Sundarbans, N Bihar, S Uttarakhand (उत्तराखंड)	Drainage first	Black colour ≠ cotton regur

Soil ↔ state match bank (classic Prelims)

Soil Group (ICAR)	NCERT Area Share	Primary Match State (Commission Anchor)	Alternate / Secondary States (If primary absent)	Why this association? (Geological / Climatic basis)	Classic Prelims Trap / Misconception
Alluvial Soil (जलोढ़ मृदा)	~40% (Class 11) ~46% (other)	Uttar Pradesh (उत्तर प्रदेश)	Bihar, West Bengal, Punjab, Haryana, Coastal deltas (AP/Odisha/TN)	Extensive Indo-Gangetic river floodplains; transported silt (<i>khadar & bangar</i>)	Trap: Bundelkhand (southern UP) is not alluvial (it has black <i>Mar/Kabar</i> & red <i>Rakar/Parua</i>); but UP as a whole holds the single largest alluvial tract (~90% of state area).
Black Soil / Regur (काली / रेगुर मृदा)	~15–16%	Maharashtra (महाराष्ट्र)	Madhya Pradesh (Malwa), Gujarat (Kathiawar), North Karnataka, Telangana	Formed <i>in situ</i> by weathering of Deccan Trap basaltic lava ; rich in titaniferous magnetite; high clay & moisture retention	Trap: Regur is strictly peninsular volcanic; never match with Himalaya or Gangetic plains. Do not mistake local black patches in eastern India (Rajmahal) as main Deccan regur.
Red & Yellow Soil (लाल एवं पीली मृदा)	~18.5% (2nd largest)	Andhra Pradesh (आंध्र प्रदेश) / Tamil Nadu	Odisha, Chhattisgarh, Chotanagpur (Jharkhand), Southern Karnataka	Formed on ancient crystalline & metamorphic granite–gneiss rocks under low-to-medium rainfall; red from ferric oxide (Fe ₂ O ₃) diffusion	Trap: Red soil is not laterite! Red soil has good tilth and responds well to fertilizers (pulses, millets, oilseeds); laterite is intensely leached, acidic, and bakes rock-hard.
Laterite Soil (लेटराइट मृदा)	~3.7–4.3%	Kerala (केरल - Malabar Coast)	Maharashtra (Western Ghats crests/Konkan if Kerala absent), Karnataka, Meghalaya (Shillong plateau)	Intense tropical leaching under alternate wet & dry seasons; silica & lime washed down (<i>eluviation</i>), Fe & Al oxides left on top; bakes like brick (<i>later</i>)	Trap: First identified in Angadipuram, Kerala (Malabar) by Francis Buchanan (1807). If Kerala is not in the options, look for Maharashtra / Karnataka (Western Ghats). Never pick northern alluvial plains (UP/Punjab) or desert (Rajasthan).
Desert / Arid Soil (मरुस्थलीय मृदा)	~4.3–4.4%	Rajasthan (राजस्थान - Western/Thar)	Northern Gujarat (Kutch fringe), Southern Haryana & Punjab	Hyper-arid to semi-arid climate, mechanical weathering; sandy texture, alkaline pH, high soluble salts, thick subsoil <i>kankar</i> (calcareous) pan	Trap: Not permanently sterile — highly productive when irrigated (Indira Gandhi Canal proved this for wheat/mustard/cotton). Do not match with wet NE states like Meghalaya!
Forest & Mountain Soil (पर्वतीय / वन मृदा)	~8.6%	Jammu & Kashmir / Ladakh / Himachal Pradesh / Uttarakhand	Sikkim, Arunachal Pradesh, higher slopes of Western & Eastern Ghats	Formed under forest cover on steep mountain slopes; thin, skeletal, immature profile (<i>azonal</i>); acidic with slow humus decomposition in cold climates	Trap: Soil depth varies sharply by relief — thin on steep slopes, deep and fertile in valley bottoms (Kashmir, Doon, Kullu valleys).
Saline & Alkaline Soil (लवणीय एवं क्षारीय / ऊसर / रेह)	~2.1%	Uttar Pradesh (Western & Central canal tracts) / Gujarat (Rann)	Punjab, Haryana, Coastal saline tracts (Sundarbans, Chilika)	High evaporation exceeding rainfall + canal over-irrigation without underground drainage → capillary rise of sodium, calcium & magnesium salts	Trap: Gypsum (CaSO₄·2H₂O) is used for alkaline/sodic soils (replaces exchangeable Na with Ca); Lime (CaCO₃) is for acidic soils (low pH). Never use lime on alkaline/usar soils!

Soil Group (ICAR)	NCERT Area Share	Primary Match State (Commission Anchor)	Alternate / Secondary States (If primary absent)	Why this association? (Geological / Climatic basis)	Classic Prelims Trap / Misconception
Peaty & Marshy Soil (पीट एवं दलदली मृदा)	Localized (~1%)	Kerala (Kuttanad & Alappuzha/Kottayam - local Kari soil)	Sundarbans (West Bengal), Coastal Odisha, Northern Bihar (Koshi floodland), Tarai (Uttarakhand)	Waterlogged conditions in humid tracts with poor drainage; slow anaerobic breakdown of dead vegetation → massive organic accumulation (humus 40–50%)	Trap: Peaty soil is heavy and jet black, but it is not regur/black cotton soil! Regur is dry/semi-arid basaltic alkaline soil; Peaty is submerged, acidic, and organic-heavy.

Local / Regional Formation	Exact Match Location	Geological / Pedological Fact	Commission Trap to Avoid
Karewa Soil (करेवा)	Jammu & Kashmir (Kashmir Valley & Bhadarwah)	Thick lacustrine (lake-bed) silt & clay deposits from Pleistocene glaciation; famous for Zafran (Saffron) , almond, and walnut orchards.	Not river alluvium; not volcanic. It is specifically lacustrine (झील निक्षेप).
Jharkhand Black Soil Pocket	Rajmahal Hills (Santhal Pargana, Jharkhand)	Derived from localized weathering of Rajmahal basaltic traps (Jurassic volcanism), forming black clayey patches.	Major Trap: Do not pick the Damodar or Subarnarekha valley as black soil! Damodar valley is a Gondwana sedimentary rift basin (sandstone/shale/coal), not basalt.
Matasi Soil (मटासी)	Chhattisgarh (Mahanadi Basin plains)	Yellowish-red sandy-loam soil on slightly elevated drained uplands; rich in silica/iron; traditional staple soil for paddy.	Regional state-exam fact (CGPCS / UPPCS); part of the <i>Bhata</i> → <i>Matasi</i> → <i>Dorsa</i> → <i>Kanhar</i> toposequence, not a separate ICAR class.
Bhabar vs Tarai Belt	Sub-Himalayan Foothills (Piedmont strip from Indus to Teesta)	Bhabar: Porous pebble-gravel-boulder strip along Shiwalik foothills where mountain streams vanish underground (coarse, dry surface, unsuited for farming). Tarai: Immediately south of Bhabar; damp, marshy, fine-silted waterlogged belt where lost streams re-emerge; rich in nitrogen & organic matter, excellent for sugarcane and paddy.	Do not invert their spatial order: Shiwalik → Bhabar (north) → Tarai (south) . Bhabar streams disappear; Tarai streams re-emerge.

Local Name	Region in UP	Core Soil Group	Distinguishing Characteristics & Crop Fit
Mar (मार)	Bundelkhand (Jalaun, Jhansi, Hamirpur)	Black Soil (Regur variant)	Highly clayey, deep black, extremely sticky when wet and cracks wide upon drying (<i>self-ploughing</i>); rich in iron & lime; prime for rabi gram, wheat, and oilseeds.
Kabar (काबर)	Bundelkhand (Banda, Hamirpur, Mahoba)	Black Soil (Coarse/compact variant)	Dark brown to blackish-grey; contains more sand than Mar; quickly hardens into unworkable stone-like clods if moisture is lost; highly sensitive to plowing window.
Parua / Padwa (परुआ)	Bundelkhand (Hamirpur, Jalaun, Yamuna ravines)	Red Soil (Sandy-loam variant)	Light reddish to yellowish-brown sandy loam; well-aerated with good drainage; low organic matter; highly productive when provided with light irrigation (mustard, wheat, jowar).
Rakar (राकर)	Bundelkhand (Sloping terrain, ravines, hill bases)	Red Soil (Gravelly/eroded variant)	Coarse, shallow, gravelly red soil formed on eroded granite knolls; low fertility, poor water retention; classified into <i>Moti Rakar</i> (gravelly) and <i>Patli Rakar</i> (fine); suited only for coarse millets (<i>bajra, sesamum</i>).

Local Name	Region in UP	Core Soil Group	Distinguishing Characteristics & Crop Fit
Bhonta / Monta (भोंटा)	Vindhyan tracts (Mirzapur, Sonbhadra, Prayagraj trans-Yamuna)	Vindhyan Red Sandy Soil	Coarse, reddish-brown stony soil formed from Vindhyan sandstone and shale weathering; low fertility, drought-prone; supports dry shrubs, pulses, and rainfed coarse grains.
Matiyar / Dhub (मटियार)	Ganga-Yamuna Doab & Eastern UP	Alluvial Clay-Loam	Fine clayey-loam alluvium with high moisture-retention capacity; prime paddy, wheat, and sugarcane belt.

Stem to type (decision map)

Diagnostic Clue / Trigger Keyword in Stem	Target Soil / Formation	Key Scientific Rationale & Mechanism	Examiner's Trap / Distractor
"Self-ploughing" (स्वतः जुताई) / deep polygonal fissures when dry, swelling & unctuous/sticky when wet	Black Soil / Regur (काली / रेगुर मृदा)	High content of montmorillonite clay mineral; swells substantially on hydration, shrinks and cracks wide on drying, causing natural soil churning.	Not alluvial clay; not peaty soil.
"High moisture retention" / retains subsoil moisture for months; sustains rainfed rabi crops long after rains cease	Black Soil / Regur (काली / रेगुर मृदा)	Extremely fine, tight argillaceous texture and high capillary water-holding capacity.	Do not pick Alluvial or Red soil (both drain significantly faster).
"Bakes like brick" upon exposure to sun and drying / alternate wet-and-dry tropical leaching	Laterite Soil (लेटराइट मृदा)	Latin <i>later</i> = brick; intense tropical leaching washes away silica and bases (<i>desilication</i>), leaving an indurated residual crust of iron and aluminium oxides.	Trap: Red soil looks red from iron diffusion, but does not bake into rock-hard building bricks.
"River-borne transported sediments" / <i>khadar</i> (newer, light, annual flood silt) vs <i>bangar</i> (older, clayey, terrace alluvium)	Alluvial Soil (जलोढ़ मृदा)	Deposited by Himalayan and Peninsular rivers; azonal soil covering ~40% (NCERT) to ~46% of India's plain area.	UP and Bihar are overwhelmingly alluvial; do not assume all plain soils are homogeneous bangar.
"Kankar nodules" (कंकड़ / <i>Calcareous nodules</i>) in lower horizons / impure calcium carbonate concretions	Old Alluvium (Bangar) OR Arid Soil	Downward leaching and precipitation of calcium carbonate (CaCO ₃) in subsoil under sub-humid to arid regimes.	<i>Khadar</i> (new alluvium) is completely devoid of kankar; kankar is characteristic of <i>Bangar</i> .
"Thick sandy texture + high soluble salts + impermeable subsoil kankar layer"	Desert / Arid Soil (मरुस्थलीय मृदा)	Mechanical disintegration dominates over chemical decay under arid conditions; high evaporation draws salts upward; kankar layer restricts root penetration.	Not permanently barren; highly productive when irrigated (e.g., Indira Gandhi Canal command area).
"Thin, skeletal, immature profile on steep slopes" / highly vulnerable to sheet erosion	Mountain & Forest Soil (पर्वतीय मृदा)	Relief and slope prevent horizon maturation; azonal/skeletal; acidic with raw un-decomposed humus in snow-bound belts; fertile in valley bottoms.	Not regur; not transported plains alluvium.
"Waterlogged organic mass" / high humus (40–50%) / dark, heavy, acidic / submerged lowlands	Peaty & Marshy Soil (पीट एवं दलदली मृदा)	Anaerobic conditions in humid wetlands prevent bacterial oxidation of dead plant matter (e.g., Kuttanad 'Kari' in Kerala).	Trap: Dark/black color is from decaying organic humus, not basaltic volcanic ash or regur titaniferous magnetite!

Diagnostic Clue / Chemical Trait in Stem	Soil Type / Problem Class	Required Soil Amendment / Remedy	Examiner's Trap
White salt efflorescence / crust on surface (<i>Reh, Kallar, Thur, Luni, Chopan</i>)	Saline Soil (लवणीय मृदा / <i>Solonchak</i>)	Leaching with fresh water + installing underground sub-surface tile drainage; salt-tolerant crops (barley, cotton).	Do not add lime or chemical precipitants; physical flushing and drainage are essential.

Diagnostic Clue / Chemical Trait in Stem	Soil Type / Problem Class	Required Soil Amendment / Remedy	Examiner's Trap
High exchangeable sodium (Na⁺) / pH > 8.5 / black alkali spots (<i>Usar</i>)	Alkaline / Sodic Soil (<i>क्षारीय मृदा / Solonetz</i>)	Gypsum (CaSO₄·2H₂O) or Iron Pyrites (FeS ₂) to replace exchangeable Na ⁺ with Ca ²⁺ .	Major Trap: Adding Lime (CaCO ₃) worsens alkaline soil! Lime is exclusively for acidic soils.
Low soil pH (< 5.5) / high acidity / strong leaching of bases	Acidic Soil (<i>अम्लीय मृदा - Laterite / Hill soils</i>)	Agricultural Lime (Quicklime CaO, Slaked lime Ca(OH) ₂ , or Limestone CaCO ₃), Dolomite, or Wood Ash.	Never prescribe gypsum for acidic soils; gypsum does not raise soil pH significantly.
Rich in Potash and Lime, but universally deficient in Nitrogen, Phosphorus & Humus	Alluvial & Black Soils of India	Supplement with NPK fertilizers (ideal Indian ratio 4:2:1); green manuring with <i>Dhaincha</i> or <i>Sunnhemp</i> .	Almost all Indian soils (except peaty/forest) are inherently deficient in Nitrogen, Phosphorus, and Humus .
Rich in Iron & Aluminium, severely depleted in Silica, Lime, Magnesia, and Potash	Laterite Soil (<i>लेटराइट मृदा</i>)	Requires heavy doses of manures, lime, and nitrogenous-phosphatic fertilizers to sustain plantation crops.	Candidates confuse Laterite with Black soil; Laterite has been leached of its bases and silica.

Crop / Vegetation Clue in Stem	Direct Target Soil	Typical Geographic Belt
Cotton (कपास) / "Black Cotton Soil" / Rainfed pulses & oilseeds	Black Soil / Regur	Deccan Trap: Maharashtra, Malwa (MP), Saurashtra (GJ), North Karnataka
Zafran / Saffron (केसर / जाफरान) / Almond & Walnut orchards	Karewa Soil (<i>करेवा</i>)	Kashmir Valley (Pampore plateau terraces) & Bhadarwah (J&K)
Cashew nut (काजू) , Tapioca, Tea, Coffee, Rubber, Cinchona	Laterite Soil (<i>लेटराइट</i>)	Western Ghats slopes (Malabar/Kerala, Coorg), Meghalaya hills
Wheat, Rice, Sugarcane, Jute, Mustard (Intensive foodgrain bowl)	Alluvial Soil (<i>जलोढ़</i>)	Indo-Gangetic Plains (UP, Bihar, WB, Punjab, Haryana)
Bajra, Jowar, Ragi, Pulses, Castor, Groundnut (Rainfed coarse crops)	Red Soil (<i>लाल मृदा</i>)	Rayalaseema (AP), Tamil Nadu plains, Chotanagpur plateau, Odisha
Below sea-level paddy farming in waterlogged backwater polders	Kuttanad Kari Soil	Alappuzha & Kottayam districts (Kerala)

Local Nomenclature / Clue Phrase	Exact Regional Setting	Associated Soil Class & Properties
Mar (मार)	Bundelkhand (Jalaun, Jhansi, Hamirpur)	Deep black clayey regur variant; extremely sticky, highly fertile for gram and wheat.
Kabar (काबर)	Bundelkhand (Banda, Hamirpur, Mahoba)	Coarser black soil variant; sets into rock-hard clods upon drying; plowing must be timed precisely.
Parua / Padwa (परुआ)	Bundelkhand (Hamirpur, Jalaun, Yamuna ravines)	Light reddish/yellowish sandy loam; responds well to light irrigation and fertilizers.
Rakar (राकर)	Bundelkhand (Sloping edges, ravine fringes, hill tops)	Shallow, gravelly, eroded red soil; low water retention; suitable only for <i>bajra</i> or sesamum.
Bhonta / Monta (भोंटा)	Vindhyan hill tracts (Mirzapur, Sonbhadra, Prayagraj trans-Yamuna)	Coarse, stony, reddish sandstone-derived soil; low fertility, rainfed crops.
Bhabar (भाबर)	Sub-Himalayan foothill strip (northern UP / Uttarakhand border)	Porous gravel-and-boulder belt; surface water sinks and flows underground; unsuited for crops.
Tarai (तराई)	South of Bhabar strip	Damp, waterlogged silty belt; underground streams re-emerge; high nitrogen & organic matter; sugarcane and paddy belt.
Bhur (भूर)	Ganga-Yamuna Doab elevated banks	Undulating, elevated sandy mounds formed by wind-blown sand deposits during dry summer months.

Local Nomenclature / Clue Phrase	Exact Regional Setting	Associated Soil Class & Properties
Matasi (मटासी)	Chhattisgarh plains (Mahanadi Basin)	Yellowish-red sandy-loam soil on drained higher sites; staple for rainfed rice.

N.1A World soil belts · phytotypes

World Zonal Soils (Dokuchaev Tradition)

Soil Type	Etymology & Literal Meaning	Climate Belt & Biome	Pedological Formation & Characteristics	Classic Prelims Match / Key Trap
Chernozem (चेरनोज़ेम)	Russian: <i>Cherniy</i> (Black) + <i>Zemlya</i> (Earth) = “Black Earth”	Temperate Grasslands (Eurasian Steppes of Ukraine/Russia, North American Prairies)	Annual death of dense fibrous grass roots; cold winters slow bacterial decay → colossal accumulation of dark humus (up to 10–16%) ; rich in calcium and moisture.	The world’s most fertile natural soil. Known as the global “Breadbasket of the World” (wheat belt).
Chestnut (चेस्टनट / भूरी स्टेपी मृदा)	Color of chestnut nut (Dark brown)	Semi-Arid Steppe Margins (drier edge of Chernozem toward deserts)	Similar to Chernozem but receives lower rainfall → lower grass biomass → less humus. The color lightens from pitch black to chestnut brown.	Transitional buffer between humid Chernozem and dry Sierozem.
Prairie (प्रेयरी / Brunizem)	French: <i>Prairie</i> = Meadow	Humid Temperate Grasslands (US Midwest Corn & Soybean Belt)	Higher rainfall than Chernozem; dense tall-grass growth yields rich humus, but higher moisture causes mild base leaching.	Transitional between Chernozem (grassland) and Podzol (forest). Highly productive.
Podzol (पोडज़ोल)	Russian: <i>Pod</i> (under) + <i>Zola</i> (ash) = “Under-ash”	Cold Temperate Coniferous (Taiga / Boreal) forests (Canada, Siberia, Scandinavia)	Acidic pine needles + heavy snowmelt and organic acids leach out iron and humus downward, leaving a bleached, ash-grey silica layer (E-horizon). Strongly acidic and nutrient-poor.	Acidic forest soil. Classic Prelims pairing: Taiga/Coniferous woodland + ash-grey leached soil.
Spodosol (स्पोडोसोल)	Greek: <i>Spodos</i> (wood ash)	Humid cold temperate climates	USDA Soil Taxonomy equivalent for the Podzol family.	Treat Spodosol and Podzol as synonymous in civil services papers.
Laterite (लेटराइट)	Latin: <i>Later</i> = Brick	Hot-Wet Tropical / Monsoon belts with alternating wet and dry seasons	Heavy monsoonal downpours wash away silica and bases (<i>desilication/eluviation</i>), leaving an insoluble crust of Iron (Fe) and Aluminium (Al) oxides . Bakes rock-hard when dry.	Intensely leached & acidic. Poor for cereal grains; ideal for cashew, tapioca, tea, coffee, rubber, cinchona .
Sierozem (सिरोज़ेम)	Russian: <i>Seriy</i> (Grey) + <i>Zemlya</i> (Earth) = “Grey Earth”	Mid-Latitude Temperate Deserts and arid scrub (Central Asia, Great Basin)	Rainfall is so low that plant cover is sparse; negligible humus accumulation → pale ash-grey tint. High evaporation draws lime/calcium salts to the surface.	High in mineral salts/lime (<i>pedocal</i>), but severely deficient in organic humus and water.
Terra Rossa (टेरा रोसा)	Italian: <i>Terra</i> (Earth) + <i>Rossa</i> (Red) = “Red Earth”	Mediterranean Karst Terrain (Italy, Greece, Spain, Croatia)	Formed exclusively on Limestone / Dolomite bedrock. Rainwater dissolves and carries away soluble CaCO ₃ , leaving behind an insoluble residual red clay mantle enriched with ferric iron.	High-Yield Prelims Pair: Limestone weathering + Mediterranean climate.

Phytotypes ↔ Ecological Substrate (Plant Habitat Classification)

Plant Label (पादप वर्ग)	Prefix Meaning	Everyday Science Anchor Word	Natural Habitat & Substrate	Typical Examples
Halophyte (लवणमृद्भिद)	Halo = Salt	Halogen (<i>Chlorine/Iodine form salts like NaCl</i>); Halite (<i>rock salt</i>)	Saline soils / Coastal salt marshes (लवणीय मृदा)	Mangroves (<i>Rhizophora</i> , Sundari), <i>Salicornia</i>

Plant Label (पादप वर्ग)	Prefix Meaning	Everyday Science Anchor Word	Natural Habitat & Substrate	Typical Examples
Cryophyte (शीतमृद्भिद)	Cryo = Extreme Cold / Ice	Cryogenic Engine (ISRO's liquid H ₂ /O ₂ rocket engine); Cryotherapy (ice therapy)	Cold, Polar, Tundra & Snowfields (शीत / टुंड्रा क्षेत्र)	Snow mosses, alpine lichens
Lithophyte (शैलोभिद)	Litho = Stone / Rock	Lithosphere (rocky crust); Paleolithic (Old Stone Age)	Bare Rock Surfaces & Cliffs (चट्टानी धरातल)	Rock-dwelling lichens, ferns, orchids
Xerophyte (मरुद्भिद)	Xero = Dry	Xerox (literally "dry copying" without liquid ink); Xerophthalmia (dry eyes)	Tropical Deserts & Arid Regimes (शुष्क मरुस्थल)	Cactus (<i>Opuntia</i>), Acacia (<i>Babool</i>), Khejri
Hydrophyte (जलोभिद)	Hydro = Water	Hydration, Hydroelectric, Hydrogen	Open Water / Aquatic (floating or submerged)	Lotus, Water Lily, <i>Hydrilla</i> , <i>Eichhornia</i>
Hygrophyte (आर्द्रतोभिद)	Hygro = Moisture / Humidity	Hygrometer (instrument to measure air humidity)	Swamps, Marshes & High Humidity (दलदली / नम स्थल)	Marsh grasses, wetland ferns, mosses
Psammophyte (बालुकोभिद)	Psammo = Sand	<i>Psammology</i> (study of sand)	Shifting Sand Dunes & Sand Fields (रेतीले टीले)	Sand binders, marram grass
Chasmophyte (दरारोभिद)	Chasmo = Fissure / Crevice	<i>Chasm</i> (deep gorge/cleft)	Rock Crevices & Fissures (चट्टानों की दरारें)	Saxifrage, cliff crevicular flora
Oxylophyte (अम्लीभिद)	Oxy = Acidic	<i>Oxygen</i> (historically "acid former")	Strongly Acidic Soils (अम्लीय मृदा)	Heather, peat moss (<i>Sphagnum</i>)
Calciphyte (चूनामृद्भिद)	Calci = Lime / Calcium	Calcium Carbonate (CaCO ₃), <i>Calcification</i>	Lime & Calcium-rich Soils (चूनायुक्त मृदा)	Calciphilous limestone flora

N.2 Soil Types

Alluvial

Chemistry	Status
Potash	Generally rich (Class 11 fact)
Phosphorus	Poor in NCERT Class 11. Some other notes say phosphoric acid is "adequate." Keep the NCERT: poor P fact.
Lime / alkalies	Variable; often adequate
Nitrogen, humus	Generally low

Belt	Region	Star crops
Upper Ganga	Punjab, Haryana, W-UP	Wheat, sugarcane, rice
Middle-lower Ganga	E-UP, Bihar, WB	Rice, jute, pulses
Brahmaputra	Assam	Rice; tea on nearby terraces
East-coast deltas	Mahanadi (महानदी), Godavari (गोदावरी), Krishna (कृष्णा), Cauvery	Rice, coconut fringe
Peninsular valleys	Narmada, Tapi flats; N Gujarat	Mixed food crops

Black (Regur / cotton soil)

Chemistry	Status
Lime, iron, magnesia, alumina	Rich
Potash	Present / variable
Nitrogen, phosphorus, organic matter / humus	Deficient — so "rich in humus" A/R reasons are false
Region	Note

Chemistry	Status
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Maharashtra (Vidarbha)	Classic / most widespread cotton regur
MP / Malwa	Deep black / "lava soil" plateau
Gujarat	Black tracts (Saurashtra–Malwa link)
Telangana / Andhra	Trap extensions; deep in upper Godavari–Krishna
N Karnataka	Cotton–jowar (ज्वार)
Parts of Tamil Nadu	Local patches (often on gneiss/schist too)
Rajmahal (Jharkhand)	Trap black
Himalaya (हिमालय)	Not a black-soil region

Red & Yellow

Chemistry	Status
Iron oxides (colour); potash often adequate	Present
N, P, humus; sometimes lime	Often deficient
Region	Note
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Tamil Nadu, Karnataka, AP/TG	Wide red loams
Western Ghat piedmont	Red loamy belt
Southern Maharashtra	Transition with black soils
Odisha, Chota Nagpur, Chhattisgarh	Upland red
Parts of MP / NE / S Middle Ganga	Local red–yellow

Feature	Red / Yellow	Laterite
Parent / climate	Crystalline rocks, often lower rain	Hot-wet intense leaching
Colour cause	Fe-oxide diffusion	Fe/Al residue after bases/silica leave
Bases	Variable	Strongly depleted
Typical crops	Millet, pulses	Cashew, tea, coffee, plantations

Laterite

Chemistry	Status
Iron oxide, aluminium compounds	Excess / rich
OM, N, phosphate, calcium, lime, silica	Poor / leached
Item	Note
-----	-----
Climate	High T + heavy rain + wet–dry
Process	Intense leaching
Building use	Bricks
Crops	Cashew, tapioca, tea, coffee, rubber, cinchona

Chemistry	Status
Belt	W Ghats / E Ghats summits, Rajmahal, Vindhya–Satpura–Malwa , Odisha, WB, Kerala, Karnataka, TN, Assam–Meghalaya hills
Not typical	Rajasthan, Uttar Pradesh plains

Alkaline (sodic / Kallar)

Feature	Saline	Alkaline (sodic)
Main problem	Soluble salts	Exchangeable Na
pH	Near-neutral to moderately high	Typically high
Structure	Variable	Dispersed, poor
Surface clue	White reh	Often darkish alkali stains
Key remedy	Leach + drain	Gypsum + leach

Problem soils (cause and remedy)

Problem	Cause	Crop effect	Remedy
Acidic	Base leaching, high rain (laterite, NE, E Himalayan foothills)	Nutrient fact-up, Al toxicity	Lime
Saline	Soluble salts, capillary rise	Germination failure	Leaching + drainage
Alkaline	Exchangeable Na, high pH	Structure collapse	Gypsum + leaching
Waterlogged	High water table / canal seepage	Root oxygen stress	Drainage; lined canals
Usar mix	Saline + sodic together	Barren patches	Gypsum + leach + drain + organics

N.3 Texture, Fertility, Erosion, Conservation

Texture	Water	Drainage	Tillage	Example
Sandy	Low	Fast	Easy	Desert
Silty	Medium–high	Medium	Moderate (नरम दल)	Some alluvium
Clayey	High	Slow	Hard when wet	Black regur
Loam	Balanced	Balanced	Favourable	Best farm texture

Particle	Diameter
Clay	less than 0.002 mm
Silt	About 0.002–0.06 mm
Fine sand	About 0.06–2 mm

Condition	pH tendency	Remedy hint
Laterite / NE humid / peaty	Acidic	Lime
Usar sodic	Strongly alkaline	Gypsum
Healthy alluvial loam	Near neutral	Maintain OM
Tea garden ideal	Slightly acidic	Avoid stagnant water + lime excess

Factor	Role
NPK (primary macros, not micros) + secondaries/micros	Direct nutrition

Factor	Role
Humus / OM	Structure, water, cation bank
pH	Controls availability
Organisms (earthworms, microbes)	Cycling, porosity
Moisture & aeration	Root function
Drainage	Prevents salinity / waterlogging

Soil Erosion Stages (The Water Erosion Ladder)

Stage / Type	Primary Agent	Physical Mechanism & Diagnostic Feature	Real-World Example & Field Signature	Exam Decision Rule
1. Splash Erosion (बूंद अपरदन)	Falling Raindrops	Kinetic impact of heavy raindrops acts like micro-bombs, smashing soil aggregates (crumbs) and detaching fine particles.	Bare plowed soil during the very first monsoon downpours.	First / Initiating stage of water erosion.
2. Sheet Erosion (परत अपरदन)	Surface Runoff (Overland flow)	Thin, uniform wash of topsoil across a broad slope; strips fertile organic humus layer without creating channels.	Gently sloping cultivated farmland after continuous heavy rains.	“The Silent Thief” — widespread, devastating, yet goes unnoticed by farmers until crop yields crash.
3. Rill Erosion (रिल / अंगुली अपरदन)	Concentrated Runoff	Runoff gathers into minute, shallow, finger-like incisions (micro-channels) across the field surface.	Sloping fallow fields and orchard floors.	Key Exam Test: Small enough that ordinary tillage/ploughing can easily erase them.
4. Gully Erosion (अवनालिका अपरदन)	Powerful Channelized Runoff	Rills deepen and widen into steep-sided trenches and gullies that cut through subsoil; fragments farmland.	Plateau margins, foothills, upper catchment of Chambal & Yamuna.	Key Exam Test: Deep channels that normal tillage CANNOT erase ; divides land into unusable fragments.
5. Ravine / Badland (बीहड़ / खड्ड भूमि)	Advanced Networked Gullies	Dense, intricate labyrinth of deep interconnected ravines and canyons; completely strips soil cover down to parent rock.	Chambal Basin (Morena, Bhind, Gwalior in MP; Dholpur in RJ; Agra–Etawah–Jalaun in UP).	Classic Badland Topography (उत्खात भूमि).
Stream Channel Erosion	Fluvial Streamflow	Under-cutting, scouring, and bank caving along permanent river corridors during peak flood stages.	Floodplains of Ganga, Brahmaputra, Kosi.	Bank erosion and river meandering.
Wind / Deflation (वायु अपरदन)	High-Velocity Wind	Rolling, saltation, and suspension of dry loose sand; dune migration burying productive lands.	Western Thar Desert (Rajasthan, southern Haryana, Kutch).	Prevalent in arid zones where vegetative cover is absent.

Soil Conservation Methods (Categorized for Effortless Recall)

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Topographic Setting
Contour Ploughing (समोच्च रेखीय जुताई)	Ploughing strictly across (along) contour lines , horizontal to the slope (never up-down). Furrows act as micro-dams that intercept runoff and let water infiltrate.	Gentle farm slopes (up to ~6° slope).
Contour / Graded Bunding (समोच्च मेड़बंदी)	Earthen embankments (<i>bunds</i>) constructed along contours to impound runoff; graded bunds gently channel excess water safely to drainage outlets.	Plains and plateau margins with moderate slopes and permeable soils.
Contour Trenching (समोच्च खाइयाँ)	Excavating continuous or staggered trenches along hill contours to catch surface runoff, arrest silt, and recharge subterranean aquifers.	Barren hill slopes and forest watersheds.

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Topographic Setting
Terracing / Bench Terracing (सीढ़ीदार खेत)	Carving steep hillsides into flat, step-like benches bounded by riser walls; transforms rapid downhill torrents into calm, contained irrigation steps.	Steep mountains & hill tracts (Himalaya, Western Ghats). <i>(Trap: Never select flat plains or Thar desert!)</i> .
Check Dams & Gully Plugs (रोक बांध / गली प्लग)	Small rock, brushwood, or concrete barriers erected across gully beds to dissipate water velocity, trap moving silt, and stabilize gully heads.	Active gullies and ravines (Chambal, Yamuna ravines).

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Farming Setting
Strip Cropping (पट्टीदार खेती)	Alternating strips of erosion-permitting crops (e.g., maize, cotton, jowar) with close-growing cover crops (e.g., grasses, pulses, clover) across the slope.	Open undulating plains and windy agricultural slopes.
Cover Cropping & Mulching (आवरण फसलें एवं पलवार)	Growing dense ground cover crops or spreading crop residue/straw over bare soil; cushions kinetic raindrop impact and halts sheet wash.	Rainfed farms, orchards, and between crop seasons.
Crop Rotation & Mixed Cropping (फसल चक्र एवं मिश्रित खेती)	Alternating tap-rooted and fibrous-rooted crops with leguminous nitrogen fixers; continuously replenishes fertility, preserves crumbs, and avoids seasonal bare exposure.	Intensive cropping systems across India.
Zero / Reduced Tillage + Stubble Retention (शून्य जुताई)	Drilling seeds directly into untilled soil through prior crop stubble; eliminates mechanical soil pulverization and conserves deep soil moisture.	Rainfed rabi crops and intensive rice-wheat tracts.
Early-Maturing Crop Varieties	Planting short-duration cultivars that establish rapid ground canopy and harvest before late-monsoon downpours.	Drought-prone and rainfed dryland belts.

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Geographical Setting
Shelterbelts / Windbreaks (रक्षक मेखला / वातरोधी वृक्ष)	Planting multiple linear rows of dense, deep-rooted trees and shrubs perpendicular to prevailing erosive winds to reduce wind velocity and deflect sandstorms.	Western Rajasthan (Thar) , coastal sandy belts, canal banks.
Sand Fences & Dune Stabilization	Erecting porous brushwood fences and planting xerophytic sand-binders (<i>Calligonum</i> , <i>Cenchrus</i> grass) to immobilize shifting sand dunes.	Active barchan and longitudinal dune fields.

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Key Focus Region
Afforestation & Reforestation (वनीकरण)	Planting native trees with deep root networks that anchor unstable regolith, encourage leaf-litter formation, and regulate watershed hydrology.	Degraded hill catchments and ravine rims.
Regulated Rotational Grazing (नियंत्रित चराई)	Limiting livestock numbers and rotating grazing pastures to prevent overgrazing, which strips grass roots and pulverizes topsoil into loose dust.	Hill pastures, grassland tracts, and scrub forests.
Agro-forestry (कृषि-वानिकी)	Integrating multipurpose timber, fruit, and fodder trees with agricultural crops and livestock on the same land unit.	Drylands and degraded farm tracts.
Checking Jhum / Shifting Cultivation	Replacing unsustainable slash-and-burn cycles with permanent settled bench terracing and horticultural plantations.	North-Eastern Hill states and parts of the Eastern Ghats.

Problem Condition	Scientific Remedy & Action	Typical Focus Tract
Alkaline / Sodic Soils (Usar)	Application of Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) or Iron Pyrites (FeS_2) to replace toxic exchangeable Na^+ with Ca^{2+} .	Canal-irrigated plains of UP, Punjab, Haryana.
Acidic Soils (Low pH)	Application of Agricultural Lime (CaCO_3), Dolomite, or Slaked Lime to neutralize soil acidity and restore base saturation.	Laterite belts (Kerala, Karnataka), humid North-East hills.
Waterlogged & Saline Soils	Installation of subterranean tile drainage + lining of canals to lower elevated water tables and halt capillary salt deposition.	Command areas of major canal irrigation projects.

Key Institutions & Milestones in Soil Conservation

Initiative / Landmark	Core Fact & Prelims Takeaway
Mitti Bachao Andolan (<i>Save the Soil Movement</i>)	Started in 1977 at Hoshangabad (now Narmadapuram), Madhya Pradesh , against waterlogging and soil salinisation caused by the Tawa Dam Project . (<i>Classic Trap: Not Thane, Mysore, or Darbhanga!</i>).
Central Soil Conservation Board (CSCB)	Established in 1953 by the Government of India to formulate soil conservation programs across different agro-climatic zones.
World Soil Day	Celebrated annually on 5 December (birthday of Thailand's King Bhumibol Adulyadej, who championed soil science).

N.4 UP Focus

UP soil belts — expanded teaching map

Group	Where	Character	Crop / tag
Bhabar	Shivalik (शिवालिक) foothill strip (~8–16 km); Saharanpur–Bijnor west	Shallow, pebbly, porous; streams disappear into fans	Poor for crops
Tarai	South of Bhabar (~15–30 km belt)	Streams re-emerge ; moist, N + OM rich, silty–clayey	Rice, wheat, sugarcane
Gangetic alluvium	West–east plains	Deep fertile; khadar/bangar/bhur	~90% of state
Bundelkhand	Jhansi (झांसी), Lalitpur, Hamirpur, Jalaun, Banda, Mahoba, Chitrakoot (चित्रकूट)	Mar/Kabar black; Parua/Rakar red	Local-name favourite
Vindhyan / Baghelkhand fringe	Mirzapur (मिर्ज़ापुर), Sonbhadra (सोनभद्र), parts of Prayagraj (प्रयागराज)–Chandauli	Red–yellow with Fe/Al; Bhonta/Monta	Pulses, millets, oilseeds
Aravali fringe	Parts of Agra	Reddish, Chambal-brought	Local fringe
Usar / Reh	W–central canal plains	Saline–alkaline crusts	Gypsum + drainage
Local name	Colour family	Trait	Core districts
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Mar	Black	Fine clayey and OM-rich	Jalaun, Hamirpur, Jhansi, Banda
Kabar	Dark to grey-black	Coarse and extremely sticky	Lalitpur, Banda, Jalaun, Hamirpur
Parua	Yellowish red	Sandy to clayey; well aerated	Throughout Bundelkhand
Rakar (pathari)	Reddish-brown	Coarse on slopes; slightly acidic	Sloping Bundelkhand

Problem	Hotspot	Fix
Usar / reh	W–central canal tracts	Gypsum + drainage + OM
Ravines / gullies	Agra–Etawah–Jalaun	Check dams, afforestation, grazing control
Wind (summer)	Agra–Mathura (मथुरा)–Etawah	Shelter belts / cover
Waterlogging	Tarai + canal seepage	Drainage; canal lining
Fertility decline	Intensive plains	Rotation, legumes, organic returns

Common Traps — Don't Fall For These

- Edaphic = climate** — FALSE. Edaphic = **soil**.

2. **Black soil in Himalaya** — FALSE. **Deccan Trap / regur**.
3. **Laterite poor in Fe/Al** — FALSE. **Rich** in Fe/Al.
4. **Every red soil is laterite** — FALSE. Laterite needs hot-wet extreme leaching.
5. **Khadar has more kankar** — FALSE. **Bangar** is the kankar-heavy fact.
6. **Gypsum for acid soils** — FALSE. **Lime** for acid; **gypsum** for alkali/sodic.
7. **Irrigation alone cures usar** — FALSE. Needs **drainage** (else more reh).
8. **Mitti Bachao = Darbhanga** — FALSE. **Hoshangabad, MP**.
9. **World Soil Day = 5 June** — FALSE. **5 December**; 5 June = Environment Day.
10. **UP = black-soil state** — FALSE. ~**90% alluvial**; black mainly Bundelkhand Mar/Kabar.
11. **Mar = Bundelkhand red** — FALSE. Mar/Kabar = **black**; Parua/Rakar = **red**.
12. **Bhabar = best rice soil** — FALSE. Pebbly/poor; streams vanish; **Tarai** re-emerges and is moist fertile.
13. **Shelter belt stops gullies** — FALSE. Shelter belt = **wind**; check dams = gullies.
14. **Terra Roxa = Indian regur** — FALSE. **Brazil** basaltic coffee soil.
15. **Contour bunding = desert margins / flood plains** — FALSE. Classic use = **hill / mountain slopes**.
16. **Earthworms are bad for agriculture** — FALSE. They are **ecosystem engineers** that raise fertility.
17. **Clay pores are large** — FALSE. Clay pores are **tiny**; that is why capillary climb and water holding are high.
18. **Organic matter cuts water-holding capacity** — FALSE. OM usually **raises** water holding; irrigation can still **salinise** land.
19. **Terra Rossa forms on granite** — FALSE. On **limestone / dolomite**.
20. **Halophytes = acidic soils** — FALSE. **Saline** soils.
21. **Earthworms have two pairs of hearts** — FALSE. Fertility statement alone is correct.
22. **Alluvial always rich in phosphorus** — FALSE. Class 11 keys **rich potash, poor phosphorus**.
23. **Karewa = Gangetic bangar** — FALSE. Kashmir **lacustrine**; saffron belt.
24. **Rann salt = only canal usar** — FALSE. Rann crust is largely **SW monsoon** salt deposit.
25. **Red soil is India's largest group** — FALSE. **Alluvial** is largest.
26. **Black soil covers 46 lakh km²** — FALSE. Keep ~**16.6%**. The 46-lakh figure cannot fit India.
27. **Alluvium is always rich in phosphorus** — FALSE. Follow **NCERT: poor P**.
28. **Himalayan soil is equal on both faces** — FALSE. **North-facing** gentler slopes hold soil; south faces are often stripped.
29. **IGC only blesses the desert** — FALSE. It can also **create usar** without drainage.
30. **Lime reclaims usar** — FALSE. **Gypsum** for alkali/usar; **lime** for acid soils.
31. **Laterite = Rajasthan / UP plain soil** — FALSE. Hot-wet Ghats / Malabar / plateau edges.
32. **Erosion ladder starts at gully** — FALSE. **Splash** → **sheet** → **rill** → **gully** → **ravine**.
33. **Black soil rich in humus** — FALSE. Suitable for cotton despite **low** N–P–humus.
34. **Tea = alkaline / regur default** — FALSE. Prefers **slightly acidic**, drained soil.

Topic 6 — Agriculture

Current Affairs (this topic)

Year	Fact	Why asked	Source
May 2026	CCEA raised MSP for 14 Kharif (खरीफ) crops, MS 2026–27; common paddy ₹2,441/q (+₹72); Grade A ₹2,461	Latest paddy MSP figure	PIB / CCEA
Static rule	MSP for 22 mandated crops ; CACP recommends, Cabinet decides; target $\geq 1.5\times$ all-India cost (Budget 2018–19)	Institution > old rupees	PIB
2025–26 Kharif	Highest MSP hike then: nigerseed, ragi, cotton, sesamum; extra push pulses/oilseeds/ Shree Anna	Diversification narrative	PIB
Ongoing	PM-AASHA (PSS via NAFED/NCCF); e-NAM ; PM-KISAN ; PMFBY	Scheme names	DA&FW
2025	Delhi bio-decomposer (fungal mix, free) for stubble	Free fungal mix for stubble-to-manure	PYQ / GNCTD

Confused Pairs

Pair	Correct	Trap	Hindi
Kharif vs rabi	Wheat = rabi ; cotton/rice = kharif	Wheat kharif / cotton rabi	गेहूँ रबी; कपास खरीफ
NSA vs GCA	Intensity = GCA/NSA × 100	Invert the ratio	सकल/शुद्ध
MSP vs FRP	Cereals/oilseeds/pulses = MSP; cane = FRP	Call cane MSP	गन्ना = FRP
CACP vs Cabinet	CACP recommends ; Cabinet decides	RBI/NITI recommends	सीएसीपी सिफारिश
Swaminathan vs Kurien	Swaminathan = GR; Kurien = milk	Swap	कुरियन = दूध
Borlaug Nobel	Peace	Agriculture	शांति नोबेल
Yellow vs Golden	Yellow = oilseeds ; Golden = horti + honey	Swap	पीला = तिलहन
Grey vs Black	Grey = fertiliser; Black = petroleum	Swap	ग्रे = उर्वरक
UP vs MH cane	UP quantity ; MH productivity	Coops explain UP yield (they don't)	UP मात्रा; MH उत्पादकता
WB vs UP jute	Jute = WB	UP–Jute	जूट = बंगाल
Kerala rubber vs GJ tea	KL–Rubber correct ; GJ–Tea / Assam–Wheat wrong	Pick GJ tea	रबर = केरल
Ahmedabad vs Mumbai	Ahmedabad in cotton region ; largest mill centre traditionally Mumbai	Ahmedabad = largest centre	कपड़ा केंद्र = मुंबई
Potato CIP	Agra (Singna)	Aligarh	आलू केंद्र = आगरा
Sultana / Gulabi / Kali Champa vs grapes	These three names are guava varieties	Pick grapes because Sultana is also a grape name	अमरूद ≠ अंगूर
Azolla vs pesticide	Azolla = biofertiliser in rice	Insecticide	एजोला = जैव उर्वरक
15 vs 127 vs 20	ACZ 15 / NARP ~127 / AER ~20	Mix counts	15 कृषि-जलवायु
Sen vs Swaminathan	Food entitlements = Amartya Sen	Swaminathan	सेन = एंटाइटलमेंट
Two wheat belts	Ganga–Satluj plains + Deccan black soil	Assam as wheat belt	दो गेहूँ पेटियाँ
White gold vs golden fibre	Cotton = white gold; jute = golden fibre	Swap nicknames	सफेद सोना / सुनहरा रेशा

Pair	Correct	Trap	Hindi
White vs tossa jute	White = <i>capsularis</i> (flood-tolerant); tossa = <i>olitorius</i> (upland, better fibre)	Swap flood / upland	सफेद / टोसा जूट
Cotton species vs staple	Four species (<i>arborescens</i> / <i>herbaceum</i> / <i>hirsutum</i> / <i>barbadense</i>); <i>hirsutum</i> = bulk; ELS = <i>barbadense</i>	"India grows only short staple"	चार प्रजाति; ELS कम
Sikkim organic	Sikkim = first fully organic state	Call any NE state first	सिक्किम जैविक
Rice rank	Textbook #2 after China ; 2024–25 Gol #1 claim	Mixing the usual rank with the 2025 headline	रैंक अलग रखें
WB rice vs PB yield	WB often volume leader; Punjab yield leader	Punjab = largest producer always	बंगाल मात्रा; पंजाब उपज
DSR vs transplant	DSR = seed in field, less water; transplant = nursery + puddle	DSR needs more standing water	डीएसआर = कम पानी
Ratoon vs rotation	Ratoon = cane from stubble ; rotation = change the crop	Call wheat a ratoon crop	रैटून = गन्ना
Mixed vs double vs inter	Mixed = crops+livestock; Double = two crops/year; Inter = same-time rows	Swap definitions	मिश्रित ≠ डबल
Parallel cropping	Wheat + mustard	Potato + rice as parallel	समान्तर = गेहूँ+सरसों
15 ACZ vs 20 AER	Planning ACZ 15 ; NBSS AER 20 ; Sengupta micro 60	Swap 15/20	15 जलवायु; 20 पारिस्थितिक
Pantnagar year	First agri uni 1960 (Nehru; later GBPUAT)	1950 / 1970	पंतनगर 1960
Randhawa vs Swaminathan	History of Indian Agriculture = Randhawa	Credit Swaminathan	रंधावा पुस्तक
Agmark Act	1937 grading mark	Call it co-op egg mark	एगमार्क 1937
Moisture 14%	FAO safe storage ≤ 14%	16 / 18 / 20%	नमी 14%
Cowpea green manure	Highest N% among common options	Pick dhaincha only	लोबिया N%
Fertigation phosphate	Avoid rock / super phosphate in fertigation	"All P fertilisers OK"	फर्टिगेशन P
Coimbatore cane	Sugarcane Breeding Institute	Lucknow only	कोयंबटूर
Pratapapur mill	First sugar mill 1903	Mawana / Balrampur first	प्रतापपुर
Contract farming	Pioneer Punjab	Pick Haryana / TN	पंजाब अनुबंध

N.1 Land Use, Farming Types, Seasons

Class	What it means
Forests	Area reported as forest
Barren and unculturable land	Rocky, desert, or otherwise unfit for crops
Permanent pastures and grazing land	Grasslands kept for grazing
Miscellaneous tree crops and groves	Orchards, groves, and tree crops outside net sown
Culturable waste	Land that could be ploughed but is currently unused
Fallow (current + other)	Cultivable land left unsown for a season or longer
Net sown area (NSA)	Land sown with crops at least once in the year

Farming types

Type	Recall
Primitive subsistence	Family food; includes jhum
Intensive subsistence	Small farms, high labour, rice/wheat
Commercial grain	Market surplus (GR wheat belts)
Plantation	Estate perennial — tea/coffee/rubber
Mixed farming	Crops and livestock together
Dairy	Milk commercial
Mediterranean (world)	Winter rain; citrus/olives/vines
Truck farming	Vegetables near cities
Dryland farming	Rainfed millets, pulses
Wetland / irrigated	Standing water or canal/tube-well
Contract farming	Punjab pioneer in India

Cropping patterns

Pattern	Recall
Mixed cropping	Two+ crops together at once
Intercropping	Fixed row ratios
Parallel cropping	Parallel rows; classic wheat + mustard
Double cropping	Two+ crops in one crop year (sequential)
Crop rotation	Sequential change on same land
Sequential cropping	One after another same year
Relay cropping	Second sown before first harvest
Ratoon cropping	From stubble (sugarcane)
Multiple cropping	More than one crop / year → intensity

Seasons

Season	Months	Typical crops
Kharif	Jun–Oct (SW monsoon (दक्षिण-पश्चिम))	Rice, maize, jowar, bajra, ragi, cotton, jute, groundnut, soybean, tur, sesame
Rabi	Oct–Mar	Wheat, barley, gram, mustard, peas, linseed
Zaid	Mar–Jun	Watermelon, cucumber, fodder, vegetables
Long / perennial	—	Sugarcane (10–18 months); tea, coffee, rubber

N.2 Food Crops

Rice

Belt	States
Lower Ganga / delta	West Bengal (multi-rice + jute)
Middle Ganga	E-UP, Bihar
Coastal east	Odisha, AP/TG, TN
Irrigated NW	Punjab, Haryana, W-UP

Belt	States
Krishna–Godavari delta	AP — “Rice Bowl of India”
Chhattisgarh plain	“Rice bowl” tag

Maize & millets (nutri-cereals)

Crop	Leading state	Note
Jowar (ज्वार)	Maharashtra	Some rabi jowar in the peninsula
Bajra (बाजरा)	Rajasthan	Lowest-rainfall cereal
Ragi	Karnataka	Ca-rich; red/lateritic soils OK

Pulses

Pulse	Season	Note
Gram / chana	Rabi	MP often among leaders
Masoor (lentil)	Rabi	Cool season
Tur / arhar	Kharif	Central India; red gram origin often tagged India
Moong, urad, moth	Kharif	Short-duration; drylands; urad can be kharif and rabi in the south
Peas	Rabi	With wheat belts

N.3 Cash, Fibre & Plantation

Oilseeds — Yellow Revolution

Crop	Season	Core state
Groundnut	Kharif	Gujarat
Mustard / rapeseed	Rabi	Rajasthan
Soybean	Kharif	Madhya Pradesh (oilseed, not pulse)
Sesame	Kharif	Drylands
Sunflower	Flexible	KA/MH/AP
Castor	Drought-hardy	GJ/RJ
Linseed	Rabi	Central/north
Safflower	Cool dry rabi	Deccan
Niger	Minor	Tribal (आदिवासी)/peninsula

Cotton (white gold)

Species	Common tag	What it gives
<i>Gossypium arboreum</i>	Desi / Old World	Short staple; drought- and pest-hardy
<i>Gossypium herbaceum</i>	Desi / Old World	Short–medium; harsh rainfed tracts (GJ / KA pockets)
<i>Gossypium hirsutum</i>	American / Upland	Bulk of India’s crop ; medium to long staple; most Bt hybrids
<i>Gossypium barbadense</i>	Egyptian / Sea Island	Extra-long staple (ELS) ; small premium pockets in the south

Class	Approx. length	India note
Short staple	≤ about 20 mm	Desi types; coarse cloth, blending, surgical cotton
Medium staple	about 20.5–24.5 mm	Large share of domestic mill use

Class	Approx. length	India note
Medium-long staple	about 25–27 mm	Modern mill demand
Long staple	about 27.5–32 mm	High-speed spinning; MSP is also quoted for a long-staple grade
Extra-long staple (ELS)	about ≥32.5 mm	Premium yarn; India still imports much ELS (Egypt (मिस्र) / USA style)

Jute (golden fibre) & mesta

Type	Botanical name	Field note
White jute	<i>Corchorus capsularis</i>	Stronger flood tolerance; classical lowland / waterlogged delta crop
Tossa jute	<i>Corchorus olitorius</i>	Softer, usually better fibre and higher yield; prefers uplands and hates prolonged flood
Mesta	<i>Hibiscus cannabinus</i> / <i>H. sabdariffa</i> (related fibre, not true jute)	Grown on drier tracts as a jute substitute; coarser fibre

Tea · Coffee · Rubber · Spices

Crop	Climate	Site	India
Tea	20–30°C; 150–300 cm	Well-drained slopes (no stagnant water)	Assam (volume), WB Darjeeling GI , Nilgiris/TN–KL; plantation
Coffee	15–28°C; 150–250 cm	Shade , well-drained	KA > KL > TN (Baba Budan / Coorg / Wayanad / Nilgiri)
Rubber	25–35°C; >200 cm	Humid equatorial; <i>Hevea brasiliensis</i> latex	Kerala (also TN, KA, A&N, Tripura pockets)
Pepper / cardamom	Hot wet Ghats	Hills	Kerala largest
Chilli	Warm Deccan	—	AP/TG strong — not Kerala-only
Turmeric / ginger	Warm	South + NE	Many GIs

Coffee species	Note
Arabica	Higher elevation, milder cup, more pest-sensitive
Robusta	Lower elevation, stronger cup, hardier

N.4 Fruits, Potato, GI

GI / variety	Place
Darjeeling Tea (दार्जिलिंग चाय)	WB hills
Basmati	North India GI belt
Alphonso	Maharashtra
Banganapalle	AP
Gir (गीर) Kesar	Gujarat
Dasheri / Langra	UP
Malabar Pepper	Kerala
Kashmiri saffron	J&K
Nashik grapes	MH
Coorg coffee / Alleppey cardamom	KA / KL GI note

N.5 Agricultural Regions & Agro-climatic Zones

System	Count	Use
Planning Commission ACZ	15 (~72–73 sub-zones)	Broad planning
ICAR NARP	~127	Research / extension
NBSS&LUP AER	~20	Soils + LGP (Length of Growing Period)

15 Agro-climatic Zones (Planning Commission)

#	Zone	Core	Agri tag
1	Western Himalayan	J&K, HP, UK	Apple, valley cereals
2	Eastern Himalayan	Assam, NE, Sikkim, N WB	Tea, rice, spices, jhum
3	Lower Gangetic Plains	West Bengal	Rice, jute, mustard
4	Middle Gangetic Plains	E-UP, Bihar	Rice–wheat, maize, cane
5	Upper Gangetic Plains	W/central UP	Wheat–rice–cane (GR)
6	Trans-Gangetic Plains	PB, HR, Delhi, parts RJ	Intensive wheat–rice
7	Eastern Plateau & Hills	CG, JH, OD uplands	Rice, millets, rainfed
8	Central Plateau & Hills	MP, RJ, Bundelkhand edge	Soybean, wheat, gram
9	Western Plateau & Hills	MH upland	Cotton, jowar, pulses
10	Southern Plateau & Hills	Interior AP/TG, KA, TN	Groundnut, millets, cotton
11	East Coast Plains & Hills	Coastal OD–AP–TN	Rice, coconut, cashew
12	West Coast Plains & Ghats	KL, coastal KA, Goa (गोवा), Konkan	Rice, coconut, spices, rubber, coffee
13	Gujarat Plains & Hills	Gujarat	Cotton, groundnut
14	Western Dry Region	W Rajasthan	Bajra, pulses, livestock
15	Islands Region	A&N, Lakshadweep (लक्षद्वीप)	Coconut, horticulture

Agro-ecological regions (NBSS&LUP ≈ 20)

Ecosystem	Example AER tags
Arid	Western Himalayas cold arid; Western Plain–Kachchh; Deccan Plateau (दक्कन पठार) arid
Semi-arid	Upper / Middle Ganga plains; Rajasthan–Gujarat plains; Deccan red–black / red sandy
Sub-humid	Eastern Plateau (Satpura (सतपुड़ा)–Mahanadi (महानदी), Bundelkhand, red–laterite (लेटराइट)); Lower Ganga; warm W Himalaya (हिमालय); Bengal Basin
Humid–perhumid	Assam–N Bengal plains; Eastern Himalayas; Purvanchal (पूर्वांचल) hills (पूर्वांचल पहाड़ियाँ)
Coastal	Eastern Coastal Plains + A&N; Western Ghats (पश्चिमी घाट) coastal plains and hills

N.6 Irrigation & Command Area Development

Type	Best terrain	Risk / note
Perennial canal	Alluvial plains (PB–HR–UP; IGC RJ). Clayey beds leak less	Waterlogging / usar
Inundation canal	Flood plains	Unreliable timing

Type	Best terrain	Risk / note
River lift	Where canal flow is irregular; common in peninsular belts; Pattiseema = Godavari→Krishna	Pumping cost
Tube well	Alluvial aquifers; GR NW + W-UP	Groundwater depletion
Dug well	Local / hard-rock	Limited yield
Tank	Peninsular hard rock	Siltation
Drip	Horti, cane, cotton, veg; 40–60% water save; less weed / erosion	High efficiency; capital cost
Sprinkler	Sandy / undulating	Wind drift

N.7 Seeds, ICAR, Universities, Pests & Practices

Schemes (practice)

Scheme	Key point
Soil Health Card	Soil fertility / balanced nutrient advice
PMKSY (incl. more crop per drop)	Irrigation access / water efficiency
PKVY (Paramparagat Krishi Vikas Yojana)	Organic / traditional farming clusters
Kisan Credit Card	Crop credit via commercial, cooperative and RRBs
e-NAM / national agri market push	Better price discovery
NFSM	Rice, wheat, pulses (later oilseeds / millets) productivity

Disease / problem	Cause
Citrus canker	Bacteria
Red rot of sugarcane	Fungus
Potato “Krishnakant”	O₂ deficiency
Wheat “Sahu”	Insect
Wheat Karnal bunt	Fungus (<i>Tilletia indica</i>)
Wheat rusts	Yellow / brown / black
Paddy Khaira	Zn deficiency (ZnSO ₄ spray)
Potato late blight	Classic potato disease
Arhar wilt	Wilt disease

N.8 Animal Husbandry

Breed	Animal	Tag
Murrah	Buffalo	Haryana / high yield
Bhadawari	Buffalo	UP / Bundelkhand note
Gir	Cattle	Gujarat
Sahiwal	Cattle	Punjab-region note; high milk
Tharparkar	Cattle	Western Rajasthan border
Deoni	Cattle	Dual-purpose
Jamunapari	Goat	UP renown; high milk

Breed	Animal	Tag
Barbari / Beetal	Goat	Other common breeds

N.9 Jhum, Organic, Precision

System	Core	Where / tool
Jhum (shifting)	Slash-burn, 1–3 yr crop, then fallow	NE (jhum); podu / penda (AP/Odisha); bewar / dahiya / mashan (MP–CG); kumari (W Ghats); waltre (SE Rajasthan); poonam (Kerala); kuruwa (Jharkhand); khil (Himalayan belt)
Organic	No synthetic agrochemicals (standards)	Compost, rotation, Azolla, Rhizobium; Sikkim = first fully organic state (declared 2016)
Precision	Site-specific inputs	GIS / GPS / sensors / drones — not “stop irrigation”

N.11 Green Revolution & Colour Revolutions

Person	Role
M.S. Swaminathan	GR in India; Evergreen idea
Norman Borlaug	HYV wheat; Nobel Peace not Agriculture
Vergheze Kurien	White / milk
L.K. Jha	Economic Administration Reforms
Morarji Desai (आमिल)	Bank social control
Amartya Sen	Food entitlements — not Swaminathan
Hiralal Chaudhuri	Blue Revolution (fisheries) tag
Revolution	Sector
-----	-----
Green	Food grains (wheat → rice)
White	Milk (Kurien / Flood / NDDDB)
Blue	Fisheries / aquaculture
Yellow	Oilseeds
Golden	Horticulture and honey
Silver	Eggs / poultry
Pink	Onion / meat / prawn (context)
Grey	Fertilisers
Black	Petroleum
Brown	Sometimes leather / cocoa / biofuel talk — weaker association
Rainbow	Integrated multi-sector
Evergreen	Sustainable productivity

Colour Revolutions — must-score set

Colour	Sector
Golden	Horticulture and honey
Grey	Fertilisers
Yellow	Oilseeds

Colour	Sector
Black	Petroleum

UP Focus (compact)

Theme	Fact
Sugarcane	Quantity leader often; productivity < MH; Sugar Bowl tag
Wheat–rice	Western UP GR (Upper Gangetic / Zone 5)
Potato	Leading producer; CIP Agra (आगरा) not Aligarh
Mango	Dasheri / Langra; Amrapali hybrid
Dairy	Large bovine base; Jamunapari goat; top milk state
Mentha oil	UP is the classic near-total producer
Not	Jute / tea / rubber leader
SAU	Pantnagar (first, 1960) ; Kanpur / Meerut

Common Traps — Don't Fall For These

1. **Wheat = kharif / cotton = rabi** — FALSE. Wheat **rabi**; cotton **kharif**.
2. **CACP / RBI / NITI decides MSP** — FALSE. CACP **recommends**; **Cabinet** decides.
3. **Cane price = cereal MSP** — FALSE. Cane = **FRP** (+ possible SAP).
4. **Borlaug Nobel = Agriculture** — FALSE. **Peace**.
5. **Kurien = Green Revolution** — FALSE. Kurien = **White / milk**; Swaminathan = GR.
6. **Yellow = horticulture** — FALSE. Yellow = **oilseeds**; Golden = horticulture **and honey**.
7. **UP–Jute / Gujarat–Tea / Assam–Wheat** — FALSE. Correct pair in that set: **Kerala–Rubber**.
8. **Ahmedabad = India's largest cotton textile centre** — FALSE. In cotton *region* yes; largest mill centre traditionally **Mumbai**.
9. **Maharashtra coops explain UP's lower cane yield** — FALSE. UP leads quantity; Maharashtra leads productivity. Cooperatives do not cause that yield gap.
10. **CIP potato centre = Aligarh** — FALSE. **Agra (Singna)**; UP is potato leader.
11. **Sultana/Gulabi/Kali Champa = grapes** — FALSE. They are **guava** varieties. The name Sultana is the grape mix-up.
12. **ACZ = 127 or 20** — FALSE. Planning ACZ = **15**; NARP ~127; AER ~20.
13. **Azolla = insecticide** — FALSE. **Biofertiliser** in flooded rice (2023).
14. **Food entitlements = Swaminathan** — FALSE. **Amartya Sen**.
15. **Tea = zaid cereal** — FALSE. **Plantation** perennial.
16. **Only one wheat belt** — FALSE. Ganga–Satluj plains **and** Deccan black-soil wheat.
17. **Cotton = golden fibre** — FALSE. Cotton = **white gold**; jute = **golden fibre**.
18. **Any NE state = first organic** — FALSE. **Sikkim** is the first fully organic state.
19. **India is always rice #1** — FALSE. Textbook rank is **#2 after China**. The **2024–25** Gol claim of first place is a separate recent claim.
20. **Punjab is always India's largest rice producer** — FALSE. It is the **yield** king; **West Bengal** often leads **volume**.
21. **DSR needs more standing water than transplanting** — FALSE. DSR is the **water-saving** method.
22. **BGREI is a Punjab second Green Revolution** — FALSE. It is an **eastern** rice-system programme.
23. **India has 20 agro-climatic and 15 agro-ecological regions** — FALSE. Counts are **swapped** (ACZ 15; AER 20).
24. **First agri university = 1950 / Ludhiana** — FALSE. **Pantnagar, 1960** (later GBPUAT).

- 25. **History of Indian Agriculture = Swaminathan** — FALSE. Author is **M.S. Randhawa**.
- 26. **Mixed farming = two crops only** — FALSE. Mixed farming = crops **plus livestock**.
- 27. **Double cropping = two crops at the same time** — FALSE. That is closer to inter / parallel cropping; double = two crops in **one year**.
- 28. **Agmark = egg co-op** — FALSE. It is a **1937** quality grading mark.
- 29. **Fertigation works best with rock phosphate** — FALSE. Avoid rock / super phosphate (precipitation).
- 30. **Blue Revolution = poultry** — FALSE. Blue = **fisheries / pisciculture**.
- 31. **Sugarcane breeding HQ = Lucknow only** — FALSE. Classic breeding institute is **Coimbatore**.
- 32. **Coffee Board HQ = Hyderabad** — FALSE. **Bengaluru**.
- 33. **Contract farming pioneer = Haryana** — FALSE. Classic pioneer is **Punjab**.
- 34. **FAO safe grain moisture = 18%** — FALSE. About **14%**.
- 35. **Seed Village = ban buying seed** — FALSE. It trains farmers to produce quality seed for self-use and neighbours at affordable cost.

Topic 7 — Natural Vegetation & Biodiversity Geography

Current Affairs (this topic)

Year	Fact	Why asked	Source
5 Jun 2026	Surha Tal / JP Narayan BS (Ballia) = India's 100th Ramsar ; UP 13th	Count + latest UP site	MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) / PIB
2025–26	UP Ramsar adds Patna (पटना) BS, Sheikha Jheel, Surha Tal (full names in N.6)	State list (राज्य सूची)	MoEFCC
Budget 2023–24	MISHTI mangrove restoration — not a 2019 start	Mangrove scheme year	Budget
Static	Olivia (Coast Guard (तटरक्षक बल), Odisha, Nov–May); coral warming 2024 PYQ	Coastal ops	ICG / PYQ
ISFR	Use the report year in the question ; 2017 = 21.54% is a frozen PYQ	Figure trap	FSI

Lakes-as-wetlands. This file keeps the **geography** facts.

Confused Pairs

Pair	Correct	Trap	Hindi
Thornthwaite vs Köppen	Vegetation = climate index → Thornthwaite	Köppen	थॉर्नथ्वेट
Evergreen vs moist deciduous	Evergreen >200 ; moist 100–200, largest area	Evergreen = largest	पर्णपाती सबसे बड़ा
Simlipal	Odisha	Madhya Pradesh (2018)	सिमलीपाल = ओडिशा
Nokrek	Meghalaya	Manipur (2025)	नोकरेक = मेघालय
Nanda Devi	Uttarakhand	Uttar Pradesh	नंदा देवी = उत्तराखंड
Botanical garden	Ex-situ	In-situ	वनस्पति उद्यान = बाह्य
Dudhwa district	Lakhimpur Kheri	Bahraich / Pilibhit	दुधवा = लखीमपुर
NP vs TR in UP	Only NP = Dudhwa; Pilibhit/Amangarh/Ranipur = TR	Extra NPs	केवल एक NP
Chandraprabha	Chandauli	Sonbhadra (2023)	चंदप्रभा = चंदौली
Hangul	Dachigam / Kashmir	Assam rhino belt	हांगुल = कश्मीर
Keibul Lamjao	Loktak, Manipur	Wular/Dal	केइबुल = लोकटक
Diversity vs latitude	Decreases equator → poles	Increases toward poles	विविधता ध्रुव की ओर घटती
MISHTI year	Budget 2023–24	2019	मिश्टी ≠ 2019
Olivia season	Nesting Nov–May	August	ऑलिविया ≠ अगस्त
Ramsar NOT UP	Rudrasagar Tripura; Sultanpur HR; Surinsar–Mansar J&K	Mark as UP	रुद्रसागर = त्रिपुरा
Ballia	Near-nil forest ; Surha Tal Ramsar	"No wetland"	बलिया = वन नहीं, रामसर है

Pair	Correct	Trap	Hindi
Forest cover vs RFA	ISFR cover ≠ legal recorded forest	Mix the two	आवरण ≠ अधिसूचित
Agasthyamalai	Kerala and TN	Only one state as if exclusive	अगस्त्यमलाई = केरल–TN
Nilgiri BR	First Indian BR (1986)	Call Nanda Devi first	नीलगिरि पहला
Hotspot count	Four Indian hotspots	Mix with 18 BRs	चार हॉटस्पॉट
Hotspot names	Himalaya + WG–Sri Lanka + Indo-Burma + Sundaland (Nicobar)	Split Himalaya east/west, or add Andaman as if it were Sundaland	निकोबार = सुंडालैंड
Wet evergreen rain	NCERT >200 cm ; many notes also use >250 for wet evergreen	Mix the two numbers in one question	200 / 250
Dry evergreen vs wet evergreen	Dry evergreen = TN coast , NE monsoon ~100 cm	Call it Western Ghat rain forest	कोरोमंडल शुष्क सदाबहार
Chir vs deodar belt	Chir = subtropical pine 1000–2000 m west; deodar = temperate	Swap belts	चीड़ निचला; देवदार ऊँचा
Keystone vs flagship	Keystone = ecosystem collapse if lost; flagship = public mascot	Call panda a keystone fact for India	कीस्टोन ≠ फ्लैगशिप
Flame of the forest	Butea monosperma / Palash / Dhak; UP state flower	Bauhinia / Jacaranda / Teak	पलाश = UP फूल
Katha wood	Khair	Sal / babool / saaja	कथा = खैर
Tree-line E vs W	Lower in Western Himalaya	Higher in west	पश्चिम ट्री-लाइन नीची
Himalaya aspect	Southern slopes denser (more rain + sun)	Northern denser	दक्षिणी ढाल सघन
Vembanad mangrove	Kerala	Karnataka	वेम्बनाड = केरल
Mahogany belt	Tropical evergreen (not Himalaya)	Call it Himalayan with fir/spruce	महोगनी ≠ हिमालय
Khejri vs Dhak	Khejri = desert social forestry; Dhak/Palash = flame of forest	Swap uses	खेजड़ी ≠ पलाश

N.1 Natural Vegetation & Forest Types

Type	Rain (cm)	Region	Trees / trait
Tropical evergreen (Selvas)	>200	W Ghats, NE, Kerala, WB, Andamans	Multilayer; rosewood, mahogany, ebony, cinchona, rubber, bamboo; no complete dry leaf-shed
Semi-evergreen	Slightly less than pure evergreen	W Ghats / NE / Andamans mix zone	Evergreen + deciduous mix
Moist deciduous	100–200	Central India, Terai (तराई)–Bhabar (भाबर), E slopes of W Ghats	Teak, sal , shisham, neem, mango; largest area ; sheds in dry season
Dry deciduous	70–100	UP, AP, MH, KA, TN; drier peninsula	Rosewood, mango, sandalwood, mahua
Thorn / scrub	<50–70	RJ, GJ, SW Punjab–Haryana, rain-shadow / W AP	Babool, ber, khair, khejri, palas, wild date; xerophytes
Montane	altitude	Himalaya; S Indian hills (sholas)	Belts, not plains rainfall ladder; deodar 1500–2500 m west
Littoral / mangrove	tidal	Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र), Godavari (गोदावरी)–Krishna (कृष्णा) deltas + coasts	Sundari; pneumatophores; saline (लवणीय)

Head	Sub-types
Moist tropical	Wet evergreen · Semi-evergreen · Moist deciduous · Littoral and swamp
Dry tropical	Dry evergreen · Dry deciduous · Thorn
Montane sub-tropical	Broad-leaved hill · Moist hill (pine) · Dry evergreen
Montane temperate	Wet temperate · Himalayan moist temperate · Himalayan dry temperate
Alpine	Sub-alpine · Moist alpine scrub · Dry alpine scrub
Share of forest area (class figures)	Type
-----	-----
About 37%	Tropical moist deciduous (largest)
About 28%	Tropical dry deciduous
About 8%	Tropical wet evergreen
About 6%	Sub-tropical moist hill (pine)
About 4%	Tropical semi-evergreen

World vegetation matches

Name	Where	Memory
Maquis	Mediterranean coast	Maquis = Mediterranean
Chaparral	California	Both start with C
Matorral	Chile	Both have "l"
Fynbos	South Africa (Cape)	Fynbos = fine bush of the Cape

Plant	Belt	Why
Epiphytes (orchids on trees)	Equatorial rainforest	Need constant wet heat and a canopy
Acacia	Sahara / arid Africa	Thorn tree of dry tracts
Baobab	Savanna	Fat-trunk tree of grassland with scattered trees
Cedars	Mediterranean	Atlas / Lebanon cedar, not equatorial forest

N.2 Type Cards

Evergreen & semi-evergreen

Note	Standard figure	Alternate note you may see
Rain	>200 cm	Wet evergreen often written >250 cm
Mean annual temperature	Above about 22°C	Some notes use about 25–27°C
Humidity	Near 70%	Some notes write >77%
Dry season	Almost none / very short	Short dry season in wet-evergreen notes

Deciduous (monsoon forest)

Type	Rain (usual)	Area	Signature trees
Moist deciduous	100–200 cm	Largest share (~ 37%)	Teak, sal , shisham, neem, mango
Dry deciduous	70–100 cm (some notes widen to 100–150)	UP, AP, MH, KA, TN; drier peninsula	Rosewood, mango, sandalwood, mahua

Type	Rain (usual)	Area	Signature trees
Whole deciduous family (wide notes)	Sometimes written 70–200 cm	Same monsoon-forest family	Split by rain, not by a different “family”

Thorn / scrub (arid / tropical thorn)

Note	Figure / trait
Rain	Often <75 cm (NCERT / many notes <70 cm)
Humidity	<50%
Temperature	About 25–30°C
Tree height	Only about 6–10 m , scattered
Dominants	Acacia and Euphorbia

Montane (altitude elevator)

Height (approx.)	Belt	What you see
Foothills	Tropical / sub-tropical deciduous	Warm lower slopes
About 1000–2000 m	Wet temperate oak–chestnut	Broad-leaf temperate start
About 2000–3000 m (southern slopes)	Chir / deodar / pine / spruce / fir	Conifer timber belt
Higher	Alpine juniper / rhododendron / birch	Scrub toward snowline
Highest usable pasture	Bugyals (alpine meadows)	Then permanent snow

Point	Eastern / Central	Western / north-western
Tree-line	About 3600–3800 m	About 3300–3600 m (lower)
Alpine climb	Climbs higher	Stops earlier
Why	Nearer the equator and the sea ; wetter	Drier monsoon side
Resource note	Wet forests	Forests are a major land resource of the western Himalayan region

Mangrove / tidal (littoral & swamp)

Mangrove pocket	State
Achra–Ratnagiri	Maharashtra
Coondapur	Karnataka
Pichavaram	Tamil Nadu
Vembanad	Kerala (not Karnataka)

Fact	Note
Flame of the forest	Butea monosperma = Dhak / Palash = UP state flower
Katha	From khair wood
Khejri	Multipurpose arid / social-forestry tree
Birch	Himalayas; writing-bark tradition
Fern	Vascular, spore-bearing; no flowers / no seeds
Oryx vs Chiru	Oryx = hot arid; Chiru = cold high-mountain steppe (स्टेपी) / semi-desert
Teak–Central India	Major teak expansion in MP–CG–Odisha; MP often leads dense deciduous area
Cinchona	Evergreen / Terai–NE match; not Chhattisgarh

N.3 Forest Distribution & ISFR

Active Forest Benchmark (For 2025 & 2026 Exams)

Parameter	Latest Figure (For 2025–26 Exams)	% of India's TGA	Trend / Change
Forest Cover	7,15,343 sq km	21.76%	Net increase across country
Tree Cover	1,12,014 sq km	3.41%	Trees outside forest (TOF)
Total Green Cover	8,27,357 sq km	25.17%	Net gain of +1,445 sq km

Metric	Official Figure	% of UP's Total Area	Notes
Forest Cover	15,045.80 sq km	6.24%	Satellite canopy ≥ 10%
Tree Cover	8,950.92 sq km	3.72%	Small patches & roadside trees
Total Green Cover	23,996.72 sq km	9.96%	UP gained +559 sq km (#2 in India!)

Metric	Official Figure	% of UK's Total Area	Notes
Forest Cover	~24,305 sq km	45.44%	Actual satellite canopy mapped
Recorded Forest Area (RFA)	~38,000 sq km	~71.05%	Legally notified forest in government records
Growing Stock Rank	Rank #1 in India	—	Highest increase in forest growing stock nationwide

N.4 Plant Acclimatization & Adaptations

Type	Hindi	Meaning	Example
Xerophyte	मरुद्भिद	Dry-adapted	Thorn forest: spines, succulence, deep roots
Hydrophyte	जलोद्भिद	Water-adapted	Lotus, hydrilla
Mesophyte	समोद्भिद	Moderate moisture	Most crops / deciduous understorey
Halophyte	लवणमृद्भिद	Salt-adapted	Mangroves
Epiphyte	अधिपादप	Grows on another plant	Equatorial orchids

Feature	Hindi	Meaning	Example
Pneumatophore	वात-मूल / श्वसन मूल	Breathing root	Mangrove roots that stick up out of mud

N.5 Protected Areas — Framework

UNESCO MAB and India's Biosphere Reserve Programme

Category	Fact
National Park	Strictest; no private rights in core idea; habitat + species
Wildlife Sanctuary	Wildlife first; some regulated rights possible
Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)	Core (strict) + buffer + transition (settlements/use) — UNESCO MAB idea; India list is national programme too
Conservation / Community Reserve (सामुदायिक आरक्षित क्षेत्र)	WLPA later additions — community/landscape gaps
Tiger Reserve	NTCA unit (core+buffer); may contain NP + WLS

High-frequency NP / species

Name	State	Extra
Dachigam	J&K	Hangul / Kashmir stag
Keibul Lamjao	Manipur	Only floating NP; Loktak ; Sangai
Dudhwa	UP	Only UP NP; Lakhimpur Kheri
Simlipal	Odisha (ओडिशा)	Not MP
Bandipur (बांदीपुर)	Karnataka	
Manas	Assam	WLS/BR/tiger landscape
Periyar (पेरियार)	Kerala	
Keoladeo	Rajasthan	Birds
Kanha	MP	Tiger
Indravati	Chhattisgarh	2020 match
Mollem	Goa	2020
Kalesar	Haryana	2020
Betla	Jharkhand	2020 (paper OCR “Betva”)

Biosphere reserves (match gold)

Biosphere	State(s)	Trap
Nilgiri (नीलगिरि)	TN, KL, KA	First Indian BR (1986)
Nanda Devi	Uttarakhand (उत्तराखंड)	Not UP
Nokrek	Meghalaya (मेघालय)	Not Manipur
Great Nicobar	A&N	
Gulf of Mannar	Tamil Nadu	Also reefs
Manas	Assam	Correct in 2025
Sundarbans	West Bengal	
Simlipal	Odisha	
Pachmarhi	MP	2021 match
Agasthyamalai	Kerala and TN	“Kerala” alone treated correct (2025)
Achanakmar–Amarkantak	MP–CG	
Panna	MP	
Kachchh	Gujarat	
Cold Desert	HP	
Seshachalam	AP	
Khangchendzonga	Sikkim	
Dibru-Saikhowa	Assam	
Dehang-Debang	Arunachal	

N.6 Uttar Pradesh — Forests, Parks, Sanctuaries, Ramsar

Name	Type	Belt	Note
Dudhwa	NP	Lakhimpur Kheri	Only UP NP

Name	Type	Belt	Note
Dudhwa TR	Tiger Reserve	Kheri + Bahraich complex	NP + Kishanpur + Katarniaghat
Pilibhit TR	Tiger Reserve	Pilibhit–Shahjahanpur (शाहजहाँपुर)	Separate Terai TR — not Dudhwa NP
Amargarh TR	Tiger Reserve	Bijnor	Corbett buffer/extension
Ranipur TR	Tiger Reserve	Chitrakoot (चित्रकूट)	Bundelkhand
Sanctuary	District	Extra	
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Katarniaghat	Bahraich	Est. 1975	
Kishanpur	Lakhimpur Kheri	Dudhwa TR	
Chandraprabha	Chandauli	not Sonbhadra	
National Chambal (चंबल)	Etawah (इटावा)–Agra	Gharial; UP–MP–RJ	
Hastinapur (हस्तिनापुर)	Meerut (मेरठ)–Bijnor–Amroha	Floodplain / swamp deer	
Kaimoor	Sonbhadra–Chandauli	Vindhyan	
Suhelwa	Balrampur–Shravasti–Gonda	Nepal fringe	
Sohagibarwa	Maharajganj	Terai	
Mahavir Swami (स्वामी)	Lalitpur	Bundelkhand	
Ranipur	Chitrakoot	TR base	
Okhla	GB Nagar fringe	Birds	

UP Ramsar (learn IN vs NOT)

#	Ramsar site (IN UP)	District / belt
1	Upper Ganga	Brijghat–Narora
2	Nawabganj	Unnao
3	Sarsai Nawar	Etawah
4	Sandi	Hardoi
5	Samaspur	Raebareli
6	Parvati Arga	Gonda
7	Saman	Mainpuri
8	Sur Sarovar (Keetham)	Agra
9	Haiderpur	Muzaffarnagar–Bijnor
10	Bakhira	Sant Kabir (कबीर) Nagar
11	Patna Bird Sanctuary	Etah
12	Sheikha / Shekha Jheel	Aligarh (अलीगढ़)
13	Surha Tal / JP Narayan Bird Sanctuary	Ballia
Name that sounds like UP	Actual state	Trap
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Rudrasagar	Tripura	Not UP
Sultanpur	Haryana	Not UP
Surinsar–Mansar	J&K	Not UP

Common Traps — Don't Fall For These

1. **Simlipal = Madhya Pradesh** — FALSE. **Odisha**.
2. **Nokrek = Manipur** — FALSE. **Meghalaya**.
3. **Nanda Devi = Uttar Pradesh** — FALSE. **Uttarakhand**.
4. **Botanical garden = in-situ** — FALSE. **Ex-situ**.
5. **Dudhwa = Bahraich / Pilibhit** — FALSE. **Lakhimpur Kheri** (2022).
6. **Pilibhit / Amangarh = National Parks** — FALSE. They are **Tiger Reserves**; only NP = Dudhwa.
7. **Chandraprabha = Sonbhadra** — FALSE. **Chandauli** (2023).
8. **Diversity increases toward poles** — FALSE. It **decreases**.
9. **MISHTI started 2019** — FALSE. Budget **2023–24**; aim = restore mangroves.
10. **Olivia in August** — FALSE. Nesting **~Nov–May**; Coast Guard, Odisha.
11. **Rudrasagar / Sultanpur in UP** — FALSE. Tripura / Haryana.
12. **Ballia has no ecology to ask** — FALSE. Near-nil *forest*; **Surha Tal** is Ramsar.
13. **Evergreen = largest Indian forest** — FALSE. **Moist deciduous**.
14. **Coral death = ocean cooling** — FALSE. **~2°C warming**.
15. **Project Tiger launched 1972 as open fact** — Open recall = **1 April 1973**; 2018 MCQ omitted 1973.
16. **Hotspots = 18** — FALSE. **Four** hotspots; **18** biosphere reserves.
17. **Nanda Devi = first BR** — FALSE. **Nilgiri (1986)** is first.
18. **Forest cover % = Recorded Forest Area %** — FALSE. ISFR cover ≠ legal RFA.
19. **Four hotspots = Eastern Himalaya + Western Himalaya + Western Ghats + Andaman** — FALSE. The four are Himalaya, Western Ghats–Sri Lanka, Indo-Burma, and Sundaland (**Nicobar**).
20. **Dry evergreen = Western Ghat rain forest** — FALSE. It is the **Tamil Nadu / Coromandel** NE-monsoon belt.
21. **Chir pine is an alpine juniper scrub** — FALSE. Chir is **sub-tropical pine** at about 1000–2000 m in the western Himalaya.
22. **Hotspots exist only in the tropics** — FALSE. Mediterranean, SW Australia and Chilean temperate forest are hotspots too.
23. **Northern Himalayan slopes denser than southern** — FALSE. **Southern** slopes denser (more rain + sun).
24. **Tree-line higher in Western Himalaya** — FALSE. Tree-line is **lower** in the west.
25. **Vembanad = Karnataka** — FALSE. **Kerala**.
26. **Flame of the forest = Teak / Bauhinia** — FALSE. **Butea monosperma (Palash / Dhak)**; UP state flower.
27. **Katha from Sal / Babool** — FALSE. From **khair**.
28. **Mahogany = Himalayan species** — FALSE. Tropical evergreen; Himalayan set = juniper, fir, spruce, deodar, birch.
29. **Cinchona grows in Chhattisgarh** — FALSE. Assam / Kerala / West Bengal evergreen belts.
30. **Birch = Nilgiris / Aravallis / Vindhya** — FALSE. **Himalayas**.

Topic 8 — Minerals, Energy & Industry

Current Affairs (this topic)

Year	Fact	Why asked	Source
2023	Li inferred resources Reasi, J&K ; India import-dependent for Li, Ni, Co	Frozen 2023 Q77	MoM / PYQ
2023	Union list (संघ सूची) of 24 critical minerals (Li, Ni, Co, REE, graphite, etc.)	Scheme/list trap	Ministry of Mines
2023–26	Critical Minerals Mission + auction of critical-mineral blocks	Living CA	PIB / MoM
2015 / 2025	ISA : India+France, HQ Gurugram ; not 38 members ; OSOWOG under ISA	2019 Q63 + 2025 Q109	ISA / MEA
Jan 2026	US announced withdrawal from ISA; ISA continues with 120+ members	Member-count trap	MEA / ET
2018	Major solar plant Mirzapur (मिर्जापुर) (UP) with France	UP energy PYQ	2020 Q46
DFC	Western Dadri (ददरी)–JNPT; Eastern Ludhiana–Dankuni — phased commissioning	Corridor syllabus	DFCCIL
Nuclear	Kakrapar / Kudankulam expansion; Narora = only UP plant	Plant–state	NPCIL

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q100

Which is the largest centre of the petrochemical industry in India?

A. Trombay B. Jamnagar C. Digboi D. Vadodara

Confused Pairs

Pair	Correct	Trap	Hindi
Iron leaders	OD + CG (2021–22 report)	JH+KA always lead	ओडिशा+छत्तीसगढ़
Bailadila	Chhattisgarh	Odisha	बैलाडीला = CG
Noamundi	Jharkhand	Odisha	नोआमुंडी = JH
Kudremukh	Karnataka (magnetite)	Odisha	कुद्रेमुख = KA
Mount Newman	Iron	Manganese	लौह अयस्क
Postmasburg	Manganese (SA)	U/mica	मैंगनीज
MALCO	Mettur, TN	Chennai	मेट्टूर
Khetri	Copper, RJ	Gold	खेतड़ी = तांबा
Panna	Diamond, MP	Gold	पन्ना = हीरा
Sukinda	Chromite, OD	Bauxite	सुकिंदा = क्रोमाइट
Zawar	Zn–Pb, RJ	Copper	जावर = जस्ता
Jaduguda	Uranium, JH	Thorium sands	जादूगुड़ा = यूरेनियम
Maolong	Meghalaya	Arunachal	मेघालय कोयला
Singrauli	MP–UP ; oldest NTPC	Northern CG	सिंगरौली ≠ CG
Ruhr	Germany	Britain/France	रुहर = जर्मनी
Kashagan	Kazakhstan	Kuwait	कशागन = कजाख

Pair	Correct	Trap	Hindi
CNG	Methane	Butane	मीथेन
Black Revolution	Petroleum	Oilseeds (Yellow)	काला = पेट्रोलियम
Shivanasamudra	Hydel	Nuclear	शिवणसमुद्र = जल
Kalpakkam	Nuclear	Hydel	कल्पक्कम = परमाणु
Ahmedabad mills	In cotton belt ; not traditionally largest centre	Largest = Mumbai	मुंबई कपड़ा केंद्र
Magnesite vs Mn	Magnesite = MgCO ₃ refractory	Manganese metal	मैग्नेसाइट ≠ मैंगनीज
Neyveli	Lignite, TN	Gondwana bituminous	नेयवेली = लिग्नाइट
ISA HQ	Gurugram	New Delhi / Paris	गुरुग्राम
LPG	Propane/butane mix	CNG = methane	एलपीजी ≠ सीएनजी
Bhilwara	Mica + Rampura-Agucha Zn	Manganese city (2020)	भीलवाड़ा ≠ Mn
Bhilai steel	Chhattisgarh	Odisha	भिलाई = CG
Firozabad	Glass	Brass (that is Moradabad)	फिरोजाबाद = कांच
Pegu Yoma	Mineral oil (Myanmar)	Tin (that is Kinta / Bangka)	पेगू योमा = तेल
Mangalore–Kanpur line	Most minerals east of line	Think west-rich	मंगलुरु–कानपुर रेखा
Northern plains minerals	Almost devoid of economic minerals	Treat Ganga plain as ore belt	मैदान ≈ खाली
Non-ferrous endowment	Poor except bauxite	India rich in all non-ferrous	बॉक्साइट अपवाद
Magnetite vs hematite	Magnetite ~ 70% Fe; hematite most used	Swap grades / uses	मैग्नेटाइट ≠ हेमेटाइट मात्रा
Chota Nagpur vs Aravalli	Chota Nagpur = richest multi-mineral	Call Aravalli richest	छोटानागपुर = समृद्ध
Mica belt	Koderma–Gaya–Hazaribagh + Nellore + Ajmer	Only “Kodarma town” as if exclusive	कोडरमा–गया–हजारीबाग
Chitradurga	Iron (Ballari–Chitradurga belt, KA)	Copper city (2020 trap)	चित्रदुर्ग = लौह
Guntur	Not an iron-ore mining centre	List with Ballari / Bailadila as iron	गुंटूर ≠ लौह
Coal rank %	Peat < 40 → lignite 40–55 → bituminous 60–80 → anthracite 80–95	Swap % or skip homes	कार्बन % सीढ़ी
Anthracite India	Scarce; J&K pockets	Call it India’s bulk coal	एन्थ्रेसाइट ≠ मुख्य
Lignite home	Neyveli (TN) (+ RJ / GJ)	Call Neyveli bituminous Gondwana	नेयवेली = लिग्नाइट
Bituminous share	~ 80% deposits; mostly non-coking	Call most Indian coal anthracite / coking	अधिकतर नॉन-कोकिंग

N.0 Framework (map first)

Rock systems (Ghatnachakra / NCERT map)

System	Character	Main minerals
Dharwar	Oldest metamorphosed belts; first studied in Karnataka	Storehouse of metallic minerals — iron, manganese, gold, copper; most mineralised system in teaching keys

System	Character	Main minerals
Gondwana	Palaeozoic–Mesozoic sediments in eastern/central valleys	About 95–99% of India's coal (Damodar, Son, Mahanadi, Godavari, etc.)
Vindhyan	Sedimentary (limestone, sandstone, shale)	Limestone , sand, china clay — cement base; not the metallic king
Cuddapah / Tertiary	Younger / basin rocks	Tertiary coal NE; petroleum in sedimentary basins (Gujarat, Assam, offshore)

How minerals occur

Mode	How it forms	Examples
Vein / lode	Molten / gaseous minerals rise into cracks of igneous and metamorphic rock, then cool	Tin, copper, zinc, lead
Bed / layer	Deposition in sedimentary strata under heat and pressure	Coal, iron ore, gypsum, potash, common salt (लवणाध्यक्ष)
Residual	Surface rock weathers; soluble parts wash away; a concentrated cap remains	Bauxite on laterite
Placer / alluvial	Hard grains collect in valley-floor / hill-base sands (not corroded by water)	Gold, silver, tin, platinum, monazite
Ocean water / ocean bed	Dissolved salts, and nodules on the sea floor	Common salt, magnesium, bromine; manganese nodules

Classification

Class	Meaning	Examples
Ferrous metallic	Contain iron / feed steel	Iron ore, manganese, chromite, nickel, cobalt
Non-ferrous metallic	Metals without iron as the main metal	Bauxite (aluminium), copper, zinc, lead, tin
Precious metals	High-value non-ferrous subset	Gold, silver, platinum
Non-metallic (inorganic)	Not used mainly as metal	Mica, limestone, gypsum, graphite, diamond, salt, magnesite
Energy / mineral fuels	Organic fossil fuels + nuclear fuels	Coal, petroleum, natural gas, uranium, thorium

Mineral belts of India

Belt	Cover	Character
North-Eastern Plateau (Chota Nagpur)	Jharkhand, Odisha, West Bengal, parts of Chhattisgarh	Richest — iron, coal, manganese, bauxite, mica, copper, chromite; often called India's Ruhr
Central belt	Chhattisgarh, Madhya Pradesh, Telangana, Andhra Pradesh, Maharashtra	Manganese, bauxite, limestone, coal, iron, mica, graphite, gems
South-Western Plateau	Karnataka, Goa, adjoining Tamil Nadu–Kerala uplands	Ferrous metals, bauxite, limestone; almost no Gondwana coal ; Neyveli lignite in Tamil Nadu; Kerala monazite sands; Goa iron
North-Western (Aravalli)	Rajasthan–Gujarat	Copper, zinc–lead, mica, building stone, gypsum, Fuller's earth , salt; Gujarat petroleum
Himalayan / Assam–Arakan	North-East and Himalayan foothills	Copper–lead–zinc pockets; Tertiary coal ; Assam oil

Master map — mineral by state / centre (revise daily)

Mineral	Main state / centre	One-line note
Iron ore	Odisha , Chhattisgarh, Karnataka, Jharkhand	Bailadila (CG); Noamundi (JH); Kudremukh / Ballari (KA)
Manganese	Odisha , Maharashtra, Madhya Pradesh	Balaghat (MP); Nagpur–Bhandara (MH)
Chromite	Odisha (Sukinda)	Stainless steel / ferrochrome
Bauxite	Odisha (Panchpatmali)	Residual laterite ore of aluminium
Copper	Khetri (RJ) ; Malanjkhand (MP); Singhbhum (JH)	India still imports copper
Zinc–lead	Zawar; Rampura-Agucha (Bhilwara) — RJ	Rajasthan near-monopoly
Gold	Hutti (working); Kolar (historic (ऐतिहासिक)) — KA	Tiny mine output vs jewellery
Mica	Koderma–Gaya–Hazaribagh ; Nellore; Ajmer (अजमेर)–Bhilwara	Dielectric sheets
Diamond	Panna (MP)	India's diamond centre
Coal (Gondwana)	Damodar–Son–Mahanadi–Godavari valleys	~ 97% of Indian coal
Lignite	Neyveli (TN)	Brown coal; pit-head power
Petroleum	Assam; Mumbai High; Gujarat; Barmer; KG / Cauvery	Sedimentary basins only
Uranium	Jaduguda (JH)	Also Tummalapalle (AP); Domiasiat (ML)
Thorium / monazite	Kerala–Tamil Nadu beach sands	Also rare earths

N.1 Metallic Minerals

Iron

Ore	Iron content	Trait
Magnetite (Fe_3O_4)	About 70%	Finest / richest ore; magnetic; electrical industry
Hematite (Fe_2O_3)	About 50–60%	Most used industrial ore by quantity
Limonite	About 40–60% (lower / variable)	Hydrated iron oxide; lesser industrial weight
Siderite	About 40–50% (lower)	Iron carbonate ore; lesser industrial weight

Belt	Note
Odisha–Jharkhand	Mayurbhanj (मयूरभंज)–Keonjhar–Sundargarh hematite; Gua–Noamundi in Singhbhum
Durg–Bastar–Chandrapur	Bailadila high-grade hematite in Chhattisgarh; Dalli–Rajhara; export via Visakhapatnam
Ballari–Chitradurga–Chikkamagaluru–Tumakuru	Karnataka magnetite–hematite; Kudremukh slurry to Mangaluru
Maharashtra–Goa	Lower grade; export via Mormugao
Mine / belt	State
-----	-----
Gurumahisani / Badampahar / Sulaipat	Odisha
Bonai / Kendujhar / Sundargarh	Odisha
Bailadila	Chhattisgarh
Dalli–Rajhara	Chhattisgarh

Belt	Note
Noamundi / Gua / Kiriburu	Jharkhand
Kudremukh	Karnataka
Donimalai / Ballari–Hospet	Karnataka
Ratnagiri belt	Maharashtra

Manganese

Point	Teaching line
Steel need	About 10 kg of manganese per tonne of steel
Rock link	Mainly associated with Dharwar rocks
Leading states	Odisha, Maharashtra, Madhya Pradesh (Karnataka is another classic belt)
Madhya Pradesh belt	Balaghat
Maharashtra belt	Nagpur–Bhandara
Odisha belt	With the iron belt of Keonjhar–Sundargarh
World centre	Postmasburg (South Africa)

Chromite

Point	Teaching line
Main Indian belt	Sukinda in Odisha (India's leading chromite belt)
Use	Stainless steel and ferrochrome
Belt family	Eastern plateau ferrous belt with Odisha iron–manganese country

Bauxite and aluminium

Point	Teaching line
Leading state	Odisha
Leading mine	Panchpatmali (Koraput, Odisha)
Other belts	Gujarat, Jharkhand, Chhattisgarh, Maharashtra
Older city tag	Katni (Madhya Pradesh)
Smelting logic	Aluminium smelting is power-hungry , so plants sit near cheap electricity as well as near ore
Company (कंपनी)	Plant
-----	-----
BALCO	Korba
HINDALCO	Renukoot
MALCO	Mettur
INDAL	Hirakud (हीराकुंड)
NALCO	Damanjodi (alumina) / Angul (smelter)

Copper

Centre	State	Note
Khetri (Khetri–Singhana belt)	Rajasthan	Classic textbook copper centre
Malanjkhand	Madhya Pradesh	Large open-pit copper mine
Singhbhum (Mosabani–Ghatsila)	Jharkhand	Eastern copper belt

Centre	State	Note
World lead	Chile (northern Andes porphyry)	Leading producer in the usual paper set

Gold

Centre	State	Note
Kolar	Karnataka	Historic goldfield; now largely closed
Hutti	Karnataka	Main working goldfield
Ramagiri	Andhra Pradesh	Smaller goldfield

Tin

Point	Teaching line
Ore	Cassiterite
Only producing state (recent IMYB)	Chhattisgarh (Dantewada belt)
Reserve tables	Haryana also appears in resource share
World centres	Kinta Valley (Malaysia); Bangka (Indonesia)

Zinc, lead and silver

Centre	State	Metal
Zawar (near Udaipur)	Rajasthan	Historic zinc–lead (silver often a by-product)
Rampura-Agucha (Bhilwara)	Rajasthan	Asia's richest zinc–lead reserve belt; Hindustan Zinc link

N.2 Non-metallic and atomic minerals

Mica

Belt	State / region	Note
Koderma–Gaya–Hazaribagh–Munger	Jharkhand–Bihar	Classic belt; Kodarma often called the “capital of mica”
Nellore	Andhra Pradesh	Southern mica centre; AP leads resource share in IMYB
Ajmer–Bhilwara	Rajasthan	Rajasthan leads recent production ; Bhilwara is also zinc country

Limestone, dolomite, gypsum, cement rocks

Rock / mineral	Use	Where (teaching centres)
Limestone	Cement; bulky → plant near quarry	UP: Chunar–Churk–Dalla ; wide Vindhyan / peninsula belts
Dolomite	Steel flux ; refractory	UP: Sonbhadra–Mirzapur ; Odisha and other plateau belts
Gypsum	Cement; reclaim usar / alkaline soils	Rajasthan; western Uttar Pradesh

Graphite and diamond

Mineral	Where	Use / note
Graphite	Odisha, Jharkhand	Crucibles, pencils, electrodes; carbon mineral
Diamond	Panna (Madhya Pradesh)	India's main diamond centre

Salt, magnesite, silica, Fuller's earth

Mineral	Chemistry / use	Where
Common salt	Food; chemicals; from sea / lakes	Sambhar Lake (RJ) inland; Gujarat coastal solar salt
Magnesite	MgCO ₃ ; refractory bricks	Tamil Nadu; Uttarakhand (उत्तराखंड)
Silica sand	Glass sand	Shankargarh (Prayagraj (प्रयागराज)) → Firozabad glass
Fuller's earth	Clay used to clean oils and grease	North-western Aravalli (अरावली) belt, Rajasthan

Other industrial minerals

Mineral	Use	Where
Kyanite / sillimanite	Refractories	Eastern plateau (Jharkhand–Odisha)
Asbestos	Older industrial fibre	Rajasthan; Andhra belts
Phosphorite	Fertilizer phosphate	Jhamarkotra (Rajasthan) ; Bundelkhand / Lalitpur

Atomic minerals and critical minerals

Mineral	Main centre	Note
Uranium	Jaduguda (Jharkhand)	Singhbhum shear; also Bhatin–Narwapahar–Turamdih
Uranium (other)	Tummalapalle (AP); Domiasiat / Mawthabah (Meghalaya)	Extra centres
Thorium / monazite	Kerala–Tamil Nadu beach sands	Also Andhra–Odisha coasts; monazite carries rare earths
Lithium	Reasi, Jammu & Kashmir	2023 inferred-resource find; India still imports Li
Nickel / cobalt	Import-dependent	Laterite / ultramafic traces; no Khetri-class mines

N.3 Coal

Two geological ages

Age	Share / quality	Where found
Gondwana coal	About 97–98% of India's coal; mostly bituminous ; usually low sulphur, high ash	River valleys of the peninsula: Damodar, Son, Mahanadi, Godavari (and Wardha)
Tertiary coal	Small share; younger; often high sulphur	Assam, Meghalaya, Nagaland, Arunachal Pradesh ; also Kalakot (J&K) in older NCERT lists

Types of coal (carbon % and Indian homes)

Type	Carbon (approx.)	Moisture / quality	Where found in India	Main use
Peat	Under about 40%	Highest moisture; first stage of coal; burns like wet wood	Limited swamp / hill pockets (not a major commercial Indian coal)	Local fuel only
Lignite (brown coal)	About 40–55%	High moisture; low grade; can ignite spontaneously	Neyveli (Tamil Nadu); Palana / Barsingsar (Rajasthan); Panandhro (Gujarat) ; also J&K / coastal Tertiary notes	Pit-head thermal power
Bituminous	About 60–80%	Most common Indian coal; dense black; high heat	Gondwana valleys — Jharkhand, West Bengal, Odisha, Chhattisgarh, Madhya Pradesh, Telangana, Maharashtra	Power + coke (if coking grade)

Type	Carbon (approx.)	Moisture / quality	Where found in India	Main use
Anthracite	About 80–95%	Hardest; lowest moisture; best quality; burns with little smoke	Very little in India — small pockets in Jammu & Kashmir	High-efficiency fuel

Major Gondwana coalfields

Coalfield	State	Valley / note
Raniganj	West Bengal	Damodar; oldest Indian field; first commercial mining 1774
Jharia	Jharkhand	Damodar; coking ; known for mine fires; largest field in many notes
Bokaro / Ramgarh / Karanpura / Giridih	Jharkhand	Damodar family
Rajmahal	Jharkhand	Eastern Jharkhand
Talcher	Odisha	Mahanadi (महानदी) valley
Ib Valley	Odisha	Mahanadi family
Korba / Mand–Raigarh	Chhattisgarh	Thermal towns
Singrauli	Madhya Pradesh–Uttar Pradesh	Son valley; not northern Chhattisgarh
Sohagpur	Madhya Pradesh	Son family
Singareni	Telangana	Godavari (गोदावरी) valley
Wardha / Kamptee / Umrer	Maharashtra	Godavari–Wardha
Coalfield	Valley	
-----	-----	
Talcher	Mahanadi	
Karanpura	Damodar (दामोदर)	
Singrauli	Son	
Singareni	Godavari	

Tertiary and lignite centres

Centre	State	Type
Makum / Naharkatiya belt	Assam	Tertiary coal
Maolong	Meghalaya	Tertiary (not Arunachal)
Namchik–Nampuk	Arunachal Pradesh (Changlang)	Tertiary
Neyveli	Tamil Nadu	Lignite (main Indian lignite)
Palana / Barsingsar	Rajasthan	Lignite
Panandhro	Gujarat	Lignite

World coalfields (by country)

Field	Country
Appalachian	USA
Lancashire	England
Ruhr	Germany
Kuzbass	Russia

Field	Country
Donetsk	Ukraine

N.4 Petroleum, gas, refineries

Fuel	Main composition	Teaching line
Natural gas / CNG	Mainly methane	Piped / compressed methane fuel
LPG	Propane–butane mix	Not the same as CNG
Petroleum	Mixture of hydrocarbons	Black Revolution = petroleum; Yellow = oilseeds

Indian fields and basins

Field / basin	Note
Digboi	Assam — oldest Indian refinery town / onshore oil
Naharkatiya / Moran / Rudrasagar	Assam–Arakan onshore
Mumbai High	Offshore Arabian Sea; India's giant offshore field
Bassein / Tapti (ताप्ती)	Western offshore gas–oil
Ankleshwar / Cambay / Kalol / Mehsana	Gujarat onshore
KG basin (including KG-D6)	East-coast gas and oil
Ravva	Krishna–Godavari offshore field
Cauvery basin	Tamil Nadu–Puducherry coast
Barmer (Mangala and related fields)	Rajasthan onshore

Refineries

Refinery	State
Nayara (Vadinar / Jamnagar belt)	Gujarat
Jamnagar	Gujarat
Koyali (Vadodara)	Gujarat
Manali	Tamil Nadu
Kochi	Kerala
Mangaluru (MRPL)	Karnataka
Numaligarh	Assam
Digboi / Guwahati / Bongaigaon	Assam
Tatipaka	Andhra Pradesh
Visakhapatnam	Andhra Pradesh
Mathura	Uttar Pradesh
Barauni	Bihar
Panipat	Haryana
Bhatinda	Punjab
Paradip	Odisha
Bina	Madhya Pradesh
Haldia	West Bengal

N.5 Power: thermal, hydro, nuclear, renewables

Type	Sources	Teaching line
Conventional	Coal, petroleum, natural gas, nuclear	Long-used commercial fuels; mostly exhaustible
Non-conventional	Solar, wind, tidal, biogas, geothermal	Cleaner / renewable set in school keys
Stored solar?	Wind, biomass	Nuclear and geothermal are not stored solar

Hydroelectricity

Hydel name	Note
Shivanasamudra	pair with Kalpakkam
Bhakra	Satluj (multipurpose)
Tehri (टिहरी)	Bhagirathi (भागीरथी)
Hirakud	Mahanadi
Koyna	Maharashtra
Sharavathi / Linganamakki	Karnataka western-ghat hydel
Nagarjuna (नागार्जुन) Sagar (नागार्जुन सागर) / Srisaillam	Krishna
Idukki	Kerala

Nuclear plants

Plant	State
Tarapur	Maharashtra (first commercial plant)
Kakrapar	Gujarat
Kudankulam	Tamil Nadu (VVER / imported reactor)
Kaiga	Karnataka
Rawatbhata (RAPP)	Rajasthan
Kalpakkam (MAPS)	Tamil Nadu
Narora	Uttar Pradesh

N.6 Industry, Regions, Corridors

Iron and steel

Plant	State
Bhilai	Chhattisgarh
Rourkela	Odisha
Bokaro	Jharkhand
Durgapur / Burnpur	West Bengal
Visakhapatnam	Andhra Pradesh
Salem	Tamil Nadu
Bhadravati / Vijayanagar	Karnataka
Jamshedpur	Jharkhand (Tata (तत) pioneer)

Industrial regions

Region	Core character
Mumbai–Pune	Cotton textiles, engineering, chemicals, automobiles
Hugli	Jute, engineering, port industries
Bangalore–Tamil Nadu belt	Diversified engineering, autos, textiles; Bengaluru IT + Chennai + Coimbatore
Ahmedabad–Vadodara–Surat (Gujarat)	Textiles, petrochemicals, engineering
Chotanagpur	Mineral-based heavy industry — India's Ruhr industrial extension
Visakhapatnam–Guntur	Coastal steel, shipbuilding, engineering
Gurgaon–Delhi–Meerut	NCR automobiles, electronics; western Uttar Pradesh link
Kollam–Thiruvananthapuram	Kerala coastal–plantation and light industry

Agencies, SEZs and corridors

Agency	Headquarters
ONGC	New Delhi
MECL	Nagpur
NMDC	Hyderabad
NALCO	Bhubaneswar

Corridor / zone	Meaning
DMIC	Delhi–Mumbai Industrial Corridor — the must-know industrial corridor
Western DFC	Dedicated freight rail Dadri–JNPT
Eastern DFC	Dedicated freight rail Ludhiana–Dankuni
CBIC	Chennai–Bengaluru Industrial Corridor
AKIC	Amritsar (ਅਮ੍ਰਿਤਸਰ)–Kolkata Industrial Corridor
VCIC	Visakhapatnam–Chennai Industrial Corridor
SEZ	Export-oriented special zone

Coalfield	State (as given in options)
1. Maolong	Arunachal Pradesh
2. Raniganj	West Bengal
3. Ramgarh	Jharkhand
4. Talcher	Odisha

N.7 Uttar Pradesh focus

City	Industry tag
Kanpur	Textile, leather
Moradabad	Brass
Aligarh	Padlocks and hardware
Firozabad	Glass
Noida–Ghaziabad	NCR engineering / electronics
Meerut	NCR belt; sports goods

City	Industry tag
Agra	Light industry and tourism, not a mineral capital
Renukoot	Aluminium (HINDALCO)
Chunar–Dalla belt	Cement

Common Traps — Don't Fall For These

1. **West of Mangalore–Kanpur is mineral-rich** — FALSE. Most major minerals lie **east** of the Mangaluru–Kanpur line.
2. **Aravalli = richest mineral belt** — FALSE. **Chota Nagpur / NE Peninsular** is richest ("India's Ruhr").
3. **Jharkhand + Karnataka always lead iron** — FALSE for **2021–22**. **Odisha + Chhattisgarh** lead.
4. **Bailadila = Odisha** — FALSE. **Chhattisgarh**. Gurumahisani = Odisha; Noamundi = Jharkhand; Kudremukh = Karnataka.
5. **Bauxite = sedimentary basin only** — FALSE. Commonly linked to **laterite** weathering; Odisha lead (**Panchpatmali**).
6. **MALCO = Chennai** — FALSE. **Mettur, Tamil Nadu**. HINDALCO = Renukoot (UP); BALCO = Korba.
7. **Maolong = Arunachal** — FALSE. **Meghalaya**.
8. **Anthracite = softest / lignite = hardest** — FALSE. Rank rises peat → lignite → bituminous → **anthracite** (hardest, highest carbon).
9. **Singrauli NTPC = northern Chhattisgarh** — FALSE. **MP–UP**; it is NTPC's oldest unit.
10. **Kashagan = Kuwait** — FALSE. **Kazakhstan**. Ruhr = **Germany**, not Britain.
11. **CNG / natural gas = butane** — FALSE. **Methane**. LPG is the propane/butane mix.
12. **Solar / wind = conventional energy** — FALSE. Conventional = coal/oil/gas/nuclear; solar–wind–tidal–biogas–geothermal = **non-conventional**.
13. **Black Revolution = oilseeds** — FALSE. Oilseeds = **Yellow**; Black = **petroleum**.
14. **Ahmedabad = largest cotton mill centre** — FALSE. In the cotton **belt** (R true); traditionally **Mumbai** is largest.
15. **Shivanasamudra = nuclear** — FALSE. **Hydel**; Kalpakkam = nuclear.
16. **ISA Secretariat = New Delhi / Paris** — FALSE. **Gurugram**. Member count is **not 38**.
17. **Nuclear / geothermal = stored solar** — FALSE. Wind and biomass are solar-linked.
18. **Copper city = Chitradurga; Mn city = Bhilwara** — FALSE. Ballari = iron; Katni = bauxite.
19. **Diu is not the first 100% solar UT** — FALSE. It **is**. First solar airport = **Cochin**, not Ahmedabad.
20. **UP = Bhilai-class steel state / no coal at all** — FALSE. Facts = **Sonbhadra** energy, **non-coking** Singrauli coal, **Mathura** refinery, **Narora** nuclear, **Shankargarh** silica. Firozabad = **glass**, Moradabad = **brass**.
21. **Pegu Yoma = tin** — FALSE. **Mineral oil**. Tin = Kinta / Bangka.
22. **Northern plains = major ore belt** — FALSE. The plains are almost **devoid** of economic minerals.
23. **India rich in all non-ferrous metals** — FALSE. **Poor except bauxite**.
24. **Hematite has more Fe than magnetite** — FALSE. Magnetite ~**70%**; hematite is the **quantity** workhorse (~50–60%).
25. **~80% Indian coal is coking** — FALSE. Most is bituminous **non-coking** (thermal).
26. **Gondwana = metallic storehouse / Dharwar = coal king** — FALSE. **Dharwar** = metallic minerals; **Gondwana** ≈ **95–99%** coal.
27. **Vindhyan = iron–gold belt** — FALSE. Sedimentary **limestone / china clay**.
28. **Tin producer = Assam / J&K** — FALSE. Only **Chhattisgarh** produces tin in recent production data.
29. **Digboi = largest petrochemical centre** — FALSE. **Oldest** oilfield/refinery centre; **Jamnagar** is the giant complex.
30. **ISPRL storages include Ankleshwar** — FALSE. **Visakhapatnam, Mangaluru, Padur**.

31. **Coal reserve order = production order** — FALSE. Proved reserves often **JH > OD > CG**; production often **CG > OD > MP**.
32. **Zawar = copper** — FALSE. **Zinc-lead** (Udaipur); copper = **Khetri / Malanjkhand**.
33. **Nathra-ki-Pal = copper of Rajasthan** — FALSE. **Iron ore**.
34. **Marble = igneous rock** — FALSE. **Metamorphic** (recrystallised limestone); **Makrana** best quality.

Topic 9 — Transport & Communication

Current Affairs (this topic)

Year	Fact	Why it matters
13 Oct 2021	PM Gati Shakti National Master Plan launched (not 2022)	Year trap with seven engines
17 Sep 2022	National Logistics Policy — complements Gati Shakti	Scheme pair
Feb 2023	Amrit Bharat: about 1,275 stations all-India; 149 of them in UP for modernisation	149 = UP stations, not new lines
2024–26	New Pamban vertical-lift railway sea bridge (RVNL)	RVNL + first vertical-lift fact
18 Mar 2024	FLY91 inaugural from Manohar Intl (Mopa, Goa (गोवा))	New airline + airport pair
2024–26	Noida International (Jewar) — UP's new greenfield intl hub	UP airport CA
DFC	Western Dadri (ददरी)—JNPT / Eastern Ludhiana—Dankuni — phased opening	Freight spine

Sagarmala / Bharatmala remain living umbrella programmes; still tests **name swap**, not project lists.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q88

Which of the following ports is on the western coastline?

A. Nhava Sheva B. Visakhapatnam C. Chennai D. Tuticorin

Q. UKPCS Prelims 2025, Q92

Which country has the highest density of railway network?

A. Canada B. U.S.A. C. Russia D. India

Confused Pairs

Pair	Correct	Trap	Hindi
NER HQ	Gorakhpur	Hajipur	गोरखपुर
SER HQ	Garden Reach, Kolkata	Cuttack	कटक नहीं
ECR HQ	Hajipur	Gorakhpur	हाजीपुर
Amrit Bharat	Station modernisation ; UP 149 stations (Feb 2023)	149 new railway lines	149 स्टेशन
Bharatmala	Highways	Ports	सड़क
Sagarmala	Port-led	Highways	पत्तन
Gati Shakti year	2021	2022	2021
Purvanchal (built)	Passes Mau; not Basti	Blind 2018 “not Mau” as today's map	मऊ हँ
Atal Tunnel	Pir Panjal	“World's longest highway”	पीर पंजाल
Igarka	Russia	China	रूस
NW-1	Ganga	Brahmaputra	गंगा
NW-3	Kerala canal	Ganga	केरल
Harbour vs port	Shelter vs shelter+facilities	Same thing	पोताश्रय ≠ पत्तन
Fibre	TIR	Refraction	पूर्ण आंतरिक परावर्तन
Motihari pipe	Amlekhganj	Kathmandu	अमलेखगंज

Pair	Correct	Trap	Hindi
Solar airport	Cochin	Delhi / Ahmedabad	कोच्चि
Paradip	Odisha	AP / WB	पारादीप = ओडिशा
Alappuzha	Kerala	TN	आलप्पुझा
NH vs rural roads	NH ~2% length but ~40% traffic	Think NH = most km	राष्ट्रीय राजमार्ग
Silchar spelling	EW corridor = Silchar / Silcher Assam	Mix with Shillong	सिलचर

N.1 Railways

Factory (फैक्टरी) / works	Place	What it builds
Rail Coach Factory	Kapurthala	Coaches
Integral Coach Factory	Perambur (Chennai)	Coaches
Chittaranjan Locomotive Works	Chittaranjan	Electric locomotives
Banaras Locomotive Works	Varanasi (वाराणसी) / Manduadih	Diesel-electric locomotives
Modern Coach Factory	Rae Bareli	Third major coach factory

Zone	Headquarters
Northern	New Delhi
North Eastern	Gorakhpur
North Central	Prayagraj
East Central	Hajipur
Eastern	Kolkata (Fairlie Place)
South Eastern	Garden Reach, Kolkata
Southern	Chennai
South Central	Secunderabad (not Hyderabad (हैदराबाद) city trap)
Western	Churchgate, Mumbai
West Central	Jabalpur
Central	Mumbai CSMT
East Coast	Bhubaneswar
North Western	Jaipur (not Jodhpur)
South Western	Hubballi
South East Central	Bilaspur
South Coast	Visakhapatnam
Metro Railway Kolkata	17th zone (2010)

N.2 Roads, Expressways, Bharatmala

Class	Who builds / maintains	Note
National Highways	Central government / NHAI	About 2% of road length but about 40% of road traffic
State Highways	State governments	Link the state capital to district headquarters
District roads	Zila Parishad (जिला परिषद) / district	Link the district headquarters to important nodes

Class	Who builds / maintains	Note
Rural roads	Local bodies; PMGSY push	About 80% of total road length; all-season village links
Border roads	BRO (from 1960)	Northern and north-eastern borders; high-altitude work
International highways	Cross-border links	Neighbour connectivity

Expressway	Note
Yamuna (यमुना)	Greater Noida–Agra
Agra–Lucknow	West–central Uttar Pradesh
Purvanchal	Lucknow–Ballia via Azamgarh and Mau
Bundelkhand	Southern Uttar Pradesh
Ganga Expressway	Meerut (मेरठ)–Prayagraj corridor (under construction / phased)
Gorakhpur Link	Connector toward Gorakhpur / NER city

N.3 Waterways

NW	River / canal	Stretch	States / note
NW-1	Ganga–Bhagirathi (भागीरथी)–Hooghly (हुगली)	Haldia–Prayagraj (about 1,620 km)	UP, Bihar, Jharkhand, West Bengal
NW-2	Brahmaputra	Dhubri–Sadiya	Assam
NW-3	West Coast Canal, Champakara and Udyogmandal canals	Kottapuram–Kollam	Kerala
NW-4	Godavari, Krishna and connected canals (including Buckingham Canal)	Kakinada–Puducherry	Andhra Pradesh–Tamil Nadu–Puducherry
NW-5	Brahmani–Mahanadi delta (डेल्टा) and East Coast Canal	Talcher–Dhamra / Geonkhali–Charbatia stretch	Odisha–West Bengal

N.4 Ports, Harbours, Maritime Trade

Coast	Major ports
West / NW	Kandla (Deendayal , Gujarat), Mumbai, JNPT, Mormugao (Goa), New Mangalore (Karnataka), Cochin (Kerala)
East / SE	Kolkata–Haldia, Paradip (Odisha), Visakhapatnam (Andhra Pradesh), Chennai, Ennore (Kamarajar), Tuticorin (VOC , Tamil Nadu)

Port–state match bank (next year can ask any)

Port	State	Tag
Kandla / Deendayal	Gujarat	Major; NW hinterland
Mumbai	Maharashtra	Major; natural harbour class
JNPT / JNPA	Maharashtra	Container flagship
Mormugao	Goa	Major; iron-ore note
New Mangalore	Karnataka	Major
Cochin / Kochi	Kerala	Major; natural harbour class
Alappuzha ★	Kerala	Non-major; 2025 Q50

Port	State	Tag
Kolkata–Haldia	West Bengal	Major; riverine
Paradip ★	Odisha (ओडिशा)	Major; 2025 Q50
Visakhapatnam	Andhra Pradesh	Major; natural harbour class
Kakinada ★	Andhra Pradesh	Non-major; 2025 Q50
Krishnapatnam	Andhra Pradesh	Non-major / private class
Chennai	Tamil Nadu	Major; often artificial harbour
Ennore / Kamarajar	Tamil Nadu	Major; N of Chennai
Tuticorin / VOC ★	Tamil Nadu	Major; 2025 Q50
Mundra / Pipavav	Gujarat	Non-major / private west coast
Dhamra	Odisha	Non-major east coast

N.7 Sagarmala, Blue Economy, Gati Shakti, NLP

Blue Economy — sector teaching card

Scheme	Focus	Trap
Sagarmala	Port-led development, coastal shipping, port connectivity, coastal communities	Not a highway scheme
Bharatmala	Highway corridors and feeders	Not ports
PM Gati Shakti NMP	Multimodal planning map of projects	Year trap
National Logistics Policy	Lower logistics cost and time	Complements Gati Shakti; does not replace it
Blue Economy	Sustainable ocean-resource use — shipping, fisheries, coastal tourism, offshore energy	Not “any coastal factory”
DFC	Dedicated rail freight hardware	Neighbour of Sagarmala, not the same scheme

N.9 Uttar Pradesh Focus

Theme	Fact
Railway HQ	North Eastern Railway — Gorakhpur
Expressway	Built Purvanchal passes Mau ; Basti is off the main alignment
Airport	Kushinagar international from 24 Jun 2020 ; Jewar = Noida International
Stations	Amrit Bharat: 149 UP stations identified for modernisation (Feb 2023), out of about 1,275 all-India
Waterway	NW-1 on the Ganga (Haldia–Prayagraj)
DFC pull	Western UP / NCR toward the Dadri–JNPT freight spine
Pipeline nodes	HBJ gas toward Jagdishpur / Auraiya; Mathura refinery product / crude complex

N.10 Tourism geography

Resort / hill-station match bank

Place	State / UT	Note
Chakrata	Uttarakhand (उत्तराखंड) (Dehradun (देहरादून))	Cantonment; Tiger Fall
Haflong	Assam	Only hill station of Assam (Dima (डिम) Hasao)
Kalimpong	West Bengal	Darjeeling hills
Kufri	Himachal Pradesh	Near Shimla
Kasauli	Himachal Pradesh	Solan
Keylong	Himachal Pradesh	Lahaul
Auli (औली)	Uttarakhand	Ski / hill station
Ooty (Udhagamandalam)	Tamil Nadu	Nilgiris
Gulmarg / Pahalgam	J&K	Valley hill stations
Chikmagalur	Karnataka	Coffee / hills
Khajjiar	Himachal Pradesh	Meadow / lake note
Banni grassland	Gujarat	Kachchh
Bugyal	Uttarakhand	Alpine meadow

City nicknames and pilgrimage geography

Fact	Correct	Trap
Pink City	Jaipur	Not Mysore
City of Lakes	Udaipur	Not Jaisalmer
City of Joy	Kolkata	Common nickname pair
City of Mahakal	Ujjain (उज्जैन)	Correct with Jaipur pair
Incredible India	MoT brand	Not "Beautiful India"
Sabarimala	Kerala (Ayyappa)	Not TN / KA / AP
Srisailem	Nallamala Hills, Andhra Pradesh	Mallikarjuna + Bhramaramba; Jyotirlinga
Omkareshwar	Mandhata island, Narmada (नर्मदा), Madhya Pradesh	Not Satmala Hills
Pushkar	Ajmer (अजमेर), Aravalli (अरावली) / Brahma temple	Not Mahadeo Hills
Biligiriranga Hills	SE Karnataka	Standing Ranganatha / Venkatesha temple
Tirupati Venkateswara	Seshachalam (Andhra Pradesh)	Not Shevaroy / Jawadhu / Biligiri as the main Tirupati hill
Udvada	Gujarat	Zoroastrian fire temple
Point Calimere	Nagapattinam, Tamil Nadu	Apex of Coromandel Coast
Konark Sun Temple	Odisha coast; UNESCO 1984 ; Narasimha Deva I	World Heritage
Champaner-Pavagarh	Panchmahal, Gujarat; UNESCO 2004	World Heritage
Mamallapuram	Pallava (पल्लव) stone town, Tamil Nadu; UNESCO	World Heritage
CST Mumbai	Victorian Gothic terminus; UNESCO	World Heritage
Ghost towns	Kuldhara, Dhanushkodi, Lakhpur	Charkhari is not a ghost town (Mahoba, UP)

N.12 World transport · ports · isopleths

Note	Fact
Busiest ocean trade route	North Atlantic (links two industrial zones)
Trans-Siberian	Long Eurasian rail; Sochi is not on the main route
Eurotunnel / Channel Tunnel	England–France
Canadian Pacific	Montreal–Vancouver
Bullet / HSR network length	China leader
Rotterdam	Rhine mouth; Europe's great seaport
Shanghai	Often #1 container harbour in recent frames
Santos	Coffee port (Brazil)
Maracaibo	Venezuela oil port
Port Nolloth	South Africa diamond port
Gwadar	Pakistan (Balochistan), Arabian Sea
Igarka	Russia (Yenisei) — not China
Alexandria	Egypt (मिस्र)

Line	What it joins (equal value)
Isobar	Pressure
Isohyet	Rainfall
Isotherm	Temperature
Isohaline	Salinity
Isobath	Depth
Isogonic	Magnetic declination
Contours	Elevation / relief

Common Traps — Don't Fall For These

1. **SER HQ = Cuttack** — FALSE. **Garden Reach, Kolkata**.
2. **NER HQ = Hajipur** — FALSE. **Gorakhpur**. Hajipur = **East Central**.
3. **Amrit Bharat = 149 new railway lines** — FALSE. It modernises **stations**. As of Feb 2023 about **1,275** stations all-India, of which **149** were in **Uttar Pradesh**.
4. **Gati Shakti launched 2022** — FALSE. **13 Oct 2021**; seven engines **true**.
5. **Bharatmala = ports / Sagarmala = highways** — SWAP. Bharatmala = **roads**; Sagarmala = **port-led**.
6. **Mau is off today's Purvanchal Expressway** — FALSE for the **built** road (Lucknow–Ballia via Azamgarh–Mau). **Basti** is not on the main alignment. The 2018 stem "will not pass" had a different official key — see that full question in the PYQ bank.
7. **Atal = world's longest highway tunnel** — FALSE as a bare claim. **Pir Panjal** is the safe fact.
8. **Igarka = China** — FALSE. **Russia**.
9. **NW-1 = Brahmaputra** — FALSE. **Ganga, Haldia–Prayagraj**. NW-2 = Brahmaputra. NW-3 = **Kerala** canal. NW-4 = Kakinada–Puducherry. NW-5 = Talcher–Dhamra.
10. **Kushinagar not international** — FALSE. Intl on **24 Jun 2020**.
11. **First solar airport = Delhi / Ahmedabad** — FALSE. **Cochin**.
12. **Optical fibre = refraction** — FALSE. **Total internal reflection**.

13. **Motihari–Kathmandu pipeline** — FALSE. **Amlekhganj, Nepal**.
14. **Paradip = Andhra / Alappuzha = Tamil Nadu** — FALSE. Paradip **Odisha**; Alappuzha **Kerala**.
15. **UP has a major seaport** — FALSE. Identity = Gorakhpur, NW-1, expressways, Kushinagar/Jewar, 149 Amrit stations.
16. **National Highways = most of India's road kilometres** — FALSE. NH are ~**2%** of length but carry ~**40%** of traffic; rural roads are ~**80%** of length.
17. **Inland waterways already move most Indian cargo** — FALSE. Share is tiny; sea still moves ~**95%** of trade by volume.
18. **IWAI = NHAI** — FALSE. IWAI (1986) = inland waterways; NHAI (1995) = national highways.
19. **Only three National Waterways exist** — FALSE. The 2016 Act declared **111**. The first five are the map you must know even if a paper has not yet asked NW-4 or NW-5.
20. **Kakinada is a major-port-state fact for Odisha** — FALSE. Kakinada = **Andhra Pradesh**; Paradip = **Odisha**.
21. **East–West Corridor = Dibrugarh–Surat / Dwarka** — FALSE. **Porbandar–Silchar**. Cross with NS at **Jhansi**.
22. **Konkan Railway = Kerala main line** — FALSE. **Roha–Mangaluru** through MH–Goa–KA.
23. **SER HQ = Bhubaneswar / SCR HQ = Hyderabad** — FALSE. SER = **Garden Reach, Kolkata**; East Coast = Bhubaneswar; SCR = **Secunderabad**.
24. **North Western Railway HQ = Jodhpur** — FALSE. **Jaipur**.
25. **Chennai = deepest natural port** — FALSE. Often **artificial**; Vizag/Paradip depth notes varies by stem.
26. **Kandla = LNG terminal** — FALSE. LNG terminals are Dahej, Hazira, Kochi, Dabhol, Ennore.
27. **Old NH-7 is still always the longest NH** — FALSE for current MoRTH data — often **NH-27 EW**; also note the **NH-44** corridor.
28. **NS–EW meet at Nagpur / Gwalior** — FALSE. **Jhansi**.
29. **First CNG train = Delhi–Agra** — FALSE. **Rewari–Rohtak**.
30. **Sethusamudram = Cape Comorin only** — FALSE. Links **Gulf of Mannar** with **Palk Bay**.
31. **City of Lakes = Jaisalmer** — FALSE. **Udaipur**. Jaipur = **Pink City**.
32. **Sabarimala = Tamil Nadu / Karnataka** — FALSE. **Kerala**.
33. **Haflong = Himachal / Kalimpong = Assam** — FALSE. Haflong = **Assam**; Kalimpong = **West Bengal**; Chakrata = **Uttarakhand**; Kufri = **HP**.
34. **Omkareshwar = Satmala / Pushkar = Mahadeo** — FALSE. Only **Srisailem–Nallamala** was correct in IAS 2015 among those three.
35. **Charkhari = ghost town** — FALSE. Ghost-town names are Kuldhara, Dhanushkodi, Lakhpat.
36. **BARC = Karnataka** — FALSE. **Maharashtra**. Thumba = **geomagnetic equator** site in Kerala.
37. **Moradabad sarees** — FALSE. Brassware; saree map = Chanderi / Banarasi / Kanjivaram.

Topic 10 — Tribes & Institutions

Current Affairs (this topic)

Year	Fact	Why it matters
15 Nov	Janjatiya Gaurav Diwas — Birsa Munda (मुंडा) birth anniversary	Tribal (आदिवासी) identity day
2023	PM-JANMAN — PVTG mission (housing, health, education, connectivity)	Living PVTG scheme
1875–2025	IMD 150 years; headquarters still New Delhi	Institute anniversary
Jun 2025	Adi Karmayogi launched by Ministry of Tribal Affairs	Ministry trap vs Social Justice / AYUSH
Static	Uttar Pradesh (उत्तर प्रदेश) ST list is still the 15-entry notification set, many of them district-limited	Match traps

Masai habitat (वास स्थान) remains **East Africa**. Do not “update” it to West Africa.

Confused Pairs

Pair	Correct	Trap	Hindi
Khasi	Meghalaya	Arunachal	मेघालय
Rengma	Nagaland	Andaman	नागालैंड
Kuki	North-East	UP	पूर्वोत्तर
Yanadi	Andhra Pradesh	Rajasthan	आंध्र
Toda / Paliyan	Tamil Nadu	Nagaland	तमिलनाडु
Birhor	Jharkhand	Islands / TN	झारखंड
Keria/Kharia	Odisha (also JH)	Random west	ओडिशा
Masai	East Africa	West Africa	पूर्वी अफ्रीका
Khirghiz	Central Asia	Sudan	मध्य एशिया
Koryak	NE Siberia	Alaska	साइबेरिया
Semang	Malaysia	Indonesia	सेमांग
Maasai	East Africa	West Africa / Zambia	मसाई
Bantu	E / C / S Africa	Sahara	बांटू
Transhumance	Seasonal herd move	Permanent emigration	ट्रांसह्यूमेंस
Tharu	Terai	Sonbhadra only	तराई
Agariya	Iron smelting	Farming-only tag	लोहा गलाना
Saharya	Lalitpur	Terai	ललितपुर
NDRI	Karnal	Lucknow / Izatnagar	करनाल
IVRI	Izatnagar (Bareilly)	Karnal	इज्जतनगर
FRI	Dehradun	Delhi / Lucknow	देहरादून
NDRI vs NDDDB	Karnal lab vs Anand co-op board	Same body	एनडीआरआई ≠ एनडीडीबी
PVTG count	75 groups	95	75 पीवीटीजी

Pair	Correct	Trap	Hindi
Fifth vs Sixth Schedule	Fifth = many Scheduled Areas; Sixth = four NE states' councils	Swap	पाँचवीं ≠ छठी

N.1 Tribes of India

State / UT	High-yield tribes
Andhra Pradesh / Telangana	Chenchu, Yanadi , Koya, Lambadi, Gond
Arunachal Pradesh	Nyishi, Apatani , Adi, Monpa — not Khasi
Assam	Bodo , Mishing, Karbi, Dimasa
Bihar	Santhal, Oraon, Munda, Tharu (north)
Chhattisgarh	Gond, Halba, Kavar, Maria and Muria of Bastar
Gujarat	Bhil, Siddi, Rathwa
Himachal (हिमाचल) / J&K–Ladakh (लद्दाख)	Gaddi, Gujjar, Kinnaura, Gujjar–Bakarwal
Jharkhand	Santhal, Munda, Oraon, Ho, Kharia, Birhor , Asur, Birjia
Karnataka	Siddi, Soliga, Koraga, Jenu Kuruba
Kerala	Irular, Kurumba, Kadar, Paniyan, Kattunayakan
Madhya Pradesh	Bhil, Gond, Baiga, Kol, Korku, Sahariya, Bharia
Maharashtra	Bhil, Gond, Warli (वारली), Katkari
Manipur	Naga (नागा) groups, Kuki
Meghalaya	Khasi, Jaintia (जयंतिया), Garo
Mizoram	Mizo (मिज़ो), Chakma, Bru / Reang
Nagaland	Angami , Ao, Sumi, Lotha, Rengma , Konyak
Odisha	Santhal, Munda, Kondh, Saora, Juang, Bonda, Kharia , Ho, Gond
Rajasthan	Bhil, Meena , Garasia, Sahariya
Sikkim	Lepcha , Bhutia, Limboo
Tamil Nadu	Toda , Kota, Irular, Paliyan , Kadar
Tripura	Tripuri, Reang, Jamatia
Uttarakhand (उत्तराखंड)	Tharu, Bhotia, Buksa, Jaunsari, Raji
Uttar Pradesh	Tharu, Buksa, Raji, Bhotia, Jaunsari , Gond, Agariya, Saharya (15 notified entries)
West Bengal	Santhal, Munda, Oraon, Toto
Andaman & Nicobar	Great Andamanese, Onge, Jarawa, Sentinelese, Nicobarese, Shompen

N.2 Tribes of Uttar Pradesh

#	Tribe	Note
1	Bhotia	Himalayan trading and herding
2	Buksa (Bhoksa)	Terai–Bhabar (भाबर) on the Bijnor fringe; PVTG
3	Jaunsari	Himalayan / Uttarakhand–UP set
4	Raji	Tiny northern community; PVTG
5	Tharu	Terai — Kheri, Bahraich, Balrampur, Shravasti, Pilibhit

#	Tribe	Note
6	Gond (Dhuria, Nayak, Ojha, Pathari, Raj Gond)	District-limited in Maharajganj, Siddharthnagar, Basti, Gorakhpur (गोरखपुर), Deoria, Mau, Azamgarh, Jaunpur (जौनपुर), Ballia, Ghazipur, Varanasi (वाराणसी), Mirzapur, Sonbhadra (सोनभद्र)
7	Kharwar	Deoria–Ballia–Ghazipur–Varanasi–Sonbhadra
8	Saharya	Lalitpur
9	Parahiya	Sonbhadra
10	Baiga	Sonbhadra
11	Pankha / Panika	Sonbhadra, Mirzapur
12	Agariya	Sonbhadra / Mirzapur — traditional iron smelting
13	Patari	Sonbhadra
14	Chero	Sonbhadra, Varanasi
15	Bhuiya	Sonbhadra
Zone	Districts	Key tribes
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Terai / north	Kheri, Bahraich, Balrampur, Shravasti, Pilibhit, Bijnor fringe	Tharu, Buksa
Vindhyan / south	Sonbhadra, Mirzapur, Lalitpur and notified east pockets	Gond, Kharwar, Chero, Baiga, Agariya, Saharya

N.6 World tribes

People	Habitat / note
Maasai	E Africa plateau — Kenya, N Tanzania, E Uganda; cattle; kraal huts
Bushman / San	Kalahari (Botswana, Namibia, parts of SA) — hunting–gathering; termites
Zulu	South Africa — agriculture + cattle; Isandlwana 1879
Semang	Malay Peninsula / Malaysia (Negrito hunter–gatherers) — not Indonesia
Sakai	Malay Peninsula / Indonesia forests — blowpipe hunting
Papuan	Papua New Guinea
Bedouin	Arabian deserts (esp. N Saudi) — pastoral nomads
Kirghiz / Kyrgyz	Central Asia (Pamir–Tianshan) — seasonal herding; yurt
Yupik	E Russia / W Alaska — dog sledge
Yukaghir	NE Siberia (Verkhoyansk / Stanovoy)
Maya	Central America — Mexico, Guatemala, Honduras
Maori	New Zealand (Polynesian)
Fulani	West Africa
Bantu	E / C / S Africa — not Sahara
Nuba	Sudan
Vedda	Sri Lanka
Berber	Morocco / N Africa
Ainu	Japan
Aeta	Philippines

People	Habitat / note
Bora (बोरा)	W Amazon (Brazil–Peru–Colombia)
Jews / Teda / Beja / Lur	Israel / Libya / Egypt (मिस्र) / Iran (ethnic match sets)

Wrong claim	Correct teaching
Semang live in Indonesia	The Semang belong to the Malay Peninsula / Malaysia
Maasai live in West Africa or Zambia	The Maasai live in East Africa (Kenya–Tanzania)
Bantu live in the Sahara	The Bantu belt is East / Central / Southern Africa , not the Sahara
Pygmy peoples are pastoral	They are hunters–gatherers of the Congo rainforest
Gypsies originally came from Egypt	Their original home in standard notes is India

Tribe / people	Habitat
1. Bushman	Kalahari Desert
2. Eskimo	Northern Canada
3. Pygmy	Congo Basin
4. Masai	West Africa

N.7 IMD and ICAR

Body	Headquarters	Role
IMD	New Delhi	Weather, climate, cyclone warnings
ICAR	New Delhi	Apex agricultural research and education
IARI	Pusa, New Delhi	Flagship ICAR institute

Common Traps — Don't Fall For These

1. **Khasi = Arunachal** — FALSE. **Meghalaya**.
2. **Rengma = Andaman native** — FALSE. **Nagaland**.
3. **Kuki = Uttar Pradesh** — FALSE. **North-East**.
4. **Yanadi = Rajasthan** — FALSE. **Andhra Pradesh**. Paliyan = **Tamil Nadu**. Keria = **Odisha**.
5. **Masai = West Africa** — FALSE. **East Africa**.
6. **Khirghiz = Sudan** — FALSE. **Central Asia**.
7. **Alaska = Koryak** — FALSE. Koryak = **north-east Siberia**.
8. **Tharu = Sonbhadra only** — FALSE. **Terai**.
9. **UP PVTG = Tharu + Gond** — FALSE. **Buxa + Raji**.
10. **Agariya = only farmers** — FALSE. Traditional **iron smelting**.
11. **Birjia / Santhal = island tribes** — FALSE. **Jharkhand / mainland**.
12. **NDRI = Izatnagar / IVRI = Karnal** — SWAP. NDRI **Karnal**; IVRI **Bareilly**.
13. **FRI = Delhi or Lucknow** — FALSE. **Dehradun**.
14. **Toda = Nagaland** — FALSE. **Tamil Nadu** (Nilgiris).
15. **IMD / ICAR HQ = a regional campus** — FALSE. Both **New Delhi**. NDRI ≠ NDDB; FRI ≠ WII.
16. **PVTG count = 95** — FALSE. There are **75** groups. Odisha has the most communities.
17. **Fifth Schedule = only four NE states** — FALSE. Those four are the **Sixth Schedule** set.
18. **PTG rename year = 1973** — FALSE. Dhebar identified PTGs in 1973. The rename to PVTG is **2006**.

19. **One all-India ST roster** — FALSE. Lists are **state/UT-wise** under Article 342.
20. **Any forest dweller = PVTG** — FALSE. PVTG is a tighter vulnerability subset with set criteria.

Topic 11 — Population Geography

Current Affairs (this topic)

Year	Fact	Why asked	Source
Census 2011	Official number set until Census 2027	Every density/SR/literacy MCQ	ORGI
NFHS-5	TFR ~ 2.0 nationally	NFHS-4 (2015–16) figure used in 2018 was 2.2	MoHFW
UNFPA	World Population Report	2021 Q79	UNFPA
~2023	UN estimate: India most populous	CA only — 2011 still 2nd in Census table	UN / UNFPA
NPP 2000	Stable population 2045	Policy year fact	MoHFW
11 July	World Population Day	2018 Q114	UN

Confused Pairs

Pair	Correct	Trap	Hindi
First census	1872 non-sync; paper 1871 (2023)	1881 as “first ever”	प्रथम जनगणना
First synchronous	1881	1872	समकालिक
2011 rank	15th / 7th after Independence	14th / 8th	15वीं
Density India	382	324 (2001)	382
Highest density state	Bihar	WB / UP / Delhi	बिहार
Lowest density	Arunachal	HP / Mizoram	अरुणाचल
Sex ratio	943	CSR 919	943
Lowest CSR (rural+urban)	Haryana	UP / J&K	हरियाणा
Literacy	74.04% , age 7+	0–6 / 18+	74.04
TFR 2.1	Children per woman	“per thousand” (that is CBR)	कुल प्रजनन दर
Natural growth	CBR – CDR	Includes migration	प्राकृतिक वृद्धि
Dividend	15–59	60+ or 0–6	जनसांख्यिकीय लाभांश
Acceleration (urban curve)	2nd stage	3rd / 4th	द्वितीय
Most urbanised religion	Jains	Christians	जैन
Kanpur / Lucknow million	1971 / 1981	1961/1971	कानपुर 1971
Shrawasti	Lowest female literacy district (UP)	Balrampur / Bahraich	श्रावस्ती
Lorenz curve	Income inequality	Illiteracy	लोरेंज
World Population Report	UNFPA	IMF / WHO / UNDP	यूएनएफपीए
Arithmetic density	Population / total area	Physiological density	अंकगणितीय घनत्व
Physiological density	Population / net sown area	Agricultural density	शारीरिक घनत्व
Natural vs induced growth	Natural = CBR–CDR; induced = migration	Same thing	प्राकृतिक ≠ प्रेरित
Immigration / emigration	In / out	Swap	आप्रवास / उत्प्रवास

Pair	Correct	Trap	Hindi
SC / ST share (2011)	SC ~ 16.6% · ST ~ 8.6%	Swap or invent	अनुसूचित जाति / जनजाति
Optimum Population	Edwin Cannan	Robbins / Dalton / Carr-Saunders as the first name	अनुकूलतम जनसंख्या
Social mal-adjustment	Henry George	Dalton's "maladjustment" formula, or Malthus	सामाजिक कुसमायोजन
Demographic Transition	Thompson (Notestein also)	Cannan / Malthus	जनांकिकीय संक्रमण
Population–food	Malthus	Marx (he criticises Malthus)	जनसंख्या–खाद्य
Geometric vs arithmetic	Population geometric ; food arithmetic	Swap the two progressions	गुणोत्तर / समानांतर
Positive vs preventive checks	Positive = death up; preventive = birth down	Swap	धनात्मक / निवारक नियंत्रण
Social capillarity	Arsène Dumont	Henry George	सामाजिक केशिका

N.1 Census of India

Phase	Years	Character
I	1901–1921	Stagnant / low growth (ends at the Great Divide)
II	1921–1951	Steady growth
III	1951–1981	Rapid growth / population explosion
IV	1981–2011	High growth with a declining rate

N.3 Population Density

Type	Formula	What it shows
Arithmetic density	Total population ÷ total area	Everyday "persons per km ² " fact
Physiological density	Total population ÷ net sown area	Pressure of people on cultivated land
Agricultural density	Agricultural population ÷ net sown area	Pressure of farm workers on cultivated land

N.13 Demographic Transition & Theories

Theory	Propounder
Optimum Population	Edwin Cannan
Social mal-adjustment	Henry George
Demographic Transition	Thompson
Population–food	Malthus

Demographic Transition — Thompson (Notestein)

Stage	Birth rate	Death rate	Growth
1 High stationary	High	High	Slow
2 Early expanding	High	Falls first	Rapid (explosion)
3 Late expanding	Falls	Low	Slows

Stage	Birth rate	Death rate	Growth
4 Low stationary	Low	Low	Stable / very slow
5 Declining (Blacker)	Very low	Low, but can exceed births	Shrinks / ages

Other scholars who appear as options

Idea	Person
Surplus population under capitalism	Karl Marx (with Engels)
Population growth can intensify farming	Ester Boserup
Social capillarity (rise in status → fewer children)	Arsène Dumont
Five-stage transition including decline	C.P. Blacker
Fertility falls as density rises	Sadler
Diet theory (better fed → lower fertility)	Doubleday
Energy to mind versus reproduction	Herbert Spencer
Optimum restated (not the first name)	Robbins / Dalton / Carr-Saunders
DTT restated (not the first name)	Notestein

N.14 Uttar Pradesh Focus

Item	Uttar Pradesh 2011	India 2011
Rank	Most populous state; largest rural population	—
Population	~ 19.98 crore (~ 16.5% of India)	121.09 crore
Density	829	382
Sex ratio	912	943
Literacy	~ 67.7%	74.04%
Urban share	~ 22.3%	31.16%
Rural share	~ 77.7%	68.84%
Decadal growth 2001–11	~ 20.23%	17.64%
Female literacy floor (district)	Shrawasti	—
Million years	Kanpur 1971 , Lucknow 1981	—
UA order	Kanpur > Lucknow > Ghaziabad > Agra	—

Common Traps — Don't Fall For These

- First census = 1881** — FALSE as "first ever." **1881 = first synchronous**; 2023 key **1871**.
- Density 324** — that is **2001**. **2011 = 382**.
- Highest density = West Bengal / Delhi as a "state"** — state fact is **Bihar**.
- Lowest density = Himachal / Mizoram** — **Arunachal**.
- CSR = 943** — 943 is **overall** SR; CSR = **919** (0–6).
- Lowest CSR = UP** — FALSE. **Haryana** (rural and urban).
- TFR 2.1 = 2.1 per thousand** — FALSE. That wording is **CBR**. TFR = per **woman**.
- Natural growth includes migration** — FALSE. **CBR – CDR** only.
- Dividend = 60+ or 0–6** — FALSE. **15–59**.
- Urbanisation acceleration = 3rd/4th stage** — FALSE. **2nd stage**.

11. **Most urbanised religion = Christians** — FALSE. **Jains**.
12. **Kanpur million in 1961 / Lucknow 1971** — FALSE. **1971 and 1981**.
13. **Largest rural population = Maharashtra** — FALSE. **UP**.
14. **Shrawasti vs Balrampur/Bahraich** — female literacy floor = **Shrawasti**.
15. **Replace 2011 with UN 2023 headcount in Prelims tables** — FALSE until **Census 2027**.
16. **Lorenz curve = illiteracy** — FALSE. It measures **income inequality**.
17. **World Population Report = WHO / UNDP** — FALSE. **UNFPA**.
18. **Highest density state = Delhi** — FALSE as a state answer. Fact **Bihar**.
19. **Nagaland = least populous** — FALSE. Negative growth $\neq$ least populous; least populous = **Sikkim**.
20. **Literacy age = 0–6 or 18+** — FALSE. Census literacy = age **7+**.
21. **Arithmetic density = population $\div$ net sown area** — FALSE. That is **physiological** density.
22. **Natural growth includes migration** as “induced natural” — FALSE. Migration = **induced**; natural = CBR–CDR.
23. **Immigration = movement out** — FALSE. Immigration = **in**; emigration = **out**.
24. **ST share = 16.6%** — FALSE. ST $\sim$ **8.6%**; SC $\sim$ **16.6%**.
25. **Skip factors of distribution because UPPCS never asked them** — FALSE. Next year can ask terrain/soil/industry pulls from the same topic.
26. **Optimum Population = Robbins / Dalton / Carr-Saunders as the first name** — FALSE. Propounder = **Edwin Cannan**. The others restated it.
27. **Social mal-adjustment = Dalton or Malthus** — FALSE. **Henry George**. Dalton’s $M = (A - O) / O$ is the optimum **gap**, not George’s land-distribution argument.
28. **Demographic Transition = Cannan** — FALSE. **Thompson** (Notestein also appears). Cannan is optimum.
29. **Population geometric / food arithmetic swapped** — FALSE. Population is **geometric**. Food is **arithmetic**.
30. **Malthusian Theory = poverty / economy / unemployment** — FALSE. It is a theory of **population**. Poverty-from-distribution is **Henry George**.
31. **Positive checks = delayed marriage** — FALSE. Positive checks **raise deaths** (famine, disease, war). **Preventive** checks lower births.
32. **Marx = geometric vs arithmetic** — FALSE. Marx criticised Malthus and treated surplus population as a product of **capitalism**.
33. **Dumont = Social Mal-adjustment** — FALSE. Dumont is **social capillarity**. George is social mal-adjustment.

Topic 12 — Human Geography (Settlements)

Current Affairs (this topic)

Year	Fact	Why asked	Source
25 Jun 2015	Smart Cities Mission + PMAY-U same day	Year pair vs JNNURM 2005	MoHUA
Jun 2015	AMRUT (~500 cities)	2025 chronology	MoHUA
1 Oct 2021	AMRUT 2.0 — water-secure towns	Successor of AMRUT	PIB / MoHUA
21 Feb 2016	SPMRM launched (not “2015 launch”)	2023 Q143	MoRD
ISAC-2020	Indore/Surat best cities; UP best state ; category winners above	2022 Q26	PIB / MoHUA
100 / 10	100 Smart Cities nationally; 10 Central in UP	2018 + 2020	MoHUA
2019	UP State Smart Cities (7 Nagar Nigams)	Does not change 2018/2020 keys of 10	UP DoUD
Mission window	Smart Cities Mission timeline extended to 31 Mar 2025	Completing works	MoHUA

Million-city **population (जनसंख्या) ranks**. Missions and settlement types

Confused Pairs

Pair	Correct	Trap	Hindi
Census town	5,000 + 400/km ² + 75% non-agri male main	Any big village / always a municipality	जनगणना नगर
Clustered village	Plains + Rajasthan water + Bundelkhand/Nagaland defence	“Thar = dispersed”	संकुल
Dispersed village	Meghalaya, UK, HP, Kerala , NE forests/hills	Ganga plain / all Rajasthan	प्रकीर्ण
Hamleted local names	panna, para, palli, nagla, dhani	A separate census town	पुरवा/दाणी
Semi-clustered	Dominant caste centre ; lower strata flanks (GJ/RJ)	Same as hamleted	अर्ध-संकुल
Class I town	≥ 1 lakh (468 in 2011; ~ 60% urban pop)	≥ 10 lakh	श्रेणी I
Metropolitan (NCERT)	10 lakh–50 lakh	10 million	महानगर
Mega city (NCERT)	> 50 lakh ; six in 2011	Only 1 crore / only Mumbai	मेगा
UN megacity	≥ 1 crore	Mixing with NCERT 50-lakh mega	UN मेगा
2011 size rank	Mumbai > Delhi > Kolkata > Chennai	Mumbai > Kolkata > Delhi (that is 2001)	2011 क्रम
Bhopal	Not on a major river bank	Agra/Patna/Kolkata are	भोपाल ≠ नदी
UP Smart Cities (papers)	10 Central listed	12/15/18; Ghaziabad	10 शहर
Ghaziabad	Not in Mission-100 / Central-10	“Is a Smart City” (only State 2019)	गाजियाबाद नहीं
SPMRM year	2016 launch	2015	2016
HRIDAY in UP	Varanasi + Mathura	Prayagraj / Ayodhya	काशी+मथुरा
AMRUT	Jun 2015	2005 (that is JNNURM)	2015

Pair	Correct	Trap	Hindi
JNNURM	2005	2015	2005
Smart Cities vs SPMRM	MoHUA vs MoRD	Same ministry	आवास vs ग्राम
Global Village	Transport + communication	UN / politics only	परिवहन+संचार
Rurbanization (options)	G.S. Ghurye	M.N. Srinivas	घुर्ये
Determinism	Nature controls man (Ratzel / Semple)	Possibilism	नियतिवाद
Possibilism	Man chooses within nature (Vidal de la Blache)	Determinism	संभावनावाद
Neo-determinism	Stop-and-go determinism (Griffith Taylor)	Pure free will	नव-नियतिवाद
Site vs situation	Local ground vs wider regional location	Same thing	स्थल ≠ स्थिति

N.0 Framework — Nature & Scope of Human Geography

Approach	Core fact	Name pair
Environmental determinism	Nature controls human life and culture	Friedrich Ratzel; Ellen C. Semple
Possibilism	Nature offers possibilities; humans choose	Vidal de la Blache
Neo-determinism / stop-and-go	Nature sets limits; man can proceed carefully	Griffith Taylor

N.1 Settlements — Rural vs Urban

Feature	Rural	Urban
Economy	Mainly primary (farm, forest, fish, village-scale mine)	Secondary / tertiary / quaternary
Density / size	Lower; hamlet to village	Compact; town to mega city
Link	Supply food and raw (रॉ) material	Supply goods, services and jobs
Census status	Village / rural	Statutory town or census town
Social life	Face-to-face primary groups	Secondary groups and specialised jobs

N.2 Types of Villages

Type	Form	Where (India) — match this
Clustered / nucleated / agglomerated	Compact houses with living area separate from fields; streets may be rectangular, radial or linear	Fertile alluvial (जलोढ़) plains ; also NE plains; Rajasthan (थार) (scarce water → cluster at well/oasis); Bundelkhand + Nagaland (defence)
Semi-clustered / fragmented	One main cluster plus groups pushed a little away	Gujarat and Rajasthan fragmentation. The dominant / land-owning community sits at the centre ; lower strata / menial workers sit on the outer flanks
Hamleted	Several physically separated units, one village name	Middle & lower Ganga, Chhattisgarh , lower Himalayan valleys
Dispersed / isolated	Isolated huts / tiny hamlets on hills, jungle, pasture slopes	Meghalaya, Uttarakhand, Himachal, Kerala ; forests & hills of North-East

Name	Region
Panna / para / palli	East / Ganga–Chhattisgarh belt

Name	Region
Nagla	Western UP / Ganga plain
Dhani	Rajasthan

Shape	Note
Linear	Road, river, canal, coast, levee
Rectangular	Planned / northern plains field grid
Circular / radial	Around tank, temple, well
Star / T / Y	Where routes meet

Belt	Note
Heavy rain (W. coast, NE)	Sloping roofs; bamboo / wood in NE
Arid Rajasthan	Flat roofs; compact around water
Himalaya (हिमालय)	Stone / slate
Ganga plain	Mud / brick; courtyard houses

N.3 Functional Classification of Towns

Class	Population	2011 fact
I (city)	≥ 1,00,000	468 towns; ~ 60% of India's urban population
II	50,000–99,999	
III	20,000–49,999	
IV	10,000–19,999	
V	5,000–9,999	
VI	< 5,000	

Type	Example
Administrative	Chandigarh, New Delhi, Bhopal, Shillong, Guwahati, Imphal, Srinagar, Gandhinagar, Jaipur, Chennai, Lucknow (लखनऊ)
Industrial	Jamshedpur, Bhilai, Durgapur, Modinagar, Barauni, Sindri, Mumbai, Madurai, Kanpur (कानपुर)
Transport / port	Kandla, Kochi, Visakhapatnam, Mughalsarai / Pt. Deen Dayal Upadhyaya Jn. , Itarsi, Katni
Commercial	Kolkata, Saharanpur , Satna
Mining	Raniganj, Jharia / Dhanbad, Digboi, Ankleshwar, Singrauli
Garrison / cantonment	Ambala, Jalandhar, Mhow, Babina, Udhampur, Meerut (मेरठ)
Educational	Roorkee, Aligarh , Pilani, Varanasi (BHU), Prayagraj
Religious / cultural	Varanasi (वाराणसी), Mathura, Amritsar (अमृतसर), Madurai, Puri (पुरी), Ajmer (अजमेर), Pushkar, Tirupati, Kurukshetra, Haridwar (हरिद्वार), Ujjain (उज्जैन)
Tourist	Nainital (नैनीताल), Mussoorie (मसूरी), Shimla, Pachmarhi, Jodhpur, Jaisalmer, Udhagamandalam (Ooty), Mount Abu, Agra (आगरा)

Layer	Fact
Ancient (~2000+ yrs)	Varanasi, Prayag (Prayagraj), Pataliputra (पाटलिपुत्र) (Patna), Madurai — religion / river / administration

Layer	Fact
Medieval (~100 towns)	Fort / court HQ — Delhi, Hyderabad, Jaipur, Lucknow, Agra, Nagpur
Modern — European foothold	First ports at Surat, Daman, Goa (गोवा), Puducherry; later Mumbai, Chennai, Kolkata ; hill stations as summer resorts; cantonments
After Independence	New capitals such as Chandigarh, Bhubaneswar, Gandhinagar, Dispur ; industrial towns such as Durgapur, Bhilai, Sindri, Barauni
Satellite towns	Old towns pulled into a metro's orbit — Ghaziabad, Gurugram, Rohtak around Delhi; Navi Mumbai / Thane around Mumbai

City	Note
New Delhi	Lutyens / Baker
Chandigarh	Le Corbusier
Bhubaneswar	Otto Koenigsberger
Jamshedpur	Company (कंपनी) / steel town (Tata (तत्))

N.4 Metropolitan Cities & Urban Hierarchy

Term	Fact (Census / NCERT)
Town	Urban centre below Class I, or loosely any urban place
City / Class I	≥ 1 lakh
Million city	≥ 10 lakh · Census 2011 = 53 UAs · largest Greater Mumbai · smallest Kota
Metropolitan (NCERT)	1–5 million (10 lakh–50 lakh)
Mega city (NCERT table)	> 5 million · six in 2011: Greater Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad · ~ 21% of urban pop
Megacity (UN)	≥ 10 million · 2011 India: Mumbai, Delhi, Kolkata
Conurbation	Once-separate cities fused (Mumbai–Thane)
Megalopolis	Chain of conurbations (Boswash type)
Primate city	One city dwarfs the national urban system — India does not have a single primate city (Mumbai vs Delhi)
NCR	Delhi + Ghaziabad, Noida, Greater Noida, Meerut (UP) + Gurugram, Faridabad (HR)

N.5 Smart Cities & Heritage Cities

Round	When	Cities
Fast-track / first UP entry	May 2016	Lucknow (first UP city on the Mission list)
Round 2	Sep 2016	Kanpur, Agra, Varanasi
Round 3	Jun 2017	Prayagraj (Allahabad (इलाहाबाद)), Aligarh, Jhansi
Round 4	2018	Moradabad, Bareilly, Saharanpur

Category	Winner in the paper's list
Culture	Indore
Governance	Vadodara
Social Aspects	Tirupati
Urban Environment	Bhopal

N.6 Smart Villages & Scheme Years

Scheme	Year / fact	Trap
SPMRM (Shyama Prasad Mukherji Rurban Mission)	Launch 21 Feb 2016 (Cabinet 16 Sep 2015); MoRD ; about 300 clusters	2015 is not the launch year (2023 NOT-matched)
Sansad Adarsh Gram Yojana (SAGY)	11 Oct 2014	
DDU-GKY	2014	
Rural Housing Interest Subsidy Scheme	2017 (that paper)	
JNNURM	2005	Confusing with AMRUT
National Urban Housing and Habitat (वास स्थान) Policy	2007	
AMRUT	June 2015 ; about 500 cities (all 1-lakh+ ULBs, capitals, HRIDAY cities and selected others)	2005
AMRUT 2.0	1 Oct 2021 — water security; all statutory towns	Calling it 2015
Jal Jeevan Mission	2019 (Har Ghar Jal)	Putting it before AMRUT
PMAY-U	25 Jun 2015 (same day as Smart Cities) — housing neighbour, year fact only	
SBM-Urban	2 Oct 2014 (with rural SBM) — sanitation neighbour	Calling it 2015 Smart Cities
DAY-NULM	Urban livelihoods (MoHUA)	Mixing with SPMRM / DDU-GKY

Scheme	Year
JNNURM	2005
National Urban Housing and Habitat Policy	2007
AMRUT	2015
Jal Jeevan Mission	2019

N.7 Uttar Pradesh Focus — Urban Centres

City	Functional / tag
Lucknow	State capital — administrative; first UP Smart City (May 2016)
Kanpur	Industrial / leather–textile; first UP million city (1971)
Prayagraj (प्रयागराज)	Ancient Prayag; educational; Smart City round 3; not HRIDAY
Varanasi	Ancient; religious–educational (BHU); HRIDAY; Smart City round 2
Agra	Medieval; tourist (Taj); Smart City round 2
Aligarh	Educational (AMU); industrial neighbour of the western UP belt; Smart City round 3
Meerut	Garrison / NCR satellite — not Central Smart-10
Ghaziabad	NCR satellite / large UA — not Central Smart-10 (State 2019 only)
Noida / Greater Noida	Planned satellite of Delhi — not Mission-100
Saharanpur	NCERT commercial town; Smart City round 4
Bareilly / Moradabad / Jhansi	Smart City round 4 / 3; Moradabad brass
Mathura	Religious; HRIDAY ; State Smart City 2019 (not Central-10)
Ayodhya (अयोध्या)	Religious; State Smart City 2019; not HRIDAY

City	Functional / tag
Singrauli belt (Sonbhadra (सोनभद्र))	Mining / energy belt — mining-town type
Mission fact	Fact
-----	-----
Central Smart Cities	10 — Ghaziabad is out of Mission-100
State Smart Cities (2019)	7 Nagar Nigams — do not change 2018/2020 keys
HRIDAY	Varanasi and Mathura only
ISAC-2020	UP is the best state
NCR (UP slice)	Ghaziabad, Noida, Greater Noida, Meerut

Common Traps — Don't Fall For These

1. **Ghaziabad = Central Smart City** — FALSE for Mission-100 / 2018–2020 keys. State-2019 list is a **different** tag.
2. **UP Smart Cities = 12/15/18** — FALSE. Papers key **10**.
3. **HRIDAY = Prayagraj / Ayodhya** — FALSE. **Varanasi + Mathura**. National total **12**, not 100.
4. **SPMRM launched 2015** — FALSE as launch year. **21 Feb 2016**.
5. **Class I town = 10 lakh** — FALSE. **1 lakh**. 10 lakh = million UA.
6. **Census town = any big village / always a municipality** — FALSE. Triple test **or** statutory town; 75% is **male main** workers.
7. **AMRUT = 2005** — FALSE. **2015**. **JNNURM = 2005**.
8. **Rurbanization = M.N. Srinivas** — FALSE in that 2022 set. **G.S. Ghurye**.
9. **Global Village = UN created it** — FALSE. **Transport + communication**.
10. **Dispersed villages = Ganga plain / all of Thar** — FALSE. Plains = clustered. **Rajasthan = clustered (water)**. Dispersed = **Meghalaya, UK, HP, Kerala, NE forests**.
11. **Smart village scheme = MoHUA Smart Cities** — FALSE. SPMRM = **MoRD**.
12. **NCERT mega city = 10 lakh or only 1 crore** — FALSE. NCERT mega = **> 50 lakh (six)**; metro = **10–50 lakh**; UN megacity = **1 crore**.
13. **ISAC Culture = Vadodara** — FALSE. Culture **Indore**; Governance **Vadodara**.
14. **Determinism = Vidal / Possibilism = Ratzel** — SWAP. Determinism = **Ratzel/Semple**; Possibilism = **Vidal**.
15. **Neo-determinism = free will without limits** — FALSE. **Griffith Taylor** stop-and-go.
16. **Site = regional location** — FALSE. Site = **local ground**; situation = region.
17. **Skip nature/scope because recent papers asked Smart Cities** — FALSE. Next year can ask approaches from the same topic.
18. **Varanasi = major seaport town** — FALSE. Religious-cultural; HRIDAY yes.
19. **2011 UA rank = Mumbai > Kolkata > Delhi** — that is **2001**. **2011 = Mumbai > Delhi > Kolkata > Chennai**.
20. **Bhopal = river-bank town like Agra/Patna** — FALSE.
21. **Hamleted = a census town** — FALSE. Local **panna/para/palli/nagla/dhani**, one village name.
22. **75% workers in primary sector = census town** — FALSE. Need **non-agriculture** male main workers.

Topic 13 — Disaster Geography

Current Affairs (this topic)

Year	Fact	Why asked	Source
Mar 2026	BIS IS 1893:2025 / Zone VI withdrawn	Do not replace II–V in the hall	The Hindu / BIS
2015–30	Sendai Framework	Successor of Hyogo	UNDRR
2019	CDRI launched by India	Infrastructure + DRR	MEA / NDMA
2016 / 2019	NDMP issued / revised	First Sendai-aligned national plan	NDMA
2023	Joshimath subsidence; Sikkim GLOF	Landslide / flood CA	NDMA / news
2021	Tauktae (Arabian Sea → Gujarat)	“Arabian Sea never hits west coast” trap	IMD
2020	Amphan (BoB → WB/Odisha)	Super cyclone east coast	IMD

Cyclone **formation** (SST ~27°C, Coriolis, low shear) and IMD wind ladder are taught in N.10. Ring-of-Fire volcano list remains a Kilimanjaro trap pair.

Confused Pairs

Pair	Correct	Trap	Hindi
Focus vs epicentre	Focus = inside ; epicentre = surface	Swapped	उद्गम / अधिकेन्द्र
Body vs surface waves	Body = through Earth (P, S); surface = along ground (Love, Rayleigh)	Treating all as one family	शरीर तरंग / पृष्ठ तरंग
P-wave vs S-wave	P = fastest, solids+liquids+gases; S = slower, solids only	S through outer core	प्राथमिक / द्वितीयक
P shadow vs S shadow	P miss ~ 103–142° ; S absent beyond ~ 103°	Same belt for both / both recorded past 105°	छाया क्षेत्र
Richter vs Mercalli	Magnitude vs intensity/damage	Same scale	रिक्टर / मर्कली
Zone V vs II	V = highest (NE, Himalaya, Kutch, A&N)	V = safest	क्षेत्र V
Latur 1993	Peninsular / Killari (Zone III)	“Only Himalaya shakes”	लातूर
Koyna 1967	Reservoir-induced	Pure Himalayan thrust	कोयना
Rajasthan / Deccan	Mostly II–III	All India Zone V	
Tsunami	Seafloor slip / landslide / volcano — Japanese harbour wave	Wind-raised waves / “tidal wave”	सुनामी
INCOIS	Tsunami warning — Hyderabad (MoES)	NDMA HQ / IMD Pune as the tsunami hub	INCOIS
Kilimanjaro	E. African Rift — not Ring of Fire	Fuji/Pinatubo/St Helens are	किलिमंजारो
Cloudburst	≥100 mm in one hour	Any heavy rain	बादल फटना
Heat wave plains	Gate ≥40°C then +4.5°C	Any hot afternoon	लू / ऊष्मा तरंग
BoB vs Arabian	BoB more cyclones	Equal / Arabian more	बंगाल की खाड़ी
Baguio	Philippines	Japan / USA	बागिओ
Willy-willies	Australia	USA	विली-विली
NDMA chair	Prime Minister	Home Minister	प्रधानमंत्री
SDMA chair	Chief Minister	Governor	मुख्यमंत्री

Pair	Correct	Trap	Hindi
Sendai vs Hyogo	2015–30 vs 2005–15	Yokohama 2015	सेंडाई
DPAP year	1973–74	1952 (that is CDP)	DPAP
Hazard vs disaster	Hazard = potential; disaster = when capacity fails	Same word	आपदा ≠ संकट बीज
Mitigation vs response	Before impact vs during/after	Swap	न्यूनीकरण ≠ अनुक्रिया
NPDM year	2009	2005 (that is DM Act)	2009 नीति
Chemical / industrial	Bhopal 1984 man-made	"Natural only India"	भोपाल
GLOF vs ordinary flood	Glacial-lake outburst (e.g. Sikkim 2023)	Any river monsoon rise	GLOF
Joshimath vs Kedarnath	Joshimath 2023 = subsidence; Kedarnath 2013 = rain–debris–flood	Same mechanism	जोशीमठ / केदारनाथ
Heat gate vs hot day	Plains need T_{max} ≥40°C then departure/actual cut	Any afternoon above 35°C	ऊष्मा तरंग
BoB peak months	May–Jun and Oct–Dec	Mid-monsoon as the only peak	चक्रवात ऋतु
NDRF year	Raised 2006 (MHA)	2005 = DM Act only	NDRF
NEC chair	Union Home Secretary	Home Minister / PM	NEC

N.1 Earthquakes — Mechanics

Seismic waves — body first, then surface

Wave	Family	Nature	Travels through	Arrival / damage
P (primary)	Body	Longitudinal / sound-like; fastest	Solids, liquids, gases	First
S (secondary)	Body	Transverse / shear; slower than P	Solids only	Second; stopped by liquid outer core
Love / Rayleigh	Surface	Along the free surface	Near-surface rock / soil	Last; usually most destructive at ground

Shadow zones

Term	Recall
Focus / hypocentre	Inside the crust
Epicentre	Surface above the focus
Richter / Mw	Magnitude (energy)
Mercalli	Intensity (damage)
Body waves	Through Earth — P and S
Surface waves	Along ground — Love and Rayleigh
P then S then surface	Arrival order
S shadow	Beyond ~ 103° (liquid outer core)
P shadow	Gap ~ 103–142° (core refraction)

N.2 Earthquakes in India

Hall earthquakes — Cause, Course, Result

Year	Event	Trap
1819	Allah Bund / Kutch	
1897	Shillong / Assam	
1905	Kangra	
1934	Bihar–Nepal (east UP felt)	
1950	Assam–Tibet (~8.6)	
1967	Koyna (MH) — reservoir-induced	"Only Himalaya"
1991	Uttarkashi	
1993	Latur / Killari — stable peninsula	Himalayan
1999	Chamoli (चमोली)	
2001	Bhuj (26 Jan)	
2004	Sumatra–Andaman (tsunami)	
2005	Kashmir / Muzaffarabad	
2011	Sikkim	
2015	Nepal Gorkha (Bihar/east UP)	

N.3 Seismic Zones of India

Zone	Risk	Where (list)
V	Very high	NE India, Andaman & Nicobar, Himalayan pockets, Kutch, N Bihar
IV	High	Rest of Himalayan belt, Delhi, N/E UP Terai
III	Moderate	Broad peninsula, most of UP, many metros, Latur–Koyna
II	Low	Large stable Deccan / western Rajasthan interiors

N.4 Tsunami

26 December 2004 — Indian Ocean mega-event

Item	Fact
Word origin	Japanese harbour wave
Trigger	Seafloor slip — quake / landslide / volcano
Not the same as	Wind waves / cyclone swell
Speed behaviour	Fast in deep water; piles up near coast
Warning hub	INCOIS, Hyderabad (MoES)
India high-risk coasts	A&N, east coast, Kerala; west coast lower but not zero
Hall event	26 Dec 2004 — Sumatra megathrust
Soft shield	Mangroves / coastal forests cut surge and wave force

N.5 Landslides

Types (coaching map)

Type	What moves
Fall	Rocks detach and freefall from a cliff
Slide	A mass moves on a distinct slip surface
Flow	Wet debris / mud moves like a fluid (debris flow)
Creep	Very slow downhill soil movement over years

N.7 Drought

Types

Type	Meaning
Meteorological	Rainfall well below normal
Hydrological	Rivers / tanks / groundwater fail
Agricultural	Soil moisture / crop stress
Socio-economic	Livelihood failure

Schemes chronology (asked order)

Scheme	Year
Community Development Programme	1952
DPAP (Drought Prone Areas Programme)	1973–74
Food for Work	1977
DDP (Desert Development Programme)	1977–78
TRYSEM	1979

N.9 Heat Waves & Cold Waves

Heat wave (IMD)

Region	Gate (Tmax)
Plains	≥ 40°C
Coast	≥ 37°C
Hills	≥ 30°C
Rule	Heat wave
-----	-----
Departure from normal	4.5–6.4°C
Actual Tmax (plains)	≥ 45°C

Confused pair

Pair	Correct
Heat wave vs any hot day	Need gate + departure / actual cut

Pair	Correct
Cold wave vs ordinary winter night	Need abnormal Tmin drop / departure
Loo	Hot dry plains wind of north India in summer

N.10 Cyclone Basins of India

IMD wind ladder (increasing intensity)

Stage	Approx. wind (km/h)
Low pressure area	< 31
Depression	31–49
Deep Depression	50–61
Cyclonic Storm	62–88
Severe Cyclonic Storm	89–117
Very Severe Cyclonic Storm	118–165
Extremely Severe Cyclonic Storm	166–220
Super Cyclonic Storm	≥ 221

Basins and seasons

Basin	Disaster fact
Bay of Bengal	Higher frequency (~4:1 vs Arabian in many decades); Odisha–AP–TN–WB–Sundarbans (সুন্দরবন) surge
Arabian Sea	Fewer storms, but they do hit Gujarat / Maharashtra
Peaks	May–Jun and Oct–Dec (retreating monsoon (लौटता मानसून)). Mid-monsoon shear suppresses BoB

Local names (match bank)

Name	Country / basin	Note
Baguio / Baguios ★	Philippines	High-yield match
Hurricane ★	USA / Atlantic–Caribbean	High-yield match
Typhoon ★	NW Pacific — China class	High-yield match
Taifu ★	Japan -facing NW Pacific name	Often paired with Japan
Willy-willies ★	Australia	Cyclone ≠ Brickfielder (ब्रिकफील्डर)
Cyclone	North Indian Ocean (India /BoB/Arabian)	IMD names
Cordonazo	West Mexico Pacific	Coaching distractor

2019 named-storm match (core four) + nearby names

Storm	Most affected (fact)
Dorian ★	Bahamas
Hagibis ★	Japan
Lekima ★	China
Mitag ★	Jeju, South Korea
Amphan (2020)	West Bengal / Odisha
Fani (2019)	Odisha

Storm	Most affected (fact)
Tauktae (2021)	Gujarat / west coast
Biparjoy (2023)	Gujarat / Arabian Sea
Odisha Super Cyclone (1999) / Phailin / Hudhud / Yaas	Classic India east-coast set

N.11 Disaster Management Framework

International and CA pairs

Body	Role
DM Act	23 Dec 2005
NDMA	Apex body; the PM is ex-officio Chair; up to 9 members
SDMA	CM chairs
DDMA	District Magistrate chairs (elected head co-chairs)
NEC	National Executive Committee — Union Home Secretary
NDRF	2006 , specialised response under MHA ; started at 8 Bn and later rose to 16 Bn
NIDM	Training / research in New Delhi (from NCDM 1995 to NIDM)
Nodal ministry	MHA (Disaster Management Division)
NDMP	1 Jun 2016 (revised 2019) — aligned to Sendai
Sendai	2015–2030 , Sendai (Japan), after Hyogo 2005–2015 and Yokohama 1994
NPDM	National Policy on Disaster Management — 2009
CDRI	Coalition for Disaster Resilient Infrastructure — India, 2019
IDDR	International Day for DRR — 13 October
NCS	National Centre for Seismology — MoES
INCOIS	Tsunami — Hyderabad
IMD	Cyclone, rainfall, heat/cold

N.12 Uttar Pradesh Focus

Institutions

Hazard	Fact
Seismic	No Zone V . Terai / north-east is IV ; Lucknow–Kanpur–Prayagraj–Varanasi–Agra belt is III
Flood	East UP — Ghaghara, Rapti, Gandak, Ganga ; trans-Ghaghara belt
Drought	Bundelkhand + Vindhyan / south-west pockets
Heat	May–Jun plains (Loo)
Cold / fog	Western UP + Terai, Dec–Jan
Cloudburst	Not a plains speciality; hills of neighbouring Uttarakhand
Biological-disaster label (2018)	Gomati (pollution) — not Yamuna/Sai/Tamsa in that paper
Framework	SDMA under CM ; DDMA under DM

Common Traps — Don't Fall For These

1. **NDMA chair = Home Minister** — FALSE. **Prime Minister**.
2. **Zone V = safest / Zone I still exists** — FALSE. **V = highest**; I abolished.
3. **Entire UP = Zone V** — FALSE. **III–IV**.
4. **Only Himalaya shakes** — FALSE. **Latur 1993, Koyna 1967**.
5. **Richter = damage scale** — FALSE. **Mercalli** = intensity.
6. **S-waves travel through the liquid outer core / both P and S recorded past 105°** — FALSE. **S** stop beyond ~103°; **P** miss ~103–142°.
7. **Surface waves arrive first / P waves do most surface damage** — FALSE. Arrival is **P → S → surface**; surface waves usually do worst ground damage.
8. **Tsunami = wind waves / cyclone swell / tidal wave** — FALSE. **Seafloor displacement**; Japanese **harbour wave**.
9. **INCOIS = NDMA HQ / IMD Pune** — FALSE. Tsunami warning = **INCOIS, Hyderabad**.
10. **Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
11. **Cloudburst = any hill rain** — FALSE. **≥100 mm/hour**.
12. **Heat wave = any day above 35°C** — FALSE. Plains **gate 40°C** + departure/actual cuts.
13. **Arabian Sea never hits India** — FALSE. **Gujarat/MH (Tauktae)**.
14. **Baguio = Japan; Willy-willy = USA** — FALSE. **PH and Australia**.
15. **Sendai = 2005–15** — FALSE. That is **Hyogo**. Sendai **2015–30**.
16. **NPDM = 2005** — FALSE. DM Act **2005**; NPDM **2009**.
17. **NDMA chair = Home Minister** — FALSE. **Prime Minister**.
18. **Hazard = disaster always** — FALSE. Disaster needs exposure + weak capacity.
19. **Skip HVRC / Sendai priorities because recent papers asked zones** — FALSE. Next year can ask the framework from the same topic.
20. **Arabian Sea never hits India** — FALSE. Gujarat–Maharashtra landfalls exist.
21. **India is disaster-free** — FALSE (2018 key).
22. **DPAP = 1952** — FALSE. **1973–74**; 1952 = CDP.
23. **Gomati vs Yamuna as 2018 biological-disaster river** — paper key **Gomati**.
24. **2004 deaths = 50 lakh / Bangladesh worst-hit by that wave path** — FALSE. Order **2–3 lakh**; Bangladesh relatively few for wave-direction reasons.
25. **Joshimath 2023 = same as Kedarnath 2013 rain wall** — FALSE. Joshimath is mainly **subsidence / slope distress**; Kedarnath is a rain–debris–flood cascade.
26. **GLOF = ordinary Ganga bank flood** — FALSE. **Glacial-lake outburst** (Sikkim Teesta / South Lhonak 2023).
27. **Cyclones need the equator / form only in mid-monsoon BoB** — FALSE. Need **Coriolis** (usually outside ~5°) and peak in **May–Jun / Oct–Dec**.
28. **Bhopal = natural earthquake disaster** — FALSE. **Man-made** chemical / industrial (1984).

Topic 14 — Earth & Universe

Current Affairs (this topic)

Year	Fact	Why asked	Source
2013–14	Mangalyaan / MOM	First Indian Mars orbiter (2024)	ISRO
2023	Aditya-L1	Solar mission — not Mars	ISRO
1987 / 1997	Montreal vs Kyoto (क्योटो)	Ozone vs climate	UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) / UNFCCC (यूएनएफसीसीसी)
2015	Paris Agreement (पेरिस समझौता)	NDC successor to Kyoto path	UNFCCC
2018	Science — Mars subglacial lake	2018 Q85	Science
2026	El Niño tendency vs Indian monsoon	IMD	

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q95

Which planet takes the longest time completing one revolution around the sun?

A. Saturn B. Uranus C. Neptune D. Jupiter

Confused Pairs

Pair	Correct	Trap	Hindi
Ozone layer	Stratosphere	Troposphere	समतापमंडल
Montreal vs Kyoto	Ozone vs climate/GHG	Kyoto "save water"	मॉन्ट्रियल / क्योटो
El Niño fish	Less plankton	Warm water → more plankton	अल नीनो
Coriolis SH	Left	Right / magnetic	कोरिऑलिस
Jet streams	Mainly westerly	"All easterly" (2024 A)	जेट धारा
TEJ	Summer easterly ~14°N	Same as STWJ	उष्णकटिबंधीय पूर्वी जेट
ITCZ	Low / rising air	High-pressure belt	आईटीसीजेड
Atmosphere heat	Long-wave terrestrial	Direct short-wave	दीर्घ तरंग
Fujita vs Saffir–Simpson	Tornado vs Hurricane	Swapped	फुजिता / सफिर–सिंपसन
Willy-Willy	Australia	USA hurricane	विली-विली
Horse latitudes	Subtropical high ~30°	Low pressure	हॉर्स लैटिट्यूड
Cumulonimbus	Intense rain / thunder	Fair-weather cumulus	कपासी वर्षा मेघ
Albedo	Reflecting power	Absorbing	अल्बीडो
GHG	Not argon	Argon as GHG	आर्गन नहीं
Saturn (that paper)	Methane	SO2 / CO	शनि
Mars	Life-candidate + 2018 lake	Jupiter lake	मंगल
Kilimanjaro	E. African Rift	Ring of Fire	किलिमंजारो
Fold vs fault	Compressional folds vs block/rift	All Himalaya = fault	वलित / भ्रंश

Pair	Correct	Trap	Hindi
Perihelion	January (NH winter)	"Closest = summer"	उपसौर
IDL	~ 180° ; west is ahead	Straight 180° only	अंतरराष्ट्रीय तिथि रेखा
IST	82°30' E, +5:30 , whole India	Only east of Mirzapur	भारतीय मानक समय
Nuclear / geothermal	Not stored solar	All energy = solar	नाभिकीय / भूतापीय
Hottest planet	Venus (greenhouse)	Mercury	शुक्र
Largest moon	Ganymede (Jupiter)	Titan / Moon	गेनीमीड
Asteroid belt	Mars–Jupiter	Earth–Mars	क्षुद्रग्रह
Solar vs lunar eclipse	New Moon vs Full Moon	Swapped	सूर्य / चंद्र ग्रहण
Roaring Forties	~ 40°S westerlies	40°N trades	गर्जन-चालीसा
Horse latitudes	~ 30° subtropical high	ITCZ	अश्व अक्षांश
Stromboli	Mediterranean / Italy	Ring of Fire Pacific	स्ट्राम्बोली
Sial vs Sima	Continents vs ocean crust	Swapped	सियाल / सीमा
Heat budget	Earth absorbs ~ 51 /100; planet albedo ~ 35 reflected	"All 100 heat the air directly"	ऊष्मा बजट
Seasons cause	23½° tilt	Perihelion distance	ऋतु
Divergent boundary	Plates apart / MOR	Convergent trench	अपसारी
Chandrasekhar limit	~ 1.44 M_☉ white dwarf	Eddington / Fowler alone	चंद्रशेखर
Milky Way shape	Barred spiral	Plain spiral only	आकाशगंगा
Earth density	Highest mean among 8	Jupiter densest	घनत्व
Albedo Mercury vs Earth	Earth higher (~0.3)	Mercury brighter albedo	अल्बीडो
Brittle star	Sea echinoderm	Pulsar / quasar class	ब्रिटल स्टार
0°–0° meet	Atlantic / Guinea	On land / India	शून्य
Cancer / Capricorn traps	Not Iran / not Bolivia	Wrong country lists	कर्क / मकर
Granite	Igneous	Metamorphic	ग्रेनाइट
Limestone →	Marble	Quartzite	संगमरमर
Sandstone →	Quartzite	Marble	क्वार्ट्जाइट
Shale →	Slate	Gneiss first	स्लेट
Magma vs lava	Below vs surface	Swapped	मैग्मा / लावा
Volcanic gas	Mostly H₂O vapour	CO ₂ / SO ₂ first	जल वाष्प
Stromboli	Active Mediterranean lighthouse	Extinct / Etna only	स्ट्राम्बोली
Kilimanjaro	Dormant (E. African Rift)	Ring of Fire active	किलिमंजारो
Aconcagua	Peak / not active volcano in GC keys	Same class as Etna	एकोंकागुआ
Focus vs epicentre	Rupture inside vs surface above	Swapped	फोकस / उपरिकेंद्र
Richter vs Mercalli	Magnitude vs intensity	Swapped / both "intensity"	रिक्टर / मर्कली
Richter +1 step	10× amplitude; ~ 32× energy	100× energy	रिक्टर ऊर्जा
S-waves	Solids only	Through liquid core	S-तरंग
Ring of Fire	Quakes + volcanoes + Pacific	Forest fires	रिंग ऑफ फायर
NW India quakes	Plate convergence	Volcano / coral	प्लेट
Tsunami word	Japanese harbour wave	Arabic / Latin	सुनामी

N.1 Origin of Universe

Universe-scale theories

Theory	Propounder	Teaching line
Nebular	Kant – Laplace	Rotating gas cloud flattened into Sun + planets
Planetesimal	Chamberlin – Moulton	Passing star pulled matter into planetesimals
Tidal	Jeans – Jeffreys	Tidal filament from a passing star broke into planets
Binary star	Russell – Lyttleton	Companion star disrupted into planetary debris
Interstellar dust	Otto Schmidt	Sun captured a gas–dust cloud that formed planets
Big Bang	Lemaître (often + Gamow)	Hot dense start ~ 13.7–13.8 bn yr ; expansion
Steady State	Hoyle – Bondi – Gold	Universe looks constant; continuous creation of matter
Pulsating	Oscillating model	Repeated expansion and contraction

Galaxy, stars and distances

Note	Fact
Expansion	Hubble — galaxies recede; redshift evidence
Evidence	Expanding space and the CMB
Galaxy	The Milky Way is a barred spiral with about 100 billion stars
Galactic year	The Sun orbits the galactic centre in about 22.5–25 crore years
Distance	One AU is about 149.6 million km . One light-year is about 9.46×10^{12} km
Star colour	Blue stars are hottest. Red stars are cooler
Star life	Nebula to main sequence to red giant to nova/supernova to white dwarf / neutron star / black hole
Chandrasekhar limit	About $1.44 M_{\odot}$ is the white-dwarf ceiling
Black hole	Massive-star remnant from which light cannot escape
Constellations	The IAU lists 88 . Saptarishi points to the Pole Star
Closest star (after Sun)	Proxima Centauri
Not space	A brittle star is a marine animal. A byte is a computing unit

N.2 Solar System

Planet desk (tilt · spin sense · moons · periods)

Planet	Axial tilt (approx.)	Rotation sense	Rotation (day)	Revolution (year)	Moons (coaching)	Must-score note
Mercury	~0°	Prograde	~ 59 Earth days	~ 88 Earth days	0	Smallest planet, fastest orbit, no moon
Venus	~177°	Retrograde	~ 243 Earth days	~ 225 Earth days	0	Hottest (~464°C); Morning/Evening Star
Earth	23½°	Prograde	~ 24 h	365¼ days	1 (Moon)	Densest mean; habitable zone
Mars	~25°	Prograde	~ 24.6 h	~ 687 Earth days	2 (Phobos, Deimos)	Red planet; day/tilt close to Earth

Planet	Axial tilt (approx.)	Rotation sense	Rotation (day)	Revolution (year)	Moons (coaching)	Must-score note
Jupiter	~3°	Prograde	~10 h	~12 Earth years	Many; Galilean Io, Europa, Ganymede, Callisto	Largest / heaviest; fastest spin
Saturn	~27°	Prograde	~10.7 h	~29.5 Earth years	Many; Titan largest of Saturn	Clearest rings
Uranus	~98°	Retrograde (sideways)	~17 h	~84 Earth years	Many	Sideways rotator; Sun rises in the west framing
Neptune	~28°	Prograde	~16 h	~165 Earth years	Many; Triton famous	Coldest / farthest in 8-planet set
Pluto (dwarf)	High / retrograde framing	Retrograde	Long	~248 Earth years	Charon, Nix, Hydra and others	IAU dwarf since 2006

N.3 Motions of Earth

Motion	Period	Effect
Rotation	~24 h (west to east)	Day and night, apparent Sun motion, and Coriolis
Revolution	365¼ days	Year; leap year every 4 years (century ÷ 400)
Orbit shape	Ellipse	Perihelion ~3 Jan (closest); Aphelion ~4 Jul (farthest)

Date	Event	Sun overhead / polar day
21 Mar / 23 Sep	Equinox	Sun over Equator; day equals night
21 Jun	NH summer solstice	Tropic of Cancer ; Arctic 24 h day
22 Dec	NH winter solstice	Tropic of Capricorn ; Arctic 24 h night

Note	Fact
Axis to orbital plane	66½° (complement of 23½°)
Circle of illumination	Great circle separating day/night; tilts with seasons
Sidereal day	~23 h 56 min (true 360° spin) vs mean solar day 24 h
Leap year	Extra day because revolution ≈ 365¼ d ; century years leap only if ÷ 400

N.4 Latitude

Parallel	Value	Note
Tropic of Cancer	23½° N	Sun overhead at NH summer solstice
Tropic of Capricorn	23½° S	Sun overhead at NH winter solstice
Arctic / Antarctic circles	66½°	90 – 23½
Heat zones	See bullets below	

N.5 Longitude, Time, Time Zones, IDL

Note	Note
Earth 360° / 24 h	Roughly 15° of longitude equals 1 hour ; 1° equals 4 minutes

Note	Note
East	Local time is ahead of Greenwich
IST	Based on 82°30' E near Mirzapur, UP
Time zones	Roughly 15° wide; countries zigzag for politics
IDL	Follows about 180° (bent around islands / Aleutians)
Prime Meridian land	Full land list below — not Niger / Nigeria as the classic wrong option
GMT places	London, Lisbon, Accra, Dublin — not Addis Ababa
Cairo	About 30° E , so about 2 h ahead of GMT

Country pair	Time zones
Russia	Most (11)
USA	6 (contiguous 4 + AK + HI)
China	1 despite width
India	1 (IST)

N.6 Atmosphere & Its Structure

Composition of dry air (by volume)

Gas	Approx. share	Teaching note
Nitrogen (N₂)	~78.08%	Most abundant gas in dry air
Oxygen (O₂)	~20.95%	Needed for respiration and combustion
Argon (Ar)	~0.93%	Main noble gas; not a greenhouse gas
Carbon dioxide (CO₂) (कार्बन डाइऑक्साइड)	~0.04% (~400 ppm class)	Trace greenhouse gas that varies with season and human activity
Neon, helium, methane, krypton, hydrogen, etc.	Trace	Very small permanent or near-permanent fractions

Permanent vs variable constituents

Type	Gases / constituents	Teaching note
Permanent	N ₂ , O ₂ , Ar (+ other rare noble gases)	Shares stay nearly constant day to day in the lower atmosphere
Variable	Water vapour, CO ₂ , O ₃ , aerosols / dust	Change with place, season, height and human activity

Vertical structure (bottom to top)

Layer	Approx. height	Temp. with height	Must-score note
Troposphere	~8 km at poles to ~18 km at the equator (avg ~13 km)	Usually falls (~6.5°C/km)	Almost all weather and most water vapour
Tropopause	Top of troposphere	Transition	Higher over the equator; lower over the poles
Stratosphere	~18–50 km	Rises because ozone absorbs UV	Ozone UV shield; dry and largely weather-free
Stratopause	~50 km	Transition	Boundary above the stratosphere
Mesosphere	~50–80 km	Falls	Coldest broad layer; meteors burn here

Layer	Approx. height	Temp. with height	Must-score note
Mesopause	~80 km	Transition	Coldest region of the atmosphere
Thermosphere / Ionosphere	above ~80 km (often framed ~80–640 km)	Rises strongly	Radio-wave reflection and aurora
Exosphere	outermost fringe	Merges into space	Very thin outer fringe

Homosphere (सममंडल) and heterosphere (विषममंडल)

Zone	Rough extent	Mixing
Homosphere	Surface to ~80–100 km	Turbulent mixing keeps major gas ratios roughly uniform
Heterosphere	Above ~80–100 km	Gases sort more by molecular weight

Heating, ozone and inversion

Idea	Correct
Main atmospheric heating	Long-wave terrestrial radiation from Earth's surface
Ozone maximum / UV shield	Stratosphere
Ozone hole (ओजोन छिद्र) focus	Antarctica in spring
Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)	Cut CFCs on the ozone path (not the Kyoto climate path)
Normal lapse (troposphere)	Temperature falls about 6.5°C per km
Temperature inversion	Temperature rises with height in a short layer

N.6A Insolation · Albedo · Heat budget

Surface	Typical albedo
Fresh snow	75–95%
Deserts	20–30%
Prairie (प्रेयरी) / grassland	10–20%
Paddy / dark wet fields	3–15%
Earth mean (planetary)	~30%

Heat budget (NCERT 100-unit picture)

Fate	Units
Reflected / scattered back to space (planetary albedo)	~35
Absorbed by the atmosphere	~14
Absorbed by Earth's surface	~51

N.6B Humidity · Dew

Humidity terms — Teaching Card

Term	Quick recall
Absolute humidity	Mass of vapour per volume of air
Relative humidity	% of saturation at that temperature
Specific humidity	Mass of vapour per mass of moist air

N.6C Clouds

Height families

Family	Typical height (mid-latitudes)	Make-up	Teaching note
High	About 5–13 km	Mostly ice crystals	Names often begin with cirro-
Middle	About 2–7 km	Water drops and/or ice	Names often begin with alto-
Low	Surface to about 2 km	Mostly water drops	Includes stratus sheets and many rain clouds
Vertical / towering	Base low; top may reach the tropopause	Strong convection	Cumulus and Cumulonimbus grow upward through layers

Form families

Form idea	Look	Classic members
Cirriform	Thin, wispy, fibrous	Cirrus, cirrostratus, cirrocumulus
Stratiform	Sheet or layer	Stratus, altostratus, nimbostratus
Cumuliform	Heap / cauliflower	Cumulus, cumulonimbus, altocumulus
Nimbus	Rain-bearing	Nimbostratus, Cumulonimbus

Main cloud types

Cloud	Family	Teaching note
Cirrus	High	Thin, wispy ice trails. Often fair-weather or approaching weather aloft
Cirrostratus	High	Thin icy sheet. May make a Sun or Moon halo
Cirrocumulus	High	Small icy ripples or “mackerel sky” patches
Alto cumulus	Middle	White or grey patches and rolls in the middle troposphere
Altostratus	Middle	Grey sheet that can dull the Sun. May thicken before steadier rain
Stratus	Low	Grey layer hugging the weather layer. Light drizzle or dull sky
Nimbostratus	Low	Thick rain sheet. Steady, often widespread rain or snow
Cumulus	Vertical	Puffy cauliflower heaps. Isolated fair-weather cumulus means unsettled tall storm not yet formed
Cumulonimbus	Vertical	Towering storm cloud. Intense rain, hail, thunder and lightning

Fog, halo and thunder links

Idea	Correct
Fog	Cloud with its base at or very near the ground
Halo around Sun / Moon	Often cirrostratus
Steady widespread rain	Often nimbostratus
Intense convective rain / hail / thunder	Cumulonimbus
Fair-weather afternoon heaps	Isolated cumulus

N.6D Tropical cyclones · Tornado · Anticyclone

Tropical cyclone

Local name	Region
Hurricane	USA / Atlantic–Caribbean
Typhoon	China Sea / western Pacific
Willy-Willy	North-west Australia
Cyclone	North Indian Ocean / India
Baguio	Philippines
Taifu	Japan

Anticyclone

System	Centre	Surface spin (NH)	Surface spin (SH)	Typical weather
Cyclone / low	Low	Anticlockwise	Clockwise	Cloud, rain, strong wind
Anticyclone / high	High	Clockwise	Anticlockwise	Clear, settled, light wind

N.8 Global Warming

Is a greenhouse gas	Not a greenhouse gas
Water vapour, CO ₂ , CH ₄ , N ₂ O, O ₃ , CFCs / HCFCs	Argon
Natural coaching set often lists CO ₂ , CH ₄ , O ₃ and N ₂ O	“Nitrogen oxide” (NO / NO ₂) in the 2020 key is not treated as that natural GHG set

Climate and ozone treaties (do not swap)

Instrument	Year	Focus	What it does
Montreal Protocol	1987	Ozone layer	Cuts ozone-depleting substances, especially CFCs and related chemicals
Kyoto Protocol	1997	Climate / GHG	Sets binding emission cuts for listed greenhouse gases and frames carbon credits
Paris Agreement (पेरिस समझौता)	2015	Climate / NDCs	Each country files Nationally Determined Contributions under the UNFCCC path

Wrong pairing	Correct
Kyoto → save water	Kyoto → climate / GHG / carbon credit
Montreal → climate mitigation (जलवायु शमन)	Montreal → ozone / CFCs
Ramsar (रामसर) → climate mitigation	Ramsar → wetlands
CITES (साइट्स) → climate mitigation	CITES → wildlife trade
Paris → ozone CFCs	Paris → climate NDCs

N.12 Pressure Belts, Planetary Winds, ITCZ

Latitude	Belt	Wind
0°	Equatorial low / ITCZ	Doldrums with rising air and rain
30°	Subtropical high	Horse latitudes; Trade winds (NE/SE) blow toward ITCZ

Latitude	Belt	Wind
60°	Subpolar low	Westerlies
90°	Polar high	Polar easterlies

Named Southern Hemisphere westerlies

Named belt	Fact
Roaring Forties	About 40°S
Furious Fifties	About 50°S
Shrieking / Screaming Sixties	About 60°S

N.13 Interior of the Earth

Boundary	Depth pair	Fact
Moho	Base of crust (~8–40 km)	Crust / mantle
Gutenberg	~2900 km	Mantle / outer core — S-waves stop (liquid)
Lehmann	~5100 km	Outer / inner core (solid iron–nickel)
Layer chemistry (names)	Fact	
----- --	-----	
Sial	Continental crust — silica + aluminium; granite; less dense than sima	
Sima	Oceanic crust — silica + magnesium; basalt; denser than sial	
Nife	Core — nickel + iron	

Shadow	Note
S-wave	None beyond ~ 103° from epicentre (liquid outer core)
P-wave	Weak/absent ~ 103–142° (refraction in the core)

N.14 Continental Drift

How the idea grew after Wegener

Later piece	Fact
Mantle convection	Holmes (~1928–29)
Sea-floor spreading	Harry Hess (1960s); paleomagnetic stripes on MOR
Plates	Lithosphere pieces; India was in Gondwana

N.15 Plate Tectonic Theory

Boundary	Motion	Result
Divergent	Apart	MOR, rift (East African), new crust
Convergent	Together	Trenches, fold mountains, volcanoes, quakes
Transform	Slide	San Andreas; quakes, little crust created

N.16A Rock Types & Rock Cycle

Family	How it forms	Key traits
Igneous (primary)	Cooling of magma / lava	No fossils; not layered like sediment
Sedimentary	Weathering → transport → deposition → lithification	Fossils common; occur in layers (strata)
Metamorphic	Any rock remade by heat + pressure + chemistry (metamorphism)	New fabric / minerals; forms deeper / hotter than surface conditions

Kind	Setting	Examples
Plutonic / intrusive	Magma cools slowly deep inside the crust	Granite , gabbro, pegmatite
Volcanic / extrusive	Lava cools at / near the surface	Basalt , volcanic breccia

Type	Examples
Mechanical (clastic)	Sandstone, shale, conglomerate; limestone often listed here in older books
Organic / carbonic	Coal; limestone / geyserite note
Chemical	Halite, potassium salts

Parent	Metamorphic product
Granite	Gneiss
Basalt	Schist / amphibolite
Shale	Slate , then phyllite, then schist at higher grade
Limestone / chalk / dolomite	Marble
Sandstone	Quartzite
Conglomerate	Gneiss (listed in some GC sheets)
Gabbro	Serpentine (GC sheet)

N.17 Volcano Types

Note	Fact
Magma	Molten rock below the surface
Lava	Magma that reaches the surface (~600–1200°C); cools to igneous rock
Most abundant volcanic gas	Water vapour (H₂O) (जलवाष्प) — then CO ₂ , SO ₂ and others
Acidic lava	High silica makes it sticky / viscous
Basic lava	Low silica makes it runny / fluid
Caldera	Large collapsed vent / crater
Geyser	Intermittent hot water + steam (e.g. Old Faithful, Yellowstone) — near magmatic heat
Fumarole / solfatara	Late-stage gas vents (a sulfurous vent is a solfatara); Katmai's "Valley of Ten Thousand Smokes" is the classic note
Type	Fact / example
-----	-----
Shield	Low-silica, runny lava; classic Hawaii pair is Mauna Loa / Mauna Kea

Note	Fact
Composite / strato	Sticky, explosive; Fuji, Vesuvius, Etna, Pinatubo, St Helens, Rainier, Cotopaxi
Cinder cone	Small, steep; Paricutin (Mexico)
Caldera	Collapse after blast; Krakatoa / Krakatau (Indonesia), Yellowstone
Fissure / flood basalt	Deccan Trap (दक्कन ट्रैप)

Status	Fact
Active	Open vent / recent eruptions — Stromboli ("lighthouse of the Mediterranean"), Etna (Sicily), Kilauea (often ranked among most active), Mauna Loa, Barren Island (India's only confirmed active — Andaman Sea), Fuji, Vesuvius, Erebus (Ross Island, Antarctica), Cotopaxi (Ecuador)
Highest active (world)	Coaching often cites Ojos del Salado (Andes, Chile–Argentina, ~6893 m). Among common options Cotopaxi may be the tallest listed
Dormant	Quiet but may revive — Kilimanjaro (Kibo dormant; Mawenzi / Shira extinct) and Damavand (Iran)
Extinct	Unlikely to erupt again — Kulal (Kenya), Chimborazo (Ecuador), Popa (Myanmar), Koh-e-Sultan (Pakistan) note

Volcano / peak	Country	Teaching note
Sabancaya	Peru	Andean volcano
Colima	Mexico	Mexican volcano match
Merapi	Indonesia	Classic Indonesian volcano
Kinabalu	Malaysia	Peak often listed with volcano/country sets
Elburz (Alborz)	Iran	Iranian range / peak match
Aconcagua	Argentina	High peak; not treated as an active volcano in standard keys
Krakatoa / Krakatau	Indonesia	Famous caldera volcano

N.17B Earthquakes · Seismic Waves · Tsunami · Scales

Earthquake terms — Teaching Card

Term	Quick recall
Focus / hypocentre	Underground rupture point
Epicentre	Surface point above the focus
Seismograph / seismometer	Instrument that records waves
Seismogram	The wave record itself

Wave	Nature	Travels through	Note
P (primary)	Longitudinal / sound-like; fastest ; arrives first	Solids, liquids, gases	Body wave
S (secondary)	Transverse / light-like; slower than P	Solids only	Body wave
Surface	Slowest; Love + Rayleigh; most damaging at surface	Near surface	Love waves are the faster surface type

Scale	Measures	Note
Richter (C.F. Richter, 1935)	Magnitude $\approx$ energy released ($\log_{10}$ amplitude)	+1 magnitude $\approx$ 10× amplitude; $\approx$ 32× energy (not 100×
Mercalli	Intensity (felt damage / effects)	Different from magnitude

Event pair	Fact
26 Dec 2004 Indian Ocean	Sumatra subduction quake. Waves reached as far as Africa. Deaths were about 2–3 lakh , not “50 lakh”. Bangladesh saw relatively few casualties because the fault trended roughly N–S while main wave energy ran roughly E–W
11 Mar 2011 Japan	A great quake and tsunami damaged Fukushima Daiichi and caused a nuclear disaster

N.18 Major Earthquake & Volcano Belts

Belt	Note
Circum-Pacific / Ring of Fire	Pacific rim arc (Chile–California–Alaska–Japan–Philippines–New Zealand)
Mid-Atlantic	Iceland, Azores — divergent; quakes + volcanoes
Alpine–Himalayan / Mid-Continental	Cape Verde–Atlantic–Mediterranean–Alps–Caucasus– Himalaya ; India’s main seismic framing
East African Rift	Kilimanjaro, Kenya — not Ring of Fire ()
Cocos plate	Between Central America and the Pacific plate

Common Traps — Don’t Fall For These

- Ozone = troposphere** — FALSE. **Stratosphere**.
- El Niño** → **more Peruvian fish** — FALSE. **Less** upwelling / plankton.
- SH deflection = right / magnetic** — FALSE. **Coriolis, left**.
- All jets are easterly** — FALSE. **Westerly**; TEJ is the easterly exception.
- ITCZ = high pressure** — FALSE. **Low**.
- Atmosphere heated by incoming short-wave** — FALSE. **Long-wave from Earth**.
- Argon is a GHG** — FALSE.
- Kyoto = ozone / save water** — FALSE. **Climate / carbon credit**. Montreal = ozone.
- Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
- Perihelion = NH summer** — FALSE. **Early January**.
- Seasons = changing distance from Sun** — FALSE. **Tilt**.
- IST only east of Mirzapur** — FALSE. **Whole India**.
- India = 6th largest / wholly tropical** — FALSE. **7th; 2.4%**; Tropic through the middle.
- Nuclear/geothermal = stored solar** — FALSE.
- Fold = Vosges; fault = Himalaya** — swapped. Himalaya = **fold**; Vosges = **block**.
- Hottest planet = Mercury** — FALSE. **Venus**.
- Largest moon = Titan / our Moon** — FALSE. **Ganymede**.
- Solar eclipse at Full Moon** — FALSE. **New Moon**. Lunar = Full.
- Roaring Forties = 40°N trades** — FALSE. **~40°S westerlies**.
- Nebular theory = Wegener / Hoyle** — FALSE. **Kant–Laplace**. Wegener = drift; Hoyle = Steady State.
- Sial = ocean floor** — FALSE. **Continents**. Sima = ocean.
- Kilimanjaro / Stromboli = same belt** — FALSE. Stromboli = **Mediterranean**; Kilimanjaro = **E. African Rift**.
- All 100 insolation units heat the air directly** — FALSE. **~35** reflected; surface **~51**; air then heated by **long-wave**.
- Seasons caused by perihelion** — FALSE. Seasons = **axial tilt**.
- Main jet streams = easterly** — FALSE. Main jets are **westerly**; TEJ is the exception.
- Skip spheres / rock cycle / plate types because recent papers asked ozone** — FALSE. Next year can ask any standard NCERT angle from this topic.

27. **Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
28. **Hottest planet = Mercury / densest = Jupiter** — FALSE. Hottest = **Venus**; densest mean = **Earth**.
29. **Stratosphere = weather layer** — FALSE. Weather = **troposphere**. Stratosphere = ozone + jet flight (little weather).
30. **Aurora = mesosphere** — FALSE. **Ionosphere / thermosphere**.
31. **Horse latitudes = low pressure** — FALSE. Subtropical **high** ~30°.
32. **Eye of cyclone is 10°C colder** — FALSE. Eye is **warmer**.
33. **Jet streams = NH only** — FALSE. Both hemispheres.
34. **Fresh snow = lowest albedo** — FALSE. Fresh snow has **highest** albedo among common options.
35. **Ganymede = Saturn's moon / largest moon = Titan** — FALSE. **Ganymede = Jupiter**; Titan = Saturn's largest.
36. **Chandrasekhar limit ≈ Eddington / Fowler name alone** — FALSE. ~**1.44 M_☉** white-dwarf limit; black-hole path above collapse residue note.
37. **Nebula = solar-system member** — FALSE. Nebulae are star-forming clouds, not planets/asteroids/comets of the Sun's family.
38. **Brittle star = pulsar class** — FALSE. Marine animal; pulsars / quasars / black holes are cosmic.
39. **Mercury albedo > Earth** — FALSE. Earth albedo higher (~0.3 vs ~0.1).
40. **Equator through Mexico / Capricorn through Bolivia** — FALSE. Check country lists carefully.
41. **0°–0° on land / in India** — FALSE. Intersection in **Atlantic** (Gulf of Guinea).
42. **Solar eclipse at Full Moon / lunar at New Moon** — SWAP. Solar = **New**; lunar = **Full**.
43. **Granite = metamorphic** — FALSE. **Igneous**; gneiss is its metamorphic child.
44. **Fossils common in granite** — FALSE. Fossils favour **sedimentary** rocks.
45. **Most volcanic gas = CO₂ / SO₂** — FALSE. **Water vapour** leads.
46. **Magma = surface melt** — FALSE. Surface = **lava**; magma is underground.
47. **Aconcagua = active like Etna** — FALSE. Peak / not active in GC keys; Etna–Vesuvius–Fuji are active.
48. **Kilimanjaro = Ring of Fire / always extinct** — FALSE. **East African Rift**; mainly **dormant** framing.
49. **No volcanoes possible in Circum-Pacific** — FALSE. That belt is the **maximum** volcanic belt.
50. **Baltic Sea = volcanic hotspot** — FALSE. Classic “eruptions do **not** occur” option among seas.
51. **Epicentre = underground rupture** — FALSE. That is the **focus/hypocentre**; epicentre is the surface point above it.
52. **Richter step = 100× energy** — FALSE. ~**10×** amplitude and ~**32×** energy; Mercalli measures **intensity**.
53. **S-waves travel through liquid outer core** — FALSE. **Solids only**; that makes the S shadow.
54. **Ring of Fire includes forest fires** — FALSE. Quakes + volcanoes around the **Pacific**.
55. **NW India quakes = coral / volcano only** — FALSE. Mainly **Indian–Eurasian plate** convergence.
56. **2004 tsunami killed 50 lakh** — FALSE. Order of ~**2–3 lakh**; Bangladesh had few deaths for wave-direction reasons.
57. **Tsunami is Latin / Arabic** — FALSE. **Japanese** harbour wave.

Topic 15 — Geomorphology & Landform Processes

Current Affairs (this topic)

Process chapter — no living scheme tag. Static PYQ surface (trellis (जालनुमा), deltas, Imphal) is the diet. Mekong/Himalaya (हिमालय) fossils do not need a yearly CA refresh.

Confused Pairs

Pair	Correct	Trap	Hindi
Weathering vs erosion	In situ vs transport	Same thing	अपक्षय / अपरदन
Granite vs basalt	Intrusive acidic vs extrusive basic	Swapped	ग्रेनाइट / बेसाल्ट
Marble	From limestone	From sandstone	संगमरमर
Quartzite	From sandstone	From limestone	क्वार्ट्जाइट
Gneiss	From granite	From shale	नाइस
Trellis	Folded structures	Dendritic on folds	जाल / जाली
Bird's-foot delta	Mississippi	Ganga / Nile	पक्षी-पाद
Arcuate delta	Nile, Ganga–Brahmaputra	Mississippi	चापाकार
Moribund (two frames)	Bengal subdivision vs Cauvery lobe — read stem	Mixing both into one key	मृतप्राय
Imphal basin	Lacustrine	Loess / glacial	झीलकृत
Barchan horns	Point downwind	Upwind	बरखान
Stalactite	Ceiling	Floor	स्टैलेक्टाइट
Stalagmite	Floor	Ceiling	स्टैलेग्माइट
U-valley vs V-valley	Glacier vs river youth	Swapped	U / V
Fjord	Glacial trough drowned	Any estuary	फियोर्ड
East vs west India mouth	Delta vs estuary	All India deltas	डेल्टा / ज्वारनदमुख
Fault-valley river	Damodar (2019)	Chambal (ravines)	भ्रंश घाटी
Antecedent vs superimposed	Cuts rising land vs inherited on cover	Swapped	पूर्ववर्ती / अधारोपित
Consequent vs subsequent	Follows slope vs follows structure	Same	अनुगामी / परवर्ती
Sill vs dyke	Parallel to beds vs cuts beds	Swapped	सिल / डाइक
Conglomerate vs breccia	Rounded pebbles vs angular	Swapped	समूहिका / ब्रेशिया
Peneplain vs pediplain	Davis humid old-age vs King arid	Same word	समप्राय मैदान / पदस्थली
Mesa vs butte	Broad flat cap vs narrow remnant	Swapped	मेसा / ब्यूट
Cuesta vs hogback	Gentle dip + steep scarp vs steep both sides	Same landform	केस्टा / हॉगबैक
Atoll shape	Horseshoe / ring around a lagoon	Triangular / rectangular / rod	एटॉल आकार
Ria vs fjord	Drowned river vs drowned glacier	Same estuary	रिया / फियोर्ड
Gorge vs canyon	Steep mountain cut vs arid stair-step	Always identical	गॉर्ज / केनियन
Endogenic vs exogenic	Interior build vs surface wear	Same	अंतर्जात / बहिर्जात

Pair	Correct	Trap	Hindi
Weathering vs denudation	In situ breakdown vs weathering+erosion	Same	अपक्षय ≠ अनाच्छादन

N.1 Rocks — Igneous, Sedimentary, Metamorphic

Class	How formed	rocks
Igneous (primary)	Cooling of magma/lava	Granite (intrusive, acidic, coarse); basalt (extrusive, basic, Deccan Trap (दक्कन ट्रैप)); gabbro, pegmatite, pumice, obsidian
Sedimentary (secondary)	Compaction of debris / chemical / organic	Sandstone, shale, limestone , conglomerate, coal, loess beds — fossils possible
Metamorphic	Heat + pressure on existing rock	See pair table

Parent	Becomes
Limestone / dolomite	Marble
Sandstone	Quartzite
Granite	Gneiss
Shale / clay	Slate → schist
Bituminous coal	Anthracite / graphite (high grade)

Body	Role
Batholith	Huge granite mass; floor not seen; mountain cores
Laccolith	Lens, convex up ; arches overlying beds
Lopolith	Saucer, concave up
Phacolith	Lens in fold crest/trough
Sill	Sheet parallel to bedding
Dyke	Wall that cuts bedding (discordant (असंगत))
Volcanic neck / plug	Filled pipe left standing

Family	Rocks
Clastic	Conglomerate (rounded), breccia (angular), sandstone, shale/mudstone
Chemical	Limestone, dolomite, gypsum, rock salt (लवणाध्यक्ष), chert
Organic / biogenic	Coal, peat, chalk, some limestone

N.2 Rock Cycle

From	Process	To
Magma	Cooling / crystallisation	Igneous
Any exposed rock	Weathering → erosion → deposition → lithification	Sedimentary
Igneous or sedimentary	Heat + pressure (no full melt)	Metamorphic
Any rock	Melting	Magma

N.3 Weathering, Erosion, Deposition

Term	Meaning
Weathering	In situ breakdown
Erosion	Pickup + transport
Deposition	Drop of load
Denudation	Weathering + erosion
Mass wasting	Gravity movement

Process	Quick recall
Frost / ice wedging	Water freezes and pries cracks
Exfoliation / unloading	Onion sheets on granite
Thermal / insolation	Desert day–night flake
Salt wedging	Crystal growth in arid pores
Block vs granular	Joint slabs vs grain-by-grain
Spheroidal	Rounded corestones

Process	Quick recall
Carbonation	Carbonic acid → limestone karst
Solution	Direct dissolve (salt, gypsum)
Oxidation	Iron rusts; red soils
Hydration	Water uptake → swell and crack
Hydrolysis	Feldspar → clay
Chelation	Organic acids leach metals

N.4 Agents of Erosion

Agent	Work words	Signature
River	Hydraulic, abrasion/corrasion, attrition, solution; load = traction · saltation · suspension · solution	V-valley → meander → delta
Wind	Deflation, abrasion, attrition	Barchan, yardang, loess
Glacier	Plucking, abrasion	U-valley, cirque, moraine
Sea	Hydraulic, abrasion, corrosion, attrition	Cliff, stack, spit
Groundwater	Solution	Karst in limestone

N.5 Fluvial Landforms (Davis cycle)

Stage	Landforms
Youth	Steep V-valley , gorge/canyon, waterfall, rapids, potholes , interlocking spurs
Mature	Meander , slip-off slope + river cliff, floodplain , natural levee , bluff, widening valley
Old	Broad floodplain, ox-bow / mortlake / meander scar, yazoo, delta , deferred tributary
Rejuvenation	Uplift / base-level fall → knickpoint , river terrace , incised meander

Fluvial forms — Teaching Card

Form	Quick recall
Meander	Outer cut cliff; inner point bar
Ox-bow	Cut-off meander lake
Floodplain	Point-bar + overbank silt
Levee	Raised silt ribbon along channel
Yazoo	Deferred tributary behind levee
Fan / bajada	Mountain-foot dump; coalescing fans
Braided	Many threads (Brahmaputra type)
Gorge vs canyon (गॉर्ज)	Narrow steep cut vs wider arid stair-step

Theory	Quick recall
Davis	Structure–process–time → peneplain + monadnock
Penck	Uplift and denudation together (no neat old-age wait)
L.C. King	Pediment retreat → pediplain

Stream origin names — Teaching Card

Stream	Follows
Consequent	Initial slope
Subsequent	Weaker beds / faults (trellis valleys)
Obsequent	Opposite to consequent (scarp face)
Resequent	Same direction as consequent, later
Insequent	No obvious control (dendritic on uniform rock)

N.6 Types of Deltas

Delta shapes — Teaching Card

Type	Shape / fact	Classic
Arcuate	Bow / fan	Nile, Ganga–Brahmaputra (Sundarbans), Rhine, Niger, Hwang Ho — not bird's-foot
Bird's-foot	Distributaries like toes	Mississippi . Ganga is not this
Cuspate	Tooth / pointed	Tiber (Italy), Ebro
Estuarine	Fills a drowned mouth	Some European / west-India type mouths
Moribund (subdivision stem)	Dying belt inside Bengal's scheme	Bengal Delta (UPPCS 2019)
Moribund (lobe / river stem)	Abandoned / ageing delta	Cauvery in many landform keys

N.7 Structural Basins & Fault Valley Drainage

Example	Type
Imphal basin	Lacustrine plain (old lake) — not loess / glacial / ordinary alluvial trap

Example	Type
Kashmir vale	Lacustrine / structural (Pir (पीर) Panjal (पीर पंजाल) wall)
Chhattisgarh / Michigan type	Structural/sedimentary basin

Plain	Agent
Alluvial	River (Indo-Gangetic)
Lacustrine	Old lake floor — Imphal , Kashmir
Glacial / outwash	Ice
Loess	Wind silt — China ; not Imphal
Lava / volcanic	Flood basalt (Deccan surface)

Structural landforms — Teaching Card

Form	Dip of beds	Side view
Cuesta	Gentle	One gentle slope + one steep scarp
Hogback	Steep	Steep on both sides

Form	Size feel	Cap
Mesa	Broad flat top	Hard cap still wide
Butte	Smaller / pillar-like	Same idea, much less area

Drainage patterns — Teaching Card

Pattern	Looks like	Control (why)
Dendritic	Tree branches	Homogeneous rock
Trellis	Lattice / garden trellis	Folded hard–soft belts
Radial	Spokes outward	Dome / volcano / Amarkantak
Centripetal	Spokes inward	Closed basin
Rectangular	Right-angle bends	Joints / faults
Parallel	Side-by-side lines	Steep uniform slope
Annular	Rings	Concentric weak beds
Deranged	Chaotic / unfinished	New glacial dump

N.8 Glaciers & Glacial Landforms

Glacier types — Teaching Card

Type	Quick recall
Cirque glacier	Armchair hollow
Valley / alpine	Tongue in a trough
Piedmont	Coalesced on a plain
Ice sheet	Greenland, Antarctica
Ice cap	Smaller dome
Ice shelf	Floating fringe

Erosional glacial landforms — Teaching Card

Form	Quick recall
Cirque / tarn	Armchair hollow; lake inside
Arête	Knife ridge between cirques
Horn	Pyramidal peak (Matterhorn type)
U-trough	Glacier (≠ river V)
Hanging valley	Higher tributary floor → waterfall
Fjord	Drowned glacial trough
Roche moutonnée	Abraded + plucked bedrock
Crag-and-tail	Hard rock + lee debris

Depositional glacial landforms — Teaching Card

Form	Quick recall
Moraine	Lateral, medial, terminal, ground
Drumlin	Streamlined till; blunt up-ice
Esker	Sinuuous subglacial sand-gravel
Kame	Ice-contact mound
Outwash / sandur	Meltwater plain beyond snout
Erratic	Far-travelled boulder
Kettle	Melt pit → lake

N.9 Aeolian Landforms

Dunes and loess — Teaching Card

Form	Quick recall
Barchan	Crescent; horns downwind
Parabolic	Crescent; horns upwind (opposite barchan)
Seif / longitudinal	Ridge parallel to wind
Transverse	Crest across wind
Star / pyramidal	Multi-directional winds
Loess	Wind silt; China — not Imphal
Erg	Sand sea

N.10 Coastal Landforms (+ karst)

Depositional coastal forms — Teaching Card

Form	Quick recall
Beach	Wave-built sand / shingle strip
Spit	Attached at one end; free tip
Bar	Closes a bay; lagoon behind

Form	Quick recall
Tombolo	Island tied to mainland
Hook	Curved spit tip
Barrier island	Offshore, parallel to coast

Reef type	Where it sits	Lagoon?
Fringing	Against the shore	Little / none
Barrier	Offshore, parallel to land	Yes — between reef and land
Atoll	Ring / horseshoe in open sea	Yes — central lagoon; island mostly gone
Coast type	Meaning	
-----	-----	
Fjord	Drowned glacial trough	
Ria	Drowned river valley (non-glacial)	
Dalmatian	Concordant drowned ranges // coast (Adriatic type)	
Emergent	Raised beach / wave-cut platform now inland	
Submergent	Ria / fjord / drowned mouth	

Common Traps — Don't Fall For These

1. **Ganga delta = bird's-foot** — FALSE. **Mississippi**. Ganga = **arcuate**.
2. **Every moribund stem = Cauvery** — FALSE. **Subdivision** wording → **Bengal**; **lobe / river match** wording → often **Cauvery**.
3. **Folds** → **dendritic** — FALSE. **Trellis**.
4. **Imphal = loess / glacial** — FALSE. **Lacustrine**.
5. **Marble from sandstone** — FALSE. **Limestone**. Quartzite = sandstone.
6. **Weathering = erosion** — FALSE. Weathering is **in situ**.
7. **Barchan horns upwind** — FALSE. **Downwind**.
8. **Stalactite on the floor** — FALSE. **Ceiling**.
9. **U-valley = youthful river** — FALSE. **Glacier**. River youth = **V**.
10. **Mekong delta = Cambodia** — FALSE. **Vietnam**.
11. **West Indian rivers = big arcuate deltas** — FALSE. **Estuaries** (Narmada/Tapi).
12. **Greater Himalaya full of marine fossils** — FALSE. That's **Lesser**; Greater is fossil-poor crystalline.
13. **Fjord = any estuary** — FALSE. **Drowned glacial trough**.
14. **Basalt = intrusive granite twin** — FALSE. Basalt = **extrusive**.
15. **Loess = river silt of Imphal** — FALSE. **Wind silt**; Imphal = lake plain.
16. **Fault-valley = Chambal** — FALSE. **Damodar**.
17. **Endogenic = only rivers** — FALSE. Endogenic = interior build (fold/fault/volcano).
18. **Weathering always transports** — FALSE. Weathering is **in situ**.
19. **Skip process framework because recent papers asked trellis/Damodar** — FALSE. Next year can ask any standard NCERT agent or pair.
20. **Sill cuts the beds** — FALSE. **Dyke** cuts; sill is **parallel**.
21. **Point bar on the concave bank** — FALSE. Concave **erodes**; convex **deposits**.
22. **Ria = fjord** — FALSE. Ria = drowned **river**; fjord = drowned **glacier**.
23. **Peneplain = King's arid pediment** — FALSE. Peneplain = **Davis**; pediplain = **King**.

24. **Marble is foliated** — FALSE. Marble/quartzite = **non-foliated**.
25. **Intertrappean = sea fossils** — FALSE. **Freshwater / land** fossils.

Topic 16 — Oceans

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Feb 2026–ongoing	After US–Israel strikes on Iran (from 28 Feb 2026), Iran moved to control / restrict the Strait of Hormuz ; tanker traffic collapsed; mines, boardings and attacks on merchant ships (नावाध्यक्ष) were reported	World's top oil–LNG chokepoint in the news	Reuters / IEA / CRS
2025 baseline	About 20 mb/d of oil products (~ 25% of seaborne oil) and ~ 20% of world LNG (mainly Qatar + UAE) normally transit Hormuz	Scale of any closure	IEA
Static geography	Hormuz joins Persian (फ़ारसी) Gulf ↔ Gulf of Oman ; shores are Iran (north) and Oman (Musandam, south) ; UAE sits on the approaches	Country / strait MCQs	Atlas
Bypass note	Only Saudi Arabia and the UAE have meaningful pipeline routes that can skip Hormuz; Iraq, Kuwait, Qatar, Bahrain and Iran lean almost entirely on the strait	Who is trapped if closed	IEA
Demand side	Most Hormuz oil goes to Asia (China, India, Japan, Korea); India is a major importer through this gate	India linkage	EIA / visualisers
Other 2026 themes	Dual blockade talk (Iran on the strait / US on Iranian ports), insurance spikes, and later escort / corridor politics kept oil prices elevated	Chokepoint = price risk	Reuters / Brookings
Static	BBNJ 2023 high-seas biodiversity treaty — India has signed; UNCLOS 12 / 24 / 200 nm unchanged	Law of the sea	UN
Static	India Arctic Policy 2022 ; Arctic Council observer since 2013 ; Deep Ocean Mission / Samudrayaan / Blue Economy (नीली अर्थव्यवस्था)	Polar + deep ocean	MEA / MoES

Confused Pairs

Pair	Correct	Trap	Hindi
Benguela	Atlantic cold (SW Africa)	Pacific	बेंगुएला
Humboldt / Peru	Pacific cold	Atlantic	हम्बोल्ट
Agulhas	Warm , Indian Ocean	Cold / Atlantic	अगुलहास
Brazil current	Warm	Cold	ब्राज़ील
California / Canary	Cold	Warm	कैलिफोर्निया / कैनरी
Kuroshio vs Oyashio	Warm Japan vs cold Kurile	Swapped	कुरोशियो / ओयाशियो
Spring vs neap	Syzygy large vs quadrature small	Straight line = small tide	वृहत् / लघु
Salinity max	Tropics belt	Equator	लवणता
Mariana	Pacific	Atlantic	मारियाना
Sunda/Java trench	Indian	Pacific	सुंडा
Molloy Deep	Arctic	Atlantic	मोलॉय
Datum line	Mean sea level reference	Date Line / 0° meridian	डेटा म रेखा
Fresh after ice	Largest remaining = groundwater	Lakes/rivers larger than GW	भूजल
Sargasso	No coastline	Has coasts like Black Sea	सारगैसो
Red Sea form	Axial trough	Ordinary eroded valley	लाल सागर
NAD	Blanket of Europe / warm	Cold Labrador clone	उत्तरी अटलांटिक ड्रिफ्ट
Halocline	Salinity jump	Thermocline (temp)	हैलोक्लाइन

Pair	Correct	Trap	Hindi
Lake Van	Very high lake salinity (~330‰)	Always pick Dead Sea	वैन झील
Diamantina	Indian Ocean	Pacific	डायमंटीना
Ninetyeast Ridge	Indian Ocean	Mid-Atlantic	नाइंटी ईस्ट
Suez	Med ↔ Red, sea-level (no climb)	Directly Atlantic–Indian	स्वेज
Panama	Atlantic ↔ Pacific + stepped chambers / Gatun	Same as Suez / no chambers	पनामा
EEZ vs territorial	200 nm vs 12 nm	Swapped	EEZ / क्षेत्रीय जल
Maitri + Bharati	Operational Antarctica	Dakshin Gangotri still open	मैत्री / भारती
Himadri	Arctic (Svalbard)	Antarctica	हिमाद्री
Guyot vs seamount	Flat drowned top vs peaked	Same	गायट / सीमाउंट
El Niño fish	Less plankton	Warm → more fish	अल नीनो
Red clay vs ooze	Deepest no carbonate vs biogenic mud	Same deposit	लाल मृत्तिका
Globigerina vs diatom	Calcareous vs siliceous (cold)	Swapped	ग्लोबिजेरिना / डायटम
Ninetyeast Ridge	Volcanic / hotspot trail	Mid-ocean spreading ridge	नाइंटीईस्ट
Carlsberg Ridge	NW Indian spreading	Atlantic MAR	कार्ल्सबर्ग
Drake vs Magellan	Open south of Cape Horn vs Chilean channels	Same strait	ड्रेक / मैगलन
Tidal bore	Funnel river wave	Ordinary spring tide	ज्वार-भित्ति
ISA / Area	Seabed beyond national zones	EEZ mining	ISA
Arctic Council	India = observer (2013)	Full member	आर्कटिक परिषद
Neritic vs hadal	Over shelf vs inside trench	Same depth word	नेरिटिक / हैडल
Territorial vs EEZ	12 nm vs 200 nm	Swapped	12 ≠ 200
India west vs east shelf	West wider (overall)	East always wider	पश्चिमी शेल्फ
Constructive vs destructive wave	Builds beach vs erodes	Same	रचनात्मक / विनाशकारी
Upwelling coast	West coast of continent	East coast rule	उत्प्लाव
Türkiye north	Black Sea	Caspian / Med	काला सागर
W→E seas ladder	Med → Black → Caspian → Aral	Black first / Aral before Caspian	मेड→ब्लैक→कैस्पियन→अराल
Baltic coast	Norway out	Norway in	बाल्टिक
Jordan sea	Aqaba / Red only	Mediterranean	जॉर्डन
Hormuz shores	Iran + Oman	UAE as both shores	होरमुज़
Persian Gulf coast	Not Oman	Oman inside Gulf	फारस की खाड़ी
Syria sea	Mediterranean	Black Sea	सीरिया
Gaza water	Mediterranean	Red / Dead Sea as coast	गाजा

N.1 Oceans & Seas

Ocean	Fact
Pacific	Largest and deepest ocean, about one-third of Earth's surface, with the Ring of Fire

Ocean	Fact
Atlantic	S-shaped basin with the Mid-Atlantic Ridge and the Telegraphic Plateau
Indian	Mostly south of Asia, on both sides of the Equator ; Greeks (Ἑρμαίνης) called it the Erythraean Sea
Southern	Surrounds Antarctica and carries the only circumpolar Antarctic Circumpolar Current
Arctic	Smallest and shallowest ocean, with the world's widest continental shelves

Sea	Note
South China Sea	Pacific marginal sea, often the deepest among common options (~5377 m)
Bering Sea	Pacific marginal sea between Asia and America
Mediterranean	Largest enclosed sea, linked to the Atlantic at Gibraltar
Black Sea	South-east Europe; Syria does not border it
Sea of Azov	Almost enclosed; opens to the Black Sea through the Kerch Strait
Tyrrhenian	Between Italy, Sicily, Sardinia and Corsica
Sargasso	North Atlantic sea with no coastline , named for Sargassum weed
Aral	Shrinking lake shared by Kazakhstan and Uzbekistan
Red Sea	Axial trough (rift) between Africa and Arabia

Direction	Sea	Easy cue
North (12)	Black Sea	Cold wall above Türkiye
South (6)	Mediterranean	Warm "middle sea" below Türkiye
West (9)	Aegean Sea	Greek-islands side
North-west pocket	Sea of Marmara	Tiny inland sea between Black and Aegean

Letter	Sea
M	Mediterranean
B	Black
C	Caspian
A	Aral

Sea / water body	Coastal countries (core list)	Classic "does NOT" trap
Mediterranean	Spain, France, Monaco, Italy, Slovenia, Croatia, Bosnia & Herzegovina, Montenegro, Albania, Greece, Türkiye, Syria, Lebanon, Israel , Egypt, Libya, Tunisia, Algeria, Morocco (+ island states)	Jordan , Chad, Iraq, Saudi Arabia
Black Sea	Russia, Ukraine, Romania, Bulgaria, Türkiye , Georgia	Syria (Med only)
Caspian (lake)	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan	Türkiye , Uzbekistan
Aral (lake)	Kazakhstan, Uzbekistan	Azerbaijan / Russia as pair
Red Sea	Egypt, Sudan, Eritrea, Djibouti, Saudi Arabia, Yemen, Israel, Jordan (Aqaba tip)	Jordan–Mediterranean
Persian / Arab Gulf	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE	Oman (faces Gulf of Oman / Arabian Sea)
Gulf of Oman	Oman, Iran, UAE (approaches)	Calling it "inside" the Persian Gulf
Baltic Sea	Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland, Sweden	Norway

Sea / water body	Coastal countries (core list)	Classic “does NOT” trap
North Sea	UK, Norway, Denmark, Germany, Netherlands, Belgium, France	Mixing with Baltic list
Arabian Sea	India, Pakistan, Iran, Oman, Yemen, Somalia, Maldives approaches	Treating Persian Gulf as identical
Bay of Bengal	India, Bangladesh, Myanmar, Sri Lanka approaches	—
South China Sea	China, Taiwan, Vietnam, Philippines, Malaysia, Brunei, Indonesia (parts)	—

Name	Where	Opens into	Note
Persian / Arab Gulf	Between Iran and Arabia	Gulf of Oman → Arabian Sea	Mouth at the Strait of Hormuz (oil chokepoint)
Gulf of Oman	SE of Hormuz	Arabian Sea	Outer side of Hormuz — not the same as the Persian Gulf
Gulf of Aden	Between Yemen and Somalia	Arabian Sea	Inner side of Bab-el-Mandeb , leading into the Red Sea
Gulf of Suez	NW arm of Red Sea (Egypt)	Red Sea	Not the Suez Canal lakes. The canal starts from this gulf toward the Mediterranean
Gulf of Aqaba	NE arm of Red Sea	Red Sea	Shared by Israel, Jordan, Egypt and Saudi Arabia, parallel to the Gulf of Suez
Gulf of Mexico	S of USA, E of Mexico	Atlantic (via Florida / Yucatan straits)	Caribbean neighbour with the Loop Current and major oil fields
Gulf of Guinea	West Africa (Nigeria–Ghana (घन) coast)	Atlantic	Warm Guinea Current and the Niger delta
Gulf of California (Sea of Cortez)	Between Baja and mainland Mexico	Pacific	Pacific arm, not Atlantic
Gulf of Alaska	S of Alaska	Pacific	Warm Alaska Current
Gulf of Thailand	Cambodia / Thailand / Vietnam	South China Sea	Shallow gulf, not part of the Indian Ocean
Gulf of Tonkin	N Vietnam / S China	South China Sea	Northern South China Sea arm
Gulf of Khambhat / Cambay	Gujarat, between Kathiawar and mainland	Arabian Sea	India’s classic tidal-energy coast
Gulf of Kutch	N of Kathiawar (Gujarat)	Arabian Sea	Second tidal site, with reefs
Gulf of Mannar	Between TN and Sri Lanka	Indian Ocean	Reefs and biosphere reserve (जैवमंडल आरक्षित क्षेत्र). Palk Strait is the narrower passage to the north
Hudson Bay	NE Canada	Atlantic via Hudson Strait	Huge inland sea, not ranked as a true ocean
Bay of Fundy	Nova Scotia–New Brunswick, Canada	Atlantic	Highest tidal range in the world (not Khambhat)
Gulf of Carpentaria	N Australia	Arafura Sea	Shallow tropical gulf
Bay of Biscay	W France / N Spain	Atlantic	Rough seas on Europe’s continental shelf
Gulf of Bothnia	Between Sweden and Finland	Baltic	Low-salinity Baltic arm
Gulf of Finland	Finland / Estonia / Russia (St Petersburg)	Baltic	Eastern Baltic arm
Bay of Bengal	E of India	Indian Ocean	Named a bay but treated as a sea; cyclone belt

N.2 Ocean Floor, Ridge, Shelf, Slope, Guyot

Step	Depth / slope	What it is	Key fact
Continental shelf	Shore to shelf break ~ 100–200 m ; slope $\leq 1^\circ$; ~ 7.5% of ocean area; avg width ~ 80 km	Drowned edge of the continent	Rich in fisheries and oil/gas . World's widest shelves are Siberian / Arctic . India's west shelf is wider than the east (Gujarat–Mumbai / Khambhat up to ~300 km; Konkan–Kerala narrower). The east shelf is narrower overall but carries big delta / Bengal Fan sediment. Oil pair: Bombay High (west) and KG basin (east)
Shelf break	~130–200 m	Sudden steepening	Marks the end of the “neritic” zone
Continental slope	~200 m down to ~ 3000 m ; 2–5°	True edge of the continental block	Cut by submarine canyons . Turbidity currents race down them
Deep-sea fan / cone	Foot of a canyon	Turbidite dump	Same idea as an alluvial (जलोढ़) fan, underwater
Continental rise	~3000–4000 m; gentler than slope	Sediment apron from shelf mud and canyon fans	Present on passive Atlantic-type margins. Often missing on active Pacific margins where a trench swallows the sediment
Abyssal plain	~ 3000–6000 m ; almost flat	Most of the ocean floor (~ 76% of ocean area in GC framing)	Hosts red clay and Mn nodules. “Abyssal hills” are small volcanic bumps on the plain
Trench (hadal)	> 6000 m (Mariana ~11 km)	Convergent plate where oceanic slab bends down	Island arcs or the Andes rise on the land side. Deepest points are in N.3
Mid-ocean ridge	Crest ~2000–3000 m (shallower than plains)	Divergent zone of new crust	Central rift valley , transform faults , fracture zones , and black smokers. Hess linked ridges to seafloor spreading

Seamount, guyot, atoll and related forms — Teaching Card

Form	Quick recall
Seamount	Peaked undersea volcano (>1000 m)
Guyot	Flat-topped drowned seamount
Volcanic island	Seamount that breaks the surface
Atoll	Ring / horseshoe reef around a lagoon
Submarine canyon	Gorge across shelf–slope
Bank	Flat shallow elevation (Dogger, Grand Banks)

Ridge	Fact
Mid-Atlantic	S-shaped; Telegraphic Plateau on the North arm
East Pacific Rise	Fast spreading
Carlsberg	NW Indian Ocean (Arabian Sea)
Mid-Indian / SW + SE Indian	Indian Ocean spreading
Ninetyeast	N–S volcanic trail in E Indian Ocean — not a classic spreading MAR

Word	Where
Neritic	Over the shelf
Bathyal	Over the slope
Abyssal	Over the plain
Hadal	Inside the trench

Deposit	Fact
Terrigenous	Land mud and sand on the shelf and slope
Red clay	Deepest abyssal deposit from volcanic and meteoric dust, with carbonate dissolved away
Globigerina ooze	Calcareous foraminifera mud at mid-depth
Pteropod ooze	Calcareous ooze, shallower than globigerina
Diatom ooze	Siliceous ooze of cold high-latitude seas
Radiolarian ooze	Siliceous ooze of tropical deep water
Manganese nodules	Potato-shaped ores on abyssal plains — classic in the Pacific ; India's claim is the Central Indian Ocean Basin

N.3 Trenches

Ocean	Deepest point
Pacific	Mariana (Challenger Deep)
Arctic	Molloy Deep
Indian	Sunda / Java Trench
Atlantic	Puerto Rico Trench

N.4 Ocean Currents (warm & cold)

Current	Type
Kuroshio (Japan)	Warm
North Pacific	Warm drift
Alaska	Warm
California	Cold
Humboldt / Peru	Cold — upwelling, rich fish. El Niño shuts this belt
East Australian	Warm
Oyashio / Kurile	Cold
Equatorial (N/S) + Equatorial Counter	

Current	Type
Gulf Stream / Florida	Warm
North Atlantic Drift	Warm drift that keeps NW Europe mild — the Blanket of Europe , helping keep Norway harbours milder and freer of ice
Canary	Cold
Labrador	Cold (meets Gulf Stream → Grand Banks fog/fish)
Brazil	Warm
Benguela	Cold — SE Atlantic , SW Africa — not Pacific
Falkland / Malvinas	Cold
Guinea	Warm

Current	Type
Agulhas	Warm current off SE Africa — the classic Indian Ocean option (not Florida, Canary or Kurile)

Current	Type
Mozambique	Warm
Somali	Seasonal (SW monsoon (दक्षिण-पश्चिम) upwelling off Somalia)
West Australian	Cold
Monsoon current	Reverses with monsoon — why N Indian Ocean current direction changes seasonally

N.5 Salinity

Belt	Salinity
Tropics of Cancer / Capricorn	Maximum in the open ocean — high evaporation, less rain
Equator	Lower than the tropics (heavy rain)
Poles	Lower (ice melt, low evaporation)

Hypersaline lake	Approx. salinity	Place note
Lake Assal	~ 348‰	Often cited as among the world's highest lake salinities
Lake Van	~ 330‰	Türkiye
Dead Sea	~ 238‰	Jordan Rift; classic high-salinity lake
Great Salt Lake	~ 220‰	Utah, USA

N.6 Tides

Tide types — Teaching Card

Type	Quick recall
Spring	Syzygy (new/full); large range
Neap	Quadrature (quarters); small range
Semi-diurnal	2 high + 2 low
Diurnal	1 high + 1 low
Mixed	Unequal highs / lows

N.7 Straits

Strait of Hormuz — must-score chokepoint

Item	Fact
Shores	Iran on the north; Oman (Musandam Peninsula) on the south
Nearby	UAE maritime zones sit on the Gulf and Arabian Sea approaches , but the classic shore pair is Iran–Oman
Width	About 21–29 nm at the narrowest teaching band; shipping lanes are only a few miles wide each
Oil	Roughly 20 mb/d / about 25% of world seaborne oil in recent IEA figures
Gas	About 20% of world LNG (mainly Qatar and UAE)
Gulf states behind the gate	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE — Oman is outside the Persian Gulf coast list
Bypass	Only Saudi Arabia and the UAE have useful pipelines that can skip Hormuz; the others are almost fully dependent on the strait

Item	Fact
Buyers	Most cargo goes to Asia (China, India, Japan, Korea)

Other high-yield straits — seas joined + shores

Strait	Joins	Shore / country cue
Hormuz	Persian Gulf ↔ Gulf of Oman	Iran–Oman
Bab-el-Mandeb	Red Sea ↔ Gulf of Aden	Djibouti / Eritrea side ↔ Yemen ; gate to the Indian Ocean for Suez traffic
Malacca	Andaman / South China Sea approaches ↔ Indian Ocean	Malaysia–Indonesia (Singapore at the tip)
Gibraltar	Mediterranean ↔ Atlantic	Spain–Morocco (UK: Gibraltar Rock)
Bosphorus	Black Sea ↔ Marmara	Entirely Türkiye
Dardanelles	Marmara ↔ Aegean	Entirely Türkiye
Bering	Arctic ↔ Pacific	Russia–USA (Alaska)
Palk	Bay of Bengal side ↔ Gulf of Mannar / Laccadive approaches	India–Sri Lanka
10° Channel	Andaman ↔ Nicobar (निकोबार)	Inside India
9° Channel (नौ डिग्री)	Minicoy ↔ Lakshadweep (लक्षद्वीप) proper	Inside India
8° Channel	Maldives ↔ Minicoy	Maldives–India
Duncan Passage	Little Andaman ↔ South Andaman	Inside India
Sunda	Indian Ocean ↔ Java Sea approaches	Sumatra–Java (Indonesia)
Torres	Coral / Arafura approaches	Australia–Papua New Guinea
Bass	Tasman Sea approaches	Australia–Tasmania
Magellan	Atlantic ↔ Pacific via Chilean channels	Chile
Drake Passage	Atlantic ↔ Pacific (open water)	South of Cape Horn — not Magellan
Dover	North Sea ↔ English Channel	UK–France
Mozambique	Indian Ocean corridor	Mozambique–Madagascar
Davis	Baffin Bay approaches	Greenland–Canada
Denmark Strait	Greenland Sea / Atlantic	Greenland–Iceland
Cook	Pacific between NZ islands	North Island ↔ South Island
Florida / Yucatan	Gulf of Mexico gates	USA–Cuba / Mexico–Cuba approaches
Formosa / Taiwan	East China Sea ↔ South China Sea	Taiwan Strait
Korea / Tsushima	Japan Sea approaches	Japan–Korea
Lombok / Makassar	Indonesian passages	Tanker alternatives when Malacca is tight
Kerch	Black Sea ↔ Sea of Azov	Russia–Ukraine zone
Otranto	Adriatic ↔ Ionian	Italy–Albania approaches
Skagerrak / Kattegat	North Sea ↔ Baltic approaches	Scandinavia–Denmark gate

N.8 Canals & Waterways

Other canals (match-list desk)

Canal	Joins	Country / note
Kiel	North Sea ↔ Baltic	Germany (Elbe estuary side ↔ Baltic). Not Mediterranean
Corinth	Ionian ↔ Aegean	Greece
Soo (Sault Ste. Marie)	Lake Superior ↔ Huron	USA–Canada. Not Superior–Michigan
Welland	Erie ↔ Ontario	Bypasses Niagara Falls
Erie Canal	Lake Erie ↔ Hudson River	Historic (ऐतिहासिक) US inland route
St Lawrence (लॉरेंस) Seaway	Great Lakes ↔ Atlantic	USA–Canada ocean door
Rhine–Main–Danube	North Sea inland system ↔ Black Sea	Europe cross-continent inland
White Sea–Baltic	Baltic ↔ White / Arctic approaches	Russia
Ludwig’s	S Germany inland waterway	Match-list filler
Volga–Don	Caspian approaches ↔ Black Sea / Azov system	Russia

N.9 UNCLOS, Territorial Waters, EEZ, Shelf

Zone	Breadth
Territorial sea	12 nautical miles — full sovereignty, with innocent passage
Contiguous zone	24 nm — customs, fiscal, immigration and sanitary control
EEZ	200 nm — sovereign rights to resources (fish, oil, wind)
Continental shelf	At least 200 nm, and may extend to 350 nm if geology allows
High seas	Beyond national zones
The Area	Seabed beyond national jurisdiction — common heritage , managed by the ISA (Kingston, Jamaica)

N.10 Important Fisheries

Ground	Why
Grand Banks (Newfoundland)	Labrador cold + Gulf Stream
Dogger Bank	North Sea shelf
Peru–Chile	Humboldt upwelling (fails in El Niño)
Japan / Kurile	Kuroshio + Oyashio mix
Iceland / Norway	Arctic/Atlantic mix
India	West coast and Wadge Bank, under EEZ rights

Common Traps — Don’t Fall For These

1. **Benguela = Pacific** — FALSE. **Atlantic**.
2. **Agulhas / Brazil = cold** — FALSE. **Warm** (2025).
3. **Syzygy = small tide** — FALSE. **Spring / large**.

4. **Salinity max at equator** — FALSE. **Tropics**.
5. **Tide every exactly 12 h 30 min** — FALSE. **~12 h 25 min**.
6. **El Niño → more Peruvian plankton** — FALSE. **Less**.
7. **Suez = Atlantic–Indian directly in 2025 codes** — FALSE. **Red + Mediterranean**.
8. **Suez lakes start at Timsah in the north** — FALSE. **Manzala** first.
9. **Mariana = Atlantic** — FALSE. **Pacific**. Puerto Rico = Atlantic.
10. **Molloy = Atlantic** — FALSE. **Arctic**.
11. **Dakshin Gangotri still operational** — FALSE. **Maitri + Bharati**.
12. **Himadri = Antarctica** — FALSE. **Arctic**.
13. **EEZ = 12 nm** — FALSE. Territorial **12**; EEZ **200**.
14. **Panama is sea-level like Suez** — FALSE. **Suez** is sea-level; Panama uses **stepped chambers** + Gatun.
15. **Kurile/Oyashio = Indian Ocean** — FALSE. **Pacific cold**. Indian MCQ = **Agulhas**.
16. **Telegraphic Plateau = Indian Ocean Ridge** — FALSE. **North Atlantic Ridge**.
17. **Guyot = peaked seamount** — FALSE. **Flat top**.
18. **Caspian = true ocean sea** — FALSE. **Lake**.
19. **Ninetyeast Ridge = Mid-Atlantic spreading** — FALSE. Volcanic trail in the **Indian** Ocean.
20. **Red clay = shallow coral mud** — FALSE. **Deepest** abyssal, carbonate gone.
21. **Diatom ooze = calcareous tropics** — FALSE. **Siliceous**, cold.
22. **Fundy = India's tidal-energy MCQ** — FALSE. World range king; India option = **Khambhat**.
23. **Drake = Magellan** — FALSE. Drake is open south of the Horn.
24. **India is an Arctic Council member** — FALSE. **Observer**.
25. **Himadri = Maitri's other name** — FALSE. Himadri = **Arctic**.
26. **ISA mines the EEZ** — FALSE. ISA = **the Area** beyond national zones.
27. **Skip ocean floor / deposits because recent papers asked currents** — FALSE. Next year can ask any standard NCERT trench, deposit or UNCLOS zone.
28. **India east shelf always wider than west** — FALSE. Coaching fact = **west wider** overall (Gujarat–Mumbai).
29. **Density ignores salinity** — FALSE. Cold + salt (+ pressure) set density and deep flow.
30. **Waves = tides** — FALSE. Waves need wind/fetch; tides need Moon/Sun gravity.
31. **Upwelling on east coasts of continents** — FALSE. Classic upwelling = **west** coasts.
32. **OMT depth is always 129 m** — FALSE. IAS key uses **~59 m** mean for SW Indian Ocean Jan–Mar; statement 1 in that stem is wrong.
33. **Fresh usable = 95% in ice of all Earth water** — FALSE. Ice is most of *fresh* water; usable total is **<1%** of all water.
34. **Rivers+lakes > groundwater** — FALSE among remaining fresh after ice. **Groundwater** is larger.
35. **Sargasso has a coastline** — FALSE.
36. **Telegraph Plateau = Indian Ocean** — FALSE. **North Atlantic Ridge**.
37. **Neap tide = new/full moon** — FALSE. New/full = **spring**. Neap = quadrature (~7–8th day).
38. **Benguela = Pacific / Indian** — FALSE. **South Atlantic**.
39. **Agulhas = cold** — FALSE. **Warm** Indian Ocean.
40. **Suez uses Panama-style stepped chambers** — FALSE. Suez is **sea-level**; Panama climbs chambers + **Gatun Lake**.
41. **Persian Gulf borders Oman** — FALSE. Oman faces **Gulf of Oman**.
42. **Soo Canal = Superior–Michigan** — FALSE. Superior–**Huron**.
43. **Great Barrier Reef = Atlantic / New Zealand** — FALSE. **Pacific**, NE **Queensland / Australia**.
44. **Coral reefs = cold coasts only** — FALSE. Tropical warm clear water.

45. **Türkiye north = Mediterranean / Caspian** — FALSE. North = **Black Sea**; south = Mediterranean.
46. **W→E = Black → Med → Caspian → Aral** — FALSE. **Mediterranean → Black → Caspian → Aral** ("My Black Cat Ate").
47. **Norway on the Baltic** — FALSE. Baltic ring excludes Norway.
48. **Jordan opens to the Mediterranean** — FALSE. Only **Aqaba / Red Sea**.
49. **Gaza faces the Red Sea** — FALSE. Gaza faces the **Mediterranean**.
50. **Hormuz shores = Iran + UAE only** — FALSE. Classic pair = **Iran + Oman** (Musandam).
51. **Syria borders the Black Sea** — FALSE. Syria is **Mediterranean**.
52. **Hormuz joins Red Sea and Mediterranean** — FALSE. That is **Suez**. Hormuz = Persian Gulf ↔ Gulf of Oman.

Topic 17 — World Rivers & Lakes

Current Affairs (this topic)

- Static map chapter — no living scheme tag.
- Lake names / dam (बांध) names do not need a yearly refresh unless a new “world’s largest” commission is officially recorded.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q94

The Grand Canyon of the U.S.A. is formed by which river?

A. St. Lawrence (लॉरेन्स) B. Missouri C. Ohio D. Colorado

Confused Pairs

Read as: **this** is correct · **that** is the usual wrong option.

	Correct	Usual wrong option	Hindi
New Orleans is on the	Mississippi	Missouri (Missouri joins Mississippi at St Louis)	न्यू ऑर्लीन्स
Hyderabad (India) is on the	Musi	Godavari / Paleru	मुसी
Mekong flows	South / south-east	South-west	मेकांग
Mekong delta is in	Southern Vietnam	Cambodia	मेकांग डेल्टा
Limpopo Capricorn	Crosses twice	Once only	लिम्पोपो
Mahaweli	Sri Lanka	India / Bangladesh	महावेली
Inselberg	Wind residual	Glacier	इंसेलबर्ग
Grand Canyon river	Colorado	Missouri / St Lawrence	ग्रैंड कैन्यन
Berlin river	Spree	Rhine	बर्लिन
Vienna river	Danube	Volga	वियना
USA–Mexico border river	Rio Grande	Colorado or Mississippi	रियो ग्रांडे
Lake Onega is in	Russia	Canada	ओनेगा
Lake Michigan lies	Wholly in the USA	Shared USA–Canada (Superior / Huron / Erie / Ontario are shared)	मिशिगन
Largest lake on Earth	Caspian (saline)	Superior	कैस्पियन
Largest fresh lake by area	Superior	Caspian / Baikal	सुपीरियर
Deepest lake + most fresh volume	Baikal	“Largest lake”	बैकाल
Longest river (tradition)	Nile	Amazon	नील
Largest discharge / basin	Amazon	Nile	अमेज़न
Highest waterfall	Angel (Venezuela)	Victoria	एंजेल
Victoria Falls is on the	Zambezi	Nile / Niagara	विक्टोरिया
Most countries on one river	Danube (~10)	Rhine	डेन्यूब

	Correct	Usual wrong option	Hindi
Busiest European inland navigation	Rhine	Danube	राइन
Budapest is on the	Danube	Rhine	बुडापेस्ट
Cologne is on the	Rhine	Danube	कोलोन
Khartoum stands at	White Nile + Blue Nile	Only one of them	खार्तूम
Congo uniqueness	Deepest + crosses Equator twice	Amazon does both	कांगो
Tanganyika vs Baikal	Tanganyika 2nd deepest / longest fresh	Tanganyika deepest	टंगाईका
Titicaca	Highest navigable	Highest of any puddle	टिटिकाका
Dead Sea	Lowest land + hypersaline	Ordinary sea	मृत सागर
Niger	Paradox (inland arc first)	Straight to Gulf	नाइजर
Volga mouth	Caspian	Black Sea	वोल्गा

N.1 World Rivers

River	Continent	Mouth	What the means by “famous”
Nile	Africa	Mediterranean	Longest river in the usual Prelims fact. White Nile from Lake Victoria + Blue Nile from Lake Tana (Ethiopia) meet at Khartoum . Cairo and Aswan sit on it.
Amazon	S America	Atlantic	Not the longest in most keys — but largest discharge and largest basin. Rises in the Andes .
Yangtze (Chang Jiang)	Asia (China)	East China Sea	Longest river in Asia . Three Gorges dam. Chongqing, Wuhan, Shanghai.
Mississippi–Missouri	N America (USA)	Gulf of Mexico	Longest system in North America. Missouri is a tributary that joins at St Louis . The mouth city is New Orleans on the Mississippi, not on the Missouri.
Ob, Yenisei, Lena	Siberia	Arctic Ocean	Three great north-flowing Siberian rivers. Angara comes north out of Lake Baikal and joins the Yenisei.
Volga	Europe (Russia)	Caspian (inland, not the ocean)	Longest river in Europe . Linked to the Black/Azov side by the Volga–Don canal.
Murray–Darling	Australia	Southern Ocean (via Murray mouth)	Australia’s main river system. Darling Range in Western Australia is a hill range , not this river.

River	Source	Mouth	Direction / countries	
Mekong	Tibetan Plateau	South China Sea at S Vietnam	China → Myanmar → Laos → Thailand → Cambodia → Vietnam. Flows south / south-east	Not SW . Delta is Vietnam , not Cambodia
Yangtze	Tibet	East China Sea	China, west → east	Asia’s longest river. Three Gorges Dam. Shanghai near the mouth
Hwang Ho / Yellow	Tibet	Bo Hai (Yellow Sea)	China	Called the “Sorrow of China”. Heavy loess silt. Arcuate delta
Amur	Mongolia–China border	Tatar Strait (Pacific)	Russia–China border river	Flows north-east / east

River	Source	Mouth	Direction / countries	
Ob (with Irtysh)	Altai	Arctic	Russia / Kazakhstan	North to the Arctic
Yenisei	Mongolia / Tuva	Arctic	Russia	The Angara leaves Baikal northward and joins the Yenisei
Lena	Highlands near Baikal	Arctic	Russia	North to the Arctic
Amu Darya	Pamir	Aral Sea	Central Asia	Aral is shrinking
Syr Darya	Tien Shan	Aral Sea	Kazakhstan / Uzbekistan	Flows north-west . With the Amu Darya it fed (and starved) the Aral
Tigris	Turkey	Shatt al-Arab → Persian (फ़ारसी) Gulf	Turkey–Iraq	Baghdad sits on the Tigris
Euphrates	Turkey	Shatt al-Arab → Persian Gulf	Turkey–Syria–Iraq	Twin of the Tigris
Jordan	Hermon / Anti-Lebanon	Dead Sea	Israel–Jordan	Ends in the Dead Sea with no ocean mouth
Irrawaddy	N Myanmar	Andaman (अंडमान) Sea	Myanmar lifeline	
Mahaweli Ganga	Sri Lanka highlands	Bay of Bengal	Sri Lanka's longest river (~335 km). The island shows radial drainage	
Salween	Tibet	Andaman Sea	China–Myanmar–Thailand	

River	Source	Mouth	Cities / dams / falls	Note
Nile	White: Victoria · Blue: Tana	Mediterranean	Khartoum (confluence), Cairo, Aswan High Dam → Lake Nasser	Arcuate delta. Traditional longest river. GERD sits on the Blue Nile in Ethiopia (Egypt (मिस्र)–Sudan dispute)
Congo / Zaire	E African highlands	Atlantic	Congo Basin rainforest	Deepest major river. Only major river that crosses the Equator twice . Second-largest discharge after the Amazon
Niger	Guinea highlands	Gulf of Guinea (Nigeria)	Inland delta in Mali, then a coastal delta	Paradox river — arcs inland / away from the nearby Atlantic before turning south-east to the Gulf
Zambezi	Zambia / Angola uplands	Mozambique Channel	Victoria Falls · Kariba · Cahora Bassa	Zambia–Zimbabwe border at the falls
Orange	Drakensberg	Atlantic	South Africa / Namibia	
Limpopo	S Africa	Indian Ocean (Mozambique)	Crosses Tropic of Capricorn twice	

River	Mouth	Cities	
Volga	Caspian	Source in the Valdai Hills . Linked by the Volga–Don canal	Longest river in Europe. Mouth is the Caspian , not the Black Sea
Danube	Black Sea	Vienna · Budapest · Belgrade	Crosses the most countries (~10)
Rhine	North Sea at Rotterdam	Basel · Strasbourg · Cologne · Bonn	Busiest European inland navigation. Flows in a rift valley
Thames	North Sea	London	
Seine	English Channel	Paris	
Po	Adriatic	N Italy	

River	Mouth	Cities	
Dnieper	Black Sea	Ukraine	
Vistula	Baltic	Warsaw	
Elbe	North Sea	Hamburg	
Rhône	Mediterranean	Lyon / S France	
Tagus	Atlantic	Lisbon	

River	Source → mouth	Cities / dams	
Amazon	Andes → Atlantic	—	Largest discharge
Paraná–Paraguay	Interior S America → Río de la Plata	Itaipu dam (Brazil–Paraguay)	
Orinoco	Venezuela / Colombia → Atlantic	—	
Mississippi	Minnesota → Gulf of Mexico	New Orleans at the mouth	Bird's-foot delta
Missouri	Rockies → joins Mississippi at St Louis	St Louis	Not the New Orleans river
Rio Grande (Río Bravo)	Rockies → Gulf of Mexico	—	USA–Mexico border (2023)
Colorado	Rockies → Gulf of California	Hoover Dam and the Grand Canyon	Not the USA–Mexico border river
Columbia	Rockies → Pacific	Grand Coulee Dam	NW USA
St Lawrence	Great Lakes → Atlantic	Montreal · Quebec	Seaway
Mackenzie (मैकेन्ज़ी)	Great Slave Lake → Arctic	—	Canada
Yukon	Canada → Bering Sea (Alaska)	—	
Murray–Darling	SE Australia → Southern Ocean	—	Main Australian system

N.2 World deltas

Shape	Meaning	Classic river
Arcuate	Bow / fan	Nile , Hwang Ho, Niger, Rhine, Ganga–Brahmaputra
Bird's-foot	Distributaries like toes	Mississippi — Nile is not this
Cuspate	Pointed tooth	Tiber (Italy)
Moribund	Abandoned / dying lobe	Cauvery (India)

N.2A River landforms (world desk)

Form	Note
Gorge	Deep narrow valley with steep walls (e.g. Chisapani on Karnali, Nepal)
Canyon	Broader/deeper eroded gorge — Grand Canyon cut by Colorado (Arizona)
V-shaped valley	Youthful vertical + lateral erosion
Meander	Looping course on gentle plains
Alluvial (जलोढ़) fan / cone	Sediment cone where a stream leaves mountains into a plain
Delta	Mouth deposit. The Ganga–Brahmaputra–Meghna is the world's largest. The Mississippi is bird's-foot . The Nile is arcuate

Form	Note
Watershed / catchment	Land unit that collects, stores and releases water to a common outlet

N.3 Cities on Rivers

City	Country	River	Trap
Budapest	Hungary	Danube	Not Rhine
Vienna	Austria	Danube	
Belgrade	Serbia	Danube (+ Sava)	
Cologne (Köln)	Germany	Rhine	Not Danube
Rotterdam	Netherlands	Rhine mouth	
Basel	Switzerland	Rhine	
Strasbourg	France	Rhine	
New Orleans	USA	Mississippi	Not Missouri
St Louis	USA	Mississippi meets Missouri	
Cairo	Egypt	Nile	
Khartoum	Sudan	White Nile + Blue Nile	Not a single-arm city
Alexandria	Egypt	Nile delta fringe	
London	UK	Thames	
Paris	France	Seine	
Rome	Italy	Tiber	
Hamburg	Germany	Elbe	
Warsaw	Poland	Vistula	
Moscow	Russia	Moskva	
Berlin	Germany	Spree	
Lisbon	Portugal	Tagus	
Baghdad	Iraq	Tigris	Not Euphrates
Shanghai	China	Yangtze	
Wuhan	China	Yangtze	
Chongqing	China	Yangtze	
Montreal	Canada	St Lawrence	
Quebec	Canada	St Lawrence	
Washington DC	USA	Potomac	
New York	USA	Hudson	
Hyderabad	India	Musi (joins Krishna (कृष्णा))	Not Godavari; not Paleru
Phnom Penh	Cambodia	Mekong	
Hanoi	Vietnam	Red River	
Yangon	Myanmar	Irrawaddy	Former capital
Madrid	Spain	Manzanares	
Asunción	Paraguay	Paraguay	
Niamey	Niger	Niger	

City	Country	River	Trap
Basra	Iraq	Shatt al-Arab	
Kinshasa / Brazzaville	DRC / Congo	Congo (Zaire)	
Bangkok	Thailand	Chao Phraya (Menam)	
Lahore	Pakistan	Ravi (ਰਾਵੀ)	
Perth	Australia	Swan	
Caracas	Venezuela	Orinoco basin note	

N.4 International / Border Rivers

River	What it divides / crosses	
Rio Grande	USA–Mexico for a long stretch	Not Colorado, not Mississippi
Colorado	Only a short US–Mexico bit near Yuma	Do not pick for “the” border river
Danube	Germany to the Black Sea through ~ 10 countries	Most international
Rhine	Switzerland–France–Germany–Netherlands	
Mekong	China, Myanmar, Laos, Thailand, Cambodia, Vietnam	
Amur	Russia–China	
Jordan	Israel–Jordan	Into Dead Sea
Zambezi	Zambia–Zimbabwe at Victoria Falls	
St Lawrence	USA–Canada	
Niagara River	USA–Canada, between Lakes Erie and Ontario	

N.5 River Navigation

River navigation — how inland waterways work

Waterway	From / to	
Rhine	Swiss Alps → Rotterdam (North Sea)	Europe’s busiest inland waterway
Danube	Germany → Black Sea	Linked to the Rhine by the Rhine–Main–Danube canal
Volga	Russia → Caspian	Volga–Don canal opens a way toward Azov / Black Sea
St Lawrence Seaway	Great Lakes → Atlantic	Ocean ships into the lakes
Yangtze	Interior China → East China Sea	China’s main inland trunk
Mississippi	Interior USA → Gulf of Mexico	USA inland trunk

N.6 Dams

Dam	River	Country	
Three Gorges	Yangtze	China	World’s largest hydro fact
Aswan High	Nile	Egypt	Creates Lake Nasser
GERD	Blue Nile	Ethiopia	Water-sharing dispute with Egypt / Sudan
Itaipu	Paraná	Brazil–Paraguay	
Kariba	Zambezi	Zambia–Zimbabwe	

Dam	River	Country	
Hoover	Colorado	USA	
Grand Coulee	Columbia	USA	
Akosombo	Volta	Ghana (घन)	
Nurek	Vakhsh	Tajikistan	
Merowe	Nile	Sudan	
Cahora Bassa	Zambezi	Mozambique	
Kurobe	Kurobe	Japan	
Grande Dixence	Dixence	Switzerland	
Enguri	Enguri	Georgia	
Xiaowan	Lancang / Mekong	China	
Zangmu	Yarlung Tsangpo / Brahmaputra	Tibet (China)	Flow concern note
Oroville	Feather	USA	

N.7 Lakes

Type	Fact / example
Tectonic / rift	Baikal, Tanganyika, Dead Sea, Malawi
Glacial	Many Canadian and Alpine lakes. The Great Lakes have glacial history (इतिहास)
Crater / caldera	Crater Lake (USA). Lonar in India is impact, not volcanic
Lagoon / coastal	Chilika-type coastal idea. Barred coastal lakes
Oxbow	Cut-off meander lakes on floodplains
Endorheic	Caspian, Aral, Chad, Dead Sea, Eyre — inland, no sea outlet
Lake	Country
-----	-----
Caspian	RU / IR / KZ / TM / AZ
Superior	USA–Canada
Baikal	Russia
Tanganyika	TZ / DRC / BI / ZM
Victoria	E Africa (TZ / UG / KE)
Michigan	USA only
Huron	USA–Canada
Erie	USA–Canada
Ontario	USA–Canada
Onega	Russia (northwest)
Ladoga	Russia
Maracaibo	Venezuela
Dead Sea	Jordan–Israel
Titicaca	Peru–Bolivia
Aral	Kazakhstan–Uzbekistan
Chad	Africa (multi-state)

Type	Fact / example
Eyre	Australia
Great Bear / Great Slave	Canada
Malawi / Nyasa	Africa rift

Lake / pair	Fact
Faguibine	Mali (W Africa) — dried / desertified
Balkhash	Kazakhstan
Finger Lakes	New York, USA
Boiling Lake	Dominica
Five Flower Lake	Sichuan, China (not Mongolia)
Red Lagoon	Bolivia
Urmia	Iran
Reindeer / Athabasca	Canada (Uranium City near Athabasca)
Plitvice	Croatia — UNESCO lake park
Great Lakes E→W	Ontario → Erie → Huron → Michigan → Superior
Cities	Duluth on Superior. Chicago on Michigan. Buffalo on Erie. Detroit on the Detroit River
Victoria	Shared by Tanzania, Uganda and Kenya (three countries). Rift-associated. White Nile outflow near Jinja
Volume ladder (fresh note)	Caspian (saline) > Baikal > Tanganyika > Superior > Malawi
Area ladder (fresh, approximate coaching order)	Superior → Victoria → Huron → Michigan → Tanganyika → Baikal → Great Bear → Malawi → Great Slave → Erie → Winnipeg → Ontario

N.8 Waterfalls

Fall	River / basin	Countries	
Angel	Churún (Caroní / Orinoco)	Venezuela	Highest on Earth
Tugela	Tugela	South Africa	Usual 2nd-height fact
Victoria	Zambezi	Zambia–Zimbabwe	Not the highest
Niagara	Niagara (Erie → Ontario)	USA–Canada	
Iguazu / Iguazu	Iguazu (Paraná system)	Brazil–Argentina	
Ribbon	Ribbon Creek	California (Yosemite)	
Browne	—	New Zealand	high fall note
Yosemite	Yosemite Creek	USA (California)	

Common Traps — Don't Fall For These

1. **New Orleans = Missouri** — FALSE. **Mississippi**. Missouri joins at **St Louis**.
2. **Hyderabad = Godavari / Paleru** — FALSE. **Musi**.
3. **Mekong = south-west** — FALSE. **South / SE**.
4. **Mekong delta = Cambodia** — FALSE. **Vietnam**.
5. **USA–Mexico = Colorado** — FALSE. **Rio Grande**.
6. **Onega = Canada** — FALSE. **Russia**.

7. **Michigan is shared like Superior** — FALSE. Michigan is **wholly USA**.
8. **Baikal = largest lake** — FALSE. **Caspian** largest; Baikal = **deepest / fresh volume**.
9. **Amazon = longest in every book** — FALSE. **Nile** is the traditional longest fact.
10. **Budapest = Rhine** — FALSE. **Danube**. Cologne = Rhine.
11. **Angel = Victoria** — FALSE. Angel = **highest**; Victoria = **Zambezi**.
12. **Three Gorges = Hwang Ho** — FALSE. **Yangtze**.
13. **Volga → Black Sea** — FALSE. **Caspian**.
14. **Congo = largest discharge** — FALSE. **Amazon**. Congo = deepest + Equator twice.
15. **Tanganyika = deepest lake** — FALSE. **Baikal**; Tanganyika is 2nd.
16. **Titicaca = highest of any lake always** — FALSE. Fact = highest **navigable**.
17. **Skip lake types because papers asked Onega** — FALSE. Next year can ask rift / endorheic / Great Lakes order.
18. **GERD = Aswan** — FALSE. GERD = Blue Nile Ethiopia; Aswan = Egypt Nile / Lake Nasser.
19. **Darling Range = Murray–Darling river** — FALSE. Darling Range = hills of **Western Australia**.
20. **Hyderabad = Godavari** — FALSE. **Musi**.
21. **Inselberg = glacier landform** — FALSE. Wind residual (with zeugen). Moraine = glacier.
22. **Grand Canyon = Missouri** — FALSE. **Colorado**.
23. **Berlin = Rhine** — FALSE. **Spree**. Vienna = Danube (not Volga).
24. **Lisbon = Tiber** — FALSE. Tiber = Rome; Lisbon = **Tagus**.
25. **Mahaweli = India** — FALSE. **Sri Lanka**.
26. **Congo Basin excludes Cameroon** — FALSE. Cameroon is in the Congo Basin set.
27. **Baikal = Ukraine** — FALSE. **Russia**.
28. **Five Flower Lake = Mongolia** — FALSE. **Sichuan, China**.
29. **Angel = Tugela as world's tallest** — FALSE. **Angel (Venezuela)** is tallest.
30. **Hoover = Tennessee** — FALSE. **Colorado**.
31. **Lake Victoria bordered by four countries** — FALSE. **Three** (TZ, UG, KE).
32. **Onega = Canada** — FALSE. **Russia**.

Topic 18 — World Landforms

Current Affairs (this topic)

None mandatory — static world-map chapter. No living scheme tag.

Confused Pairs

Pair	Correct	Trap	Hindi
Andes vs Himalaya	Andes = longest fold chain; Himalaya = highest	Swapped	एंडीज़ / हिमालय
Kilimanjaro vs Fuji	Kilimanjaro = E African Rift ; Fuji = Ring of Fire	Kilimanjaro in Pacific	किलिमंजारो
Pyrenees vs Alps	Pyrenees = Spain–France ; Alps = Switzerland / C Europe	Alps on Spain–France MCQ	पिरेनीज़
Apennine vs Balkan	Apennine = Italy ; Balkan = Bulgaria	Swapped	एपिनाइन / बाल्कन
Atlas vs Hoggar	Atlas high peak Toubkal = Morocco ; Hoggar = Algeria	Both “North Africa” dump	अटलास / होगार
Gobi vs Sahara	Gobi = cold , Mongolia+China; Sahara = hot , Africa	Gobi in Kazakhstan	गोबी
Gibson vs Sonoran	Gibson = Australia ; Sonoran = USA	Gibson–Brazil	गिब्सन
Pampas vs Campos	Pampas = Argentina ; Campos = Brazil	Pampas–Brazil/Chile	पम्पास
Prairie vs Steppe	Prairie = N America; Steppe = Eurasia	Names swapped	प्रेयरी / स्टेपी
Savanna vs hot desert	“Land of Big Games” = Savanna , not Sahara	Desert = safari trap	सवाना
Maquis vs Chaparral	Maquis = Med basin ; Chaparral = California	Same shrub, wrong coast	माकी / चैपरेल
Fynbos vs Matorral	Fynbos = South Africa ; Matorral = Chile	Swapped SH west coasts	फिनबॉस
Taiga vs Tundra	Taiga = conifer forest ; Tundra = treeless cold	Both “cold vegetation”	टैगा / टुंड्रा
Borneo vs Java	Borneo = Sunda shelf , not volcanic dump; Java = volcanic arc	“All Indonesia = volcano”	बोर्नियो
Greenland vs New Guinea	Greenland = largest island ; New Guinea = 2nd	Australia is a continent	ग्रीनलैंड
Med vs monsoon	Med rain = winter ; monsoon rain = summer	Med summer rain	भूमध्यसागरीय
Pamir vs Tibet	Pamir = phrase Roof of the world ; Tibet = highest large plateau	Swapped	पामीर / तिब्बत
Hot desert coasts	West coasts + cold currents + subtropical high	East-coast deserts	पश्चिमी तट
Chernozem	Temperate grassland / steppe–prairie black soil	Equatorial laterite	चेरनोज़ेम
Residual mountains	Worn leftovers of old folds (Appalachians type)	Young Himalaya	अवशिष्ट
Continent area	Asia largest; Australia smallest	Africa largest	महाद्वीप
Most countries	Africa (~54)	Europe / Asia	अफ्रीका
Guyana	South America	Africa (Gabon/Guinea set)	गुयाना
Highest mean elevation	Antarctica	Asia	अंटार्कटिका
Longest range	Andes	Himalaya / Rockies	एंडीज़
Sierra Nevada	Block mountain	Young fold with Alps	सिएरा नेवादा
Death Valley	Rift + extreme heat (California)	Cold Siberia only	डेथ वैली
Telegraphic Plateau	N Atlantic Ridge	Indian Ocean	टेलीग्राफिक

Pair	Correct	Trap	Hindi
Thar	Most populated desert	Sahara densest people	थार
Downs vs tropical	Downs = temperate (Australia)	Called tropical	डाउन्स
Selvas vs grassland	Selvas = Amazon rainforest	Treated as grassland	सेल्वास
Veld vs Downs	Veld = South Africa ; Downs = Australia	Veld–Australia	वेल्ड / डाउन्स
Faroe	Sheep Islands (Denmark)	Canary / Madeira	फेरो
Honshu	Japan's largest ; Tokyo	Hokkaido largest	होन्शू
Falklands	S Atlantic	Indian Ocean	फॉकलैंड
Galápagos	Ecuador	Peru	गैलापागोस
Socotra	Yemen	Oman	सोकोत्रा
Majuli	Brahmaputra river island	Amazon	माजुली

N.0A Continents

Rank	Continent	Area pair (km ²)	% of Earth's land (Philip's-type)	Countries (GC sheet)
1	Asia	~44.6 million	~29.8%	~48
2	Africa	~30.1 million	~20.3%	~ 54 (most of any continent)
3	North America	~24.5 million	~16.2%	~23
4	South America	~17.8 million	~11.8%	~12 (Brazil largest)
5	Antarctica	~14.2 million	~9.2%	—
6	Europe	~9.9 million	~6.8%	~49
7	Australia	~7.7 million	~5.9%	Oceania ~14 in wider lists

N.1 Mountains

Family	Force / story	World examples
Fold	Compression (orogeny)	Himalaya, Alps, Andes, Rockies, Atlas, Pyrenees, Zagros. The Appalachians are an old worn fold
Block / fault / rift	Tension / uplifted blocks	Vosges, Black Forest, Sierra Nevada; African rift shoulders (Kilimanjaro, Kenya, Rwenzori)
Volcanic cone	Magma	Fuji, Etna, Cotopaxi, Mauna Loa — location table in N.4
Residual / dome	Old highland left after long denudation; or magma push (dome)	Residual leftovers of old folds (Appalachian-type worn highlands); Black Hills / laccolith dome

Continent	Peak	Range	Country fact
Asia	Everest (8848 m)	Himalaya	Nepal–China
Asia (2nd)	K2 / Godwin Austen	Karakoram (काराकोरम)	Pakistan–China
South America	Aconcagua	Andes	Argentina — highest outside Asia
North America	Denali / McKinley	Alaska Range	USA
Africa	Kilimanjaro	—	Tanzania (not Kenya)
Europe	Elbrus	Caucasus	Russia

Continent	Peak	Range	Country fact
Alps (W Europe)	Mont Blanc	Alps	France–Italy
Australia (mainland)	Kosciuszko	Snowy Mts	Australia
Oceania (often asked)	Puncak Jaya / Carstensz	Sudirman	Indonesia (New Guinea)
Antarctica	Vinson	Ellsworth	

Range	Where	Highest / note
Andes	W South America, ~7,000 km	Longest fold chain. Chimborazo (Ecuador). Copper Chile–Peru. Rain-shadow → Atacama + Patagonia
Himalaya	India–Nepal–Bhutan–China	Highest . Young Tertiary. Everest / Kanchenjunga (कंचनजंगा)
Karakoram (काराकोरम)	PK–India–China	K2 . Not a Himalaya “range name” swap
Rockies	W Canada–USA–Mexico fringe	Young cordillera. Not the Appalachians
Appalachians	Eastern USA	Old fold; worn; coal fields
Alps	France–Switzerland–Austria–Italy	Mont Blanc . Switzerland is the classic Alps country. The foehn is a warm downslope wind on the north side
Pyrenees	Spain–France (Andorra)	Not Alps
Apennines	Italy	Spine of the peninsula
Balkans	Bulgaria / SE Europe	Not Carpathians
Carpathians	Romania / Slovakia / Ukraine	Transylvania arc — do not place on Bulgaria
Caucasus	Russia–Georgia–Azerbaijan	Elbrus is Europe’s high point
Urals	Russia	Conventional Europe–Asia divide
Atlas	Morocco–Algeria–Tunisia	Highest peak Toubkal is in Morocco (not Algeria)
Drakensberg	South Africa / Lesotho	SE African escarpment
Ethiopian Highlands	Ethiopia	Blue Nile / Lake Tana source side
Rwenzori	Uganda–DRC	Mt Stanley — country fact Uganda
Hoggar / Ahaggar	Algeria	Sahara highland. Not Atlas
Tibesti	Chad	Sahara highland
Zagros / Elburz	Iran	Zagros is the west fold. Elburz is the north range, with Damavand
Tien Shan / Altai	C Asia	Syr Darya / Ob headwaters
Kunlun	N of Tibet	Tibet’s northern wall
Great Dividing Range	Eastern Australia	Old; not Himalaya-young; ~3700 km; Mt Kosciuszko
Southern Alps	South Island, New Zealand	Aoraki / Mt Cook
Darling Range	SW coast of Australia (Western Australia)	Not NE Australia
Arakan Yoma	Myanmar	India–Myanmar hills: Naga, Chin, Patkai / Lushai; peak Natma Taung / Mt Victoria — Khasi is Meghalaya (মেঘালয়) , not this border trio
White Mountains	New Hampshire / Maine, USA	
Black Mountains	North Carolina, USA (Appalachian)	Not Black Forest
Black Forest	Germany (Feldberg); east of Rhine ; Vosges west in France	Block mountain

Range	Where	Highest / note
Salt Range	Pakistan	Block mountain note
Allegheny	USA (Appalachian)	
Cantabrian	Spain	
Mackenzie (मैकेंज़ी)	Canada	
Jura	France–Switzerland	
Dinaric Alps	Balkans coast belt	
Pennines	Europe (UK)	Residual / old with Appalachians & Aravallis

Peak	Country	Do not confuse with
Kilimanjaro	Tanzania	Kenya (Mt Kenya is a different peak); not Ring of Fire
Toubkal	Morocco	Hoggar
Stanley	Uganda	Kilimanjaro
Hoggar	Algeria	Atlas / Toubkal
Chimborazo	Ecuador	Peru/Chile Andes dump
Aconcagua	Argentina	Chile
Cotopaxi	Ecuador	Volcanic Andes

N.2 Plateaus

Kind	Meaning	World example
Intermontane	High block between fold ranges	Tibet (Himalaya–Kunlun); Altiplano (Andes)
Piedmont	At the foot of mountains	Appalachian piedmont; Patagonian edge
Continental / shield	Old worn-down block	Deccan (दक्कन), Brazilian, African, Western Australian
Lava / volcanic	Flood basalt	Columbia (USA), Deccan Trap (दक्कन ट्रैप) cover
Plateau	Where	Note
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Tibetan Plateau	China / Tibet	Highest large plateau. Headwaters: Indus (सिंधु), Sutlej (सतलुज), Brahmaputra (ब्रह्मपुत्र), Mekong, Yangtze, Hwang Ho
Pamir	Tajikistan knot	Phrase “ Roof of the world ”. Knot of Himalaya–Karakoram–Hindu Kush–Tien Shan
Deccan	Peninsular India	Old gneiss + Deccan Trap lava in the west. Not Brazil
Malwa / Chotanagpur (छोटानागपुर)	India	Chotanagpur = Jharkhand coal–iron table
Brazilian	Brazil	Coffee on Terra Roxa ; iron. Separate from Andes
Guiana	Venezuela–Guyana–N Brazil	
Patagonian	S Argentina	Rain-shadow plateau east of Andes
East African / Ethiopian	Africa	Rift lakes; Blue Nile from Ethiopia
African (south)	SA / Botswana / Namibia	High veld sits on it
Colorado	SW USA	Grand Canyon (Colorado River) cut into it
Columbia	Washington–Oregon–Idaho	Lava plateau

Kind	Meaning	World example
Colorado Plateau ≠ Columbia	—	Canyon vs lava — do not swap
Mexican (Mesa Central)	Mexico	
Anatolian	Türkiye	Interior of Asia Minor
Iranian	Iran	Dasht-e-Kavir / Lut on it
Mongolian	Mongolia	Gobi on the dry side
Altiplano	Bolivia–Peru	Highest inhabited Andean plateau; Lake Titicaca
Meseta	Spain	Interior tableland of Iberia
Loess Plateau	N China	Wind-silt; Hwang Ho yellow colour
Kimberley / Western Plateau	Australia	Old shield
Ozark	USA	Domed plateau note
Shan	Myanmar	Fluvial / tableland note
Telegraphic Plateau	North Atlantic Ridge (Newfoundland–Ireland note)	Not Indian Ocean

N.2A Valleys

Valley types and named valleys — Teaching Card

Place	Quick recall
Death Valley	California rift; heat; salt pan; ~–86 m
Panjshir	NE Afghanistan (Hindu Kush)
Taylor Valley	Antarctica — “Valley of the Dead”
Silicon Valley	California chip / tech belt
Valley of the Kings	Egypt (Luxor / Nile west bank)
Great Artesian Basin	Australia groundwater basin
Great Basin	USA endorheic interior
Blind valley / doline	Karst limestone

N.3 Plains

How plains form — Teaching Card

Kind	How formed	Example
Alluvial	River dump	Indo-Gangetic, Nile, Mesopotamia, Hwang Ho, Mississippi
Glacial / till	Ice-sheet	North European Plain, Prairie of Canada–USA
Coastal	Sea + river	Atlantic coastal USA; eastern India
Lacustrine	Old lake floor	Imphal-type locally; some interior basins
Plain	Where	Note
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Indo-Gangetic	India–Pakistan–Bangladesh	Alluvium; wheat–rice–cane

Kind	How formed	Example
North / Great European	France → Poland → Russia	Wheat–dairy; glacial + river
Great Plains / Prairie	USA–Canada	Temperate grassland overlay
Pampas	Argentina (into Uruguay)	Temperate grassland and a plain. Not Chile, not Brazil
Llanos	Orinoco (Venezuela–Colombia)	Tropical grassland plain
Manchurian	NE China	Soy / wheat
West Siberian	Russia	Ob–Yenisei lowland
Mesopotamian	Iraq	Tigris–Euphrates
Nile valley / delta (डेल्टा)	Egypt	Alluvium in a desert
Po valley	N Italy	Alpine alluvium
Hungarian (Alföld)	Hungary	Puszta grassland sits here
Canterbury	New Zealand	Downs / sheep
Nullarbor	S Australia	Limestone plain, not a grassland name

N.4 Volcanoes (world locations)

Belt	Share of great volcanoes	names
Circum-Pacific / Ring of Fire	~80%	Fuji, Pinatubo, Mayon, Apo, St Helens, Rainier, Popocatépetl, Andes cones, Krakatoa, Tambora
Mid-Atlantic	Divergent	Iceland (Hekla), Azores
Alpine–Himalayan / Med	Collision	Etna, Vesuvius, Stromboli (Italy)
East African Rift	Tension	Kilimanjaro, Kenya — not Ring of Fire
Volcano	Country	Belt / extra
-----	-----	-----
Fuji	Japan	Pacific. Classic cone
Pinatubo	Philippines	Pacific. 1991 eruption tag
Mayon	Philippines	Pacific. Perfect cone
Apo	Philippines	Highest PH; 2021 list
St Helens	USA (Washington)	Cascades / Pacific. 1980
Rainier	USA	Cascades / Pacific. 2021 list
Popocatépetl	Mexico	Pacific
Paricutin (paper: Pericutine)	Mexico	Cinder cone; born 1943. 2021 list
Cotopaxi	Ecuador	Andes / Pacific rim
Krakatoa / Krakatau	Indonesia	Sunda arc / Pacific
Tambora	Indonesia	1815; “year without summer”
Mauna Loa / Mauna Kea	Hawaii (USA)	Hot-spot shield , not a collision arc
Etna	Italy (Sicily)	Med. 2021 list. Not Pacific
Vesuvius	Italy	Pompeii AD 79
Stromboli	Italy	“Lighthouse of the Mediterranean”
Kilimanjaro	Tanzania	Rift . 2024 “which is NOT Ring of Fire”

Belt	Share of great volcanoes	names
Mt Kenya	Kenya	Rift. Lower than Kilimanjaro
Erebus	Antarctica	Ross Island; southern active
Hekla	Iceland	Mid-Atlantic

N.5 Islands

Rank	Island	Note
1	Greenland	Largest island; Denmark politically
2	New Guinea	Indonesia (Papua) + Papua New Guinea
3	Borneo	Indonesia, Malaysia, Brunei
4	Madagascar	Africa's large island
5	Baffin	Canada (Nunavut)
6	Sumatra	Indonesia
7	Honshu	Japan's main island
8	Victoria	Canada (Arctic)
9	Great Britain	England–Scotland–Wales
10	Ellesmere	Canada (Arctic)

Kind	How made	Examples
Continental	Piece of a continent, shallow shelf	Greenland, Borneo, New Guinea, Madagascar , Britain, Sri Lanka
Volcanic arc / hot-spot	Magma	Java, Iceland, Hawaii , Japan (Honshu mixed arc), Sicily (Etna)
Coral / atoll	Reef on a sinking volcanic peak or shelf	Maldives, Lakshadweep (लक्षद्वीप), many Pacific atolls
Coral reef (not an island rank)	—	Great Barrier Reef (Australia) is a reef , not a "largest island" option
Island	Fact	
-----	-----	
Greenland	Largest island. Denmark. Ice sheet. Inuit	
New Guinea	2nd. Split Indonesia (Papua) + Papua New Guinea	
Borneo (Kalimantan)	3rd. Three countries: Indonesia, Malaysia (Sabah/Sarawak), Brunei. Sunda shelf — not volcanic deposition (2022 Only 1)	
Madagascar	4th. Africa's big island; unique fauna (प्राणीजात); not three-country split	
Baffin	Canada / Arctic	
Sumatra	Indonesia; Sunda arc is volcanic — contrast with Borneo	
Java	Indonesia; dense, volcanic	
Sulawesi	Indonesia; K-shaped	
Honshu	Japan's main island. Hokkaido / Kyushu / Shikoku = the other three	
Victoria	Canada / Arctic; after Honshu in the area ladder	

Kind	How made	Examples
Great Britain	England–Scotland–Wales (Ireland is separate)	
Ellesmere	Canada / Arctic; after Great Britain in the top-ten set	
Iceland	Mid-Atlantic Ridge; volcanic	
Hawaii	Pacific hot-spot chain	
Sri Lanka	Continental; Palk Strait / Gulf of Mannar vs India	
Tasmania	S of Australian mainland	
Sicily / Sardinia / Corsica / Crete / Cyprus	Med islands. Sicily has Etna	
New Zealand	North Island + South Island; Cook Strait	

Island / group	Note
Nuuk	Capital of Greenland; elevation ~ 1–3 m
Diego Garcia	Largest of Chagos (Indian Ocean); UK/US base
Socotra	Yemen ; UNESCO natural heritage 2008 ; Guardafui / Arabian Sea mouth — not Oman
Seychelles / Mauritius / Madagascar	SW Indian Ocean set
Falklands	South Atlantic (UK), ~650 km E of S America — not Indian Ocean
Azores / Madeira / Cape Verde	Atlantic (Azores/Madeira = Portugal; Cape Verde = sovereign)
Bahamas	Atlantic / Lucayan
Fiji	Pacific Melanesia; Viti Levu + Vanua Levu hold most people
Melanesia	New Guinea, Bismarck, Solomon, Vanuatu, New Caledonia, Fiji (not Gilbert/Marshall (मार्शल)/Society = Micronesia/Polynesia)
Galápagos	Pacific volcanic; Ecuador (not Peru)
Grenada	Caribbean; “Island of Spices”
Balearic	Spain; W Mediterranean ; Palma de Mallorca
Faroe	Denmark; Sheep Islands ; N Atlantic
Macau	Handed to China 1999
Majuli	Largest river island; Brahmaputra (ব্রহ্মপুত্র), Assam
Japan N→S	Hokkaido → Honshu → Shikoku → Kyushu ; Tokyo on Honshu (largest)
Indonesia W→E	Sumatra → Java → Bali (बलि) → Lombok
NZ South Island	Larger than NZ North / Java / Newfoundland among common options
Australia	Island continent (smallest continent)
Polynesia	Tonga, Tuvalu, Tahiti — Tinian (Marianas) is not Polynesia
Saint Martin	Caribbean salt-pan / lagoon coast pair
Indonesia island count	~ 17,500 islands (highest among common MCQ options; Sweden note for absolute (निरपेक्ष) world count)

N.6 Deserts

Kind	Why dry	Examples
Hot subtropical	Descending air ~ 30° (horse latitudes)	Sahara, Arabian, Australian, Kalahari, Thar

Kind	Why dry	Examples
Coastal / fog	Cold current + stable air	Atacama (Peru/Humboldt), Namib (Benguela)
Rain-shadow	High range steals moisture	Patagonia (Andes), Atacama (Andes), Thar (Aravalli leeward)
Cold interior	Far from sea, winter freeze	Gobi, Taklamakan , Great Basin, Colorado Plateau dry side
Desert	Country / belt	
-----	-----	-----
Sahara	N Africa (many states)	Largest hot desert. Hoggar / Tibesti are highland islands in it
Arabian	Arabian Peninsula	2nd large hot desert. Rub' al-Khali = Empty Quarter
Gobi	Mongolia + China (Inner Mongolia, Gansu, Xinjiang fringe)	Cold . 2025: Only 1 and 2. Not Russia, not Kazakhstan
Great Sandy	NW Australia	Smallest of the 2023 four
Great Victoria	S/W Australia	Largest Australian desert by many lists
Gibson	Australia (between Victoria and Sandy)	Not Brazil (2018)
Simpson	Central Australia	Red dunes
Sonoran	USA + Mexico	2018 correct pair. Cactus desert
Mojave	USA	Death Valley
Chihuahuan	USA–Mexico	
Taklamakan	China (Xinjiang)	Cold; Tarim basin. 2018 correct pair
Karakum	Turkmenistan	2018 correct pair. Not Kazakhstan's default dump
Kyzylkum	Uzbekistan–Kazakhstan	Red sand; separate from Karakum
Kalahari	Botswana / South Africa / Namibia	Bushman / San. Semi-desert
Namib	Namibia coast	Fog; among the oldest
Atacama	N Chile (into Peru)	Among the driest on Earth
Patagonia	Argentina	Cold rain-shadow. Not Chile's Atacama
Thar	India–Pakistan	Pleistocene–recent sands
Dasht-e-Lut / Kavir	Iran	Hot interior; Lut = extreme surface-temperature renown (NASA Aqua note)
Great Basin	USA	Cold interior
Rub' al-Khali	Arabian Peninsula	Largest contiguous sand sea
An Nafud	Saudi Arabia	

N.7 Grasslands

Local name	Where	Climate	Economy tag
Prairie	USA–Canada (Great Plains)	Temperate	Wheat / maize / cattle; chernozem-type soils
Pampas	Argentina (+ Uruguay)	Temperate	Cattle / wheat
Veld (High / Middle / Low)	South Africa	Temperate–subtropical	Maize, gold belt overlay
Downs	Australia	Temperate	Sheep / wheat
Canterbury Downs	New Zealand	Temperate	Sheep

Local name	Where	Climate	Economy tag
Steppe	Ukraine–S Russia–Kazakhstan–Mongolia	Semi-arid BS	Wheat / nomadism; classic chernozem
Puszta	Hungary	Temperate	Hungarian Alföld fact
Campos	Brazil (plateau)	Tropical / temperate mix	Cattle. Not Pampas
Llanos	Venezuela–Colombia (Orinoco)	Tropical	Cattle
Savanna / Sudan type	Africa (also N Australia, Brazilian edges)	Tropical Aw	Land of Big Games
Sahel	S of Sahara	Semi-arid fringe	Not the same as savanna core
Gran Chaco	Paraguay–N Argentina–Bolivia	Dry woodland/grass	Not Pampas

N.8 Vegetation (world belts)

Belt	Where	Plant picture
Equatorial rainforest (Af)	Amazon, Congo, Malaysia–Indonesia	Multilayer, epiphytes , hardwood (mahogany, ebony). Brazil name = Selva
Tropical monsoon	India, mainland SE Asia, N Australia	Teak, sal; sheds in dry season
Savanna	Sudan belt, Campos, Llanos, N Australia	Grass + scattered trees; baobab
Hot desert / thorn	Sahara, Thar, Australia	Acacia , cactus, xerophytes
Mediterranean sclerophyll	5 west-coast patches (N.9)	Olive, vine, cork, cedar ; shrubs below
China type / laurel	E coasts ~25–40°	Camphor, tea in hills; summer rain forest
Temperate deciduous	W Europe, E USA, N China	Oak, beech, maple
Taiga / boreal	Canada, Scandinavia, Siberia	Conifer — spruce, fir, pine, larch. Not treeless
Tundra	Arctic fringe, high Andes/Tibet alpine	Treeless — moss, lichen, dwarf shrub. Cryophyte steppe = Arctic tundra (2024)
Ice	Greenland, Antarctica	None

Name	Region
Maquis	Mediterranean Sea coast
Fynbos (paper: Fynbus)	South Africa (Cape)
Chaparral	California
Matorral	Chile

Plant	Region
Epiphytes	Equatorial
Acacia	Sahara / thorn
Baobab	Savanna
Cedars	Mediterranean

Brazil pair	Meaning
Selva	Equatorial rainforest

Brazil pair	Meaning
Terra Roxa	Fertile dark basaltic coffee soil — not a forest name. Parallel to Deccan fertility, not the same as Indian regur (रेगुर) as a name

N.9 World Climatic Regions

Type (Köppen (कोपेन))	Where	Rain timing	Vegetation / crop
Equatorial (Af)	Amazon, Congo, Malaysia	Year-round convection	Selva; not a citrus belt
Savanna / Sudan (Aw)	Africa belt, Campos, Llanos, N Australia	Summer wet, winter dry	Grass + trees; baobab; Land of Big Games
Tropical monsoon	India, mainland SE Asia	Summer monsoon	Deciduous teak/sal
Hot desert (BWh)	Sahara, Arabia, Thar, Australia	Rare	Xerophytes
Cold desert (BWk)	Gobi, Taklamakan	Rare / snow	Sparse
Steppe (BS)	Ukraine to Mongolia; rain-shadow edges	Scanty	Short grass
Mediterranean (Cs)	30–45° west coasts — five patches below	Winter cyclonic	Olive, citrus, vine, cork, wheat, cedar, maquis
China / warm-temperate east coast	E China, SE USA, NSW, Natal	Summer rain	Laurel / tea hills
British / W European (Cfb)	Britain, NW France, W Norway	Year-round cyclonic	Deciduous oak-beech
Taiga / boreal	50–70°N	Modest, summer max	Conifers
Tundra (ET)	Arctic coast	Low	Moss/lichen; permafrost (स्थायी हिम मृदा)
Ice cap (EF)	Greenland interior, Antarctica	—	None

Common Traps — Don't Fall For These

- Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
- Pampas = Brazil** — FALSE. **Argentina**. Brazil = **Campos**.
- Gibson = Brazil** — FALSE. **Australia**.
- Gobi in Kazakhstan / Russia** — FALSE for the 2025 code. **Mongolia + China**.
- Sahara smaller than Gobi** — FALSE. Sahara is the giant.
- Borneo = volcanic island** — FALSE. **Sunda shelf**; three countries.
- Maquis = California** — FALSE. **Chaparral**. Maquis = Med basin.
- Fynbos = Chile** — FALSE. **South Africa**. Chile = **Matorral**.
- Land of Big Games = Sahara** — FALSE. **Tropical savanna**.
- Mediterranean rain in summer** — FALSE. **Winter**.
- Taiga = treeless** — FALSE. That's **tundra**.
- Selva = Brazilian soil** — FALSE. Selva = **forest**; soil = **Terra Roxa**.
- Pyrenees = Italy** — FALSE. **Spain–France**. Italy = **Apennine**.
- Toubkal = Algeria** — FALSE. **Morocco**. Hoggar = Algeria.
- Etna = Pacific** — FALSE. **Italy / Mediterranean**.
- Australia = largest island** — FALSE. Australia is a **continent**; largest island = **Greenland**.
- Hot deserts only on east coasts** — FALSE. Classic set = **west** coasts + cold currents + Horse Latitudes.

18. **Pamir = highest large plateau** — FALSE. That is **Tibet**; Pamir = **Roof of the world** phrase.
19. **Chernozem = equatorial laterite** — FALSE. Chernozem = temperate grassland / steppe–prairie.
20. **Skip mountain families because papers asked country matches** — FALSE. Fold / block / residual can all appear.
21. **Atlas = France/Spain** — FALSE. That is **Pyrenees**. Atlas peak Toubkal = **Morocco**.
22. **Pamir = highest large plateau** — FALSE. Phrase is Pamir; **Tibet** is the highest large plateau.
23. **Colorado Plateau = lava** — FALSE. **Columbia** is lava; Colorado holds the **Grand Canyon**.
24. **Aconcagua = Chile** — FALSE. **Argentina**. Chimborazo = Ecuador.
25. **Africa = fewest countries / Australia = largest continent** — FALSE. Africa has the **most countries**; Asia is largest by area; Australia is **smallest**.
26. **Guyana = Africa** — FALSE. **South America**.
27. **Highest mean elevation = Asia** — FALSE. **Antarctica**.
28. **Sierra Nevada = young fold like Alps** — FALSE. **Block** mountain.
29. **Telegraphic Plateau = Indian Ocean** — FALSE. **North Atlantic Ridge**.
30. **Death Valley = cold Siberia** — FALSE. Hot **rift** in **California**.
31. **Thar = emptiest desert** — FALSE. **Most densely populated** desert.
32. **Europe has the worst desertification** — FALSE. **Least** among continents in GC framing.
33. **Downs = tropical grassland** — FALSE. Downs are **temperate** (Australia).
34. **Veld = Australia** — FALSE. Veld = **South Africa**; Australia = Downs.
35. **Selvas = grassland** — FALSE. Selvas = Amazon **tropical rainforest**.
36. **Savanna forest check = burrowers + soil only** — FALSE. Lead trio = **fire + grazers + seasonal rain**.
37. **Falklands = Indian Ocean** — FALSE. **South Atlantic**.
38. **Galápagos = Peru** — FALSE. **Ecuador** (volcanic Pacific).
39. **Socotra = Oman** — FALSE. **Yemen** (UNESCO 2008).
40. **Japan N→S = Hokkaido–Kyushu–Honshu** — FALSE. **Hokkaido** → **Honshu** → **Shikoku** → **Kyushu**.
41. **Borneo = volcanic dump like Hawaii** — FALSE for standard/UPPCS. **Sunda shelf**; three countries.
42. **Tinian = Polynesia** — FALSE. Marianas / Micronesia set.

Topic 19 — World Regional Geography

Current Affairs (this topic)

None mandatory — static political-map chapter.

Confused Pairs

Pair	Correct	Trap	Hindi
Nobi / Kanto	Japan	Korea	नोबी / कांतो
Esperanto	Artificial world language	Mountain / seaport	एस्पेरान्तो
Bahasa	Indonesia	Thailand	बहासा
Tamil abroad	Singapore official set	Myanmar major	तमिल
Sunni Triangle	Baghdad, Tikrit, Ramadi	Basra (Shia south)	सुन्नी त्रिभुज
Pontic vs Zagros	Pontic = N Turkey ; Zagros = W Iran	Swapped on W→E list	पोंटिक / ज़ाग्रोस
Kara Kum vs Kyzylkum	Kara Kum = Turkmenistan	Kazakhstan / Tajikistan	काराकुम
Tashkent vs Ashgabat	Tashkent = Uzbekistan ; Ashgabat = Turkmenistan	Swapped stans	ताशकंद
Bishkek vs Dushanbe	Bishkek = Kyrgyzstan ; Dushanbe = Tajikistan	Swapped	बिश्केक
Darling vs Great Dividing	Darling = SW Australia; GDR = east	Darling on east coast	डार्लिंग रेंज
N Australia climate	Tropical	Temperate (2025 R)	उत्तरी ऑस्ट्रेलिया
Cape Verde capital	Praia	Bamako (Mali)	केप वर्डे
Madeira	NE Atlantic (Portugal)	Caribbean	मदेरा
Maitri / Bharati	Both operational Antarctica	Dakshin Gangotri still open	मैत्री / भारती
W Europe rain	All months (westerlies)	Mediterranean winter-only	पश्चिमी यूरोप
Philippines cane/coconut	Spanish and Americans	British / Dutch	फिलीपींस
Palestine vs Israel map	West Bank + Gaza as units	"No such region on map"	फिलिस्तीन
Seoul vs Pyongyang	Seoul = South ; Pyongyang = North	Swapped	सियोल / प्योंगयांग
Sumatra vs Java	Sumatra is west of Java	Java first on W→E list	सुमात्रा / जावा
Bali vs Lombok	Bali west of Lombok (Wallace Line between them)	Lombok before Bali	बाली / लॉम्बोक
Mekong delta	S Vietnam	Cambodia	मेकांग डेल्टा
Bolivia vs Uruguay	Bolivia = landlocked ; Uruguay has Atlantic	Uruguay landlocked	बोलीविया
Igarka	Russia (Yenisei)	China	इगारका
Casablanca vs Rabat	Rabat = capital	Casablanca	रबात
UK vs Great Britain	UK = GB + N Ireland ; GB = Eng+Scot+Wales	Ireland in UK	यूके
Ethiopia vs Eritrea	Ethiopia landlocked ; Eritrea has Red Sea	Ethiopia still coastal	इथियोपिया
Bolivia vs Paraguay	Both landlocked S America	Only one of them	पराग्वे
NZ vs Tasmania	NZ = separate country, capital Wellington	NZ is an Australian state	न्यूज़ीलैंड
Continent size	Asia largest; Australia / Oceania smallest inhabited	Europe largest	महाद्वीप
Double landlocked	Uzbekistan / Liechtenstein	Any coastal state	दोहरा स्थलरुद्ध
38th vs 49th parallel	Korea DMZ vs Canada–USA	Swapped	38° / 49°
Capital ≠ mega-city	Ankara / Canberra / Brasília / Abu Dhabi	Istanbul / Sydney / Rio / Dubai	राजधानी

N.1 Asia (overview + East + South)

East Asia	Capital	Physical fact
China	Beijing	Tibet; Hwang Ho / Yangtze; Manchurian plain NE; Tropic through south; Shanghai on Yangtze mouth
Japan	Tokyo	Islands Hokkaido–Honshu–Kyushu–Shikoku (N→S). Kanto is the Tokyo plain. Nobi is the Nagoya plain. These are not in Korea (2021)
Mongolia	Ulaanbaatar	Landlocked; Gobi to the south
Taiwan	Taipei	Off Fujian; not Caribbean
North / South Korea	See N.5	Peninsula — not the Japanese plains

South Asia (India's neighbourhood as a *region*)

Country	Capital	Fact
India	New Delhi	Peninsula; Tropic
Pakistan	Islamabad	Indus (सिंधु); Karachi ≠ capital
Nepal	Kathmandu	Landlocked ; Himalaya (हिमालय)
Bhutan	Thimphu	Landlocked
Bangladesh	Dhaka	Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र) delta
Sri Lanka	Sri Jayawardenepura Kotte / Colombo	Island; Palk Strait
Maldives	Malé (މާލެ)	Atolls; equator-near
Afghanistan	Kabul	Landlocked hinge to West/Central Asia; Hindu Kush

N.2 West Asia

Order	Range	Where
1	Pontic	North Turkey , Black Sea coast
2	Zagros	Western Iran (Iraq border)
3	Hindu Kush	Afghanistan
4	Karakoram (काराकोरम)	Pakistan–China

Country	Capital	Extra
Turkey	Ankara	Pontic + Taurus; Istanbul is not the capital
Iran	Tehran	Zagros; Elburz / Damavand
Iraq	Baghdad	Tigris; Basra on the Shatt
Saudi Arabia	Riyadh	Rub' al-Khali; Mecca / Jeddah west
Syria	Damascus	
Lebanon	Beirut	
Jordan	Amman	East of the Jordan R. / Dead Sea
Yemen	Sanaa	Bab el-Mandeb
Oman	Muscat	
UAE	Abu Dhabi	Dubai is not the capital

Country	Capital	Extra
Qatar	Doha	
Kuwait	Kuwait City	
Bahrain	Manama	
Israel	Jerusalem (claimed)	Mediterranean west

N.3 Central Asia

Country	Capital	Desert / extra
Uzbekistan	Tashkent	Aral / Amu–Syr irrigation
Tajikistan	Dushanbe	Pamir knot
Kyrgyzstan	Bishkek	Tien Shan
Turkmenistan	Ashgabat	Kara Kum / Karakum desert (2022)
Kazakhstan	Astana (Nur-Sultan was the 2019–22 name)	Steppe (स्टेपी); not Kara Kum

N.4 South-East Asia

Country	Capital	Fact
Indonesia	Jakarta	Thousands of islands; equator. Port–country pair Jakarta–Indonesia is correct (2023)
Malaysia	Kuala Lumpur	Peninsula + Sabah/Sarawak on Borneo
Singapore	Singapore	City-state; Malacca approach
Brunei	Bandar Seri Begawan	North Borneo
Philippines	Manila	Luzon / Visayas / Mindanao; Spanish then US
Thailand	Bangkok	Never colonised; Chao Phraya
Myanmar	Naypyidaw (Yangon = old/port)	Pegu Yoma; Bay of Bengal
Laos	Vientiane	Landlocked ; Mekong
Cambodia	Phnom Penh	Mekong passes through — delta is not here
Vietnam	Hanoi (Ho Chi Minh City = south)	Mekong delta = southern Vietnam (2021)
Timor-Leste	Dili	East of Indonesia’s Timor

N.6 Palestine

Piece	Fact
Gaza Strip	Mediterranean coast; borders Egypt + Israel
West Bank	West of Jordan River; borders Jordan + Israel
Jordan River	Sea of Galilee → Dead Sea (no ocean outlet)
Dead Sea	Israel–Jordan; lowest land, saline (लवणीय)
Jerusalem	Claimed capital; Old City holy sites
Ramallah	West Bank administrative centre often in news

Extra map piece	Fact
Golan Heights	NE; borders Syria

Extra map piece	Fact
Sinai	Egyptian peninsula; Suez Canal on its west; Gaza at its NE corner
Occupied / disputed labels	Learn where Gaza, West Bank, Golan sit — do not write a politics essay

N.7 Europe

Belt	Who	Note
Western Europe climate	Britain, NW France, Low Countries	Westerlies + rain in all months (2024 Both)
Mediterranean Europe	Spain, S France, Italy, Greece	Winter rain — not the “all months” fact
British Isles	UK + Ireland	The UK is Great Britain plus Northern Ireland . Great Britain is England, Scotland and Wales. The Republic of Ireland is not in the UK. Capital London
Iberia	Spain, Portugal	Iberian Peninsula
Benelux	Belgium, Netherlands, Luxembourg	Brussels, Amsterdam (government also The Hague), Luxembourg. Rotterdam is the Netherlands port (2023 correct pair)
Nordic / Scandinavia	Norway, Sweden, Denmark; Finland + Iceland often bundled	Oslo, Stockholm, Copenhagen, Helsinki, Reykjavik
Baltic states	Estonia, Latvia, Lithuania	Tallinn, Riga, Vilnius
Alps	Switzerland, Austria, parts of FR/IT/DE	Bern, Vienna
Balkans	Greece, Bulgaria, Serbia, Croatia, Albania, N Macedonia, Bosnia, Montenegro, Kosovo, Romania/Slovenia fringe	Bulgaria holds the Balkan range name
East-Central	Poland, Hungary, Czechia, Slovakia, Ukraine	Warsaw, Budapest, Prague, Kyiv
Russia	Moscow	Europe + Asia; Igarka (Yenisei) is in Russia , not China

Port–country	Right?
Rotterdam – Netherlands	Yes
Montevideo – Uruguay	Yes
Jakarta – Indonesia	Yes
Igarka – China	NO — Igarka is Russia (Yenisei, Siberia)

N.8 Africa

Belt	Countries / fact
Maghreb	Morocco, Algeria, Tunisia, Libya (+ Western Sahara disputed)
Sahel	Senegal–Mauritania–Mali–Burkina–Niger–Chad–Sudan fringe south of Sahara
Horn	Ethiopia, Somalia, Eritrea, Djibouti . Ethiopia became landlocked after Eritrea (1993) took the Red Sea
West Africa	Nigeria (Abuja, not Lagos as capital), Ghana (ਝੜ) (Accra), Senegal (Dakar), Mali (Bamako)
Central	DRC (Kinshasa), Congo-Brazzaville, Cameroon, CAR
East	Kenya (Nairobi), Tanzania (Dodoma; Dar es Salaam = port), Uganda (Kampala, landlocked)

Belt	Countries / fact
Southern	South Africa (Cape Town / Pretoria / Bloemfontein — three capitals), Namibia, Botswana (landlocked), Zimbabwe (landlocked), Zambia (landlocked), Mozambique (coast)
Islands	Madagascar (4th largest island), Cape Verde (Praia), Mauritius, Seychelles, Comoros. Madeira / Canary = Atlantic, not “African Caribbean”

Country	Capital
Morocco	Rabat (not Casablanca)
Tunisia	Tunis
Algeria	Algiers
Libya	Tripoli
Cape Verde (Cabo Verde)	Praia
Mali	Bamako

N.9 North America

Country	Capital	Physical spine
Canada	Ottawa (not Toronto/Vancouver)	Shield, Prairies, Rockies, Arctic islands; Baffin ; 49th parallel with USA
USA	Washington DC (not New York)	Rockies west, Appalachians east, Great Plains, Mississippi, Great Lakes. Alaska (Arctic/Pacific) + Hawaii (Pacific islands) are states
Mexico	Mexico City	Plateau; Yucatan; Gulf of Mexico
Greenland	Nuuk	Denmark ; largest island , not a US state

N.10 South America

Country	Capital	Coast?	Regional fact
Brazil	Brasília (not Rio / São Paulo)	Atlantic	Selva + Terra Roxa (2025 both). Amazon
Argentina	Buenos Aires	Atlantic	Pampas; Aconcagua
Chile	Santiago	Pacific	Atacama; not landlocked
Peru	Lima	Pacific	Andes; Titicaca with Bolivia. Not landlocked (2022 trap)
Ecuador	Quito	Pacific	Equator; Galápagos
Colombia	Bogotá	Caribbean and Pacific	
Venezuela	Caracas	Caribbean	Llanos / Orinoco
Bolivia	Sucre (const.) / La Paz (seat)	Landlocked	2022 answer. Lost Pacific in 19th c.
Paraguay	Asunción	Landlocked	The other S American landlocked state
Uruguay	Montevideo	Atlantic	Port pair 2023 correct. Not landlocked
Guyana	Georgetown	Atlantic	Former British Guiana
Suriname	Paramaribo	Atlantic	Not landlocked (2022 trap)
French Guiana	Cayenne	Atlantic	France overseas — often in “Guianas”

N.11 Australia

Coast	What sits there
South-west	Darling Range (Perth / WA) — 2020
East	Great Dividing Range ; Sydney, Brisbane
North-east	Great Barrier Reef; tropical Queensland
North	Tropical (monsoon / savanna (सवाना)) — not temperate
South	Melbourne, Adelaide; more temperate
Tasmania	South of Bass Strait

Extra	Fact
Outback	Dry interior
Great Artesian Basin	Groundwater under the east-central plains
Murray–Darling	SE basin; not the Darling Range
Uluru / Ayers Rock	NT
Tropic of Capricorn	Cuts WA–NT–QLD

New Zealand (Oceania pair — always asked with Australia)

Note	Note
Capital	Wellington (not Auckland)
Islands	North Island + South Island; Cook Strait
Alps	Southern Alps on South Island
Not	An Australian state. Tasmania is the Australian island

N.12 Antarctica

Station	Fact
Dakshin Gangotri	First Indian (1983/84). Not operational now (supply base / abandoned as permanent station)
Maitri	1988/89, Schirmacher Oasis — operational
Bharati	2012, Larsemann Hills — operational

N.13 Caribbean

Group	Islands
Greater Antilles	Cuba, Jamaica, Hispaniola (Haiti + Dominican Republic), Puerto Rico
Lesser Antilles (names)	Grenada, Montserrat, Anguilla , Barbados, Trinidad (S edge), Antigua
NOT Caribbean	Madeira (Portugal, NE Atlantic off Morocco), Canary Islands (Spain, Atlantic), Azores, Cape Verde , Bermuda (N Atlantic)
Island	Capital / extra
-----	-----
Cuba	Havana; largest Caribbean island
Jamaica	Kingston

Group	Islands
Haiti / Dominican Rep.	Port-au-Prince / Santo Domingo — one island, two countries
Puerto Rico	US territory
Trinidad and Tobago	Port of Spain; near Venezuela
Bahamas	Atlantic archipelago — often not the “Caribbean Sea ” MCQ

N.14 Landlocked · Equator · Capitals · Boundaries (map diet)

Line	Note
38th parallel	Approx. Korea DMZ / North–South Korea
49th parallel	Much of Canada–USA border
Durand Line	Afghanistan–Pakistan
Radcliffe Line	India–Pakistan (1947)
McMahon Line	India–China (eastern sector)
Maginot / Siegfried	Historic France–Germany fortification names

Country	Capital	Famous non-capital
Australia	Canberra	Sydney
Brazil	Brasília	Rio / São Paulo
Turkey	Ankara	Istanbul
Nigeria	Abuja	Lagos
Tanzania	Dodoma	Dar es Salaam
South Africa	Pretoria / Cape Town / Bloemfontein	Johannesburg
USA	Washington DC	New York
Canada	Ottawa	Toronto
Morocco	Rabat	Casablanca
UAE	Abu Dhabi	Dubai
New Zealand	Wellington	Auckland
Pakistan	Islamabad	Karachi
Myanmar	Naypyidaw	Yangon
Bolivia	Sucre / La Paz	—
Netherlands	Amsterdam	Rotterdam (port); The Hague = government/ICJ

N.15 World languages

Note	Fact
Largest by L-1	Often Mandarin Chinese
Largest by total (L-1 + L-2) in recent Ethnologue frames	Often English
Hindi	Huge L-1 + L-2; classic “top world languages” set
Spanish	Official across much of Latin America

Country / place	Language
Singapore	Official set includes Malay (national), English , Mandarin , Tamil
Chile / Colombia / Cuba	Spanish official
Republic of Congo	Not Spanish (French / local)
Myanmar	Burmese (not Tamil as major)
Indonesia	Bahasa Indonesia (Bahasa ≠ Thailand)
Denmark / Netherlands / China	Danish / Dutch / Mandarin
Pakistan	Urdu + English official; Punjabi widely spoken but not the sole “official = Punjabi” story
Bangladesh	Stronger ethnic/linguistic homogeneity than Pakistan in classic A/R stems

Common Traps — Don’t Fall For These

1. **Nobi/Kanto = Korea** — FALSE. **Japan**.
2. **Basra in the Sunni Triangle** — FALSE. **South Iraq**.
3. **Northern Australia = temperate** — FALSE. **Tropical**. Interior dry is still true.
4. **Darling Range = east coast** — FALSE. **South-west**. East = Great Dividing Range.
5. **Kara Kum = Kazakhstan** — FALSE. **Turkmenistan**.
6. **Bishkek = Kazakhstan** — FALSE. **Kyrgyzstan**.
7. **Cape Verde = Bamako** — FALSE. **Praia**. Bamako = Mali.
8. **Madeira = Caribbean** — FALSE. **Atlantic / Portugal**.
9. **Dakshin Gangotri still operational** — FALSE. **Maitri + Bharati**.
10. **W Europe = Mediterranean winter rain only** — FALSE. Rain in **all months**.
11. **Philippines plantations = Dutch** — FALSE. **Spanish and Americans**.
12. **Istanbul = capital of Turkey** — FALSE. **Ankara**.
13. **Borneo = volcanic + two countries** — FALSE. **Three countries**, continental shelf.
14. **Iraq borders Israel** — FALSE.
15. **Greenland = US state** — FALSE. **Denmark**.
16. **Java is west of Sumatra** — FALSE. **Sumatra → Java → Bali → Lombok**.
17. **Mekong delta = Cambodia** — FALSE. **Southern Vietnam**.
18. **Uruguay / Peru / Suriname = landlocked** — FALSE. **Bolivia** (and Paraguay).
19. **Igarka = China** — FALSE. **Russia**.
20. **Ethiopia has a Red Sea coast** — FALSE since Eritrea. **Landlocked**.
21. **Wellington is in Australia** — FALSE. **New Zealand**. Canberra = Australia.
22. **Equator through India** — FALSE.
23. **Paraguay has a seacoast** — FALSE. Landlocked with Bolivia.
24. **Casablanca / Istanbul / Sydney / Rio / Auckland = capitals** — FALSE. Rabat / Ankara / Canberra / Brasília / Wellington.
25. **Europe is the largest continent** — FALSE. **Asia** is largest; Australia/Oceania is smallest inhabited.
26. **Double-landlocked = Bolivia** — FALSE. Bolivia is landlocked; **Uzbekistan / Liechtenstein** are the double-landlocked pair.
27. **49th parallel = Korea** — FALSE. Korea ≈ **38th**; 49th ≈ Canada–USA.
28. **Skip capitals map because recent papers asked Australia A/R** — FALSE. Next year can ask any Lucent capital or landlocked trap.

Topic 20 — World Agriculture

Current Affairs (this topic)

None mandatory. If a paper quotes a **year** (e.g. 2016 coffee), freeze **that** ranking. Do not chase every FAO revision in the teaching home.

Confused Pairs

Pair	Correct	Trap	Hindi
Plantation vs food grain	Tea (also coffee, rubber, cocoa, cane, oil palm, banana)	Wheat / rice / maize as "plantation"	बागान
Producer vs exporter (rice/wheat)	China often #1 volume ; not the classic exporter	China = wheat/rice export king	उत्पादक / निर्यातक
Brazil vs Vietnam coffee	Brazil #1 ; Vietnam #2 (2016 paper)	Vietnam first	ब्राज़ील / वियतनाम
Arabica vs Robusta	Brazil/Colombia highland = arabica; Vietnam = robusta bulk	Swapped	अरेबिका / रोबस्टा
Cocoa vs coffee Africa	Cocoa = Côte d'Ivoire + Ghana ; coffee East Africa too (Ethiopia/Uganda)	Latvia as cocoa	कोकोआ
Kerala vs Assam	Rubber Kerala ; tea volume Assam	Assam rubber / Gujarat tea	केरल / असम
Kinta Valley	Tin (Malaysia)	Rubber / tea / coffee	किंटा घाटी
Amazon origin vs SE Asia crop	Rubber and cocoa originated in Amazon; bulk crop now elsewhere	"Origin = Malaysia / Ghana"	अमेज़न
Jhum vs plantation	Jhum = shifting subsistence	Jhum = tea estate	झूम
Ladang vs milpa	Ladang = Malaysia ; milpa = Mexico	Swapped	लदांग / मिल्पा
Roca vs chena	Roca = Brazil ; chena = Sri Lanka	Swapped	रोका / चेना
Cane vs beet	Cane = tropics (Brazil); beet = Europe	Beet in Kerala	गन्ना / चुकंदर
Citrus belt	Mediterranean	Equatorial / Himalayan valleys as <i>world fact</i>	नींबू वर्गीय
Oil palm vs rubber	Oil palm volume = Indonesia + Malaysia	Oil palm = Kerala only	तेल ताड़
Black tea vs green tea	Black = fermented ; green = not	Swapped	काली / हरी चाय
Philippines cane	Spanish and Americans	Dutch (Indonesia) / British	फिलीपींस
Intensive vs extensive	Intensive = small + labour + monsoon rice; extensive = huge + machines + Prairie wheat	Swapped	सघन / विस्तृत
Producer vs exporter (coffee/tea)	Brazil coffee #1 volume; Kenya often tea exporter story	Same word	
Soybean triangle	USA–Brazil–Argentina	Only India	सोया
Milk volume vs dairy export	India = milk volume ; NZ/Netherlands = export dairy	India = cheese ship king	दूध
Golden Crescent	AFG–Iran–Pakistan	Includes Iraq	गोल्डन क्रिसेंट
Fazenda vs Chena	Brazil plantation vs Sri Lanka shifting	Fazenda = jhum	फाज़ेंडा
Mocha vs Kona	Yemen vs Hawaii	Swapped	मोचा / कोना

Pair	Correct	Trap	Hindi
Olericulture	Vegetables	Call it horticulture-only	शाककृषि
Von Thünen vs Whittlesey	Thünen = distance rings from market; Whittlesey = world type map	Same theory	

N.1 Agriculture (world systems)

Split	Meaning	picture
Subsistence	Family food; little surplus	Wet-rice monsoon Asia; shifting tropics
Commercial	Surplus for market / export	Prairie wheat, plantation, dairy NZ
Intensive	High labour (or capital) per hectare ; small farms	Monsoon rice; Dutch dairy; Japanese rice
Extensive	Low labour per hectare; huge farms, machines	Canadian Prairie, Australian Downs, Pampas
Sedentary	Same fields year after year	Most of Eurasia
Nomadic / shifting	People or plots move	Sahara herders; Amazon/Congo jhum

System	Climate / where	What is grown / raised	Note
Mixed farming	W Europe, US Midwest	Crops + livestock on the same farm	Fodder rotation; manure; not "only wheat"
Dairy farming	NW Europe, Great Lakes, NZ, Denmark, Netherlands	Milk, butter, cheese	Mild climate + fodder + urban (नगरीय) market; New Zealand is classic grass-export dairy
Mediterranean	30–45° west coasts (five patches)	Citrus, vine, olive , winter wheat, vegetables	Winter rain + dry bright summer (2020 citrus answer C)
Livestock ranching	Pampas, W USA Prairie, Australian Downs, S African Veld	Cattle / sheep, commercial	Not nomadic herding (that is subsistence)
Nomadic herding	Sahara–Arabia–C Asia–tundra (ढुँढ़ा)	Camel, sheep, yak, reindeer	Moves with pasture; low density
Truck / market gardening	Urban fringes	Vegetables, fruit, flowers	High value, perishable, near city
Horticulture	Med + hill + greenhouse	Fruit, veg, flowers	Viticulture means grape / wine
Plantation	Humid tropics	Tea, coffee, cocoa, rubber, cane, oil palm, banana, coconut, spices	Humid-tropic estate monoculture
Shifting / slash-and-burn	Humid tropics	Millet, maize, cassava, upland rice	See shifting-cultivation card below
Collective / cooperative	Former USSR kolkhoz; Israel kibbutz; Denmark dairy co-ops	Shared marketing / land	Secondary example

Term	Recall
Viticulture	Grapes (wine) — France, Italy, Spain, California, Chile, Australia, Cape
Sericulture	Silk — China volume; India mulberry + tussar/eri/muga
Pisciculture	Fish rearing
Apiculture	Bees / honey
Floriculture	Flowers (Netherlands auction)

N.2 Whittlesey agricultural regions

Type	Core belt
Nomadic herding	Sahara, Arabia, Gobi, tundra, C Asian drylands
Livestock ranching	Pampas, W USA, Australia, Veld
Shifting cultivation	Amazon, Congo, SE Asia hills, NE India, C America
Rudimentary sedentary tillage	Parts of tropics where plots are fixed but tools simple
Intensive subsistence — rice dominant	E / SE / S Asia deltas and coasts
Intensive subsistence — without paddy rice	N China, interior India wheat-millet, Nile
Commercial plantation	Caribbean, NE India-SL tea, Malaysia-Indonesia rubber/palm, Brazil coffee, W Africa cocoa
Mediterranean agriculture	Five west-coast patches (Med proper, California, C Chile, Cape, SW Australia)
Commercial grain farming	Prairies, Pampas, Downs, Ukrainian-Russian steppe
Mixed crop-and-livestock	W Europe, US Midwest Corn Belt edge
Subsistence crop-and-livestock	Parts of highlands / poorer temperate belts
Commercial dairy	NW Europe, Great Lakes, NZ
Specialised horticulture / market gardening	W Europe urban belt, California, Mediterranean coasts

Patch	Where
1	Mediterranean Sea basin (Spain, Italy, Greece, Levant (लेवेंट), Maghreb)
2	California
3	Central Chile
4	Cape of South Africa
5	SW Australia (Perth side)

N.3 World cereals

Wheat

Belt	Where
Prairies	Canada + USA
Pampas	Argentina
Downs	Australia
Steppe	Ukraine / S Russia / Kazakhstan
N European plain	France (export), Germany
Indo-Gangetic rabi	India (volume, mostly domestic)

N.4 Fibres, sugar, oilseeds

Sugarcane vs sugar beet

	Sugarcane	Sugar beet
Climate	Tropical; long season 10–18 months ; frost-free	Temperate; 4–7 months

	Sugarcane	Sugar beet
Rain / water	~75–150 cm or irrigation	Moderate (नरम दल) rain; Europe plains
World centre	Brazil (also ethanol from cane)	France, Germany, Russia, Ukraine, USA
Asia names	India, China, Thailand, Pakistan	Not a Kerala crop

Oilseeds and tree oils

Crop	Climate / belt	Usual leaders
Soybean	Warm temperate	USA, Brazil, Argentina (export triangle). China = big user
Groundnut	Tropical / subtropical	China, India, Nigeria, USA
Rapeseed / canola	Cool temperate	Canada, China, India, EU
Sunflower	Temperate	Russia, Ukraine, Argentina
Oil palm	Equatorial, high rain	Indonesia + Malaysia (volume). Nigeria origin story, not today's tonnage king
Olive	Mediterranean winter-rain	Spain, Italy, Greece, Tunisia

N.5 Plantation crops (the system)

Feature	Fact
Climate	Tropical / subtropical; no severe frost
Organisation	Estate (large), often colonial origin
Capital + labour	High capital; hired labour (not only family)
Crop	Monoculture for export
Processing	Factory (फैक्टरी) on or near the estate (tea factory, rubber smoke-house, sugar mill)
Classic list	Tea, coffee, cocoa, rubber, sugarcane, banana, oil palm, coconut, spices
Not plantation	Wheat, rice, maize as the <i>predominant</i> system

Crop	India	Do not write
Tea	Assam (volume), WB Darjeeling, Nilgiri (नीलगिरि)	Gujarat — Tea
Coffee	Karnataka > Kerala > Tamil Nadu (shade, Western Ghats (पश्चिमी घाट))	Assam as coffee king
Rubber	Kerala (Kottayam–Ernakulam)	Assam — Wheat; UP — Jute
Spices	Kerala pepper (काली मिर्च) + cardamom	
Coconut	Kerala–TN–Karnataka coast	Philippines is the <i>world</i> Spanish–US story
Banana	India = huge producer ; Ecuador = classic exporter	India = world's banana-ship king

Place	Crop	Credit / trap
Philippines	Coconut + sugarcane	Spanish and Americans (2023) — not Dutch
Indonesia / Java	Cane, coffee, rubber	Dutch (Hollanders)
Malaya	Rubber estates	British
NE India / Ceylon	Tea	British
West Africa	Cocoa	Shifted from Amazon after colonial planting
Brazil	Coffee	Portuguese / highland Terra Roxa

N.6 Tea

Type	Processing	Where the puts it
Black	Fully fermented	India, Sri Lanka, Kenya
Green	Not fermented	China, Japan
Oolong	Semi-fermented	Taiwan / SE China
CTC	Crush–tear–curl (granular, tea-bag)	Indian plains/Assam bulk
Orthodox	Whole-leaf	Darjeeling, Ceylon high-grown
Region	Fact	
-----	-----	
China	Usual largest producer ; original home; green tea	
India	Assam (Brahmaputra valley — volume), Darjeeling (GI, hill, orthodox), Dooars/Terai (तराई), Nilgiri (नीलगिरि), Kangra, Munnar	
Sri Lanka	" Ceylon tea"; central highlands; high / mid / low grown	
Kenya	Leading African name; often among top exporters ; highland	
Indonesia, Vietnam, Bangladesh, Turkey	Also revise	

N.7 Coffee

Type	Character	Where
Arabica (<i>C. arabica</i>)	Milder aroma; higher hills (~600–1600 m)	Brazilian highlands, Colombia , Ethiopia, India's Ghats
Robusta (<i>C. canephora</i>)	Stronger, more caffeine, hardier, lower altitude	Vietnam , Uganda, lowland tropics
Liberica	Secondary example	W Africa pockets

Country	Note
Brazil	Usual #1 . São Paulo–Minas–Paraná highlands. Terra Roxa soil. Mostly arabica
Vietnam	#2 in the 2016 fact. Robusta bulk. Central Highlands. Trap = putting it first
Colombia	Mild arabica ; third in 2016 four (चातुर्याम)
Indonesia	Sumatra / Java / Sulawesi; fourth in 2016 four
Ethiopia	Origin story of arabica — not the 2016 tonnage king
India	Karnataka (Kodagu , Chikmagalur , Hassan) > Kerala (Wayanad) > Tamil Nadu (Nilgiri). Shade often silver (रूपक) oak

N.8 Cocoa

Country	Fact
Côte d'Ivoire (Ivory Coast)	World's usual #1
Ghana (घन)	#2 belt; quality name
Cameroon	Major (2021 list)
Nigeria	West African belt
Indonesia, Ecuador, Brazil	New World / SE Asia

Country	Fact
Latvia	NOT a cocoa country (2021) — Baltic temperate forest , not equator

N.9 Rubber

Country	Fact
Thailand	Usual world leader in recent textbook years
Indonesia	Large
Vietnam	Fast growth
Malaysia	Classic estate region; not Kinta Valley
India	Kerala (Kottayam–Ernakulam); also TN, KA, Andamans, NE pockets
China	Growing consumer-producer
Brazil	Origin; wild/native story, not the estate king now

N.10 Other plantation / tropical cash crops

Crop	Climate	World centre	Trap
Oil palm	Equatorial, high rain	Indonesia + Malaysia	Latvia / Kerala-only
Banana	Hot, wet, frost-free	India = huge producer ; Ecuador (also Philippines, Costa Rica) = export	Equating producer with exporter
Coconut	Coastal tropics	Indonesia, Philippines, India	Philippines development = Spanish and Americans
Pepper (काली मिर्च)	Hot, wet vine	Vietnam / India / Indonesia (ranks move)	"Only Kerala in the world"
Cardamom	Humid hills	Guatemala + India (Kerala–Karnataka)	
Clove / nutmeg	Equatorial islands	Indonesia (Maluku / "Spice Islands")	
Tobacco	Warm, frost-careful	China, Brazil, India, USA	Not a slope-tea crop
Citrus	Med winter-rain + some subtropics	Spain, Italy, California, Brazil (orange juice)	World belt name = Mediterranean (2020)

N.11 Livestock, dairy, wool, silk

Item	Note
Cattle numbers	India often #1 inventory; then Brazil, China, USA
Beef export	Brazil, Australia, USA — not "India exports most beef because it has most cattle" as a blind fact
Dairy	India leads milk volume (domestic). New Zealand , the Netherlands and Denmark lead commercial export dairy. Do not treat India as the cheese-ship king
Sheep / wool	Australia (Merino), plus China and New Zealand. Wool is not cotton
Pigs	China
Ranching map	Pampas cattle, Australian Downs sheep, Veld, W USA
Silk	China volume; India second-story (mulberry + wild silks)

N.12 Shifting cultivation

Local name	Where
Jhum / jhoom	North-East India
Podu	Andhra / Odisha hills
Bewar / dahia	Madhya Pradesh
Penda	Chhattisgarh / Bastar
Kuruwa	Jharkhand
Kumari	Western Ghats / Kerala hills
Khil	Himalaya (हिमालय)
Valre / waltre	SE Rajasthan
Bringa / koman	Odisha pockets

Local name	Where
Ladang	Malaysia
Caingin	Philippines
Tamrai	Thailand
Chena	Sri Lanka
Humah	Java
Milpa	Mexico / Central America
Roca	Brazil
Masole	Congo basin
Ray	Vietnam
Fang	Equatorial Africa

N.13 World Green Revolution and FAO (short)

Note	Note
HYV package	Seeds + water + fertiliser + pesticide — same logic worldwide
Wheat	Norman Borlaug ; CIMMYT Mexico ; Nobel Peace (1970) — not “Nobel in Agriculture”
Rice	IRRI, Los Baños, Philippines
Spread	Mexico wheat into India/Pakistan; Philippines IR8 rice across Asia
FAO	Food and Agriculture Organization; HQ Rome ; World Food Day 16 October
Golden rice	Vitamin A (correct 2018 pair)

N.14 GC extras — cash, livestock, terms

Crop / item	Usual coaching leader / note
Rice + wheat volume	China often leads both
Rice area	India often has the largest harvested area
Hybrid rice	Especially associated with China

Crop / item	Usual coaching leader / note
IRRI / Rice Gene Bank	Los Baños, Philippines
Sugarcane	Brazil leads
Sugar beet	Russia and France lead beet sugar
Ukraine sugar	Almost entirely from beet
Natural rubber	Thailand (with Indonesia close behind)
Coffee	Brazil and Vietnam lead; Robusta is an African species also grown in India
Tea leaves	China leads
Tobacco	China leads
Silk	China leads
Bananas	India leads
Milk volume	India leads
Castor	India often leads
Coconut	Indonesia leads
Maize	USA leads
Livestock inventory	China often leads total headcount

Note	Fact
Mocha coffee	Yemen (port (बंदरगाह) name)
Kona coffee	Hawaii (USA)
Sacramento–San Joaquin	California — grapes / citrus
Sugar Bowl	Classic tag for Cuba
Fazenda	Brazil plantation (coffee) — not shifting cultivation
Taungya	Agroforestry (कृषि वानिकी) / forest–farm system in Myanmar
Alley cropping	Crops in alleys between tree rows
Golden Crescent	Illicit opium — Afghanistan, Iran, Pakistan (not Iraq)
Shahtoosh	Tibetan antelope / Chiru hair (Tibet / Ladakh (लद्दाख) story)
Catechu (kattha)	From Khair (<i>Acacia catechu</i>) wood
Groundnut native	Brazil / Mato Grosso; Gambia's classic peanut economy
Alfalfa / Lucerne	Fodder legume (Fabaceae)
Heroin	From opium

Term	Recall
Apiculture	Bees / honey
Viticulture	Grapes (viniculture = wine side)
Pomology	Fruits
Olericulture	Vegetables only
Horticulture	Fruits + vegetables + flowers
Green Box (AoA)	Allowed support — research, pest control, training, marketing

Common Traps — Don't Fall For These

1. **Wheat/rice/maize = plantation** — FALSE. **Tea**.
2. **Vietnam #1 coffee 2016** — FALSE. **Brazil**, then Vietnam.
3. **Latvia = cocoa** — FALSE. **Côte d'Ivoire / Ghana / Cameroon**.
4. **Kinta Valley = rubber** — FALSE. **Tin**.
5. **Gujarat = tea / Assam = wheat / UP = jute** — FALSE. **Kerala = rubber**.
6. **Jhum = tea estate** — FALSE. **Shifting subsistence**.
7. **Ladang = Mexico** — FALSE. **Malaysia**. Milpa = Mexico.
8. **Citrus = equatorial** — FALSE. **Mediterranean**.
9. **Philippines cane = Dutch** — FALSE. **Spanish and Americans**.
10. **Rubber originated in Kerala** — FALSE. **Amazon**; Kerala is India's producer.
11. **Cocoa originated in West Africa** — FALSE. **Amazon**; Africa is production.
12. **Colombia first in 2016 coffee four** — FALSE. **Third**.
13. **Oil palm = Latvia** — FALSE. **Indonesia + Malaysia**.
14. **Shifting = intensive Ganga rice** — FALSE.
15. **Brazil lost all coffee** — FALSE. Still the 2016 #1.
16. **China = rice/wheat export king** — FALSE. Volume **producer**; exporters often others.
17. **Borlaug Nobel = Agriculture** — FALSE. **Peace**.
18. **Green tea = fermented** — FALSE. **Black** is fermented.
19. **India = banana export king because it produces most** — FALSE. **Ecuador** ships; India eats.
20. **Ranching = nomadic herding** — FALSE. Ranching is **commercial**.
21. **Von Thünen = Whittlesey** — FALSE. Rings vs world-type map.
22. **India = dairy export king because milk volume #1** — FALSE. Volume ≠ NZ/Netherlands export story.
23. **Soy = only Asian crop belt** — FALSE. USA–Brazil–Argentina triangle.
24. **Skip systems because papers asked tea/coffee** — FALSE. Next year can ask any Lucent system or crop belt.

Topic 21 — World Minerals & Energy

Current Affairs (this topic)

None mandatory. Producer ranks and “leading country” freeze to the **paper’s year**. Do not chase every USGS revision in the teaching home.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q83

Which country is the leading producer of gold in the world in the year 2023?

A. China B. U.S.A. C. Canada D. South Africa

Confused Pairs

Pair	Correct	Trap	Hindi
Ruhr	Germany	Britain / France / Russia	रुहर
Kuzbass	Russia	Ukraine / Germany	कुजबास
Donetsk	Ukraine coal	Kyiv / Kharkiv as the coal answer	दोनेत्स्क
Appalachian	USA	England	अप्लेशियन
Lancashire	England	USA / Germany	लंकाशायर
Mount Newman	Iron , Australia	Mn / Cu / bauxite	माउंट न्यूमैन
Krivoy Rog	Ukraine iron	Russia	क्रिवोय रोग
Normandy iron	France	Germany (2020 wrong pair)	नॉर्मंडी
Karaganda	Kazakhstan (coal renown; 2020 iron list accepted)	Kuwait oil	कारागांदा
Chile copper	Andes porphyry , N Chile	“Chile has no copper”	चिली तांबा
Kinta Valley	Tin	Rubber / tea / coffee	किंटा
Pegu Yoma	Mineral oil	Tin (that is Kinta / Bangka / Tenasserim)	पेगू योमा
Kashagan	Kazakhstan	Kuwait	कशागन
Burgan	Kuwait	Kashagan dump	बुर्गान
Haft Kel	Iran	Iraq	हफ्त केल
Zubair	Iraq	Iran	जुबैर
Dhahran	Saudi Arabia	Kuwait	धहरान
Methane	Natural gas and CNG	LPG (propane/butane)	मीथेन
Nuclear vs solar	Nuclear not stored solar	Marking only wind as non-solar	नाभिकीय
Iodine	Chile	Japan / USA / China as the 2018 fact	आयोडीन
Postmasburg	Manganese , South Africa	Uranium / mica / bauxite	पोस्टमासबर्ग
Weipa vs Newman	Weipa = bauxite ; Newman = iron	Swap	वीपा / न्यूमैन
Witwatersrand	Gold , South Africa	Postmasburg Mn dump	विटवाटरसैंड
Sudbury	Canada Ni + Cu	Tin / oil	सडबरी

Pair	Correct	Trap	Hindi
North Field / South Pars	Qatar / Iran — same Gulf gas giant	Two unrelated fields	नॉर्थ फील्ड
OPEC vs IAEA	Both Vienna ; OPEC = oil, IAEA = nuclear	Swap jobs	वियना
Brass vs bronze	Brass = Cu+Zn; bronze = Cu+Sn	Swap	पीतल / कांसा
Morocco	Phosphate reserves	Oil / copper	मोरक्को
Broken Hill	Australia Pb–Zn	Iron / tin	ब्रोक्न हिल
Lithium Triangle	Chile–Argentina–Bolivia (brines)	Brazil in the triangle	लिथियम
Rare earths	China dominates processing	“India has zero REE”	दुर्लभ मृदा
German silver	Cu–Ni–Zn (no Ag)	Contains silver	जर्मन सिल्वर
Peace Pipeline	Iran–Pakistan gas	Russia–Europe only	पीस पाइपलाइन
Uranium City	Canada	Australia / USA	यूरेनियम सिटी
Kinta Valley	Tin (Malaysia)	Rubber	किंटा घाटी
Carajás vs Newman	Both iron ; Carajás = Brazil, Newman = Australia	Swap countries	काराजास
Producer vs exporter (coal/iron)	China often huge volume ; Australia/Indonesia ship	China = export king	

N.1 Coal

Type	Carbon	Character	Region
Peat	Lowest	First stage; high water	Ireland, Russia, Canada bogs
Lignite (brown coal)	Low	Soft; thermal only	Germany (Lusatia), Australia, Neyveli-type
Bituminous	Medium–high	Steam and coking	Ruhr, Appalachian, Damodar (दामोदर), Kuzbass
Anthracite	Highest	Hardest; cleanest burn	Pennsylvania (USA); South Wales

Role	Usual names
Volume miners	China (eats most of it), India, Indonesia, USA, Australia, Russia, South Africa
Exporters	Indonesia, Australia, Russia, USA, South Africa
Trap	China is not the classic export king — it is the volume consumer

Coalfield	Country	Note
Appalachian	USA	Eastern USA (not the Rockies as the name)
Pennsylvania anthracite	USA	Highest-rank US coal
Interior / Illinois	USA	Bituminous
Lancashire	England	Historic English coal–textile belt
Yorkshire / Northumberland / Midlands	England	Same island; do not place on the USA
South Wales	UK	anthracite
Ruhr	Germany	Not Britain / France / Russia
Saar	Germany–France border	Also revise
Lusatia / Rhineland	Germany	Lignite
Silesia (Upper)	Poland	

Coalfield	Country	Note
Kuzbass / Kuznetsk	Russia	Siberia
Pechora	Russia	Arctic
Kansk–Achinsk	Russia	Lignite / brown coal
Donets (Russian side)	Russia	Do not swap with Ukraine Donetsk in a “country” MCQ
Donetsk / Donbas	Ukraine	2023 largest among Kyiv / Kharkiv / Zaporizhzhia
Karaganda	Kazakhstan	Coal city; 2020 iron list also tagged it
Shanxi / Shaanxi / Inner Mongolia	China	Volume belt
Hunter Valley (NSW)	Australia	Export
Bowen (Queensland)	Australia	Export
Witbank / Highveld	South Africa	
Damodar — Jharia, Raniganj	India	Jharia = coking; Raniganj = West Bengal

N.2 Iron ore

Ore	Colour	Fe	Fact
Hematite	Red	~70%	Bulk ore of world trade (पण्माध्यक्ष)
Magnetite	Black	~72%	Highest grade ; magnetic
Limonite	Yellow-brown	~60%	Hydrated; lower grade
Siderite	Grey-brown	~40%	Iron carbonate; poorest of the four (चातुर्यामि)

Role	Usual names
Volume miners	Australia, Brazil, China , India, Russia
Exporters	Australia (Pilbara) and Brazil (Minas / Carajás)
Trap	China mines a lot and still imports . India is a miner, not the Pilbara-style ship king

Area	Country	Note
Mount Newman	Australia	2023. Pilbara / Hamersley
Hamersley Range	Australia	Same belt
Itabira	Brazil	Minas Gerais; old “iron quadrangle”
Carajás	Brazil	Giant Amazon-side mine
Cerro Bolívar	Venezuela	Also revise
Mesabi Range	USA	Lake Superior / Minnesota
Labrador Trough	Canada	Quebec–Labrador
Lorraine	France	Minette ore; not Germany
Normandy	France	Not Germany (2020 wrong pair)
Pyrenees	France	2020 accepted pair
Kiruna / Gällivare	Sweden	Magnetite; Arctic rail to Narvik
Bilbao / Basque	Spain	
Salzgitter / Siegerland	Germany	Real German iron — do not write Normandy
Cleveland / Scunthorpe	UK	Historical
Krivoy Rog / Kryvyi Rih	Ukraine	2020 correct pair. Not Russia

Area	Country	Note
Kursk Magnetic Anomaly	Russia	
Magnitogorsk / Urals	Russia	Steel town on the ore
Karaganda	Kazakhstan	Coal city; 2020 iron list treated as matched
Anshan / Wuhan	China	
Sishen	South Africa	
Liberia / Mauritania	W Africa	Classic export names
Odisha–Chhattisgarh–Jharkhand–Karnataka	India	Bailadila, Noamundi, Kudremukh — India-map detail

N.3 Copper

Place	Country	Note
Chuquibambilla	Chile	Open-pit giant; Atacama / N Chile
El Teniente	Chile	One of the world's largest underground Cu mines
Escondida	Chile	Modern volume name
Andes (Peru)	Peru	2nd Andes producer
Katanga (Shaba)	DRC	African copper-cobalt
Copperbelt	Zambia	Ndola–Kitwe belt
Bingham Canyon	USA	Utah
Butte	USA	Montana; historical
Sudbury	Canada	Nickel + copper together
Norilsk	Russia	Ni–Cu–PGE
Grasberg	Indonesia	Cu + gold (Papua)
China	China	Huge user ; also mines
Khetri / Malanjkhand / Singhbhum	India	India-map fact

N.4 Tin

Place	Country	Note
Kinta Valley	Malaysia	2020 PYQ. Rubber grows in Malaysia; Kinta still = tin
Bangka / Belitung	Indonesia	
Yunnan / Guangxi	China	Usual volume leader in modern lists
Potosí / Andes lodes	Bolivia	Highland tin
Andes	Peru	
Jos Plateau	Nigeria	Older textbook name
Cornwall	UK	Historical; worked out
Rondônia (राष्ट्रीय अन्वेषण एजेंसी)	Brazil	Also revise

N.5 Petroleum

Field / area	Country	Trap
Ghawar	Saudi Arabia	Often tagged world’s largest conventional field
Dhahran	Saudi Arabia	2024 correct pair
Abqaiq / Safaniya	Saudi Arabia	Safaniya = offshore
Ras Tanura	Saudi Arabia	Export terminal , not a field name swap
Burgan	Kuwait	Kuwait’s giant. Not Kashagan
Kirkuk / Mosul	Iraq	North Iraq
Rumaila / Zubair / Basra	Iraq	South Iraq. Zubair = 2024 correct
Haft Kel	Iran	2024 correct. Not Iraq
Masjid-i-Suleiman	Iran	First Middle-East commercial well (1908 story)
Agha Jari / Gachsaran	Iran	
Kashagan	Kazakhstan	Caspian offshore. Kuwait–Kashagan = WRONG (2024)
Tengiz	Kazakhstan	Onshore Caspian
Baku	Azerbaijan	Caspian; Nobel/old Russian oil story
West Siberia / Samotlor	Russia	Volume belt
Volga–Urals	Russia	Older Russian belt
North Sea	UK + Norway	Ekofisk (NO), Forties/Brent (UK)
Prudhoe Bay	USA (Alaska)	
East Texas / Permian / Gulf of Mexico	USA	
Maracaibo / Orinoco	Venezuela	Heavy oil / sands story
Cantarell	Mexico	
Campos / pre-salt	Brazil	
Hassi Messaoud	Algeria	
Bonny / Niger Delta (डेल्टा)	Nigeria	
Dukhan	Qatar	
Murban	UAE	
Daqing	China	
Minas	Indonesia	
Seria	Brunei	
Yenangyaung / Chauk	Myanmar	With Pegu Yoma oil (2022)
Mumbai High / Digboi / Ankleshwar	India	India-map fact

N.6 Natural gas

Role	Usual names
Reserves	Russia, Iran, Qatar, USA, Turkmenistan
Production	USA, Russia, Iran, China, Qatar, Australia, Norway
LNG export	Qatar, Australia, USA

Field / area	Country	Note
Urengoy / Yamburg / Medvezhye	Russia	West Siberia
North Field	Qatar	Same giant reservoir as Iran's South Pars
South Pars	Iran	Persian (फ़ारसी) Gulf; world's largest gas accumulation with Qatar
Groningen	Netherlands	Historic European giant
Troll / Ekofisk gas	Norway	North Sea
Hugoton / Marcellus shale	USA	
Hassi R'Mel	Algeria	
Galkynysh	Turkmenistan	
Shah Deniz	Azerbaijan	Caspian; pipelines to Turkey/EU
NW Shelf / Gorgon	Australia	LNG
North Sea	UK + Norway	Oil and gas
KG / Mumbai High gas	India	India-map fact

N.7 Energy resources

Project	River	Country
Three Gorges	Yangtze	China
Itaipu	Paraná	Brazil–Paraguay
Grand Coulee	Columbia	USA
Hoover	Colorado	USA
Aswan High	Nile	Egypt (मिस्र)
Guri	Caroní	Venezuela
Churchill Falls	Churchill	Canada
Sayano–Shushenskaya	Yenisei	Russia

Country	Fact
Iceland	Power + heating
New Zealand	Rotorua / Taupo
Philippines	Ring of Fire
USA	The Geysers, California
Italy	Larderello (first plant)

Body	HQ
OPEC	Vienna
IAEA	Vienna
IEA	Paris
IRENA	Abu Dhabi

N.8 Other world minerals

Place	Country
Postmasburg / Kalahari	South Africa (2020 — not U / mica / bauxite)
Moanda	Gabon
Groote Eylandt	Australia
Nikopol	Ukraine
Odisha / Maharashtra	India

Place	Country
Weipa / Gove / Darling Range	Australia
—	Guinea (reserves renown)
Jamaica / Suriname	Caribbean–Guiana
Trombetas	Brazil
Les Baux	France (name origin)
Odisha–Gujarat–Jharkhand	India

Place	Country
Witwatersrand	South Africa
Kalgoorlie	Australia
Yukon / Klondike	Canada
Homestake	USA (historical)
Grasberg	Indonesia (with copper)
—	China (modern volume)

Mineral	World centre
Sheet mica	India (Jharkhand–Andhra–Rajasthan) classic
Graphite	China; India; Sri Lanka (vein)
Sulphur	USA, Mexico, Poland; volcanic Italy/Japan
Platinum	Bushveld (South Africa); Norilsk
Cobalt	DRC (with Katanga copper)
Lithium	Australia hard-rock (spodumene); Lithium Triangle brines = Chile–Argentina–Bolivia (Brazil is not in the triangle)
Rare earths (REE)	China dominates mining and especially processing ; India has monazite sands but not China's refining fact
Tungsten	China ; Bolivia
Mercury	Almadén (Spain) historical
Asbestos	Russia, Kazakhstan; Canada historical

N.9 GC extras — silver German · oil · uranium city

Note	Fact
German silver	Cu–Ni–Zn alloy — no silver (name from appearance)
Silver volume	Often Mexico #1 in recent frames
Aluminium ore	Bauxite (≠ haematite / magnetite / limonite)

Note	Fact
Primary AI volume	Often China first in recent frames
Tin	Mostly placer ; Kinta Valley (Malaysia) = tin
Brent crude	Major crude class; sourced from North Sea ; low sulphur — not “zero sulphur”
Gasohol	~90% gasoline + ~10% ethanol; USA classic producer/consumer
Peace Pipeline	Iran–Pakistan gas pipeline name
First commercial oil (1857 register)	Romania (Drake 1859 Titusville is the other classic “first well” story)
Crude reserves	Often Venezuela #1 proved in IMYB frames; Saudi #2
SE Asia oil	Indonesia
Venezuela	Oil-dominated export economy
Ras Tanura	Refinery — Saudi Arabia
Pegu Yoma (Myanmar)	Mineral oil pair
Uranium City	Canada (Athabasca)
U production	Often Kazakhstan #1; reserves Australia (Olympic Dam)
Radium	From pitchblende (uranium ore)
Rare earths	China dominates; export curbs + electronics demand
Lithium Afghanistan find	Compared with oil discovery in Iran
Chile	Nitrate / iodine (Atacama caliche)
Diamond volume	Often Russia
Cobalt (~¾ of batteries)	DRC
Camphor	Tree native to China/Japan (<i>Cinnamomum camphora</i>)

Common Traps — Don’t Fall For These

1. **Ruhr = Britain / France / Russia** — FALSE. **Germany**.
2. **Kuzbass = Ukraine** — FALSE. **Russia**. Donetsk = Ukraine.
3. **Appalachian = England** — FALSE. **USA**. Lancashire = England.
4. **Mount Newman = manganese** — FALSE. **Iron**, Australia. Mn = **Postmasburg**.
5. **Germany–Normandy iron** — FALSE. Normandy = **France**.
6. **Chile has no copper** — FALSE. 2022 **leading** + Andes **porphyry**.
7. **Kinta = rubber** — FALSE. **Tin**.
8. **Pegu Yoma = tin** — FALSE. **Mineral oil**.
9. **Kuwait = Kashagan** — FALSE. Kashagan = **Kazakhstan**. Kuwait = **Burgan**.
10. **LPG = methane / CNG** — FALSE. CNG = methane; LPG = propane+butane.
11. **Iodine = Japan 2018** — FALSE. **Chile**.
12. **Kyiv = Ukraine’s coal centre** — FALSE. **Donetsk**.
13. **Nuclear / geothermal = stored solar** — FALSE.
14. **2022 steel #1 = India** — FALSE. **China**.
15. **Haft Kel = Iraq** — FALSE. **Iran**. Zubair = Iraq. Dhahran = Saudi.
16. **Mount Newman = bauxite** — FALSE. **Iron**. Weipa = bauxite.
17. **Postmasburg = gold** — FALSE. **Manganese**. Witwatersrand = gold.
18. **Pegu Yoma = Kinta tin** — FALSE. Pegu = **oil**; Kinta = **tin**.
19. **North Field = Russian Urengoy** — FALSE. **Qatar** (pair = Iran South Pars).

- 20. **OPEC HQ = Riyadh / Geneva** — FALSE. **Vienna**.
- 21. **Lithium Triangle includes Brazil** — FALSE. Chile–Argentina–Bolivia.
- 22. **Rare earths = Chile only** — FALSE. Processing fact = **China**.
- 23. **Carajás = Mount Newman** — FALSE. Carajás = Brazil iron; Newman = Australia iron.
- 24. **Skip non-PYQ fields because papers asked Ruhr/Kinta** — FALSE. Next year can ask any Lucent field.

Topic 22 — World Industries

Current Affairs (this topic)

None mandatory.

Confused Pairs

Pair	Correct	Trap	Hindi
Osaka	Cotton textile	Auto / ship	ओसाका
Detroit	Automobile	Cotton	डेट्रॉइट
Cuba	Cigar	Ship	क्यूबा
St. Petersburg	Shipbuilding	Cotton	सेंट पीटर्सबर्ग
Igarka	Russia (Yenisei timber)	China	इगारका
Rotterdam	Netherlands	Germany / Belgium	रॉटरडैम
Montevideo	Uruguay	Argentina	मोंटेवीडियो
Suez waters	Mediterranean + Red Sea	Atlantic as a canal bank	स्वेज
Suez cut	~ 7,000 km India–Europe	5,000 / 10,000	7,000 किमी
Suez lakes N→S	Manzala → Timsah → Great Bitter → Little Bitter	Reverse / skip Timsah	मंज़ाला पहले
Panama	Atlantic (Caribbean) ↔ Pacific	Med ↔ Red (that is Suez)	पनामा
Kiel	North Sea ↔ Baltic	Med	कील
Chinook	Warm dry, Rockies	Cold wind	चिनूक
Foehn	Alps , warm dry	Australia	फॉन
Mistral	Southern France (Rhône)	Australia	मिस्ट्रल
Shamal	Arabia / Persian Gulf	Austria	शमाल
Brickfielder	Australia	France	ब्रिकफील्डर
Santa Ana	California	Spain	सांता आना
Harmattan	West Africa	Alps	हरमट्टन
Bora	Adriatic (cold)	Warm Chinook dump	बोरा
Suez vs Panama	Suez = no facts, Med–Red; Panama = facts , Atlantic–Pacific	Swap	स्वेज / पनामा
Duisburg	Inland Rhine port, Germany	Netherlands / sea entrepôt	ड्यूसबर्ग
Willy-willy	Australian cyclone	Local wind like Brickfielder	विली-विली
Akron	Tyres / rubber (USA)	Steel	एक्रॉन
Toulouse	Airbus	Boeing (Seattle)	टूलूज़
Khamsin	Egypt	Australia	खामसिन
Osaka vs Detroit (Japan)	Osaka = Manchester (cotton); Nagoya = Detroit (auto)	Swap nicknames	
Footloose vs steel	Footloose = electronics / R&D; steel = ore + coal	Electronics on coalfield only	
Silicon Valley	California Bay Area electronics / IT	Detroit auto dump	सिलिकॉन वैली

N.1 Industrial location factors

Factor	Industry
Ore + coking coal	Iron and steel (Ruhr, Pittsburgh, Damodar (दामोदर))
Sugarcane / timber	Sugar mill, sawmill — bulky, perishable
Limestone	Cement — bulky
Cheap hydel	Aluminium , electro-chemicals
Cheap / skilled labour	Cotton textiles, garments, watches
Market	Bread, printing, consumer goods
Break-of-bulk (port, lake, rail)	Oil refining, flour, lake-steel
Agglomeration	Auto parts around Detroit / Nagoya / Turin
Government / SEZ	Planned estates
Capital + R&D	Pharma, electronics

Industry	Typical location
Iron–steel	Coalfield or ore or lake/port mixing both (Mesabi ore + Appalachian coal → Pittsburgh–Great Lakes)
Cotton textile	Humid labour districts (Lancashire, Osaka, Mumbai–Ahmedabad (अहमदाबाद))
Wool	Sheep hinterland + mill town (Yorkshire)
Shipbuilding	Deep estuary (ज्वारनदमुख) (Glasgow, St. Petersburg, Yokohama, Busan, Belfast)
Automobiles	Market + parts cluster (Detroit, Nagoya, Wolfsburg, Turin)
Oil refining	Port / pipeline end (Rotterdam, Houston, Abadan, Jamnagar)
Aircraft	Tech + large airfield (Seattle Boeing; Toulouse Airbus)
Films	Hollywood / Los Angeles
Watches	Skilled Alps–Jura (Switzerland)
Electronics	Footloose — Silicon Valley, East Asia coasts

N.2 Industrial regions

Region	Country	Fact
Ruhr	Germany	Coal, steel, heavy engineering; Essen–Dortmund–Duisburg
Saar	Germany–France border	Coal–steel
Lorraine	France	Iron ore + steel
Sambre–Meuse	Belgium	Coal–steel
Randstad	Netherlands	Rotterdam port + manufacturing
Upper Silesia	Poland	Coal–steel
Catalonia	Spain	Barcelona textiles / diversified
Lancashire	England	Cotton — Manchester–Liverpool
Yorkshire	England	Wool — Leeds–Bradford
Midlands / Black Country	England	Birmingham engineering
South Wales	UK	Coal–steel (historical)
Clydeside / Scottish Lowlands	UK	Glasgow shipbuilding
Po Basin	Italy	Milan–Turin–Genoa

Region	Country	Fact
Moscow–Nizhny	Russia	Diversified
Donbas	Ukraine	Coal–steel
Kuzbass	Russia	Coal–steel, Siberia
Urals / Magnitogorsk	Russia	Ore + steel

Region	Fact
New England	Early cotton; now diversified
New York–Philadelphia	Port, chemicals, publishing
Pittsburgh–Lake Erie	Steel — Appalachian coal + Mesabi ore via lakes
Chicago–Gary	Steel, machinery, meat-packing
Detroit	Automobiles
Birmingham (Alabama)	Southern steel
Gulf Coast	Petrochemicals (Houston)
Southern California	Aerospace, films, LA–Long Beach port
Silicon Valley	Electronics
Windsor–Ontario	Canada auto (next to Detroit)

Region	Fact
Keihin	Tokyo–Yokohama — Japan’s largest
Hanshin	Osaka–Kobe — cotton, ship, port
Chukyo	Nagoya — automobiles
Kitakyushu / N Kyushu	Japan — steel
Setouchi	Inland Sea, Japan — heavy industry
Shanghai–Nanjing / Yangtze	China — port + diversified
Pearl River Delta	Guangzhou–Shenzhen–Hong Kong
Beijing–Tianjin	China — north coast
Manchuria / Anshan–Shenyang	China — steel
Wuhan	China — steel on Yangtze
Seoul–Incheon	South Korea — auto, port
Busan	South Korea — ship + port

Region	Fact
São Paulo–Santos	Brazil
Buenos Aires	Argentina
Witwatersrand	South Africa — gold + industry
SE Australia	Sydney–Melbourne
Mumbai–Pune (पुणे)	India — cotton, engineering
Hugli	India — jute, engineering
Chotanagpur (छोटानागपुर)	India — minerals, steel

N.3 Industrial cities

Centre	Industry
Osaka	Cotton textile
Detroit	Automobile
Cuba (Havana)	Cigar
St. Petersburg	Shipbuilding

Nickname	City	Fact
Manchester of Japan	Osaka	Cotton
Detroit of Japan	Nagoya (Chukyo)	Automobiles
Pittsburgh of Japan	Kawasaki	Steel / heavy
Russian Manchester	Ivanovo	Cotton textiles

City	Industry
Manchester	Cotton
Ahmedabad	Cotton ("Manchester of India")
Leeds / Bradford	Wool
Lyon	Silk
Milan	Textiles / fashion
Dundee	Jute (historical)
Sheffield	Steel / cutlery
Solingen	Cutlery (Germany)
Essen	Krupp steel / engineering
Birmingham (UK)	Engineering
Pittsburgh	Steel
Chicago	Meat-packing, steel, machinery
Minneapolis	Flour
Milwaukee	Brewing
Akron	Rubber / tyres
Hollywood / Los Angeles	Films
Glasgow / Belfast / Gdansk	Shipbuilding
Yokohama / Busan	Shipbuilding
Turin / Wolfsburg / Nagoya / Stuttgart	Automobiles
Magnitogorsk / Anshan / Jamshedpur	Steel
Zurich / Jura	Watches
Dresden / Limoges	Porcelain
Toulouse	Airbus aircraft
Seattle	Boeing aircraft
Bournville	Chocolate
Havana	Cigar (Cuba row in 2022)
Abadan / Houston	Oil refining

City	Industry
Johannesburg	Gold
Coimbatore	Cotton / engineering (India)

N.4 Ports and seaports

Type	Recall	Example
Outport	Deep gate for up-river city	Cuxhaven, Tilbury, Le Havre, Bremerhaven
Entrepôt	Re-export hub	Singapore, Rotterdam, Hong Kong
Oil port	Crude / product	Ras Tanura, Houston, Rotterdam, Abadan
Container	Box trade	Shanghai , Singapore, Ningbo, Shenzhen, Busan, Rotterdam
Naval	Fleet base	Portsmouth, Pearl Harbor, Vladivostok
Packet	Short ferry	Dover–Calais
Inland / river	Sea-going on river/lakes	Igarka , Duisburg, Chicago, Kolkata

Port	Country
Rotterdam	Netherlands — Rhine mouth; Europort; 2023 correct
Antwerp	Belgium
Hamburg	Germany
Duisburg	Germany — largest inland port
London / Southampton	UK
Marseille	France
Genoa / Trieste	Italy
Piraeus	Greece
Gdansk	Poland

Port	Country
Shanghai / Ningbo / Shenzhen / Guangzhou	China — container volume
Hong Kong	China
Singapore	Singapore — Malacca entrepôt
Yokohama / Tokyo / Kobe / Nagoya	Japan
Busan / Incheon	South Korea
Jakarta / Tanjung Priok	Indonesia — 2023 correct
Manila	Philippines
Bangkok / Laem Chabang	Thailand
Ho Chi Minh	Vietnam
Colombo	Sri Lanka
Karachi / Gwadar	Pakistan
Chittagong	Bangladesh
Yangon	Myanmar
Jebel Ali (Dubai)	UAE
Bandar Abbas	Iran

Port	Country
Basra / Umm Qasr	Iraq
Aden / Djibouti	Red Sea–Aden gate
Mumbai / JNPT / Chennai / Kolkata / Kandla / Paradip / Vizag	India

Port	Country
New York / Houston / LA–Long Beach / New Orleans	USA
Vancouver / Montreal / Halifax	Canada
Santos	Brazil (São Paulo’s gate)
Buenos Aires	Argentina
Montevideo	Uruguay — 2023 correct; not Buenos Aires
Callao	Peru
Valparaíso	Chile
Durban / Cape Town	South Africa
Lagos	Nigeria
Mombasa	Kenya
Alexandria / Port Said / Suez	Egypt
Sydney / Melbourne / Fremantle	Australia
Auckland	New Zealand
Igarka	Russia — Yenisei timber; not China
Vladivostok / Novorossiysk / Murmansk	Russia

N.5 Trade routes

Canal / seaway	Links	Extra
Kiel	North Sea ↔ Baltic	Germany; not Med
Corinth	Ionian ↔ Aegean	Greece; ship canal
Soo (Sault Ste. Marie)	Lake Superior ↔ Huron	USA–Canada
Welland	Erie ↔ Ontario	Bypasses Niagara
St Lawrence (लॉरेंस) Seaway	Great Lakes ↔ Atlantic	USA–Canada
Rhine–Main–Danube	North Sea ↔ Black Sea inland	Europe
Volga–Don	Caspian system ↔ Black Sea	Russia

Passage	Gates
Malacca	Indian Ocean ↔ South China Sea (Singapore)
Sunda / Lombok	Alternate Indonesia routes
Hormuz	Persian (फ़ारसी) Gulf oil
Bab el-Mandeb	Red Sea ↔ Gulf of Aden
Gibraltar	Med ↔ Atlantic
Bosporus / Dardanelles	Black Sea ↔ Med
Dover	English Channel
Bering	Arctic ↔ Pacific

Passage	Gates
Magellan	S America (inshore of Cape Horn)
Cape of Good Hope	South of Africa — pre-Suez / ULCC tankers / Suez blockage
Cape Horn	South of America — pre-Panama

Common Traps — Don't Fall For These

1. **Osaka = automobile** — FALSE. **Cotton.** Detroit = auto.
2. **Cuba = shipbuilding** — FALSE. **Cigar.** St. Petersburg = ships.
3. **Igarka = China** — FALSE. **Russia.**
4. **Suez = Atlantic + Med** — FALSE. **Red Sea + Med.**
5. **Suez cut = 10,000 km** — FALSE. **~7,000 km.**
6. **Suez lakes start at Little Bitter** — FALSE. North door = **Manzala.**
7. **Panama = Med-Red** — FALSE. That is Suez. Panama = Atlantic-Pacific.
8. **Kiel = Med** — FALSE. **North Sea-Baltic.**
9. **Mistral = Australia** — FALSE. **France.** Brickfielder = Australia.
10. **Shamal = Austria** — FALSE. **Arabia.**
11. **Chinook is cold** — FALSE. Warm **and dry.**
12. **Foehn = Rockies** — FALSE. **Alps.** Chinook = Rockies.
13. **Rotterdam = Germany** — FALSE. **Netherlands.**
14. **Montevideo = Argentina** — FALSE. **Uruguay.**
15. **Lancashire = wool** — FALSE. **Cotton.** Yorkshire = wool.
16. **Hanshin = Tokyo** — FALSE. Osaka-Kobe. Keihin = Tokyo-Yokohama.
17. **Suez has Panama-style facts** — FALSE. Suez is **sea-level.**
18. **Willy-willy = Brickfielder** — FALSE. Willy-willy is a **cyclone.**
19. **Duisburg = Netherlands sea port** — FALSE. **German inland** Rhine port.
20. **Toulouse = Boeing** — FALSE. **Airbus.** Boeing = Seattle.
21. **Manchester of Japan = Nagoya** — FALSE. **Osaka.** Nagoya = Detroit of Japan.
22. **Footloose = steel on coal only** — FALSE. Electronics / R&D flexible location.
23. **Silicon Valley = Detroit** — FALSE. California electronics / IT.
24. **Skip non-PYQ cities/winds** — FALSE. Next year can ask any Lucent pair on this map.

Topic 23 — Political & Map-Based Geography

Current Affairs (this topic)

None mandatory. 12 / 24 / 200 nm are static.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q89

What is the correct latitudinal extent of the mainland of India?

A. 6 degrees 45' N to 37 degrees 6' N B. 6 degrees 4' N to 36 degrees 7' N C. 8 degrees 4' N to 36 degrees 12' N D. 8 degrees 4' N to 37 degrees 6' N

Confused Pairs

Pair	Correct	Trap	Hindi
Territorial vs EEZ	12 nm vs 200 nm	Swapped	क्षेत्रीय जल / EEZ
Contiguous	24 nm	12 or 200	सन्निकट
Shelf vs EEZ	Shelf may go to 350 nm ; EEZ stops 200	Shelf = 12	महाद्वीपीय मग्नतट
McMahon	India–China (Arunachal)	Pakistan / Nepal / Myanmar	मैकमोहन
Durand	Pakistan–Afghanistan	India–China	दूरंड
Radcliffe	1947 India–Pak / Bangladesh	McMahon	रैडक्लिफ
Longest coast	Gujarat	AP / MH / Kerala	गुजरात
Longest land border	Bangladesh	China / Pakistan	बांग्लादेश
Shortest land border	Afghanistan	Bhutan	अफ़ग़ानिस्तान
Suez waters	Med + Red Sea	Atlantic	स्वेज
Panama	Atlantic ↔ Pacific	Med–Red	पनामा
Kiel	North Sea ↔ Baltic	Med	कील
Hormuz	Persian Gulf ↔ Gulf of Oman	Malacca	होरमुज़
Gibraltar	Med ↔ Atlantic	Bosporus	जिब्राल्टर
Bosporus	Black Sea ↔ Marmara	Gibraltar	बोस्फोरस
Mistral / Köppen Cs	Med winter rain	Rain all year (that is Cfb W Europe)	भूमध्य
Köppen vs Thornthwaite	Thornthwaite = “vegetation is true index of climate”	Köppen letter codes	थॉर्नथ्वेट / कोपेन
Bolivia	Landlocked	Uruguay / Peru / Suriname	बोलीविया
Nobi / Kanto	Japan	Korea / China	जापान
Igarka	Russia	China	इगारका
Cape Verde capital	Praia	Bamako (Mali)	प्राइया
India area rank	7th largest; 2.4% of world	6th; wholly tropical	भारत क्षेत्र
Tropic of Cancer / UP	Passes 8 states — not UP	UP on the Tropic	कर्क रेखा
Double landlocked	Uzbekistan , Liechtenstein	Nepal / Bolivia as “double”	दोहरा स्थलरुद्ध

Pair	Correct	Trap	Hindi
Kazakhstan	Largest landlocked by area	Has ocean coast	कज़ाखस्तान
Lesotho	Enclave inside South Africa	Coastal SA dump	लेसोथो
49th Parallel	USA–Canada (long stretch)	USA–Mexico	49वीं
38th Parallel	North–South Korea	India–China	38वीं
Horn of Africa	Djibouti, Eritrea, Ethiopia, Somalia	Includes Sudan	हॉर्न ऑफ अफ्रीका
Slovenia capital	Ljubljana	Bratislava (Slovakia)	ल्युब्लियाना
Gold Coast	Old name of Ghana	Dutch Guiana / Suriname	गोल्ड कोस्ट
Great Britain vs UK	GB = England+Wales+Scotland	Includes Northern Ireland	ग्रेट ब्रिटेन
Greenland	Denmark politically; N America geographically	Independent continent	ग्रीनलैंड
Caspian littoral	Five states; not Armenia/Iraq	Armenia on Caspian	कैस्पियन
Oceania	Melanesia/Micronesia/Polynesia + Australia	Indonesia included	ओशियानिया
Laos	Only SE Asia landlocked	Cambodia / Thailand	लाओस
Rio Grande	USA–Mexico	USA–Canada	रियो ग्रांडे
Baikonur	Kazakhstan	Ukraine	बैकोनुर
Midnight Sun vs Rising Sun	Midnight Sun = Norway (Arctic); Rising Sun = Japan	Swapped	मध्यरात्रि सूर्य / उगता सूर्य
Morning Calm vs Rising Sun	Morning Calm = Korea ; Rising Sun = Japan	Korea = Japan	सुबह की शांति
Thousand Lakes	Finland	Switzerland / Netherlands	हज़ार झीलें
Bird Continent	South America	Australia	पक्षी महाद्वीप
Mistress of Eastern Sea	Sri Lanka	India / Myanmar	पूर्वी सागर की स्वामिनी
Manchester of the East	Osaka	Tokyo / Nagoya	पूर्व का मैनचेस्टर
Pearl of Siberia	Baikal	Great Bear	साइबेरिया का मोती
Venice of the North	St. Petersburg (usual Russia key)	Moscow / Vladivostok	उत्तर का वेनिस
City of Smoke	Chicago	London / Kolkata	धुएँ का शहर
Oil Capital of Europe	Aberdeen	Liverpool / Belfast	यूरोप की तेल राजधानी
Land of Promise	Mindanao	Java / Cuba	वादा द्वीप
Ninety East Ridge	Indian Ocean	Pacific / Atlantic	नाइंटी ईस्ट रिज

N.3 Boundary lines

Other lines (one home)

Line	Note
49th Parallel	Long USA–Canada stretch (not USA–Mexico)
38th Parallel	North Korea–South Korea armistice story
Maginot Line	Historical France–Germany fortification (WWII note)
Oder–Neisse	Germany–Poland post-WWII border
Hindenburg Line	WWI German defence line (Europe) — not an India border
Mannerheim Line	Historic Finland / Russian front note (coaching “Poland–Russia” trap — learn as Mannerheim, not Radcliffe)

Line	Note
17th Parallel	Historical Vietnam divide
Rio Grande	Forms much of USA–Mexico border (with Colorado water-sharing note)
Alpine / Little Maginot	France–Italy framing

Central Asia capitals (match bank)

Country	Capital	Trap
Uzbekistan ★	Tashkent	—
Tajikistan ★	Dushanbe	Not Ashgabat
Kyrgyzstan ★	Bishkek	—
Turkmenistan ★	Ashgabat	Not Dushanbe
Kazakhstan	Astana (Nur-Sultan phase)	Almaty is not the capital
Afghanistan	Kabul	Neighbourhood
Iran	Tehran	Neighbourhood

N.5A Country ranks · regions · dependent territories · old names

Region	Members / pair
Horn of Africa	Djibouti, Eritrea, Ethiopia, Somalia — not Sudan
Balkans	Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Greece, Kosovo, Montenegro, North Macedonia, Romania, Serbia, Slovenia, and European Türkiye (Eastern Thrace). Austria is not Balkan.
Scandinavia (geo)	Norway, Sweden, Denmark — Finland often cultural/historical set but MCQ may exclude Finland
Central America	Belize, Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama — not Sierra Leone (W Africa)
Oceania	Melanesia / Micronesia / Polynesia + Australia — Indonesia not Oceania
Arabian Peninsula	Saudi Arabia, Yemen, Oman, UAE, Bahrain, Qatar, Kuwait (+ Iraq fringe) — Syria not
UAE emirates	Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al-Quwain, Fujairah, Ras Al-Khaimah
Gaza Strip	Borders Egypt and Israel (Mediterranean strip)
Caspian shores	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan — not Armenia / Iraq
Dead Sea	Israel / West Bank / Jordan — not Lebanon
Afghanistan neighbours	Iran, Pakistan, China, Tajikistan, Uzbekistan, Turkmenistan (+ India SE note) — not Russia / Azerbaijan / Kyrgyzstan
Myanmar neighbours	India, Bangladesh, China, Laos, Thailand — not Vietnam/Cambodia/Malaysia as the trio
Israel land borders	Lebanon, Syria, Jordan, Egypt
Chile	Longest N–S extension among common traps (W South America)
Japan	Largest among common “island without land border” options vs NZ / Philippines / Cuba
Indonesia land borders	Malaysia, Papua New Guinea, Timor-Leste — not Brunei

Place	Owner / note
Greenland	Denmark politically; geographically N America ; largest island

Place	Owner / note
Christmas Island	Australia (Indian Ocean)
Martinique	France (Caribbean)
Falklands / St Helena / Gibraltar / Pitcairn / BIOT	UK
Puerto Rico / American Samoa / US Virgin / Midway	USA
Canaries / Ceuta / Melilla / Balearics	Spain
Azores / Madeira	Portugal
Cook / Niue / Tokelau / Ross Dependency	NZ association
Chechnya	Republic of Russia (Caucasus; capital Grozny)
Aleutian	USA; Bear Norway; Franz Josef Russia

Old	New
Siam	Thailand
Formosa	Taiwan
Mesopotamia (मेसोपोटामिया)	Iraq
Burma	Myanmar
Gold Coast	Ghana (घन)
Dutch Guiana	Suriname (not Ghana)
Southern Rhodesia	Zimbabwe (Harare was Salisbury)
Abyssinia	Ethiopia
Basutoland	Lesotho
Bechuanaland	Botswana
Zaire	DR Congo
Nyasaland	Malawi

N.5B Geographical sobriquets

Sobriquet	Place
Land of the Midnight Sun	Norway (Arctic set)
Bird Continent / Continent of Birds	South America
Mistress of the Eastern Sea / Pearl of the Indian Ocean	Sri Lanka
Gateway to Asia	Singapore
Land of White Elephants	Thailand
Land of the Rising Sun / Nippon	Japan
Land of Thousand Lakes / Garden of Lakes	Finland
Land of Morning Calm	(South) Korea
Dark Continent	Africa
Land of Thunderbolts	Bhutan
Land of Golden Pagodas	Myanmar
Land of Kangaroos	Australia
Britain of the South	New Zealand

Sobriquet	Place
Key of the Mediterranean	Gibraltar
Holy Land	Palestine / Jerusalem
Sick Man of Europe	Türkiye
Playground of Europe	Switzerland
Island of Cloves	Zanzibar
Pearl of the Antilles / Sugar Bowl of the World	Cuba
Island of Pearls / Pearl of the Persian (फ़ारसी) Gulf	Bahrain
Gift of the Nile	Egypt
Land of Canals	Pakistan (coaching tag)
Blue Grass State	Kentucky (USA)
Land of Promise	Mindanao (Philippines)

Sobriquet	Place
Venice of the North	St. Petersburg (usual Russia key). Amsterdam, Bruges, and Stockholm also carry canal “Venice” nicknames in some lists.
Gateway to the West	Istanbul
City of Canals / Queen of the Adriatic / City of Dreams	Venice
City of Seven Hills / Eternal City	Rome
Manchester of the East / Nation’s Kitchen	Osaka
City of the Golden Gate	San Francisco
City of Smoke / City of Smokers	Chicago
Paris of South America	Buenos Aires
Empire City	New York
Oil Capital of Europe	Aberdeen
Pearl of Siberia	Lake Baikal (Russia)
Roof of the World	Pamir (Pamir Knot — southern edge of Central Asia)

Sobriquet	Place
City of Golden Temple (स्वर्ण मंदिर)	Amritsar (अमृतसर)
Land of Five Rivers	Punjab
Gateway of India / City of Seven Islands	Mumbai
Queen of the Arabian Sea	Kochi (Cochin)
Spice Garden of India	Kerala
Switzerland of India	Kashmir
Twin Cities	Hyderabad (हैदराबाद)–Secunderabad
Manchester of India	Ahmedabad (अहमदाबाद)
Venice of the East	Alappuzha (Alleppey)
Sorrow of Bengal	Damodar (दामोदर)
Sorrow of Bihar	Kosi (कोसी)
Sorrow of China	Huang He (Hwang Ho)

Sobriquet	Place
Queen of Chotanagpur (छोटानागपुर)	Netarhat (Jharkhand)
Queen of Mountains (India)	Mussoorie (मसूरी)
World's Bread Basket	Prairies of North America

N.8 Straits

Other straits (list)

- Bab el-Mandeb — Red Sea and Gulf of Aden (Yemen–Djibouti/Eritrea).
- Palk — India and Sri Lanka, with Adam's Bridge.
- 10° Channel (दस डिग्री) — Andaman and Nicobar. 8°/9° Degree — Lakshadweep / Maldives side.
- Dover — North Sea and English Channel (UK–France).
- Magellan — Atlantic and Pacific inshore (Chile); Cape Horn is the open-ocean south.
- Torres — Australia and Papua New Guinea. Bass — Tasmania and Australia. Cook — NZ islands.
- Florida / Yucatán — doors of the Gulf of Mexico.
- Mozambique — Africa and Madagascar.
- Tsushima / Korea / Taiwan — East Asia.
- Davis / Denmark — Greenland–Canada / Iceland Arctic doors.

N.10 Mountains (map)

Range–region match bank (next year can ask any)

Range / highland	Region / country fact	Tag
Pontic ★	Northern Turkey (Black Sea side)	2024 W→E
Zagros ★	Western Iran	2024 W→E
Hindu Kush ★	Afghanistan (into N Pakistan)	2024 W→E
Karakoram (काराकोरम) ★	N Pakistan / Ladakh / China border	2024 W→E; K2
Alps	Europe (France–Switzerland–Italy–Austria)	Mont Blanc
Atlas	NW Africa (Morocco–Algeria–Tunisia)	
Andes	South America (spine)	Aconcagua in Argentina
Rockies	North America (W USA–Canada)	
Appalachians	Eastern USA	Older fold belt
Urals	Russia — Europe / Asia conventional divide	
Caucasus	Russia / Georgia / Azerbaijan belt	Elbrus (Europe peak)
Drakensberg	South Africa / Lesotho	
Great Dividing Range	Australia (east)	
Pamir	Tajikistan / “Roof of the World” knot	
Tien Shan	Central Asia / China–Kyrgyzstan	
Kunlun	North of Tibetan Plateau (China)	
Altai	Russia / Mongolia / China / Kazakhstan	
Ethiopian Highlands	Ethiopia / Horn of Africa	

Highest peak by continent (match bank)

Continent	Peak	Country fact / trap
Asia	Everest	Nepal–China
South America	Aconcagua	Argentina
North America	Denali	USA (Alaska)
Africa	Kilimanjaro	Tanzania (not Kenya)
Europe	Elbrus	Russia / Caucasus
Australia (mainland)	Kosciuszko	Australia
Antarctica	Vinson Massif	Also revise

Common Traps — Don't Fall For These

1. **EEZ = 12 nm** — FALSE. Territorial **12**; EEZ **200**.
2. **McMahon = Pakistan** — FALSE. **China**.
3. **Durand = India–China** — FALSE. **Pak–Afghan**.
4. **Longest coast = AP / Kerala** — FALSE among 2018 options. **Gujarat**.
5. **Longest land border = China** — FALSE. **Bangladesh**.
6. **Sri Lanka = land neighbour** — FALSE. **Maritime**.
7. **Suez = Atlantic** — FALSE. **Med + Red Sea**.
8. **Panama = Med–Red** — FALSE. Atlantic–Pacific.
9. **Kiel = Suez cousin on Med** — FALSE. North Sea–Baltic.
10. **Hormuz = Malacca** — FALSE. Hormuz = Gulf oil mouth.
11. **Nobi/Kanto = Korea** — FALSE. **Japan**.
12. **Igarka = China** — FALSE. **Russia**.
13. **Bolivia has a Pacific coast in that 2022 four** — FALSE. **Landlocked**.
14. **"Vegetation is true index of climate" = Köppen** — FALSE. **Thornthwaite** for that quote; Köppen = letter-code classes.
15. **Mediterranean = rain all year** — FALSE. **Winter** rain. All-year = **W Europe Cfb**.
16. **Turkmenistan = Dushanbe** — FALSE. **Ashgabat**. Dushanbe = Tajikistan.
17. **Yangon = Myanmar capital** — FALSE. **Naypyidaw**.
18. **Cape Verde = Bamako** — FALSE. **Praia**. Bamako = Mali.
19. **India = 6th largest / wholly tropical** — FALSE. **7th**; Tropic through middle; north is extra-tropical. **UP is not on the Tropic**.
20. **Uzbekistan has a sea** — FALSE. It is **double landlocked**.
21. **Kazakhstan has an ocean coast** — FALSE. Largest landlocked by area; Caspian ≠ ocean access.
22. **Lesotho = double landlocked Asia** — FALSE. Enclave in South Africa.
23. **49th Parallel = USA–Mexico** — FALSE. USA–Canada stretch.
24. **Skip non-PYQ atlas pairs** — FALSE. Next year can ask any Lucent capital, strait, or line.
25. **Horn includes Sudan** — FALSE. Horn = Djibouti, Eritrea, Ethiopia, Somalia.
26. **Slovenia = Bratislava** — FALSE. Ljubljana. Bratislava = Slovakia.
27. **Ghana = Dutch Guiana** — FALSE. Gold Coast → Ghana; Dutch Guiana → Suriname.
28. **Great Britain includes Northern Ireland** — FALSE. That makes the **United Kingdom**.
29. **Greenland = independent continent / only Europe** — FALSE. Largest island; Denmark politically; N America geographically.

- 30. **Armenia / Iraq on Caspian** — FALSE. Caspian five exclude both.
- 31. **Indonesia = Oceania** — FALSE. SE Asia.
- 32. **Longest world coast = Australia / Brazil** — FALSE. **Canada**.
- 33. **Baikonur = Ukraine** — FALSE. **Kazakhstan**.
- 34. **Durand = Bangladesh–India** — FALSE. Pak–Afghan. Bangladesh–India = Radcliffe.
- 35. **Norway = Rising Sun** — FALSE. Rising Sun = **Japan**. Norway = **Midnight Sun**.
- 36. **Japan = Morning Calm** — FALSE. Morning Calm = **Korea**.
- 37. **Bird Continent = Australia** — FALSE. **South America**.
- 38. **Mistress of Eastern Sea = India** — FALSE. **Sri Lanka**.
- 39. **Manchester of the East = Tokyo** — FALSE. **Osaka**.
- 40. **Pearl of Siberia = Great Bear** — FALSE. **Baikal**.
- 41. **Venice of the North = Moscow** — FALSE. **St. Petersburg**.
- 42. **City of Smoke = London** — FALSE in that GC key. **Chicago**.
- 43. **Ninety East Ridge = Pacific** — FALSE. **Indian Ocean**.

Uttarakhand — Topic 1 — Location, Relief & Structure

Current Affairs (this topic)

- Peak / glacier / Char Dham corridor news can refresh district tags (Nanda Devi (नंदा देवी)–Chamoli (चमोली), Tehri (टिहरी) dam (टिहरी बांध) landscape).
- Keep MBT–MCT zone wording stable — mismatch stems recycle every year.

Quick Revision — Spine Only

Head	Correct
Location	Central Himalaya (हिमालय); between Himachal (हिमाचल) (W), Nepal/Tibet (E/N), UP plains (S)
Lat–long	Roughly 28°43'–31°27' N · 77°34'–81°02' E
South → north belts	Tarai → Bhabar (भाबर) → Shiwalik → Dun (दून) → Lesser Himalaya → Great Himalaya (Trans/Tethys farther north)
Between MBT and MCT	Lesser Himalayan zone
South of MBT	Shiwalik (शिवालिक) zone
North of MCT	Central crystalline / Higher Himalaya approach
Duns	Longitudinal valleys in Shiwalik belt (e.g. Dehradun (देहरादून))
Highest peak	Nanda Devi (नंदा देवी) (~ 7816–7817 m , Chamoli (चमोली)); Kamet (कामेत) (~ 7756 m) next among major UK peaks
Divisions	Garhwal (गढ़वाल) (west) · Kumaon (कुमाऊँ) (east)

Confused Pairs

Pair	Correct
MBT–MCT belt vs Shiwalik	Lesser Himalaya ≠ Shiwalik (Shiwalik is south of MBT)
Tarai last in N→S list	Tarai is southernmost — south→north starts with Tarai
Bhabar vs Tarai	Bhabar = porous gravel, streams sink ; Tarai = streams reappear , marshy, fertile
Great Himalaya vs Tethys	Tethys / Trans-Himalaya lies farther north — a different question family
UK vs HP relief vocabulary	Same thrust names; apply to Uttarakhand districts in state papers

Common Traps

1. **MBT–MCT = Shiwalik** → Lesser Himalaya.
2. **South→north ends with Tarai** → Tarai is first from the south.
3. **Central crystalline between MBT and MCT** → that belt is north of MCT.
4. **Bhabar = marshy fertile belt** → that is Tarai; Bhabar is porous gravel.
5. **Nanda Devi = Uttarkashi only** → Chamoli (Garhwal high Himalaya).

Uttarakhand — Topic 2 — Climate & Drainage

Quick Revision — Spine Only

Head	Key Fact
Longest River in UK	Kali (Sharda (शारदा)) — 252 km; source: Lipulekh (लिपुलेख) / Kalapani; international boundary with Nepal
Ganga Formation	Bhagirathi (भागीरथी) (205 km) + Alaknanda (अलकनंदा) (195 km) unite at Devprayag (देवप्रयाग) to form the Ganga (गंगा)
West \rightarrow East Rivers (2025)	Yamuna (यमुना) \rightarrow Bhilangana \rightarrow Alaknanda \rightarrow Gori Ganga (UKPCS 2025 Q109)
Panch Prayag (प्रयाग) (N \rightarrow S)	Vishnuprayag (विष्णुप्रयाग) \rightarrow Nandaprayag \rightarrow Karnaprayag (कर्णप्रयाग) \rightarrow Rudraprayag (रुद्रप्रयाग) \rightarrow Devprayag
Largest Glacier in UK	Gangotri (गंगोत्री) Glacier (Uttarkashi (उत्तरकाशी)) — 30 km long, 2 km wide; source of Bhagirathi
Largest Glacier in Kumaon (कुमाऊँ)	Milam (मिलाम) Glacier (Pithoragarh (पिथौरागढ़)) — 16 km long; source of Gori Ganga
2nd Longest Glacier	Pindari (पिंडारी) Glacier (Bageshwar (बागेश्वर)) — source of Pindar (पिंडर) river; discovered by G. W. Traill
Highest Shiva Temple	Tungnath (Rudraprayag) — at 3,680 m; part of Panch Kedar (arms worshipped)
Mystery / Skeleton Lake	Roopkund (Chamoli (चमोली), 5,029 m) — human skeletons from 9th cent. CE (Raja (राजा) Jasdhaval)
Deepest Lake in Kumaon	Naukuchiatal (Nainital (नैनीताल)) — 9-cornered lake; approx. 40 m deep
Largest Lake in Kumaon	Bhimtal (Nainital) — triangular lake with an island restaurant in the center

River Systems of Uttarakhand: Comprehensive Master Matrix

River System	Source Glacier / Origin	Length in UK	Major Tributaries & Confluences
Kali (Sharda)	Lipulekh / Kalapani (Pithoragarh)	252 km (Longest)	Kuti Yankti (Gunji), Eastern Dhauliganga (Tawaghat), Gori Ganga (Jauljibi (जौलजीबी)), Saryu (Pancheshwar), Ladhia (Chuka).
Bhagirathi (भागीरथी)	Gaumukh (गौमुख) (Gangotri Glacier, Uttarkashi)	205 km	Rudraganga (Gangotri), Jadhganga/Jahanvi (Bhaironghati), Asiganga (Uttarkashi), Bhilangana (Old Tehri).
Alaknanda (अलकनंदा)	Satopanth & Bhagirath Kharak (Chamoli)	195 km	Saraswati (सरस्वती) (Mana (माना) / Keshavprayag), Western Dhauli (धौली) (Vishnuprayag), Nandakini, Pindar, Mandakini.
Western Ramganga (रामगंगा)	Dudhatoli range (Pauri (पौड़ी)/Chamoli/Almora (अल्मोड़ा))	155 km	Gagas, Binau, Deogadh. Traverses Corbett National Park into UP.
Kosi (कोसी) (Kaushiki)	Dharpani Dhar (Kausani, Almora)	168 km	Drains Almora, Someshwar, Ramnagar; joins Ramganga in UP.
Yamuna (यमुना)	Yamunotri (यमुनोत्री) Glacier (Bandarpunch (बंदरपूँछ), Uttarkashi)	136 km	Rishiganga, Hanuman Ganga, Kamal, Tons (टौंस) (Dakpathar), Giri, Asan (Rampur Mandi (मंडी)).
Tons (Supin)	Bandarpunch / Har-Ki-Doon (Uttarkashi)	148 km	Formed by confluence of Rupin and Supin at Netwar. Carries twice the water volume of the Yamuna!
Saryu	Sarmool (Bageshwar)	146 km	Gomti (Bageshwar), Panar (Rameshwar), Eastern Ramganga (Muawani). Meets Kali at Pancheshwar.

River System	Source Glacier / Origin	Length in UK	Major Tributaries & Confluences
Mandakini (मंदाकिनी)	Chaurabari Glacier (Kedarnath, Rudraprayag)	72 km	Vasukiganga (Sonprayag), Madhu Ganga. Meets Alaknanda at Rudraprayag.
Bhilangana	Khatling Glacier (Tehri Garhwal)	—	Balganga, Dharmaganga. Submerged in Tehri Dam (टिहरी बांध) reservoir.

Master Table of High-Altitude Lakes & Tals

Lake / Tal	District & Altitude	High-Yield Prelims Characteristics
Roopkund	Chamoli (चमोली) (5,029 m)	“Mystery Lake” / “Skeleton Lake” : Contains hundreds of 9th-century human skeletal remains (linked to King Jasdhaval of Kannauj (कन्नौज) and Queen Vallabha (वल्लभ)); situated on the Nanda Devi Raj Jat (नंदा देवी राज जात) route.
Hemkund (हेमकुंड) (Lokpal (लोकपाल))	Chamoli (4,329 m)	Sacred lake surrounded by 7 snow-clad peaks ; holy Gurudwara of Guru (गुरु) Gobind Singh (गुरु गोबिंद सिंह) and ancient Lakshman Temple; source of Lakshmanganga.
Nainital Lake	Nainital (नैनीताल) (1,938 m)	Pear-shaped lake discovered by P. Barron in 1841; divided into <i>Mallital</i> and <i>Tallital</i> ; surrounded by 7 hills (China Peak, Naina Peak, Ayarpatta, etc.).
Naukuchiatal	Nainital	Deepest lake in Kumaon (~40 m); possesses 9 corners ; haven for migratory birds.
Bhimtal	Nainital	Largest lake in Kumaon region; triangular shaped; features a picturesque island restaurant.
Dodital	Uttarkashi (उत्तरकाशी) (3,024 m)	6-cornered lake ; mythologically associated with the birth of Lord Ganesha; origin of the Asiganga river; famous for Himalayan Golden Trout.
Sahasratal	Tehri Garhwal	Group of hundreds of glacial tarns situated on the northern ridge of Khatling glacier; largest tal in Garhwal.
Deoria Tal	Rudraprayag (रुद्रप्रयाग) (2,438 m)	Famous for crystal-clear reflection of the Chaukhamba (चौखंबा) peaks ; mythologically identified with the Yaksha-Yudhisthira dialogue.

Confused Pairs & Common Traps

Confused Pair	Critical Distinction
Longest River vs Deepest River	Longest river = Kali (252 km) ; River with greatest water volume = Alaknanda .
Gangotri Glacier vs Milam Glacier	Gangotri = Largest in Uttarakhand (Uttarkashi, 30 km); Milam = Largest in Kumaon (Pithoragarh, 16 km).
Vishnuprayag vs Devprayag	Vishnuprayag = Alaknanda + Western Dhauliganga ; Devprayag = Alaknanda + Bhagirathi (creates the Ganga).
Roopkund vs Dodital	Roopkund = Skeleton lake (Chamoli, 5,029 m); Dodital = 6-cornered lake of Ganesha (Uttarkashi).
Tungnath vs Kedarnath	Tungnath = Highest Shiva Temple (3,680 m, arms); Kedarnath = Supreme Jyotirlinga Dham (hump).

Uttarakhand — Topic 3 — Vegetation & Wildlife

Current Affairs (this topic)

- New sanctuary notifications or UNESCO / BR news can refresh fauna (प्राणीजात) lists; keep district–sanctuary pairs stable.
- NP foundation years (Corbett 1936, Rajaji (राजाजी) 1983, Gangotri (गंगोत्री) 1989, Govind 1990) are high-recycle traps.

Quick Revision — Spine Only

Head	Correct
Forest richness	Large forest share; belts from tropical foothill to alpine — relief + climate explain it
Altitude stack	Foothill sal → chir → oak–rhododendron → deodar / fir → alpine birch–juniper → bugyal
Deodar	Abundant western Himalayan temperate belt 1500–2500 m ; highest among deodar / pine / sal / teak
Birch	Himalayan alpine / moist alpine scrub; writing-bark tradition
Lisa	Resin product from pine ; turpentine after distillation; rural UK livelihood
Khair	Sub-tropical forests (Shiwalik (शिवालिक) / western Himalaya (हिमालय) scrub); also yields katha
Nanda Devi BR	UNESCO WHS (with Valley of Flowers (फूलों की घाटी)); fauna includes snow leopard, musk deer, blue sheep
Askot sanctuary	Pithoragarh (पिथौरागढ़)
Binsar sanctuary	Almora (अल्मोड़ा)
Govind Pashu Vihar / NP	Uttarkashi (उत्तरकाशी)
Nandhaur sanctuary	Nainital (नैनीताल)
Corbett NP	1936 (India's first NP tradition)
Rajaji NP	1983
Gangotri NP	1989
Govind NP year	1990 (not 1989)

Confused Pairs

Pair	Correct
Askot vs Binsar districts	Pithoragarh vs Almora — classic swap
Govind 1989 vs 1990	1990 is correct; 1989 pairs with Gangotri
Nanda Devi fauna	Snow leopard + musk deer + blue sheep (bharal) all in
A true but R not explain	UKPCS 2025 treats R as explaining A for forest richness
Lisa vs Burans	Lisa = pine resin; not Burans / deodar / sheesham
Khair forest type	Sub-tropical (not sub-alpine / cool temperate)
Deodar vs chir height	Deodar temperate 1500–2500 m ; chir sub-tropical pine lower (~ 900–2000 m)
Birch location	Himalayas — not Nilgiris / Aravallis / Vindhya

3.4 Sanctuary–district map

Sanctuary	District
Askot	Pithoragarh
Binsar	Almora
Govind	Uttarkashi
Nandhaur	Nainital

Common Traps

1. **Govind NP = 1989 → 1990.**
2. **Askot = Almora** → Pithoragarh.
3. **R does not explain forest richness** → UKPCS keys that it does.
4. **Lisa from Burans / Deodar** → from **pine**.
5. **Khair = sub-alpine / temperate** → **sub-tropical**.
6. **Birch = Nilgiris** → Himalayas.
7. **Highest tree among options = pine / sal** → **deodar**.
8. **Vembanad = Karnataka** → Kerala.

Uttarakhand — Topic 4 — Minerals, Power & Industry

Current Affairs (this topic)

- Watch new SIIDCUL allotments, hydropower clearances, and any revised mineral-district keys in state releases.
- Tehri (टिहरी) and Alaknanda-belt project news often doubles as power CA.

Quick Revision — Spine Only

Head	Correct
Dolomite	Dehradun (देहरादून), Pithoragarh (पिथौरागढ़)
Soapstone (soap stone)	Chamoli (चमोली), Bageshwar (बागेश्वर)
Graphite	Almora (अल्मोड़ा), Nainital (नैनीताल) (correct pair in UKPCS 2025)
Silica sand	Dehradun, Nainital, Tehri Garhwal (गढ़वाल) — not Pauri (पौड़ी) Garhwal
Power	Hydropower on Bhagirathi (भागीरथी)–Alaknanda (अलकनंदा)–Yamuna (यमुना) systems; Tehri (टिहरी) as flagship
Industry sketch	Forest-based, tourism services, pharma/IT pockets in plains & Dehradun belt

Confused Pairs

Pair	Correct
Silica sand–Pauri	Wrong in the 2025 question — use Dehradun / Nainital / Tehri
Soapstone–Pithoragarh	Soapstone card is Chamoli–Bageshwar
Dolomite–Almora	Dolomite card is Dehradun–Pithoragarh
Power vs mineral	Do not answer hydropower questions with mineral districts

Common Traps

1. **Silica sand = Pauri** → wrong.
2. **Soapstone = Dehradun–Pithoragarh** → that is dolomite’s pair.
3. **All Himalayan districts have the same minerals** → learn the named pairs.
4. **Graphite = Chamoli–Bageshwar** → that is soapstone; graphite is Almora–Nainital.

Uttarakhand — Topic 5 — Agri, AH & Irrigation

Current Affairs (this topic)

- Fresh land-use releases can shuffle Haridwar (हरिद्वार) vs Udham Singh Nagar (ऊधम सिंह नगर) absolute (निरपेक्ष) ranks — keep both names and prefer the Series B letter when that paper is cited.
- Horticulture and dairy scheme news can appear as CA without rewriting the hill–plain crop map.

Quick Revision — Spine Only

Head	Correct
Hill farming	Terraced cereals, millets, pulses; orchards in mid-hills
Granary belt	Tarai / plains districts lead in cropped area
Highest agri land (2025)	Haridwar (हरिद्वार) (Series B provisional key A)
Cultivated-area leader	Official Series B (2025) answer is Haridwar ; many notes also name Udham Singh Nagar (ऊधम सिंह नगर) — learn both names
Animal husbandry	Supplementary to agriculture (milk, wool, manure)
Sheep / pastures	Environmental-degradation narrative appears in many notes — 2025 Series B accepted Only statement 1
Irrigation	Canals and tubewells in Tarai; springs / guls / limited canals in hills

Confused Pairs

Pair	Correct
Haridwar vs US Nagar on agri area	Tarai/plains lead cropped area vs hills. Exact rank varies by source. For 2025 Series B , mark Haridwar ; many books still name Udham Singh Nagar as the Tarai rival
AH as main vs supplementary	Correct reading is supplementary occupation
Q102 both vs only 1	Series B Only 1 — do not mark both unless an amended key says so
Hill vs plain irrigation	Tubewell–canal plains ≠ spring-fed hill systems

Common Traps

- Blindly marking Both on Q102** → Series B is Only 1.
- Always US Nagar for max agri land** → 2025 key is Haridwar.
- AH replaces agriculture in hills** → it supplements.
- Same irrigation system statewide** → plains ≠ hills.

Uttarakhand — Topic 6 — Population, SC/ST & Settlements

Quick Revision — Spine Only

Indicator	Uttarakhand State Value	Highest District	Lowest District
Total Population	1,00,86,292 (1.01 Crore)	Haridwar (हरिद्वार) (18.90 Lakh)	Rudraprayag (रुद्रप्रयाग) (2.42 Lakh)
Decadal Growth (2001–11)	18.81% (National: 17.70%)	Udham Singh Nagar (ऊधम सिंह नगर) (33.45%)	Pauri (पौड़ी) Garhwal (गढ़वाल) (-1.41% negative!)
Negative Growth Districts	Pauri (-1.41%) & Almora (अल्मोड़ा) (-1.28%)	Severe out-migration / Palayan	Creates "Ghost Villages" (Bhootiya Gaon)
Population Density	189 persons / sq km	Haridwar (801)	Uttarkashi (उत्तरकाशी) (41)
Sex Ratio	963 (National: 943)	Almora (अल्मोड़ा) (1142)	Haridwar (880)
Child Sex Ratio (0–6)	890 (National: 919)	Almora (922)	Pithoragarh (पिथौरागढ़) (816)
Overall Literacy Rate	78.82% (Rank: 17th in India)	Dehradun (देहरादून) (84.25%)	Udham Singh Nagar (73.10%)
Male Literacy	87.40%	Rudraprayag (93.90%)	Haridwar (81.04%)
Female Literacy	70.04%	Dehradun (78.54%)	Uttarkashi (59.74%)
Scheduled Caste (अनुसूचित जाति) (SC) %	18.76% of total pop	Bageshwar (बागेश्वर) (27.73%)	Dehradun (14.5%)
Scheduled Tribe (अनुसूचित जनजाति) (ST) %	2.89% of total pop	Udham Singh Nagar (7.46%)	Rudraprayag (0.15%)
Urban (नगरीय) Population %	30.23% (Rural: 69.77%)	Dehradun (55.52%)	Bageshwar (3.49%)

Census 2011 Master District Rankings (All 13 Districts)

Rank	Population	Density (per \text{km}^2)	Sex Ratio (females/1000)	Literacy Rate (%)
1	Haridwar (18.90 L)	Haridwar (801)	Almora (1142)	Dehradun (84.25%)
2	Dehradun (16.96 L)	Dehradun (549)	Rudraprayag (1114)	Nainital (नैनीताल) (83.88%)
3	US Nagar (16.48 L)	US Nagar (425)	Pauri Garhwal (1103)	Chamoli (चमोली) (82.65%)
4	Nainital (9.55 L)	Nainital (225)	Bageshwar (1090)	Pithoragarh (82.25%)
5	Pauri Garhwal (6.87 L)	Tehri Garhwal (170)	Tehri Garhwal (1077)	Pauri Garhwal (82.02%)
6	Tehri Garhwal (6.18 L)	Pauri Garhwal (129)	Pithoragarh (1020)	Rudraprayag (81.30%)
7	Almora (6.22 L)	Bageshwar (116)	Chamoli (1015)	Almora (80.47%)
8	Pithoragarh (4.83 L)	Almora (198)	Uttarkashi (958)	Champawat (चंपावत) (79.83%)
9	Chamoli (3.91 L)	Champawat (147)	Champawat (980)	Bageshwar (80.01%)
10	Uttarkashi (3.30 L)	Rudraprayag (122)	Nainital (934)	Tehri Garhwal (76.36%)
11	Bageshwar (2.59 L)	Pithoragarh (68)	US Nagar (920)	Uttarkashi (75.81%)
12	Champawat (2.59 L)	Chamoli (49)	Dehradun (889)	Haridwar (73.43%)
13	Rudraprayag (2.42 L)	Uttarkashi (41)	Haridwar (880)	US Nagar (73.10%)

Confused Pairs & Common Traps

Confused Pair	Critical Distinction
Almora vs Haridwar (Sex Ratio)	Almora = highest sex ratio (1142); Haridwar = lowest sex ratio (880).
Haridwar vs Uttarkashi (Density)	Haridwar = highest density (801); Uttarkashi = lowest density (41).
Pauri vs Udham Singh Nagar (Growth)	US Nagar = highest decadal growth (+33.45%); Pauri Garhwal = lowest/negative (-1.41%).
Highest SC % vs Highest ST %	Highest SC % = Bageshwar (27.73%) ; Highest ST % = Udham Singh Nagar (7.46%) .
Highest Overall Literacy vs Male Literacy	Highest overall = Dehradun (84.25%) ; Highest male = Rudraprayag (93.90%) .

Uttarakhand — Topic 7 — Transport, Tourism & Hazards

Quick Revision — Spine Only

Head	Key Fact
Silkyara–Barkot Tunnel	4.531 km long ; cuts Gangotri (गंगोत्री)–Yamunotri (यमुनोत्री) highway distance by >20 km (UKPCS 2025 Q51)
Airport Mismatch (2025)	Gauchar is in Chamoli (चमोली) (NOT Rudraprayag (रुद्रप्रयाग)!) (UKPCS 2025 Q103)
Airports in UK	Jolly Grant (Dehradun (देहरादून)), Pantnagar (US Nagar), Naini Saini (Pithoragarh (पिथौरागढ़)), Chinyalisaur (Uttarkashi (उत्तरकाशी)), Gauchar (Chamoli)
Pass: Darma & Byans	Sinla Pass (Pithoragarh) — connects Darma valley to Byans valley
Pass: Pithoragarh & Tibet	Lipulekh (लिपुलेख) Pass (5,334 m) — traditional Kailash-Mansarovar Yatra route
Pass: Pithoragarh & Bageshwar (बागेश्वर)	Traill’s Pass (5,312 m) — crossed in 1830 by Commissioner G. W. Traill
Pass: Uttarkashi & HP	Shringkanth Pass (connects Uttarkashi to Kinnaur/HP)
Largest Bugyal in UK	Bedni Bugyal (Chamoli) — velvet alpine meadow on the Nanda Devi (नंदा देवी) Raj Jat (नंदा देवी राज जात) route
Butter Festival Bugyal	Dayara Bugyal (Uttarkashi) — celebrated for Anduri Utsav / Butter Festival
State Climate Council (SCCC)	Established in 2011 (Forest & Environment (पर्यावरण) Dept drafts UAPCC, NOT Revenue Dept) (UKPCS 2025 Q106)
Malpa Landslide Tragedy	17–18 August 1998 (Pithoragarh); killed 200+ pilgrims including dancer Protima Bedi
Kedarnath (केदारनाथ) Flash Deluge	16–17 June 2013 — Chaurabari glacial lake breach; Mandakini (मंदाकिनी) river deluge
Tapovan/Rishiganga Disaster	7 February 2021 (Chamoli) — rock-ice avalanche wiped out Rishiganga & Tapovan-Vishnugad hydel plants

Mountain Passes of Uttarakhand: Master Reference Table

Pass Name	Elevation	Connects / Geographic Span	Key Prelims Features
Lipulekh Pass	5,334 m	Pithoragarh \longleftarrowrightarrow Tibet (China)	Landmark pass on the ancient border trade (पण्थाध्यक्ष) and annual Kailash-Mansarovar Yatra ; tri-junction with Nepal.
Mana (माना) Pass (Dungri-La)	5,608 m	Chamoli \longleftarrowrightarrow Tibet	One of the highest motorable roads in the world (NH-58 terminal); ancient Indo-Tibetan trade link.
Niti (नीति) Pass	5,068 m	Chamoli \longleftarrowrightarrow Tibet	Last pass in the Dhauliganga (धौलीगंगा) valley; closed following the 1962 war.
Sinla Pass	5,495 m	Darma Valley \longleftarrowrightarrow Byans Valley	Inter-valley high pass within Pithoragarh district.
Traill’s Pass	5,312 m	Bageshwar \longleftarrowrightarrow Pithoragarh	Connects Pindari Glacier to Milam valley; discovered in 1830 by Commissioner G. W. Traill.
Kalindi Pass	5,950 m	Uttarkashi \longleftarrowrightarrow Chamoli	Severe glaciated mountaineering pass linking Gangotri/Gaumukh (गौमुख) directly to Badrinath.
Shringkanth Pass	1,500 m	Uttarkashi \longleftarrowrightarrow Himachal (हिमाचल) Pradesh	Ancient transit corridor connecting Rawain to Kinnaur.
Thang-La Pass	5,360 m	Uttarkashi \longleftarrowrightarrow Tibet	Second highest pass in the Garhwal (गढ़वाल) Himalaya (हिमालय); accessed via Jadhganga valley.

Pass Name	Elevation	Connects / Geographic Span	Key Prelims Features
Chorhoti Pass	5,364 m	Chamoli \longlefttrightarrow Pithoragarh	Connects Niti valley to the Kumaon borderland.
Sunderdhunga Pass	5,520 m	Bageshwar \longlefttrightarrow Chamoli	Traverses high glacial ridgelines between Maikotli and Panwali peaks.

Alpine Meadows (Bugyals) of Uttarakhand

District	Prominent Bugyals	High-Yield Prelims Characteristics
Chamoli	Bedni Bugyal, Auli (औली), Gorson, Ali, Panar, Bagji, Chitrakantha, Vedini	Bedni Bugyal: Largest bugyal in Uttarakhand; famous for Bedni Kund and rare <i>Brahma Kamal</i> (ब्रह्मकमल) flowers. Auli Bugyal: Premier skiing destination with 4 km ropeway (<i>Cable Car</i>) from Joshimath.
Uttarkashi (उत्तरकाशी)	Dayara, Kush Kalyan, Panwali Kantha, Manji (मंजी), Devdamini, Har-ki-Doon	Dayara Bugyal: Vast 28 sq km meadow; venue of the famous Anduri Utsav (Butter & Milk Festival) celebrated on Bhado Sankranti. Panwali Kantha: Home to blooming red, pink, and white rhododendrons; long trekking trail.
Rudraprayag (रुद्रप्रयाग)	Chopta, Tungnath, Madhyamaheshwar, Kasni Khark	Chopta: Hailed as the “Mini Switzerland of Uttarakhand” ; base camp for Tungnath and Chandrashila treks.
Pithoragarh (पिथौरागढ़)	Khalia Top, Chhipla Kot, Byans Bugyal, Ralam, Betulidhar	Khalia Top: Near Munsyari; panoramic views of Panchachuli (पंचाचूली) peaks. Chhipla Kot: Sacred alpine meadow housing the Chhipla Kedar shrine.
Tehri (टिहरी)	Kotli Bugyal, Masartal Bugyal, Sahasratal pastures	Alpine grazing grounds used by Gujjars and local sheep herders.

Natural Hazards & Historical Disaster Timeline

Date / Year	Calamity & Location	Geological Nature & Human/Infrastructure Toll
20 October 1991	Uttarkashi Earthquake (6.6 M)	Severe tectonic tremor along the MCT; over 768 fatalities ; flattened thousands of traditional stone houses across Bhatwari and Dunda.
17–18 August 1998	Malpa Landslide (Pithoragarh)	Catastrophic rockfall and debris avalanche in the Kali (काली) valley wiped out the village of Malpa overnight. 221 dead , including a batch of 60 Kailash-Mansarovar pilgrims and renowned classical dancer Protima Bedi .
29 March 1999	Chamoli Earthquake (6.8 M)	Epicentre near Chamoli/Gopeshwar; over 100 fatalities ; extensive damage to Gopeshwar, Chamoli, and Rudraprayag.
16–17 June 2013	Kedarnath Cloudburst & GLOF (Mandakini Deluge)	Massive multi-day torrential rains coupled with the sudden breach of Chaurabari Lake (Gandhi (गांधी) Sarovar) sent a gigantic wall of mud and boulders down the Mandakini river. Devastated Kedarnath temple township, Rambara, Gaurikund, and Sonprayag. Over 5,700 fatalities ; largest Himalayan disaster of modern times. Led to establishment of State Disaster Response Force (SDRF).
7 February 2021	Chamoli / Tapovan Flash Flood (GLOF & Rockslide)	A massive hanging glacier and rock mass collapsed from the Rongti / Nanda Ghunti peak , creating a colossal flash flood in the Rishi Ganga and Dhaulī (धौली) Ganga rivers. Completely destroyed the Rishi Ganga Hydel Project (13.2 MW) and severely damaged NTPC’s Tapovan-Vishnugad Project (520 MW) . Over 204 dead/missing workers trapped inside tunnels.

Date / Year	Calamity & Location	Geological Nature & Human/Infrastructure Toll
January 2023	Joshimath Land Subsidence	Severe land sinking, foundational shifting, and deep wall fissures appeared across 9 municipal wards of Joshimath, built upon ancient landslide moraine debris. Hundreds of families evacuated.

Confused Pairs & Common Traps

Confused Pair	Critical Distinction
Gauchar vs Rudraprayag	Gauchar airstrip is in Chamoli district (trap distractor: Rudraprayag).
Silkyara Length	Exact length is 4.531 km (trap distractors: 3.531 km, 6.531 km).
Sinla Pass vs Lipulekh Pass	Sinla Pass = internal pass connecting Darma and Byans valleys; Lipulekh = international pass to Tibet.
Dayara Bugyal vs Bedni Bugyal	Dayara = Butter Festival / Anduri Utsav (Uttarkashi); Bedni = largest bugyal / Brahma Kamal (Chamoli).
Malpa (1998) vs Kedarnath (2013)	Malpa = Landslide / rockfall on Kali river (Protima Bedi); Kedarnath = GLOF / cloudburst on Mandakini.
SCCC Year & Department	SCCC was created in 2011 ; UAPCC is managed by the Forest & Environment Department (NOT Revenue Dept).